



CAPSTONE PROJECT REPORT

Report 4 – Software Design Document

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I. Record of Changes

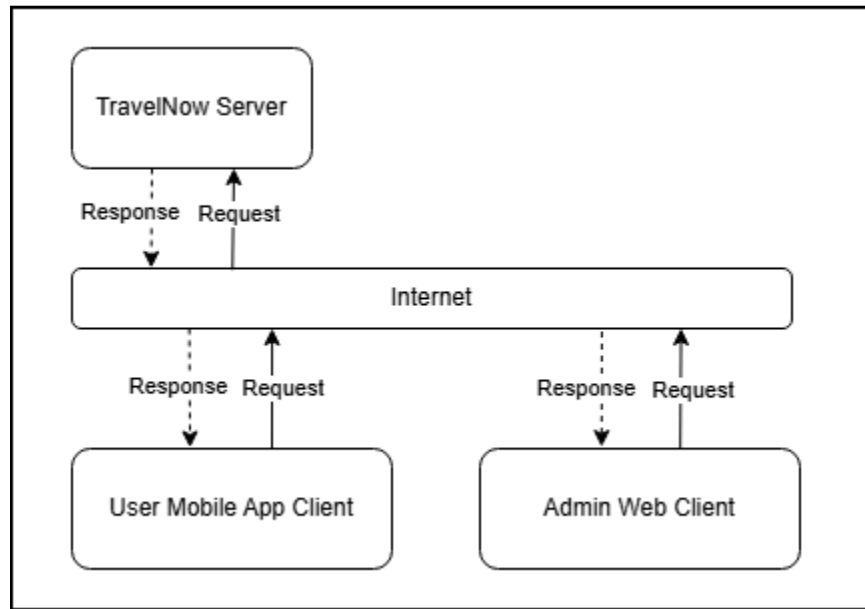
Date	A* M, D	In charge	Change Description

*A - Added M - Modified D - Deleted

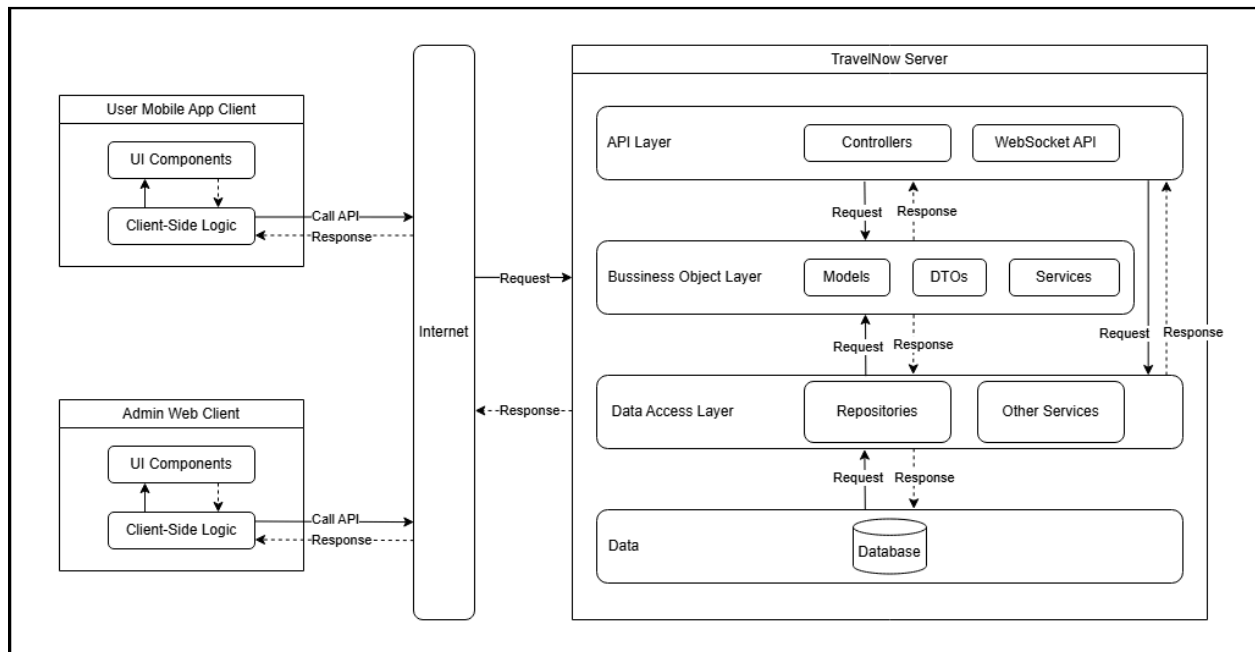
II. Software Design Document

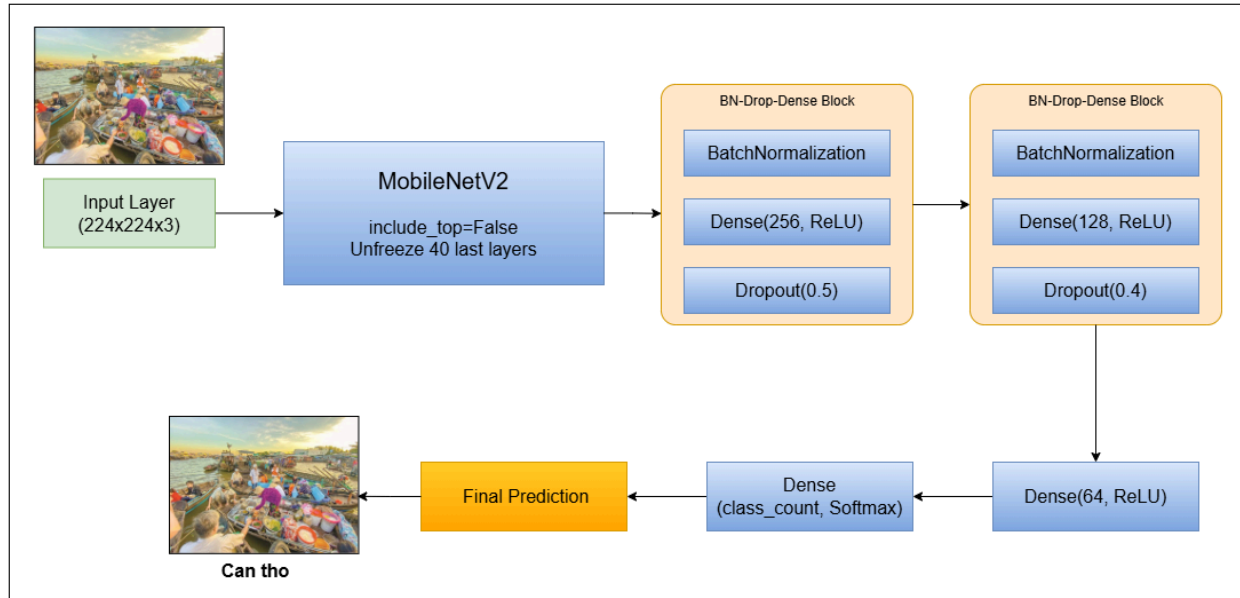
1. System Design

1.1 Overall Architecture



1.2 System Architecture





The proposed model architecture is based on a fine-tuned MobileNetV2 backbone, pretrained on the ImageNet dataset. The top classification layers of MobileNetV2 are removed (`include_top=False`), and global average pooling is applied to obtain a compact feature representation. To enable more specialized feature extraction for the target task, the last 40 layers of the MobileNetV2 backbone are unfrozen for fine-tuning.

A custom classification head is appended to the base model, consisting of a series of fully connected layers with batch normalization and dropout regularization to reduce overfitting. The head includes:

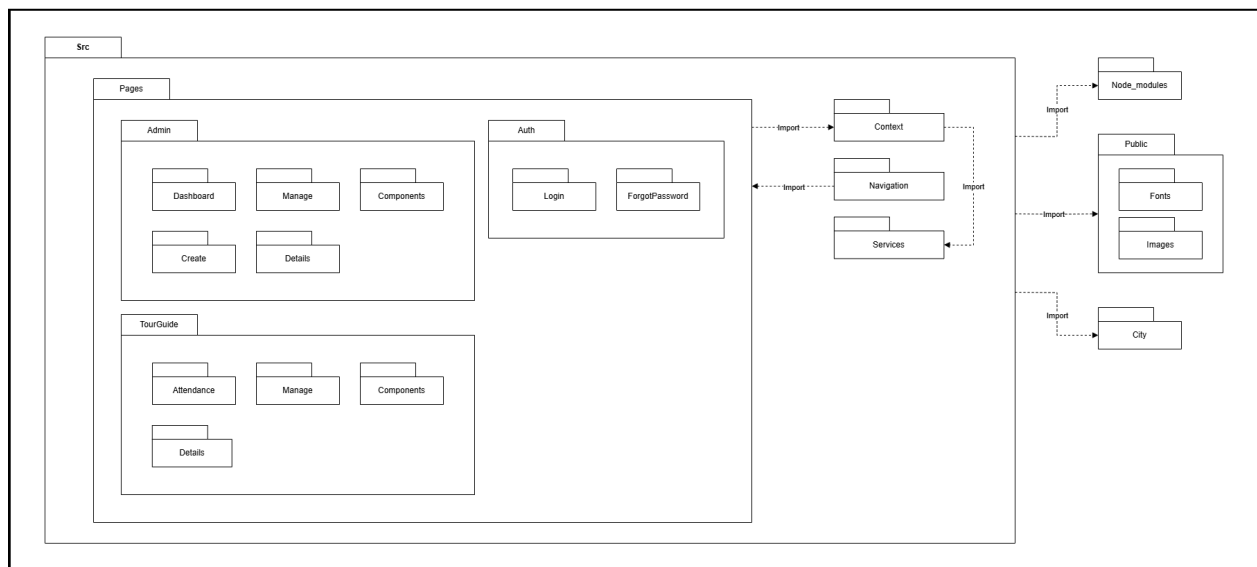
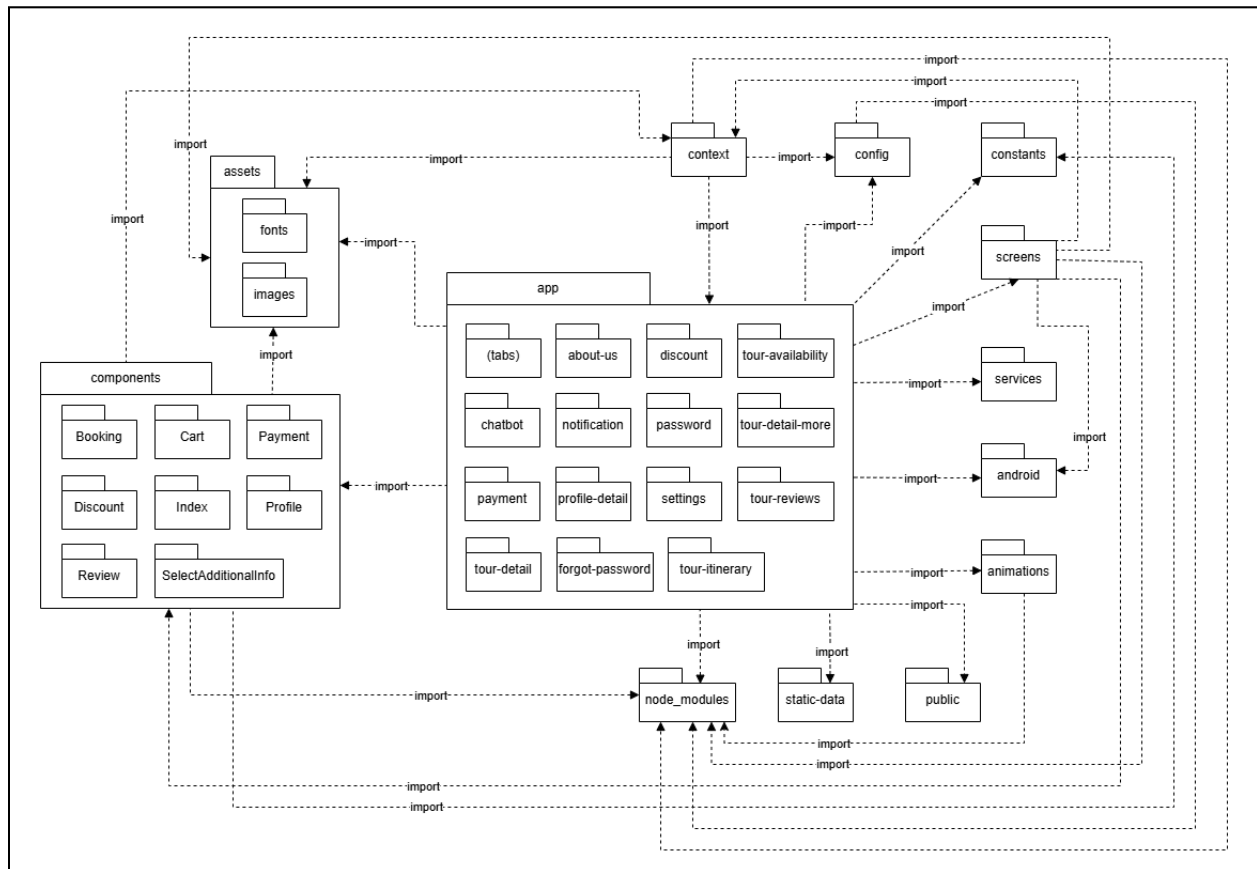
- A BatchNormalization layer to stabilize and accelerate training
- A Dense layer with 256 units and ReLU activation
- A Dropout layer with a rate of 0.5
- Another BatchNormalization layer
- A Dense layer with 128 units and ReLU activation
- A Dropout layer with a rate of 0.4
- A Dense layer with 64 units and ReLU activation
- A final Dense layer with softmax activation for multi-class classification (`class_count` outputs)

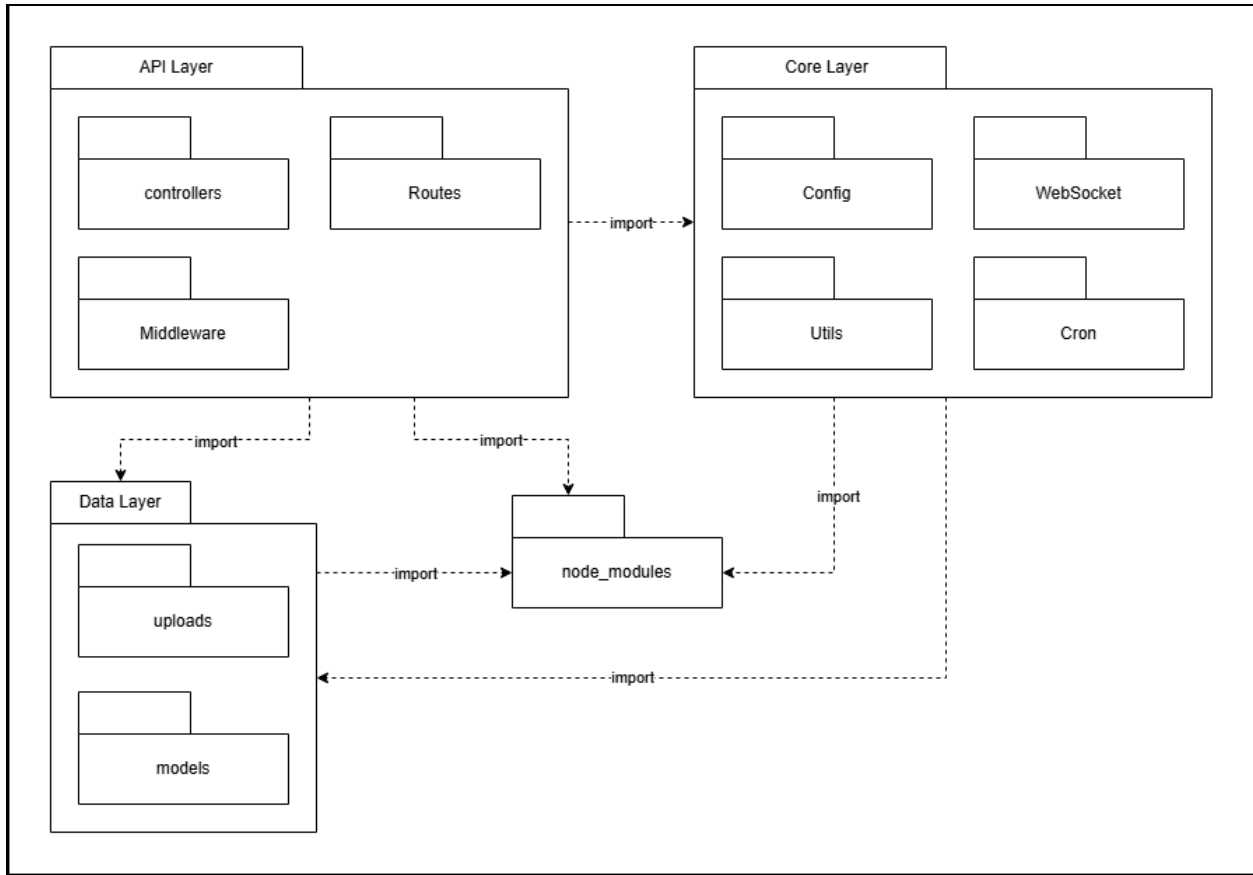
The model is compiled as a single end-to-end trainable architecture, combining the strengths of transfer learning and deep feature extraction with regularization techniques to improve generalization performance.

The dataset used in this study was self-collected and consists of 3,246 images across 16 classes. The dataset was divided into training, validation, and testing sets in the ratio of 80:10:10, respectively. The proposed image classification model achieved an accuracy of 89.42% on the test set. This shows that the model performs well on new, unseen images. By combining MobileNetV2 with custom dense layers and

using fine-tuning, the model was able to learn important features and make accurate predictions. The result proves that this approach is effective for solving image classification tasks.

1.3 Package Diagram





Code Package

a. User mobile app client code packages

No	Package	Description
01	assets	Contains static resources used in the application, such as images, fonts, and other media files.
02	fonts	Stores font files and configurations for custom typography used in the application.
03	images	Holds image files used throughout the application, such as icons, backgrounds, or illustrations.
04	components	Contains reusable UI components that can be shared across different parts of the application.
05	Booking	Manages booking-related logic, components, or models specific to the booking process.
06	Cart	Handles cart functionality, including adding, removing, and managing items for purchase.
07	Payment	Contains logic and components for processing payments and integrating payment gateways.

08	Discount	Manages discount-related features, such as applying promo codes or calculating reductions.
09	Index	Serves as the entry point or main file for initializing or organizing the application structure.
10	Profile	Handles user profile management, including display and editing of user information.
11	Review	Contains functionality for submitting, displaying, and managing user reviews.
12	SelectAdditional Info	Manages components or logic for selecting additional information, such as preferences or options during booking.
13	context	Stores context-related files for state management across the application (e.g., React Context).
14	config	Contains configuration settings for the application, such as API keys, environment variables, or themes.
15	constants	Holds constant values or enumerations used throughout the application for consistency.
16	screens	Contains the main screen or page components that represent different views in the application.
17	services	Manages business logic and API calls to interact with backend or external services.
18	android	Contains Android-specific configurations or files for a mobile application.
19	animations	Stores animation definitions, libraries, or utilities for enhancing UI interactions.
20	node_modules	Directory for third-party dependencies and libraries installed via package managers (e.g., npm).
21	static-data	Holds static data files or mock data used for testing or initial setups.
22	public	Contains publicly accessible files, such as HTML templates, favicon, or static assets for web deployment.
23	app	Main package for the application, including core logic, initialization, or top-level components.
24	(tabs)	Manages tab navigation or tab-related components for organizing content.
25	about-us	Contains components or content related to the "About Us" section of the application.
26	discount	Handles discount-specific UI or logic (potentially overlapping with package 08, depending on scope).
27	tour-availability	Manages availability checking and display for tours or similar offerings.
28	chatbot	Contains logic and components for implementing a chatbot feature.

29	notification	Handles notification-related functionality, such as push notifications or in-app alerts.
30	password	Manages password-related features, such as reset or change password workflows.
31	tour-detail-more	Contains additional details or extended information for tour-related content.
32	payment	Manages payment-specific UI or logic (potentially overlapping with package 07).
33	profile-detail	Handles detailed profile views or editing options for users.
34	settings	Contains components and logic for managing application or user settings.
35	tour-reviews	Manages the display and submission of reviews specific to tours.
36	tour-detail	Contains components or logic for displaying detailed tour information.
37	forgot-password	Handles the forgot password workflow, including UI and logic for recovery.
38	tour-itinerary	Manages the display tour itineraries.

b. Web client code packages

No	Package	Description
01	Src	Contains the source code of the application, including pages, components, and logic.
02	Pages	Holds the main page components of the application, such as Admin and TourGuide sections.
03	Admin	Manages admin-related pages like Dashboard, Manage, Create, and Details for administrative tasks.
04	Auth	Handles authentication-related pages such as Login and ForgotPassword for user access.
05	TourGuide	Manages booking-related logic, components, or models specific to the booking process.
06	Context	Manages shared state and data across the application using context.
07	Navigation	Contains navigation logic and components for routing within the application.
08	Services	Provides reusable services and API calls for the application.
09	Node_modules	Stores external libraries and dependencies required by the application.
10	Public	Stores publicly accessible static assets used by the application.
11	Fonts	Stores font files and configurations for custom typography used in the application.
12	Images	Holds image files used throughout the application, such as icons or backgrounds.
13	City	Contain Vietnam's cities data.

c. Server code packages

No	Package	Description
01	API Layer	Contains logic and definitions for handling API requests and responses.
02	controllers	Manages request handling and business logic for specific API endpoints.
03	Routes	Defines API routes and maps them to corresponding controllers or handlers.
04	Middleware	Contains functions for processing requests, such as authentication or logging.
05	Core Layer	Holds the core logic and shared functionality central to the application.
06	Config	Stores configuration settings, such as database connections or API keys.
07	WebSocket	Manages real-time communication logic and configurations for WebSocket connections.
08	Utils	Contains utility functions and helpers used across the application.
09	Cron	Handles scheduled tasks and cron jobs for automated processes.
10	Data Layer	Manages data access and interactions with databases or external data sources.
11	uploads	Stores files uploaded by users, such as images.
12	models	Defines data schemas and structures for interacting with the database.
13	node_modules	Directory for third-party dependencies installed via package managers (e.g., npm).

2. Database Design

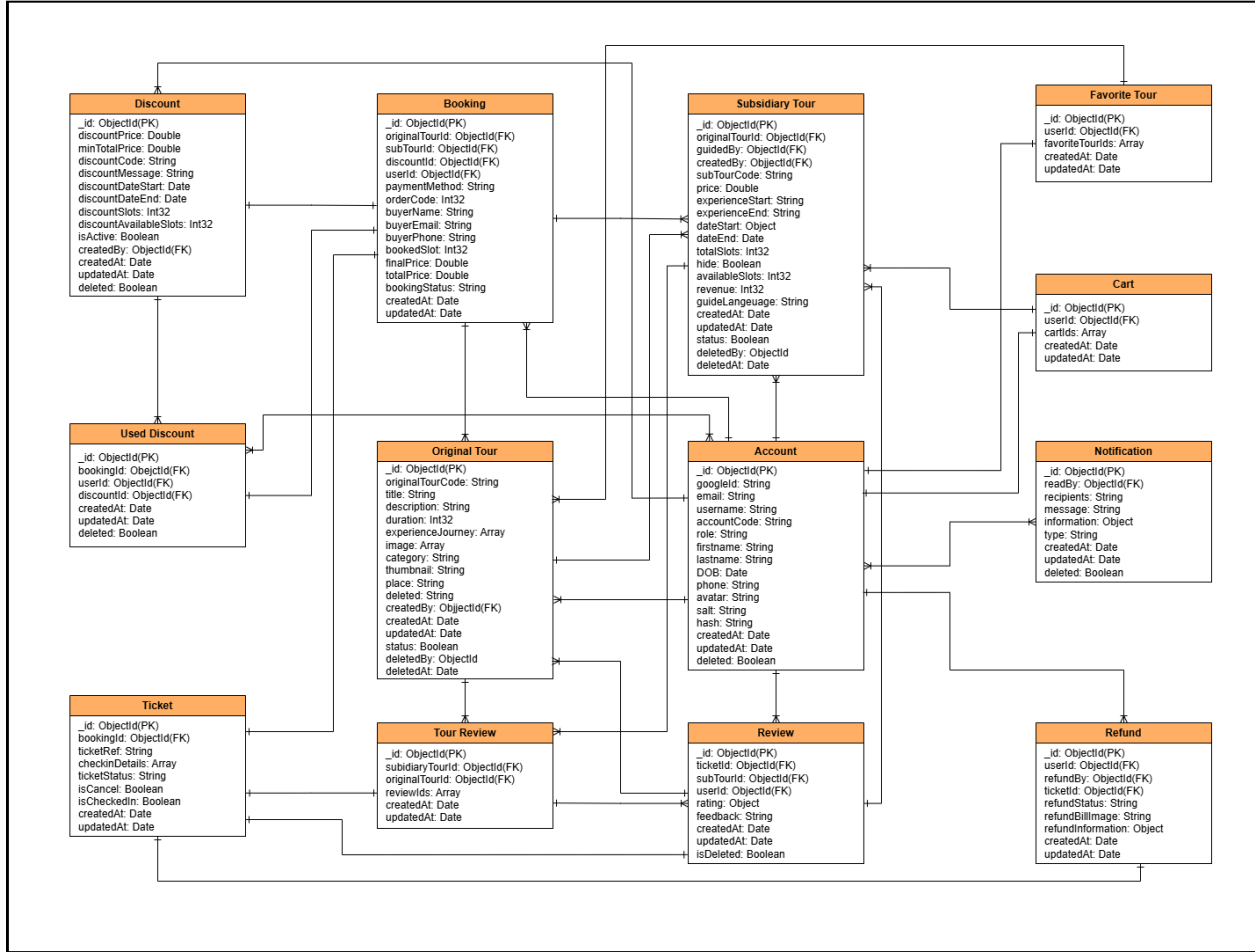


Table Descriptions

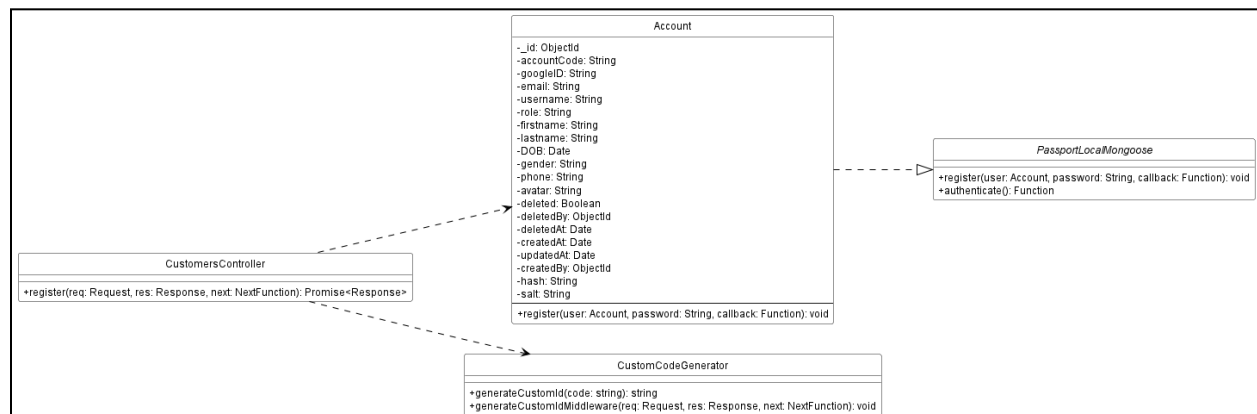
No	Table	Description
01	Account	Stores user account details, including authentication credentials. - Primary keys: <code>_id</code> - Foreign keys: <code>deletedBy</code> → <code>Account._id</code>
02	Booking	Manages tour bookings made by users. - Primary keys: <code>_id</code> - Foreign keys: <code>userId</code> → <code>Account._id</code> , <code>originalTourId</code> → <code>OriginalTour._id</code> , <code>subTourId</code> → <code>SubsidiaryTour._id</code> , <code>discountId</code> → <code>Discount._id</code>
03	Cart	Stores items users add to their cart before booking. - Primary keys: <code>_id</code> - Foreign keys: <code>userId</code> → <code>Account._id</code>
04	Discount	Stores discount codes and promotions. - Primary keys: <code>_id</code> - Foreign keys: <code>createdBy</code> → <code>Account._id</code> , <code>deletedBy</code> → <code>Account._id</code>
05	Used Discount	Stores discount have used by users.

		- Primary keys: <code>_id</code> - Foreign keys: <code>userId</code> → <code>Account._id</code> , <code>originalTourId</code> → <code>OriginalTour._id</code> , <code>subTourId</code> → <code>SubsidiaryTour._id</code>
06	Favorite Tour	Manages users' favorite tours. - Primary keys: <code>_id</code> - Foreign keys: <code>userId</code> → <code>Account._id</code> , <code>favoriteTourIds</code> → <code>OriginalTour._id</code>
07	Refund	Handles refund requests. - Primary keys: <code>_id</code> - Foreign keys: <code>userId</code> → <code>Account._id</code> , <code>refundBy</code> → <code>Account._id</code> , <code>ticketId</code> → <code>Ticket._id</code>
08	Notification	Stores notifications related to various actions. - Primary keys: <code>_id</code> - Foreign keys: <code>readBy.userId</code> → <code>Account._id</code>
09	Original Tour	Stores details about main tour packages. - Primary keys: <code>_id</code> - Foreign keys: <code>createdBy</code> → <code>Account._id</code> , <code>deletedBy</code> → <code>Account._id</code>
10	Subsidiary Tour	Stores details about specific tour dates and slots. - Primary keys: <code>_id</code> - Foreign keys: <code>originalTourId</code> → <code>OriginalTour._id</code> , <code>createdBy</code> → <code>Account._id</code> , <code>guidedBy</code> → <code>Account._id</code> , <code>deletedBy</code> → <code>Account._id</code>
11	Ticket	Stores booking tickets. - Primary keys: <code>_id</code> - Foreign keys: <code>bookingId</code> → <code>Booking._id</code>
12	Review	Stores customer reviews for a tour. - Primary keys: <code>_id</code> - Foreign keys: <code>ticketId</code> → <code>Ticket._id</code> , <code>subTourId</code> → <code>SubsidiaryTour._id</code> , <code>userId</code> → <code>Account._id</code>
13	Tour Review	Aggregates reviews for a tour. - Primary keys: <code>_id</code> - Foreign keys: <code>subTourId</code> → <code>SubsidiaryTour._id</code> , <code>reviewIds</code> → <code>Review._id</code>

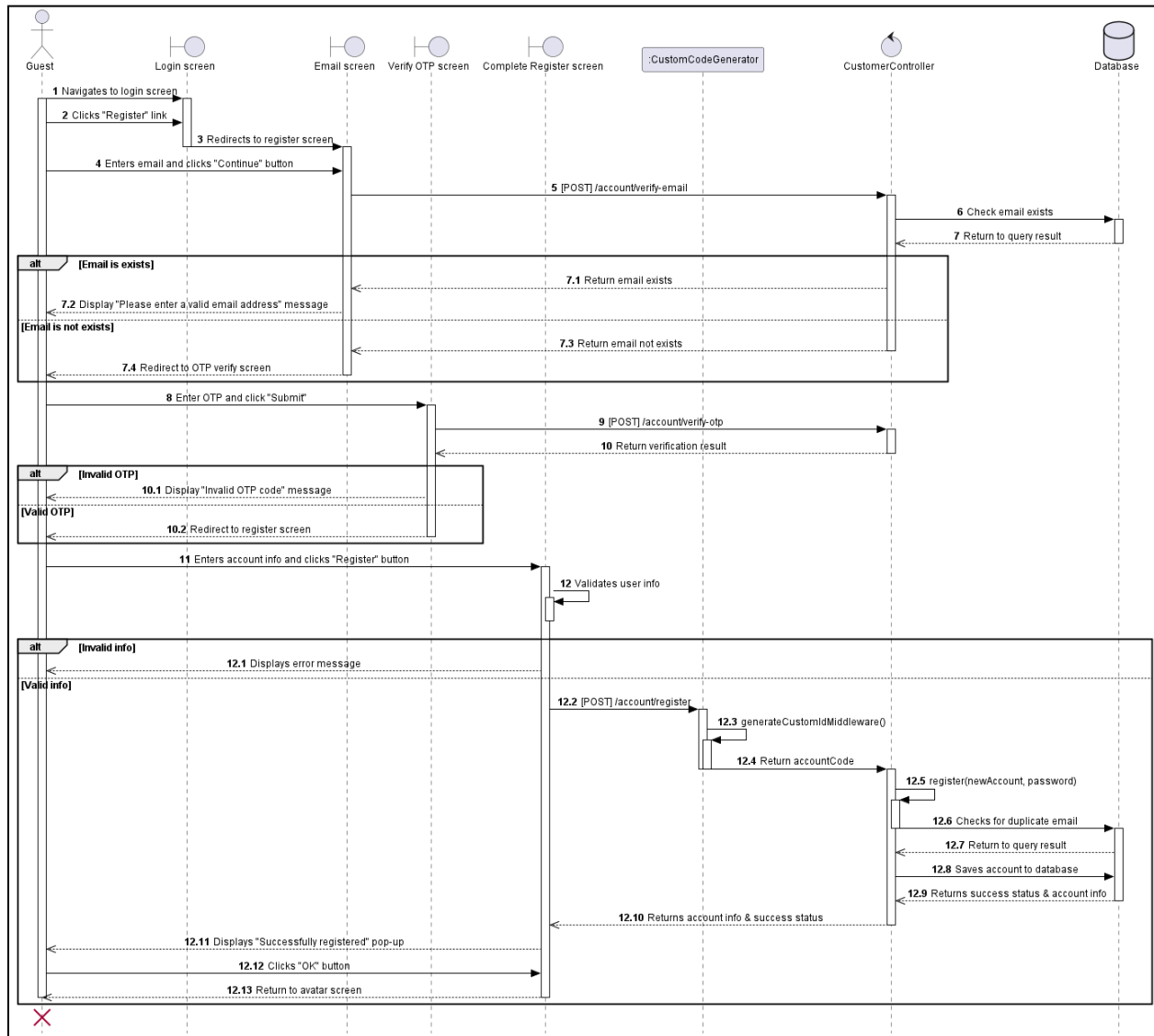
3. Detailed Design

3.1 Register Customer Role

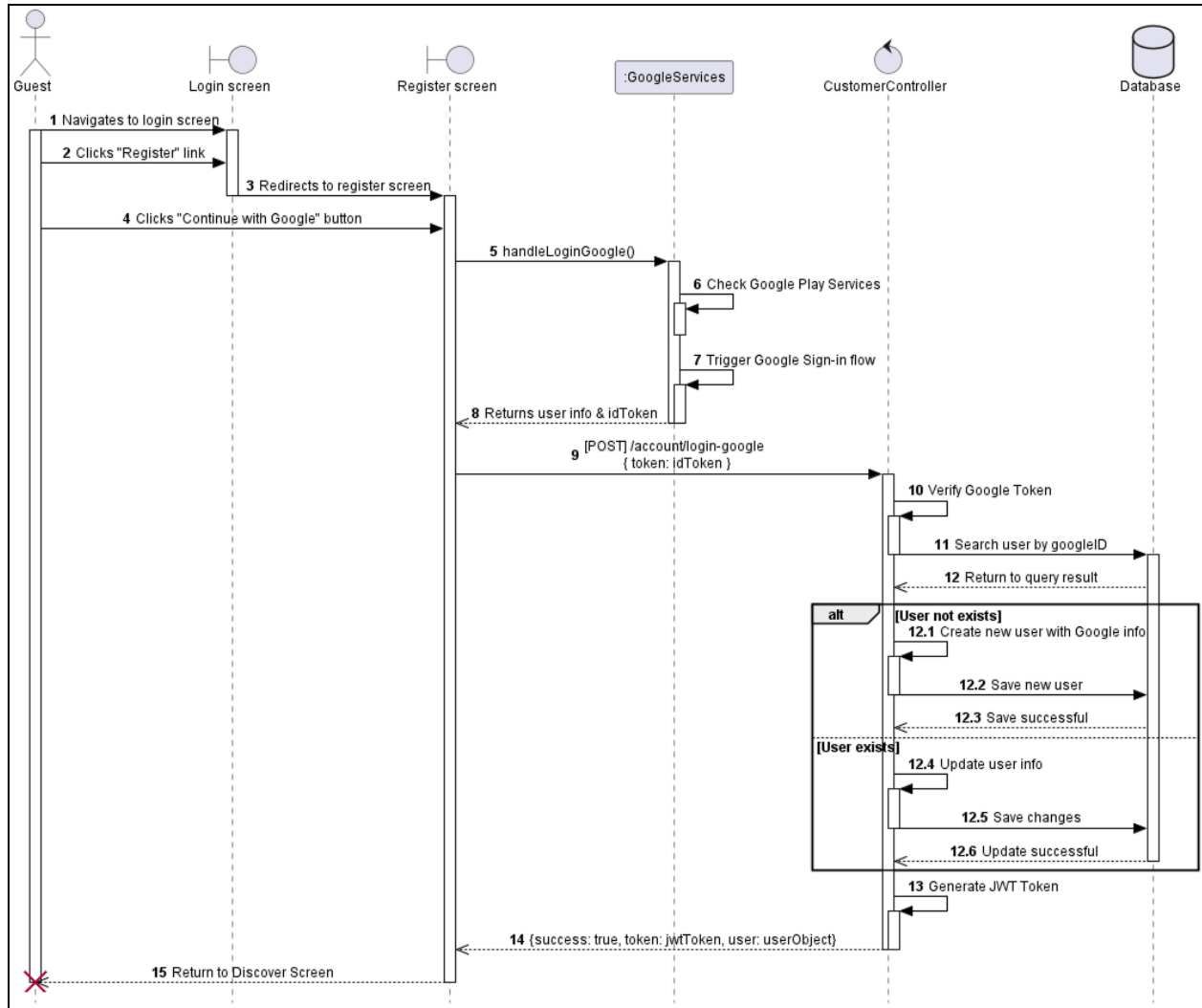
a. Class Diagram



b. Sequence Diagram(s)

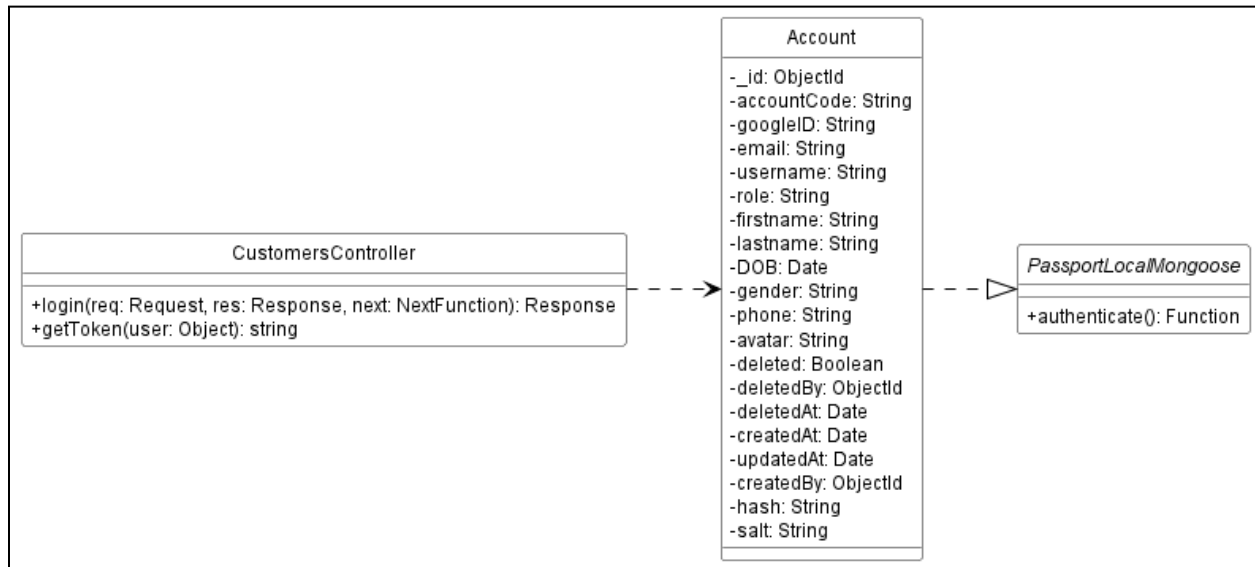


* The sequence diagrams for Register Customer Role with Google:

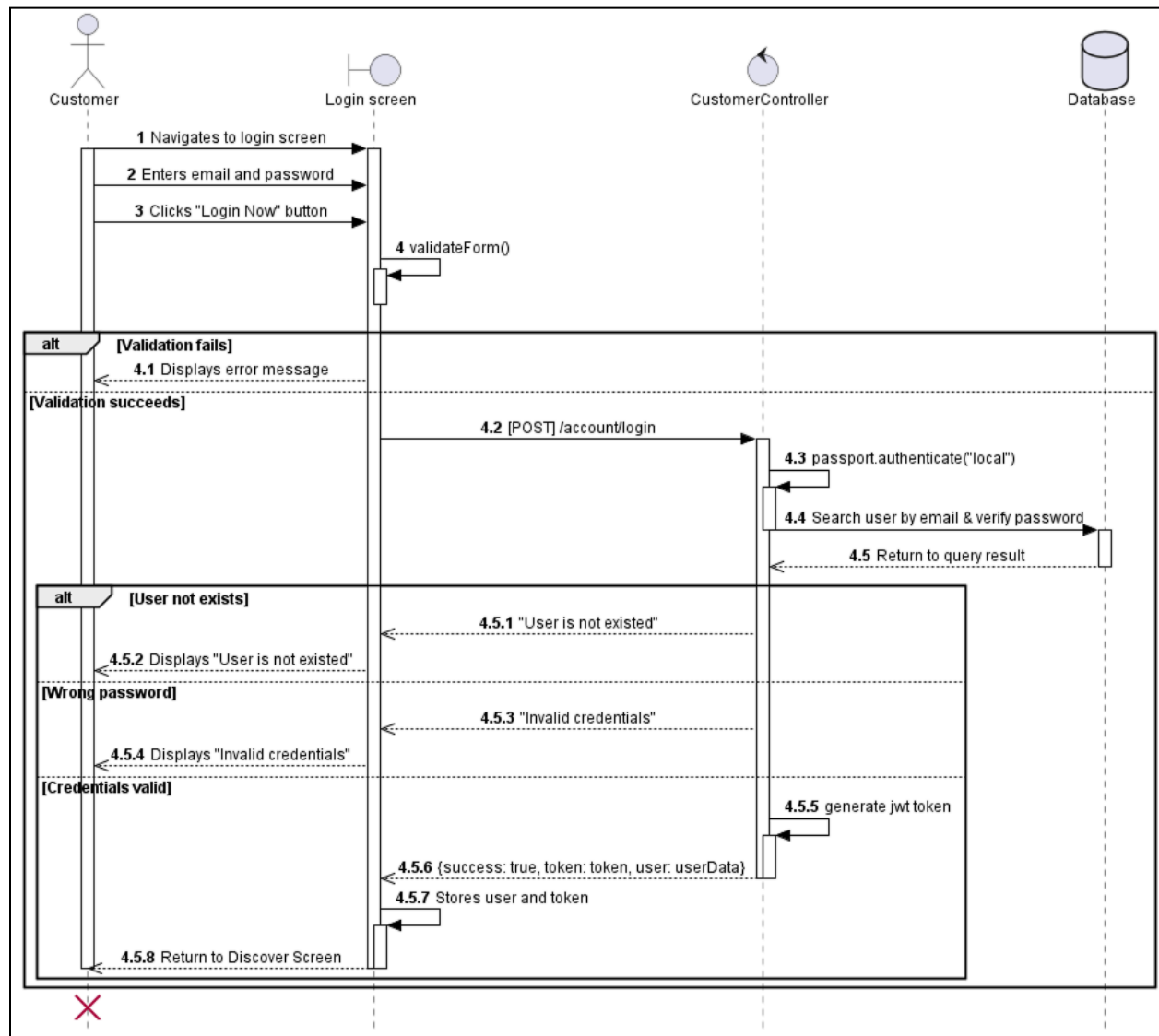


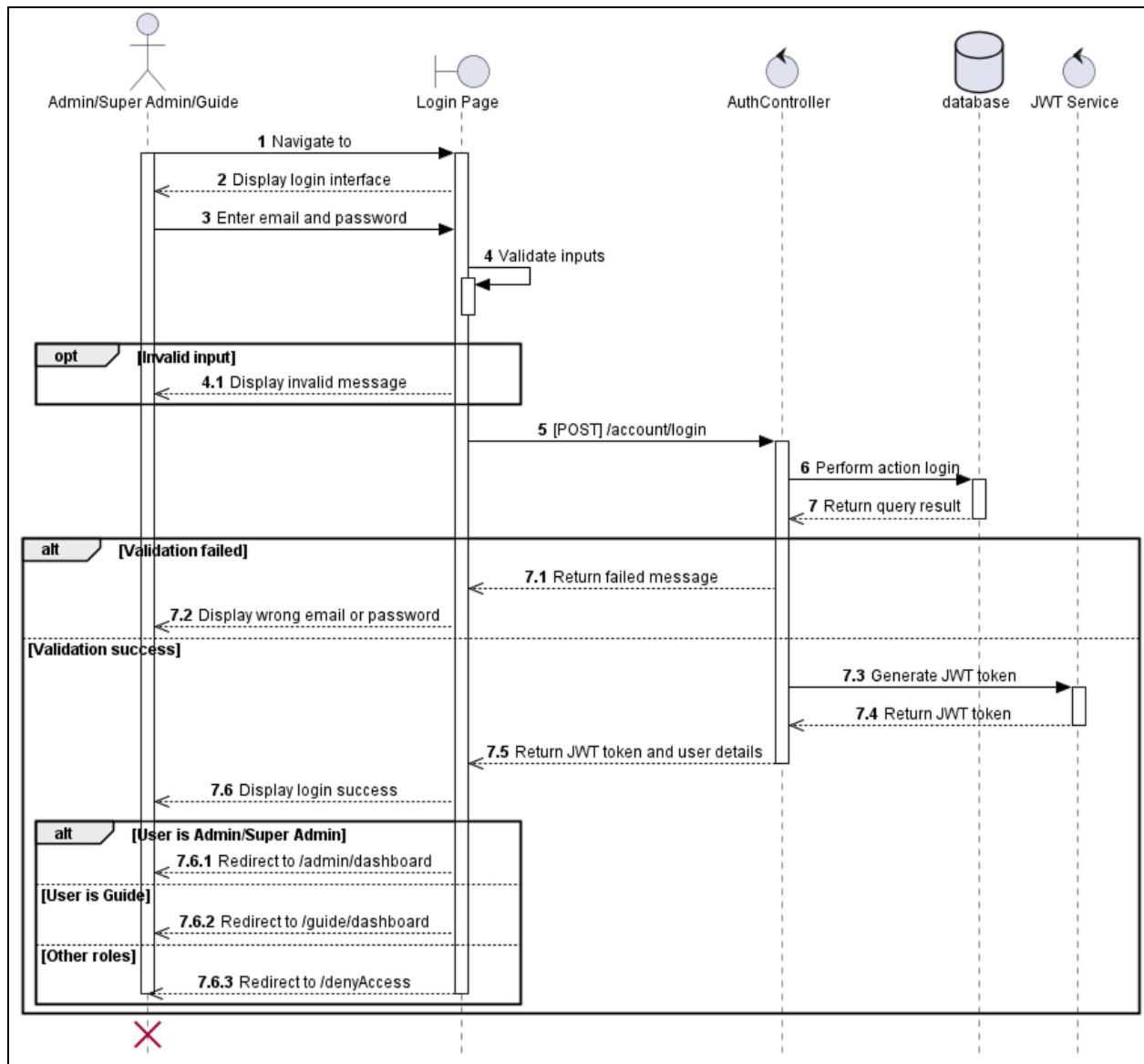
3.2 Login

a. Class Diagram

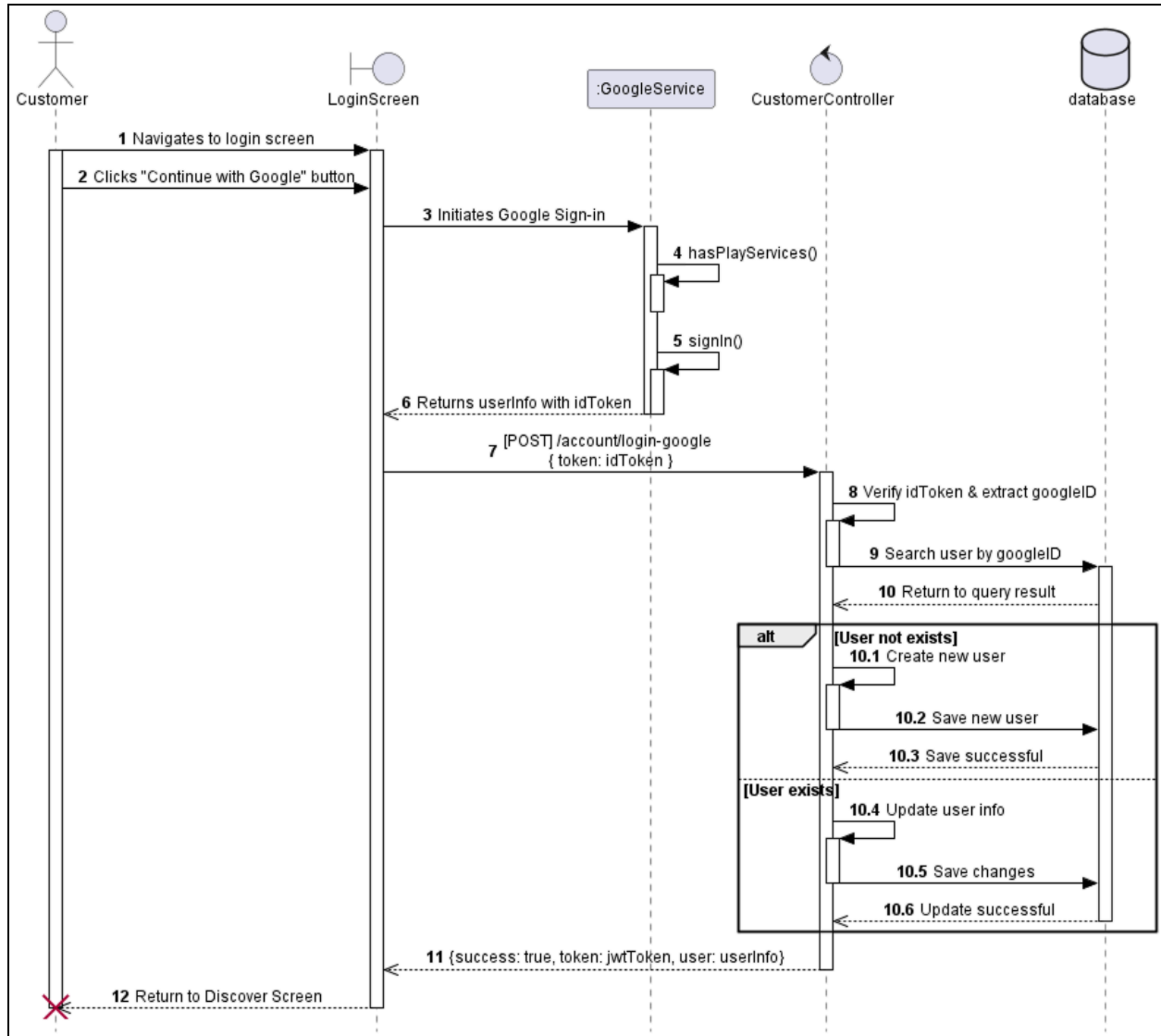


b. Sequence Diagram(s)



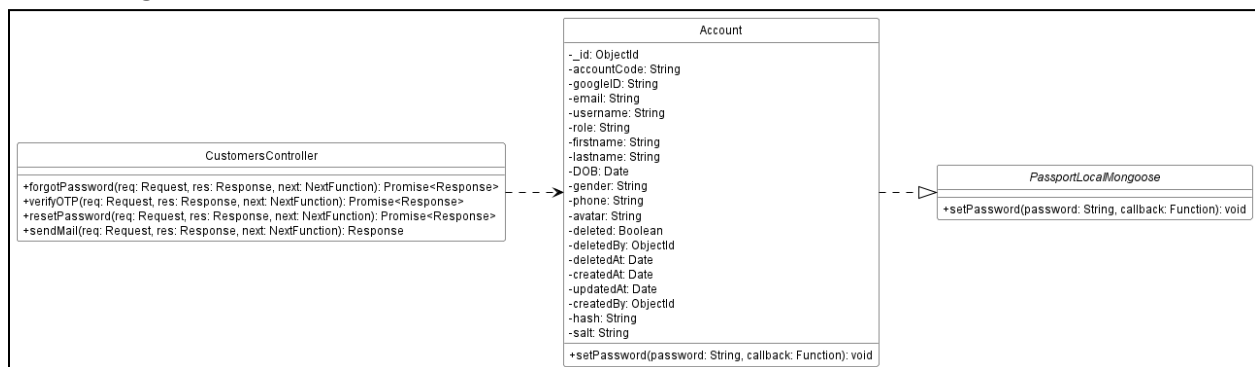


* The sequence diagrams for Login with Google is described below:

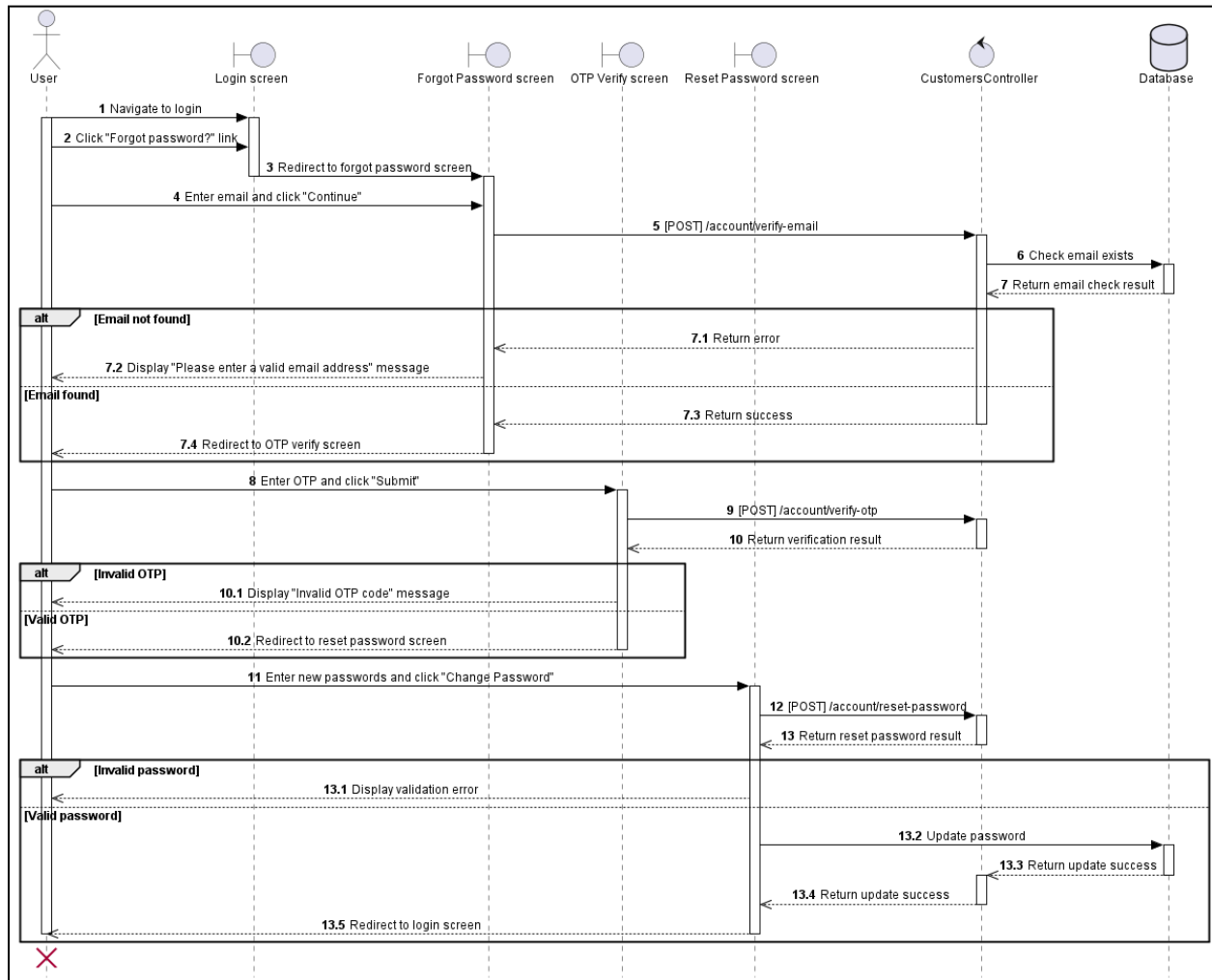


3.3 Forgot Password

a. Class Diagram



b. Sequence Diagram(s)

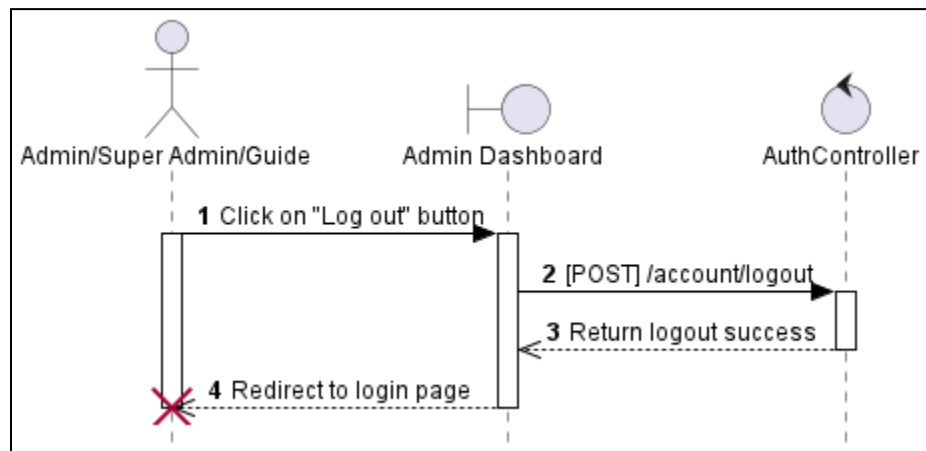
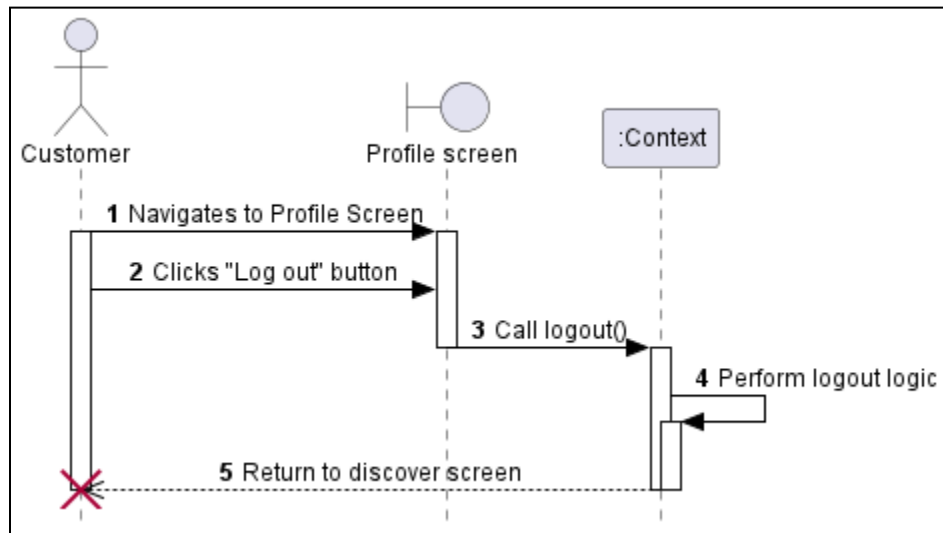


3.4 Log out

a. Class Diagram

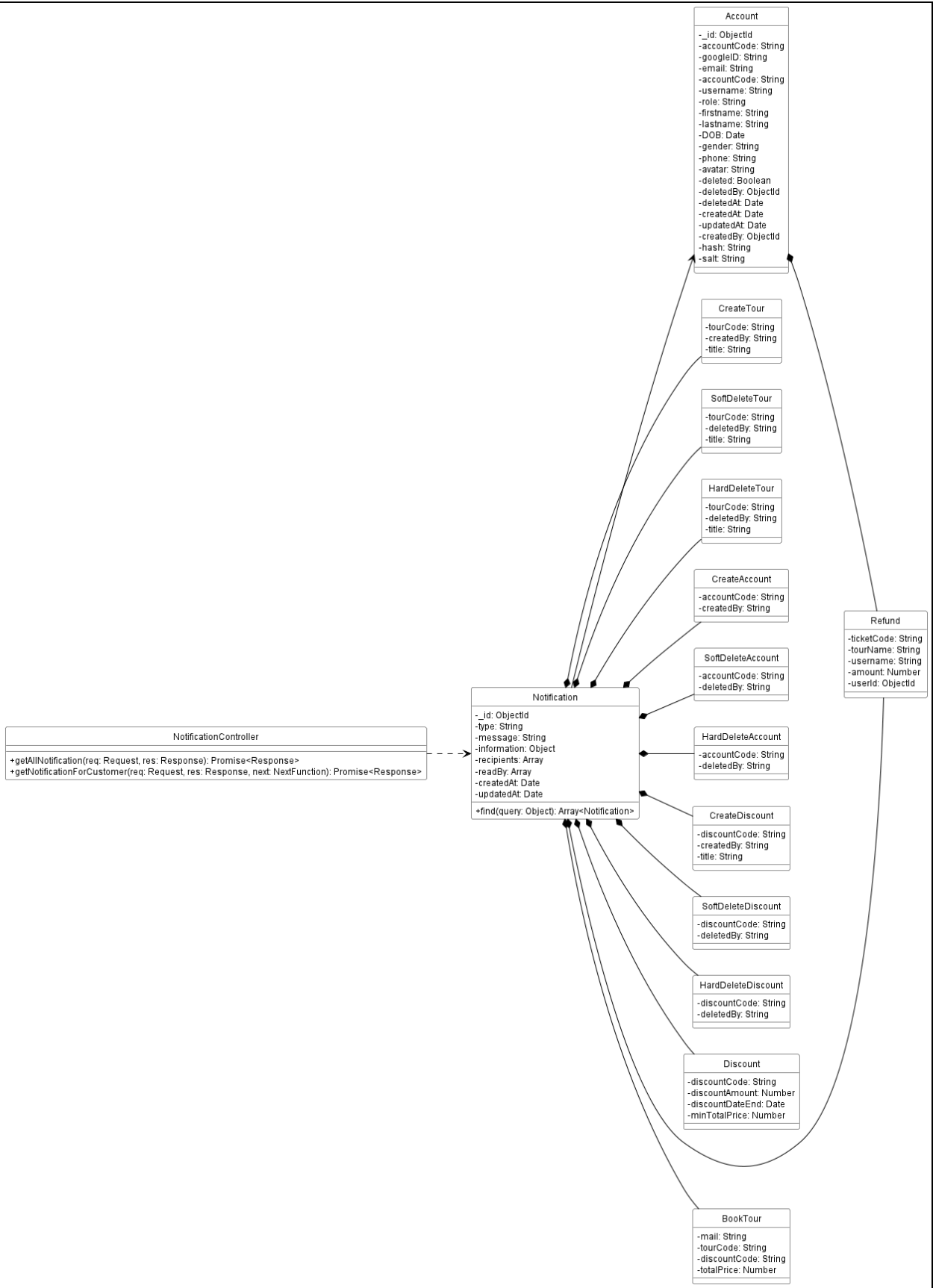
This function does not have an associated class diagram.

b. Sequence Diagram(s)



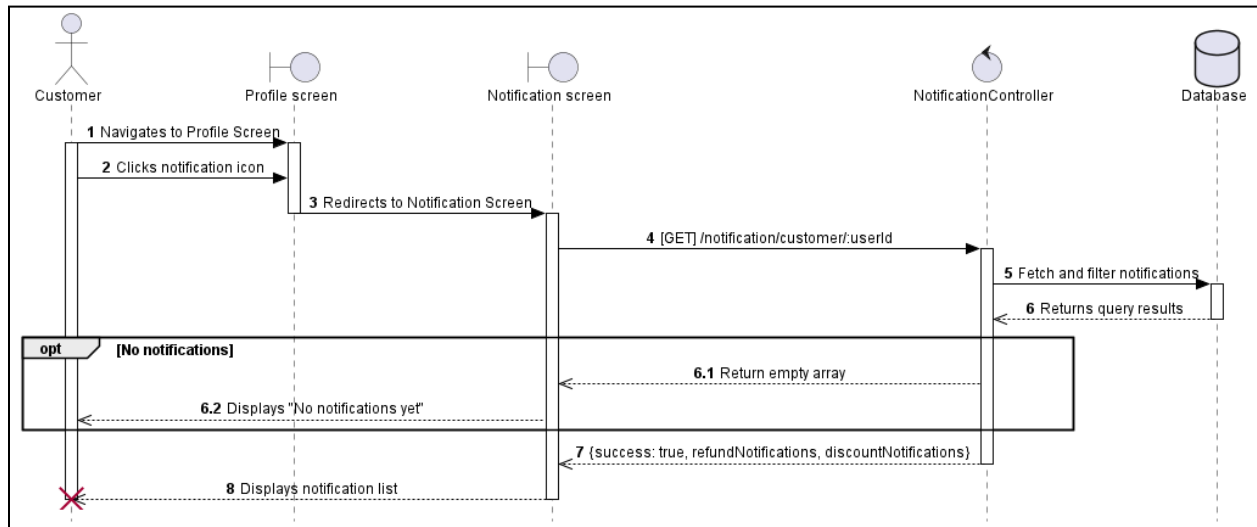
3.5 View Notifications

a. Class Diagram

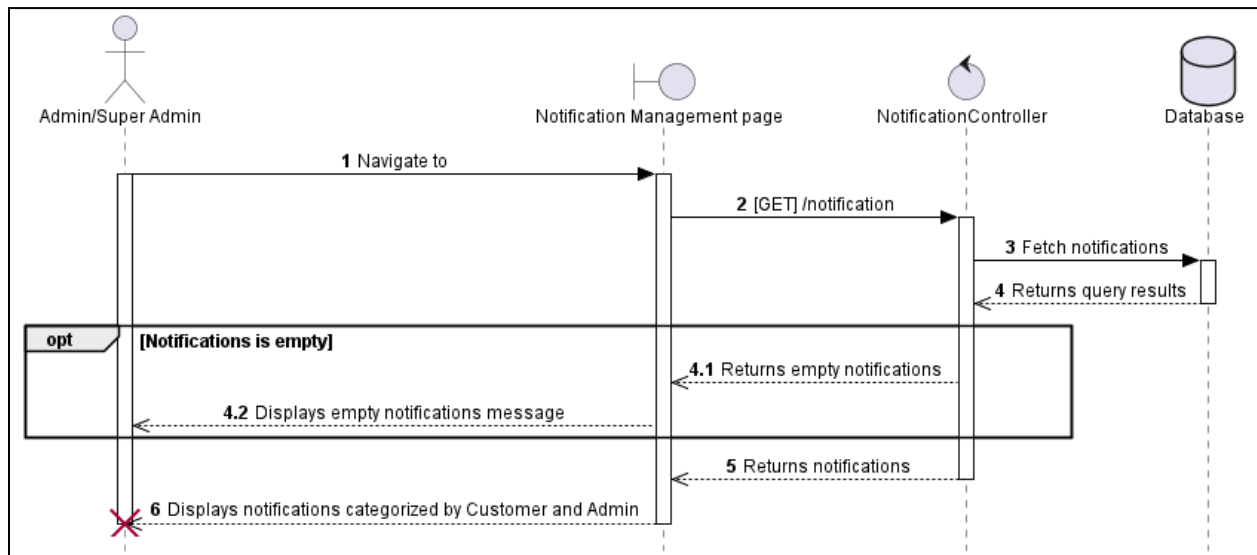


b. Sequence Diagram(s)

* Sequence diagram of View Notifications function on App:

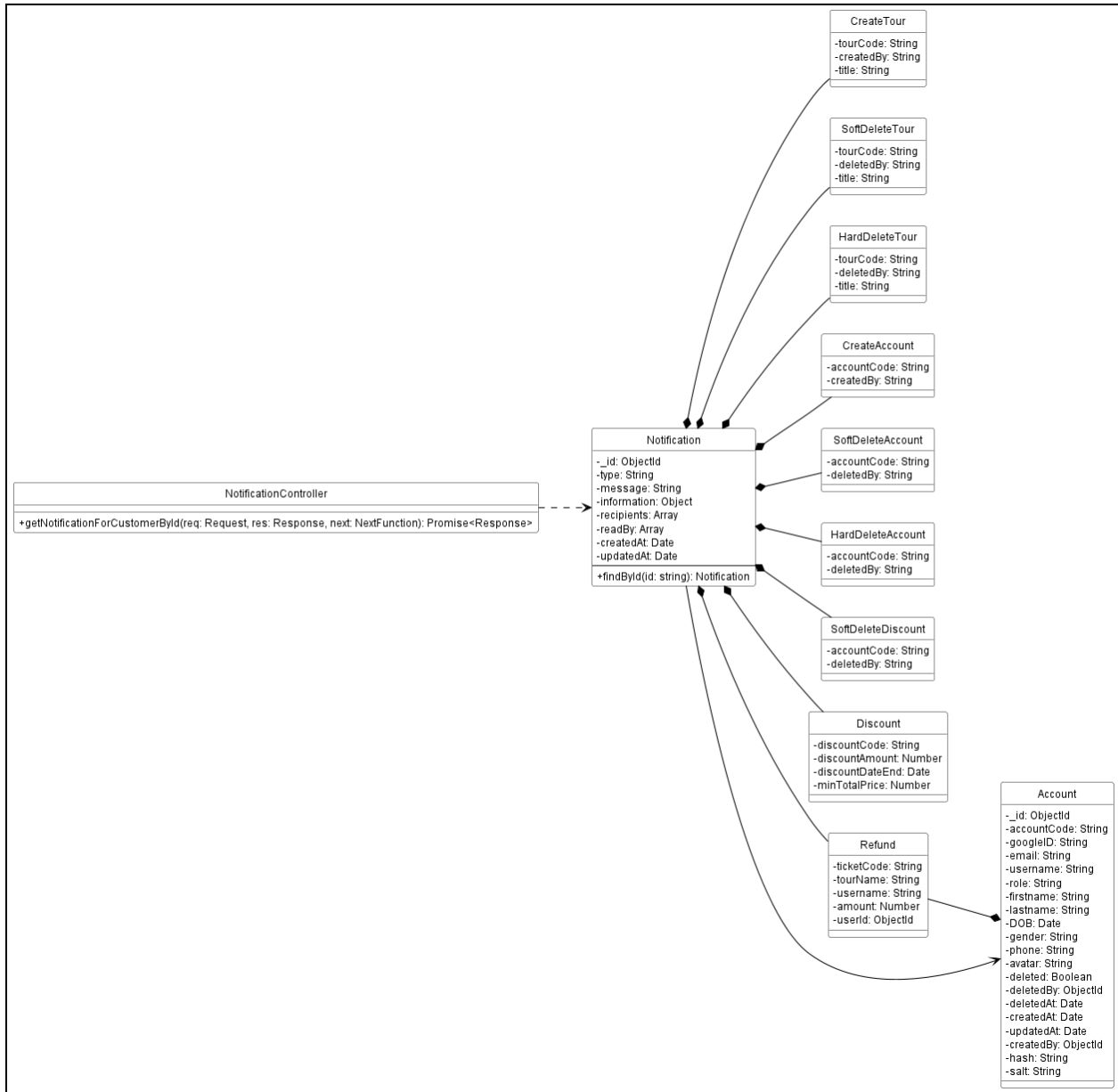


* Sequence diagram of View Notifications function on Web

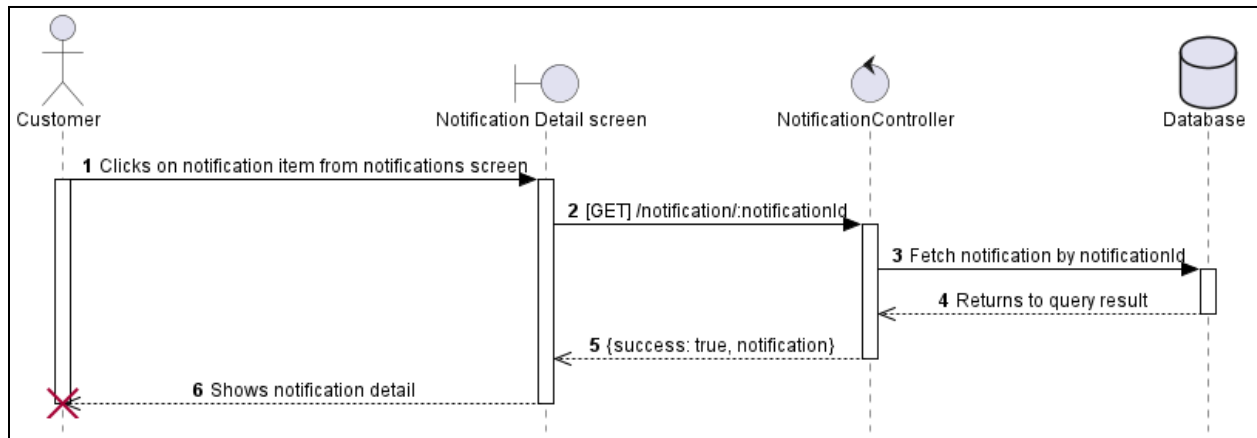


3.6 View Notification Details

a. Class Diagram

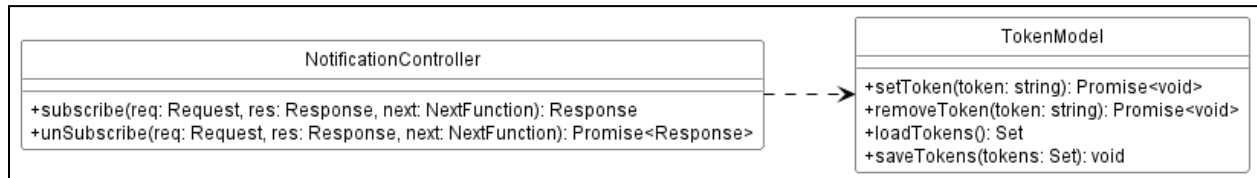


b. Sequence Diagram(s)

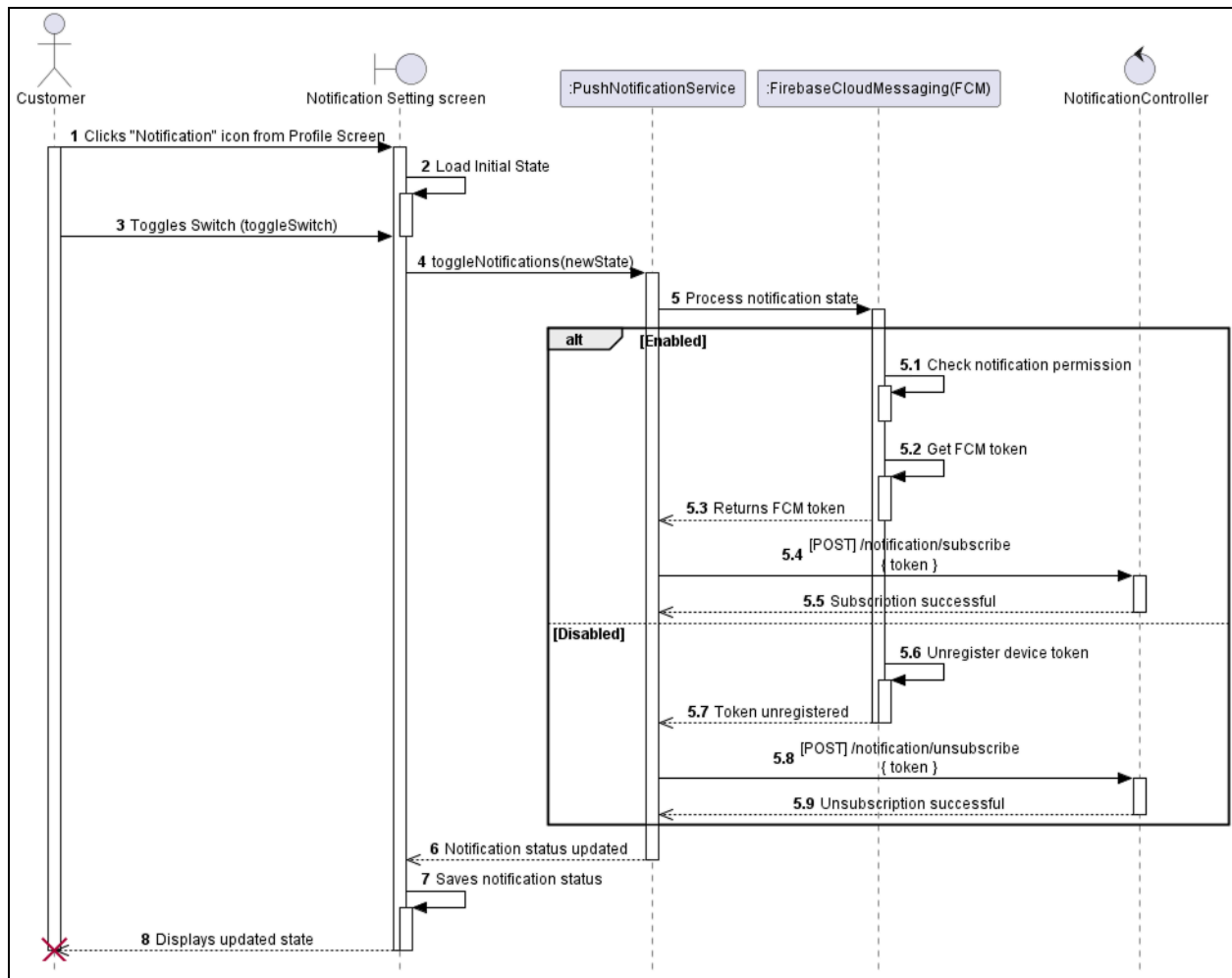


3.7 Setting Notification

a. Class Diagram

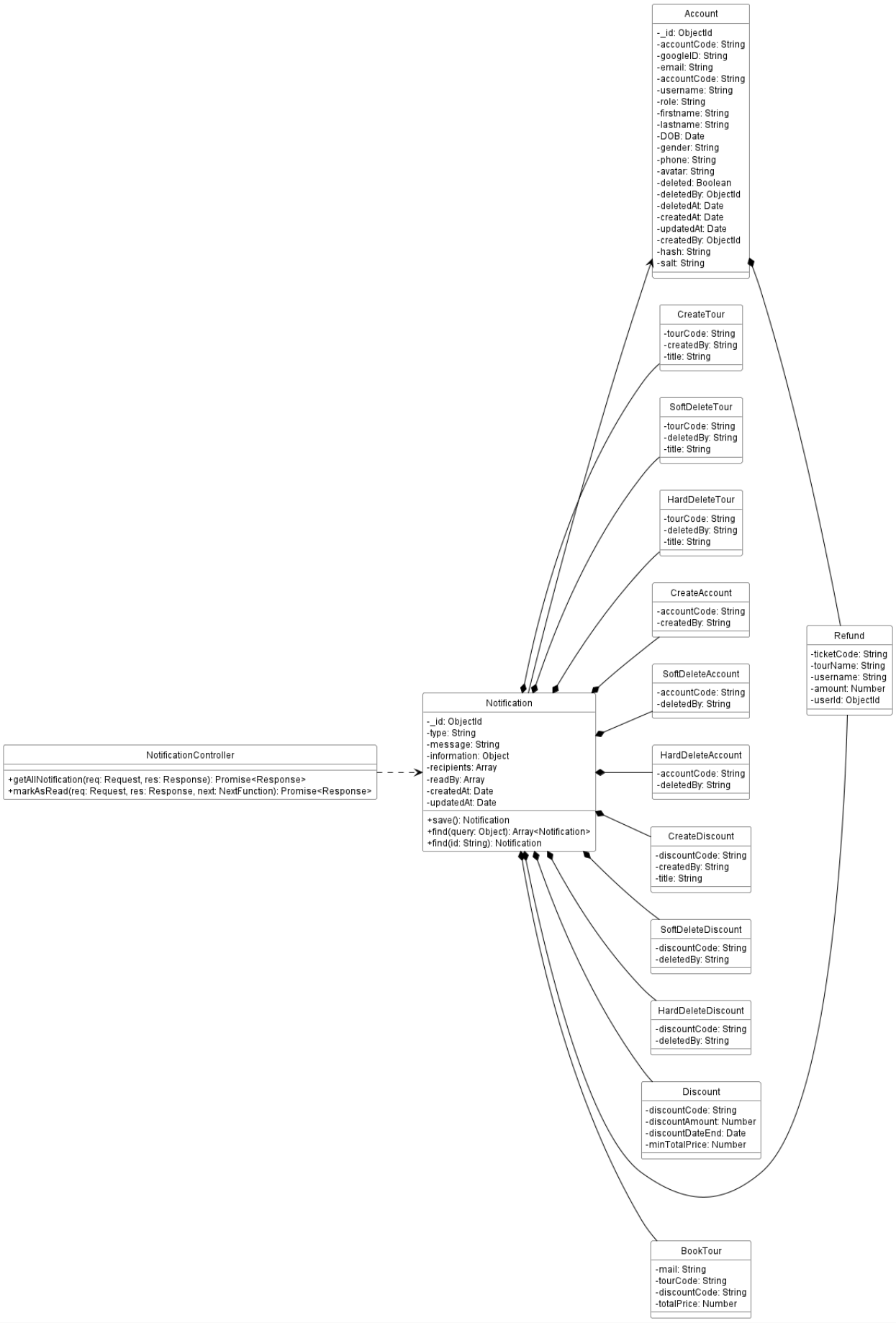


b. Sequence Diagram(s)

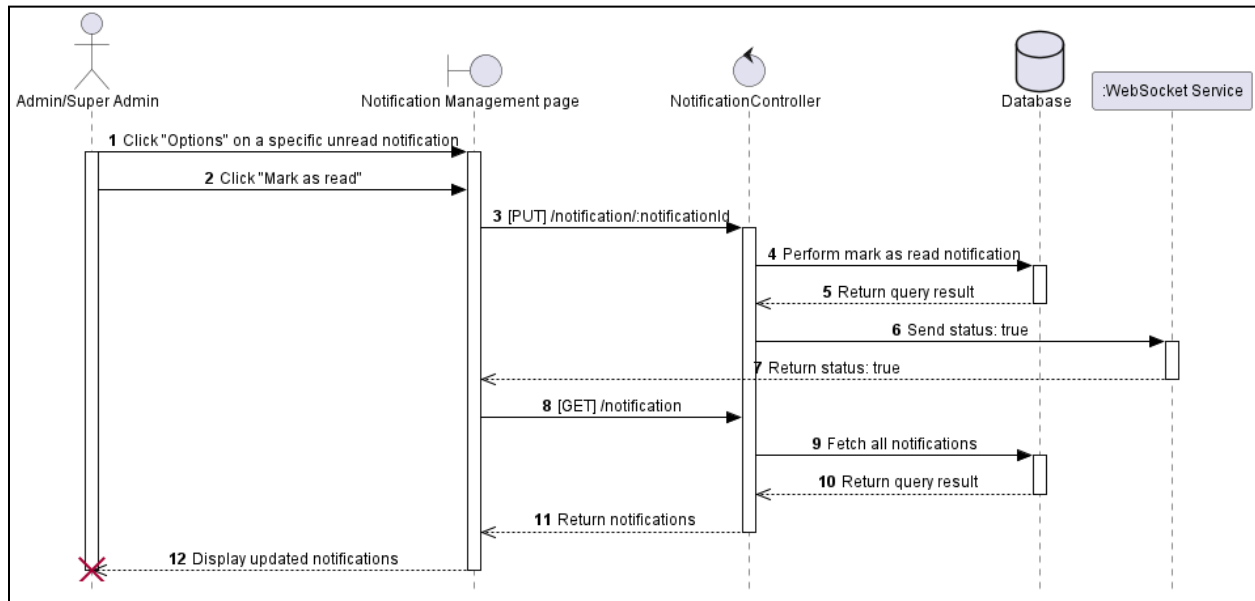


3.8 Mark As Read

a. Class Diagram

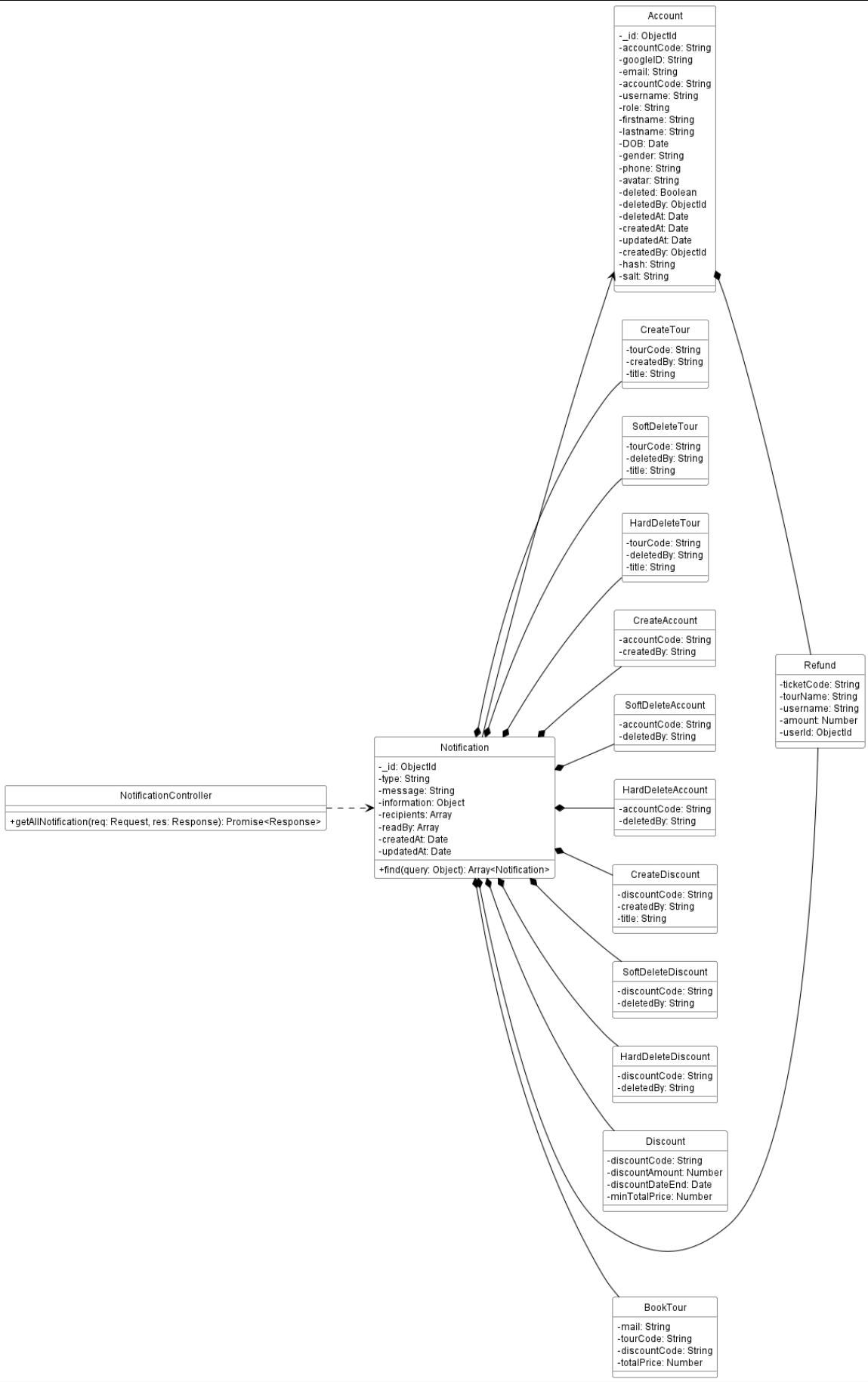


b. Sequence Diagram(s)

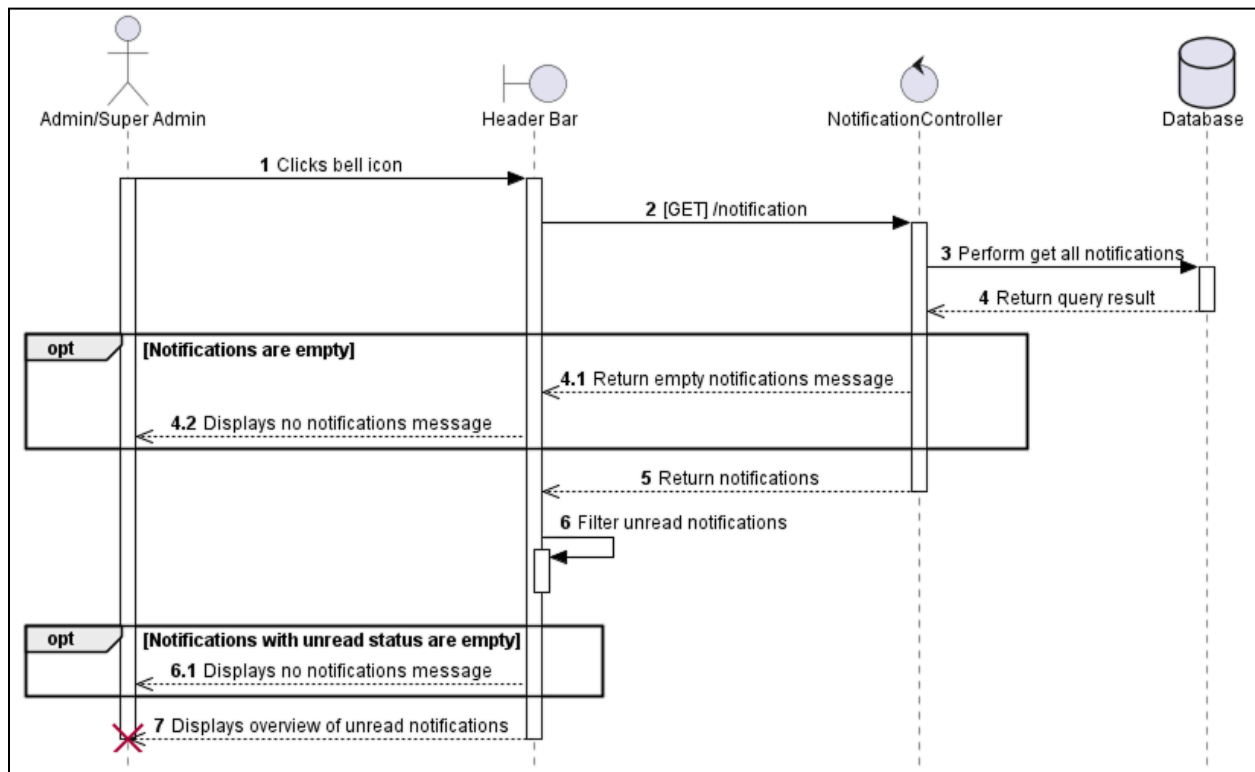


3.9 Overview Notification

a. Class Diagram

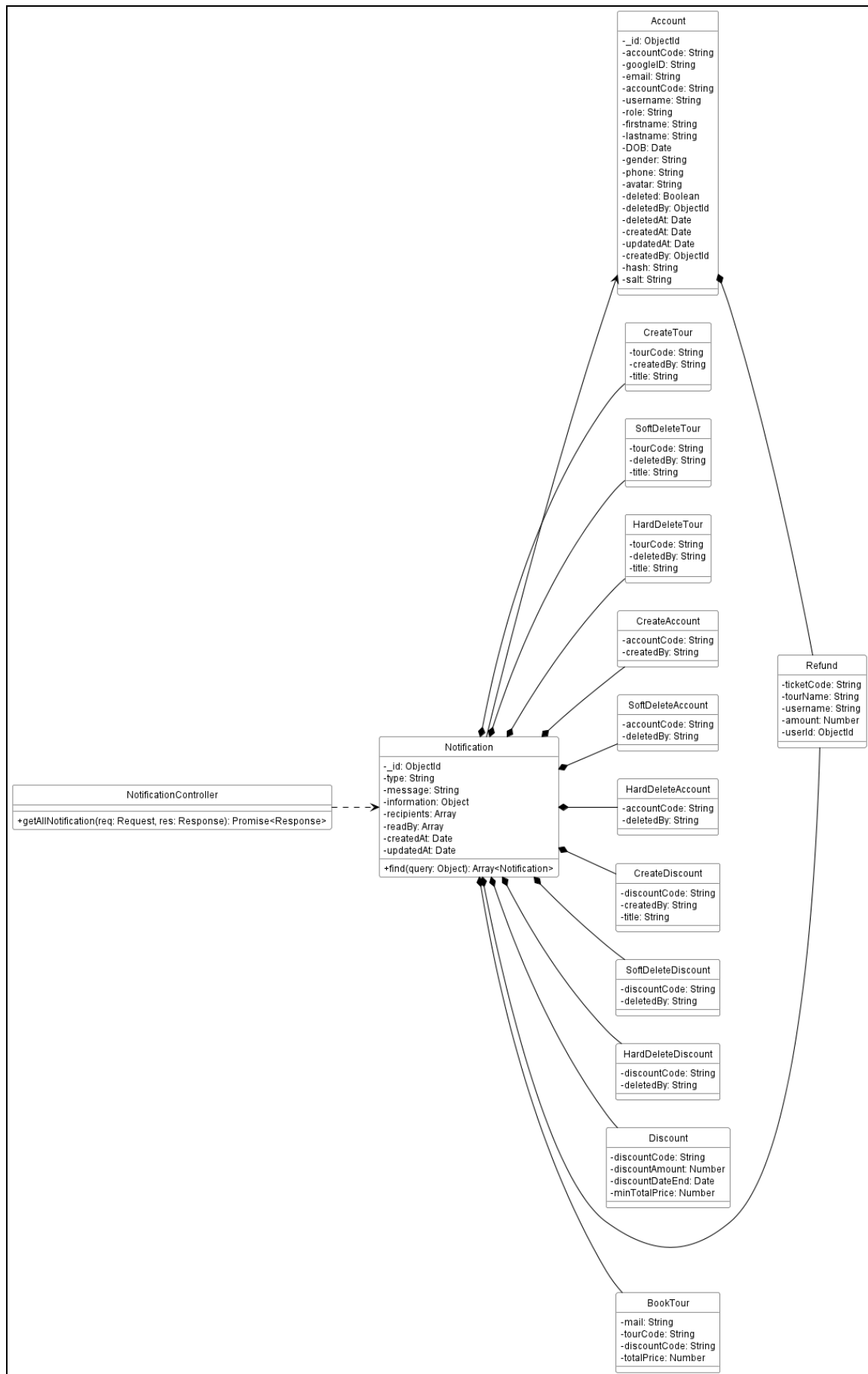


b. Sequence Diagram(s)

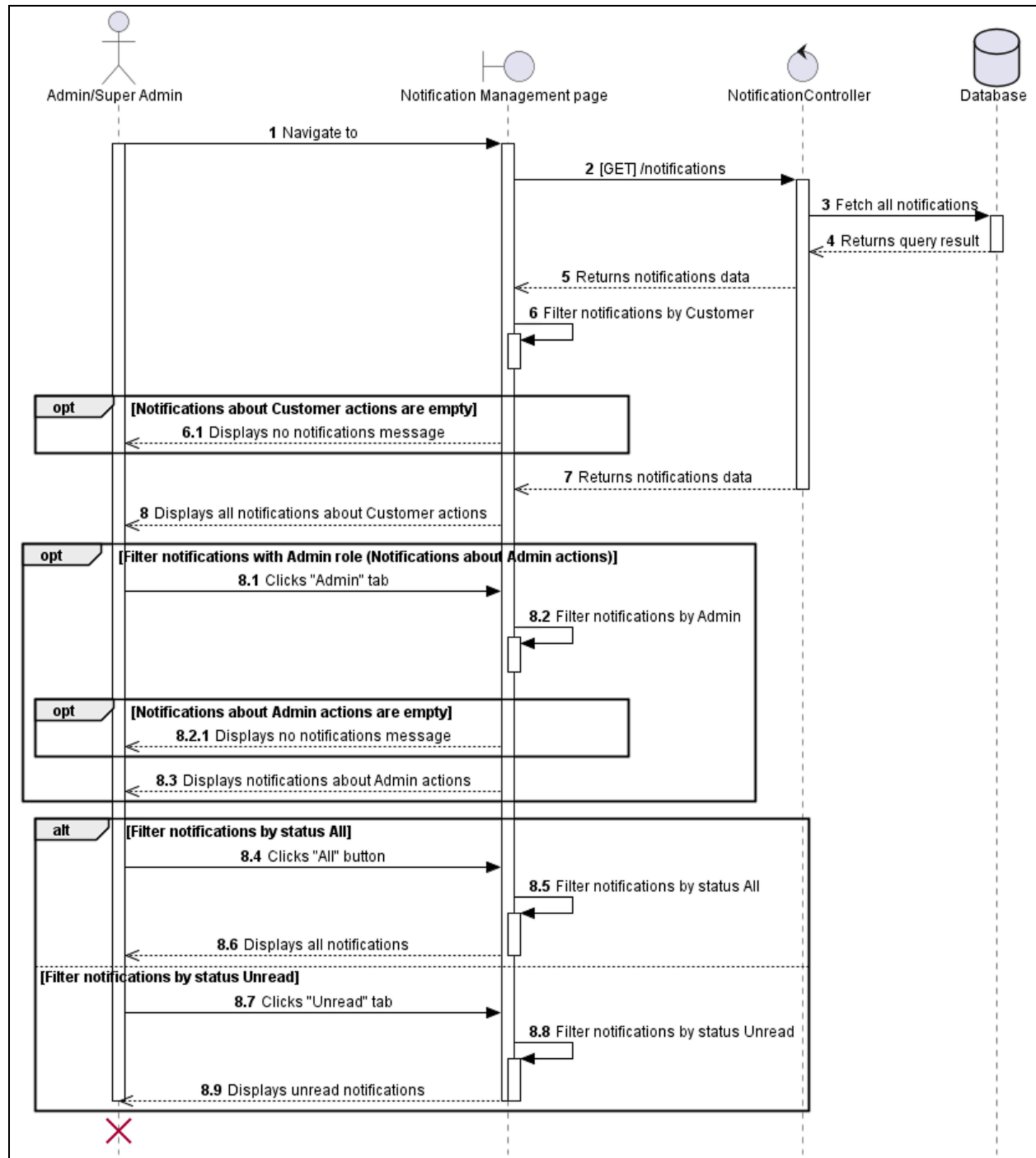


3.10 Filter Notifications

a. Class Diagram

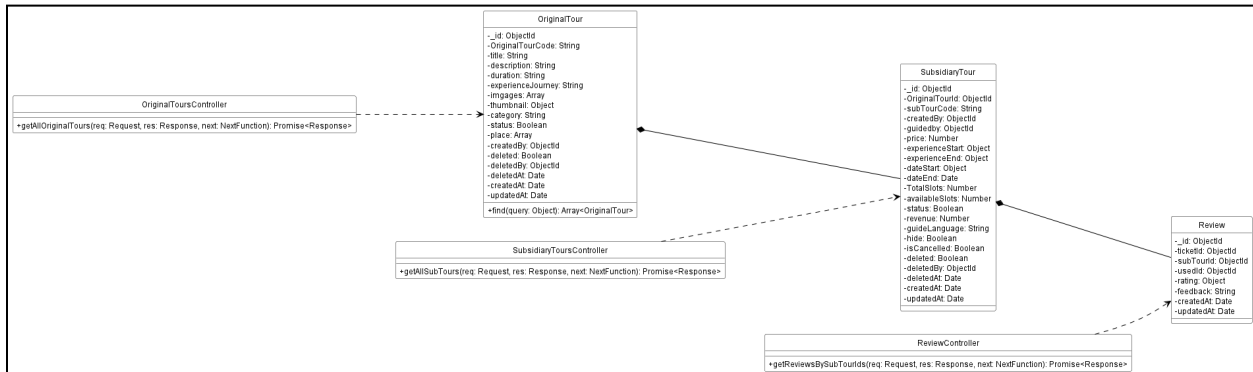


b. Sequence Diagram(s)

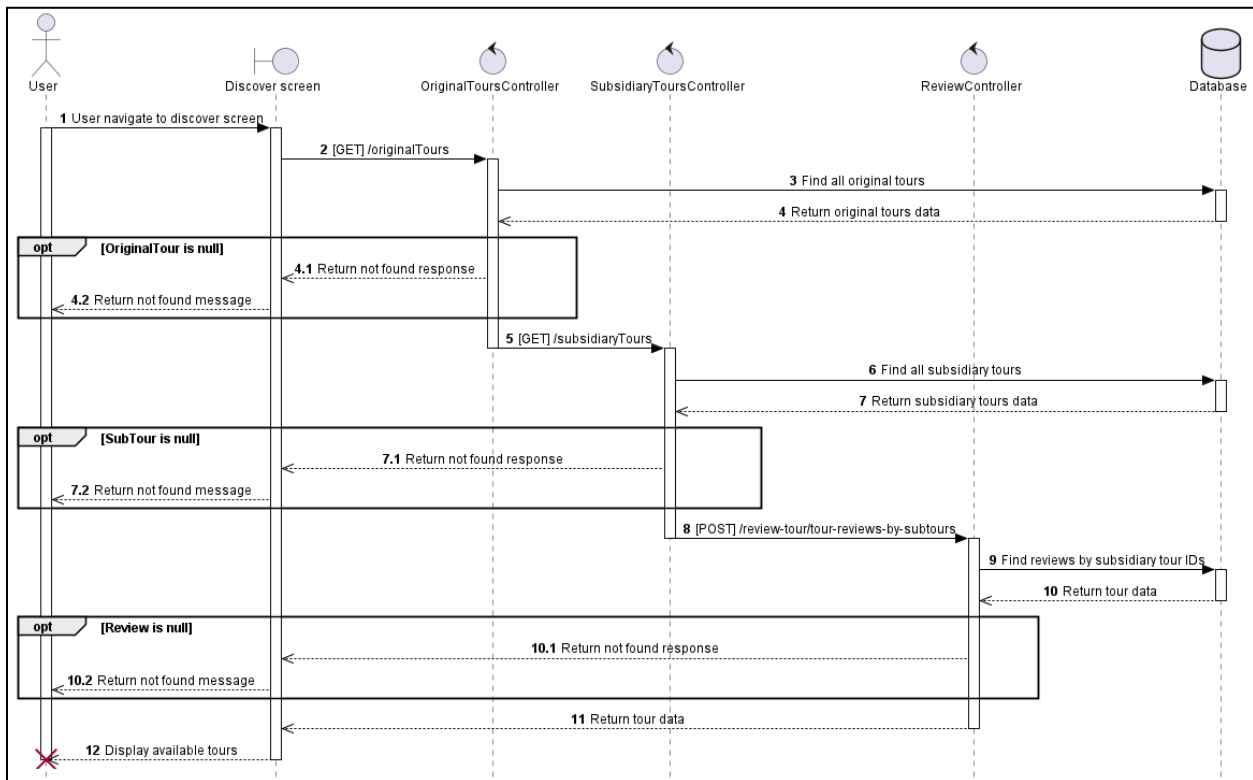


3.11 View Tour's List

a. Class Diagram



b. Sequence Diagram(s)

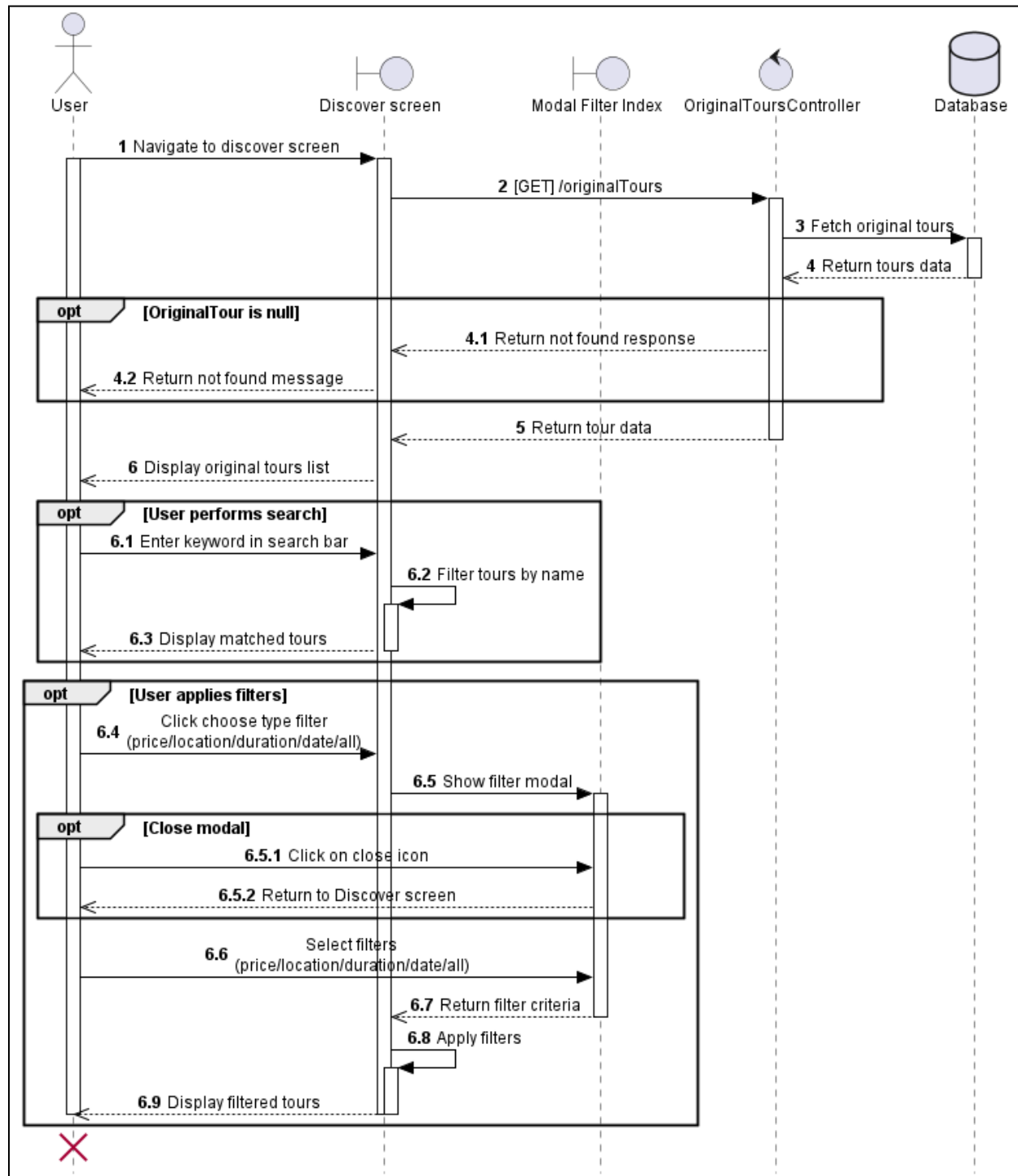


3.12 Filter & Search Tour

a. Class Diagram

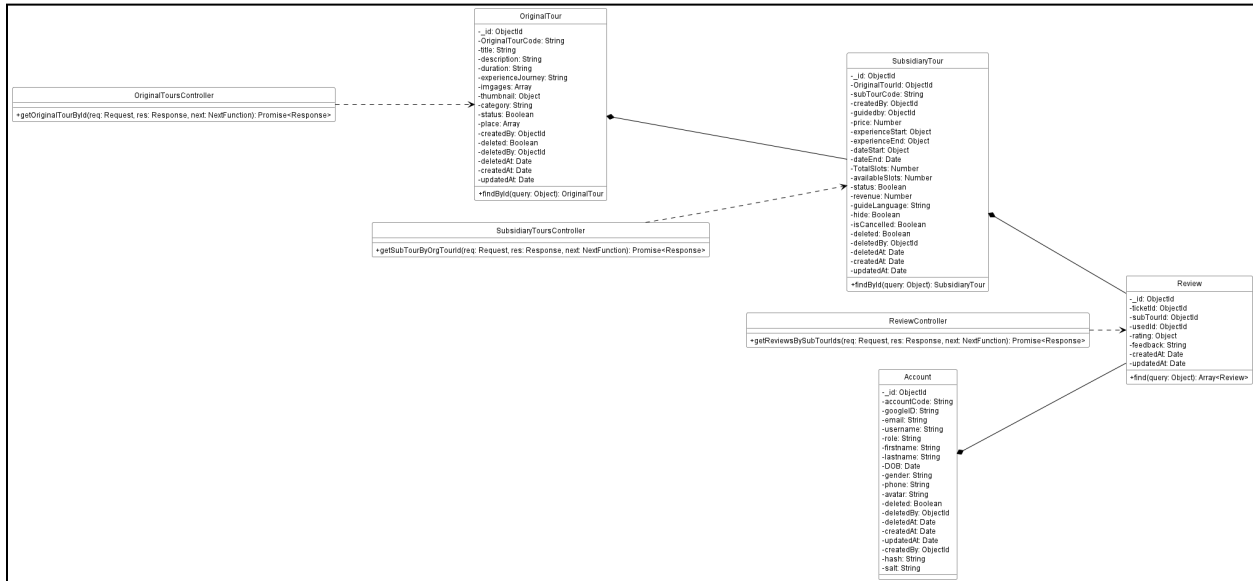
This function does not have an associated class diagram.

b. Sequence Diagram(s)

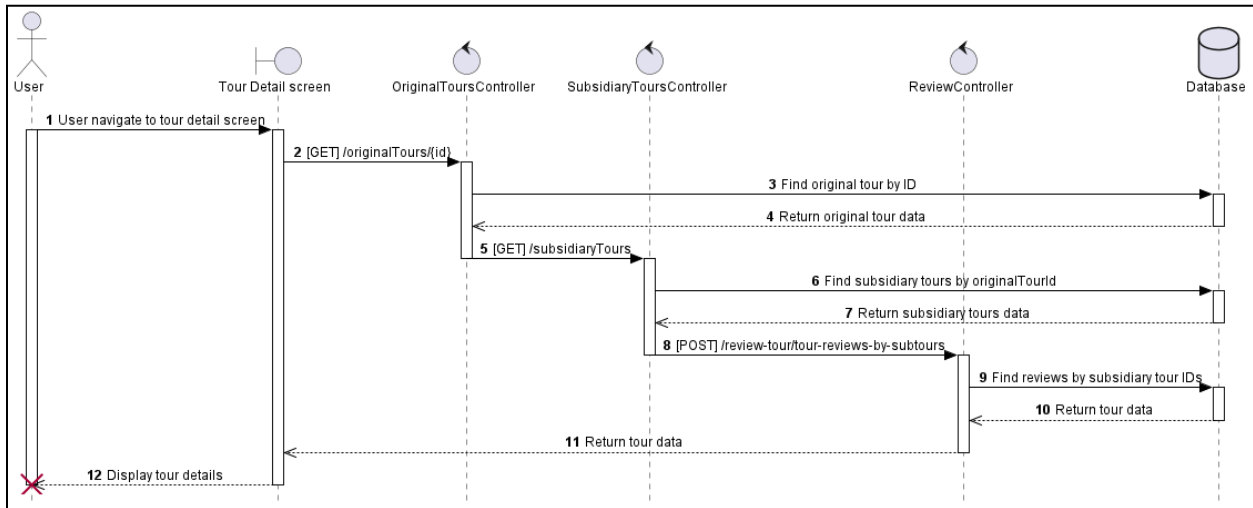


3.13 View Tour's Detail

a. Class Diagram

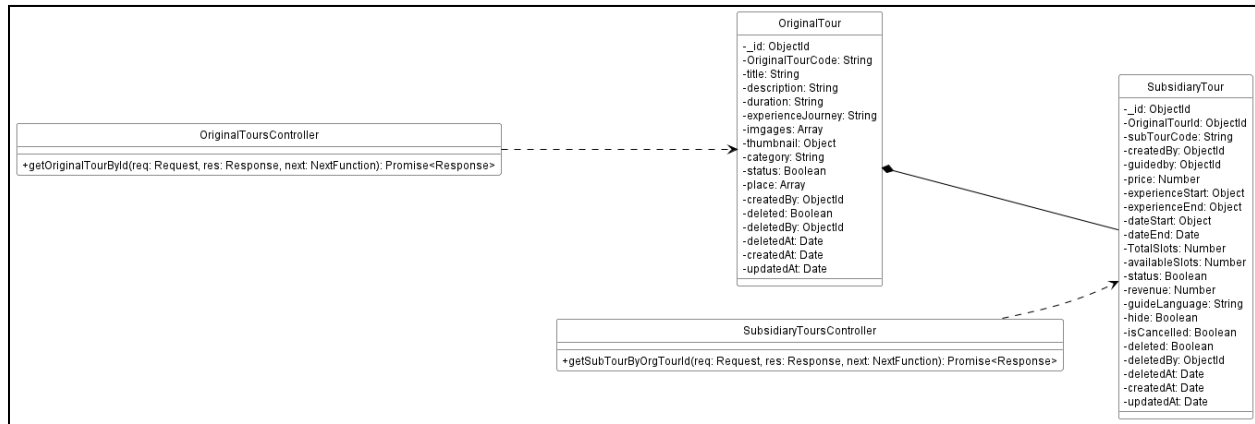


b. Sequence Diagram(s)

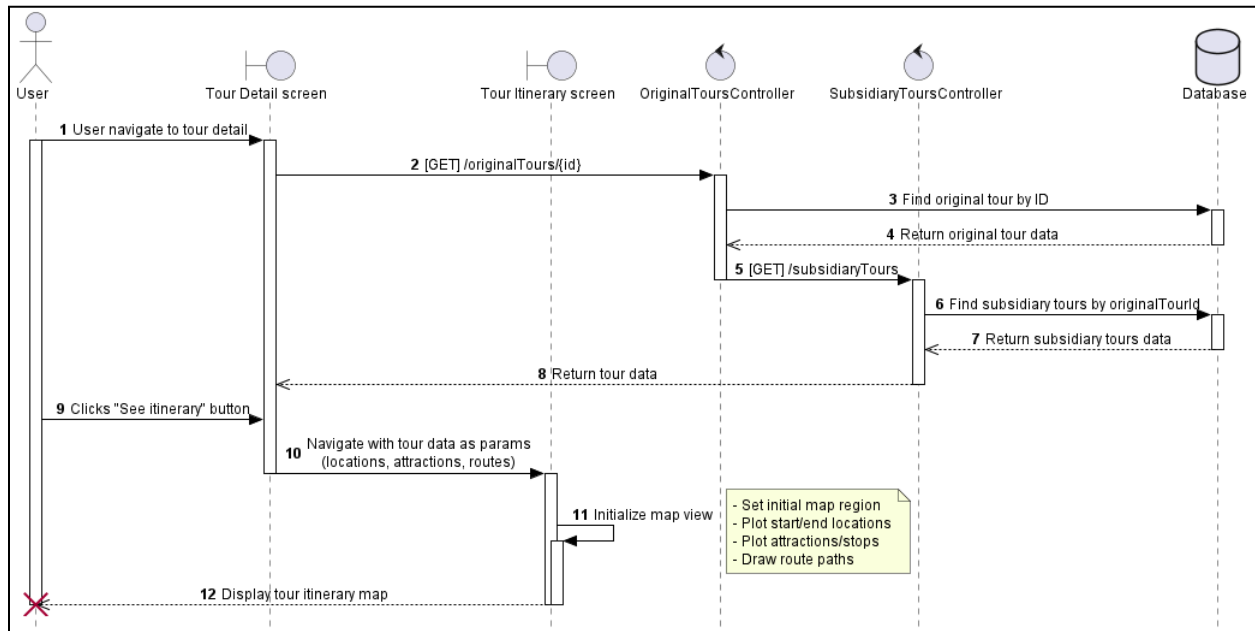


3.14 View Tour Itinerary

a. Class Diagram

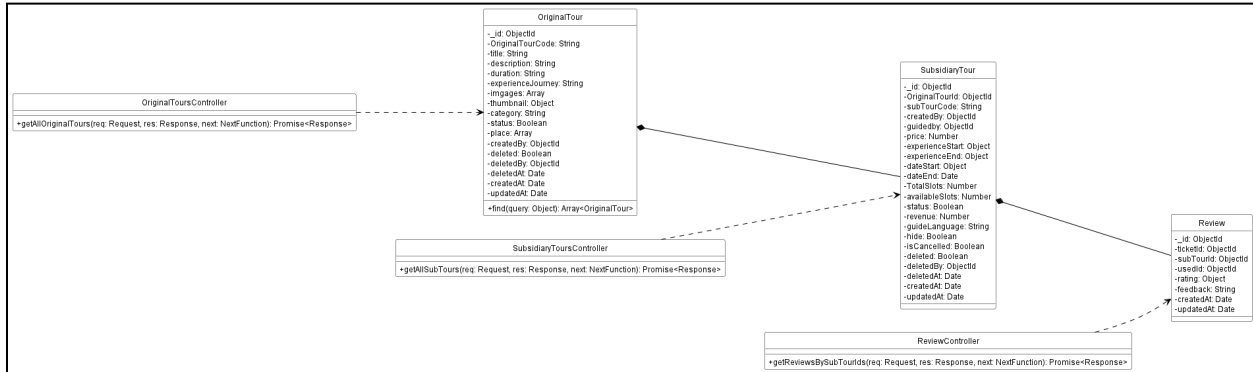


b. Sequence Diagram(s)

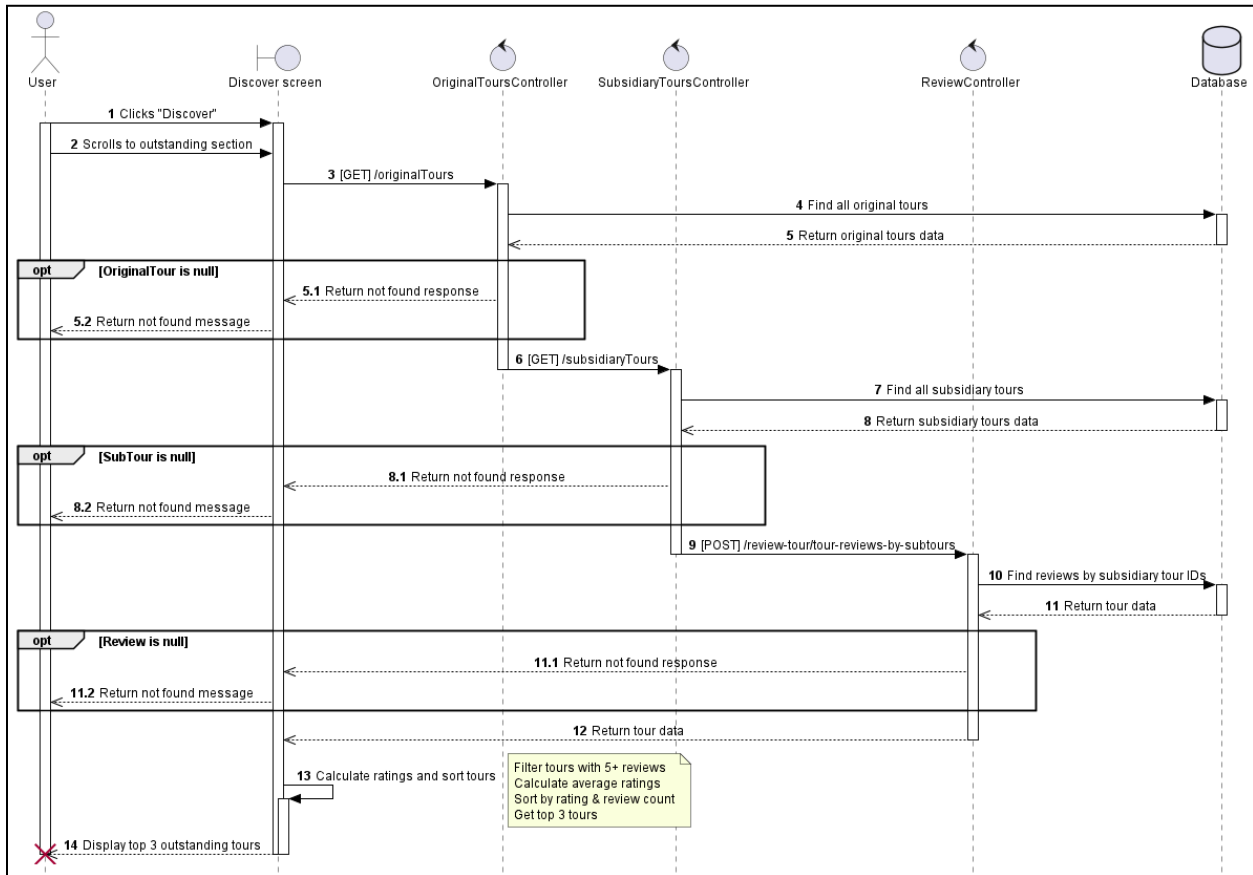


3.15 View Outstanding

a. Class Diagram

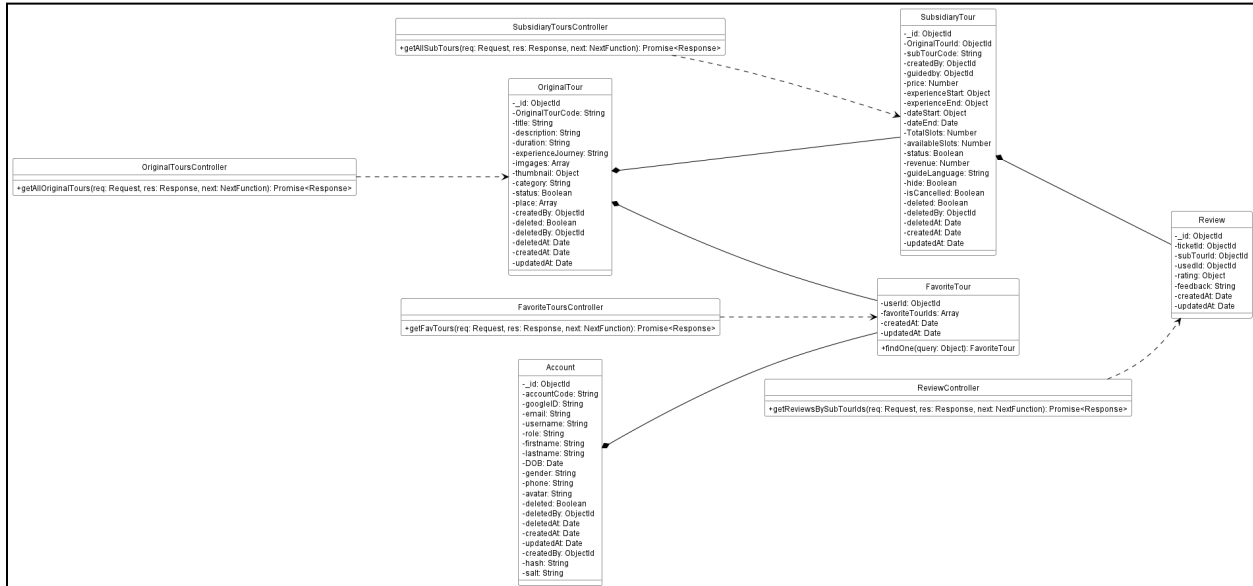


b. Sequence Diagram(s)

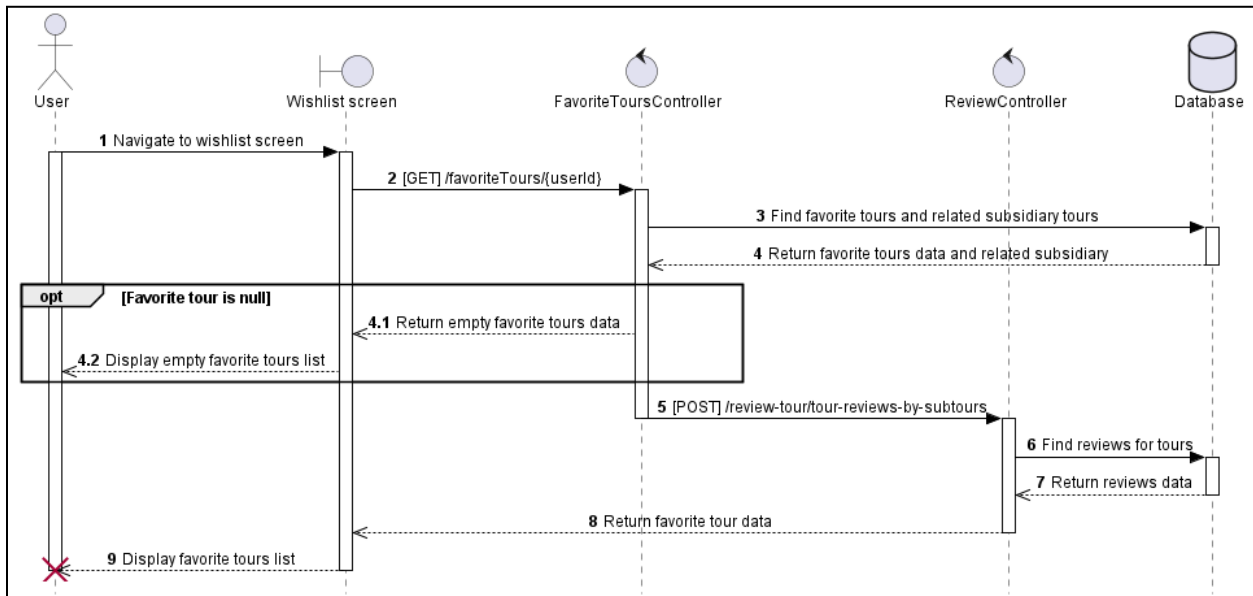


3.16 View Favorite Tour

a. Class Diagram

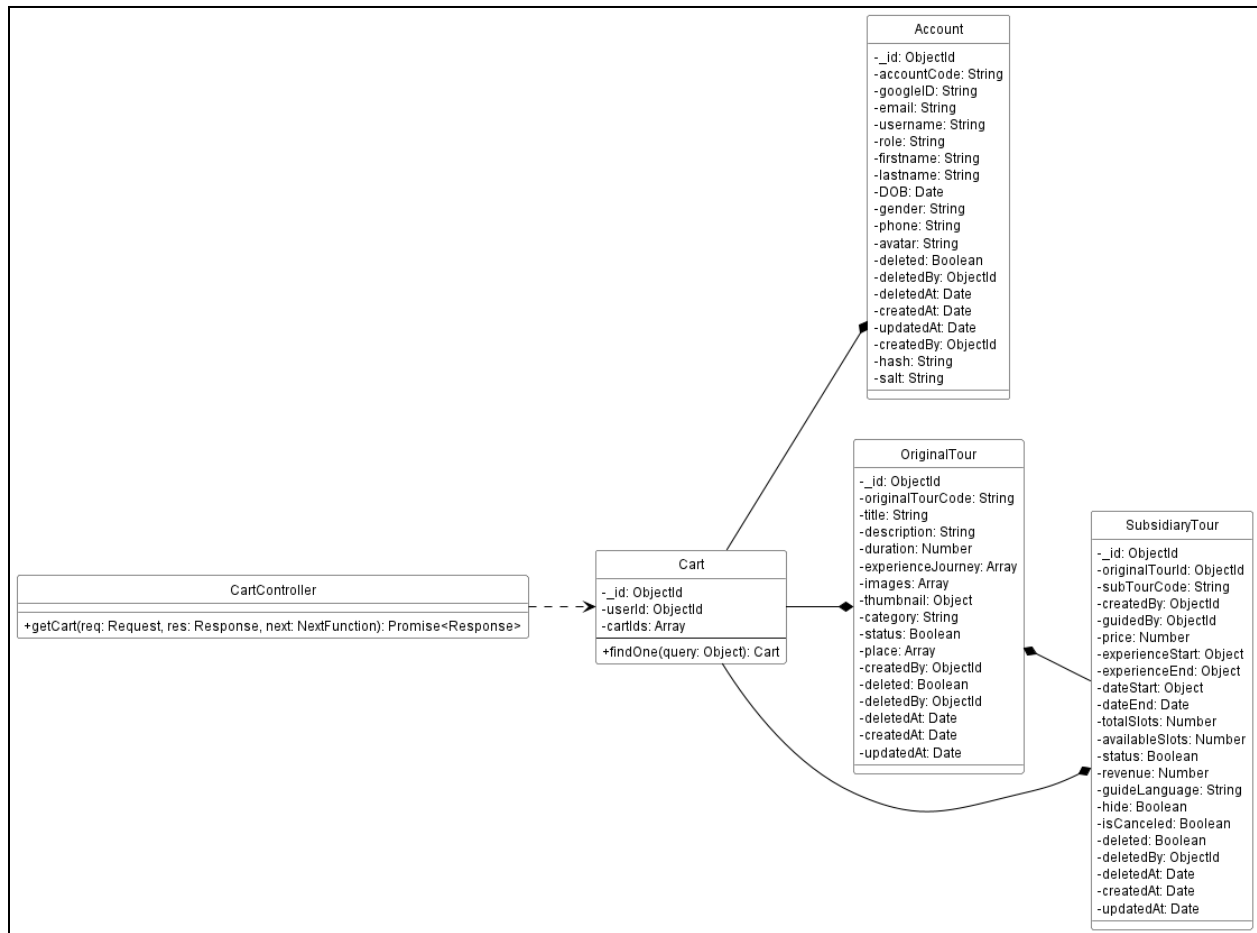


b. Sequence Diagram(s)

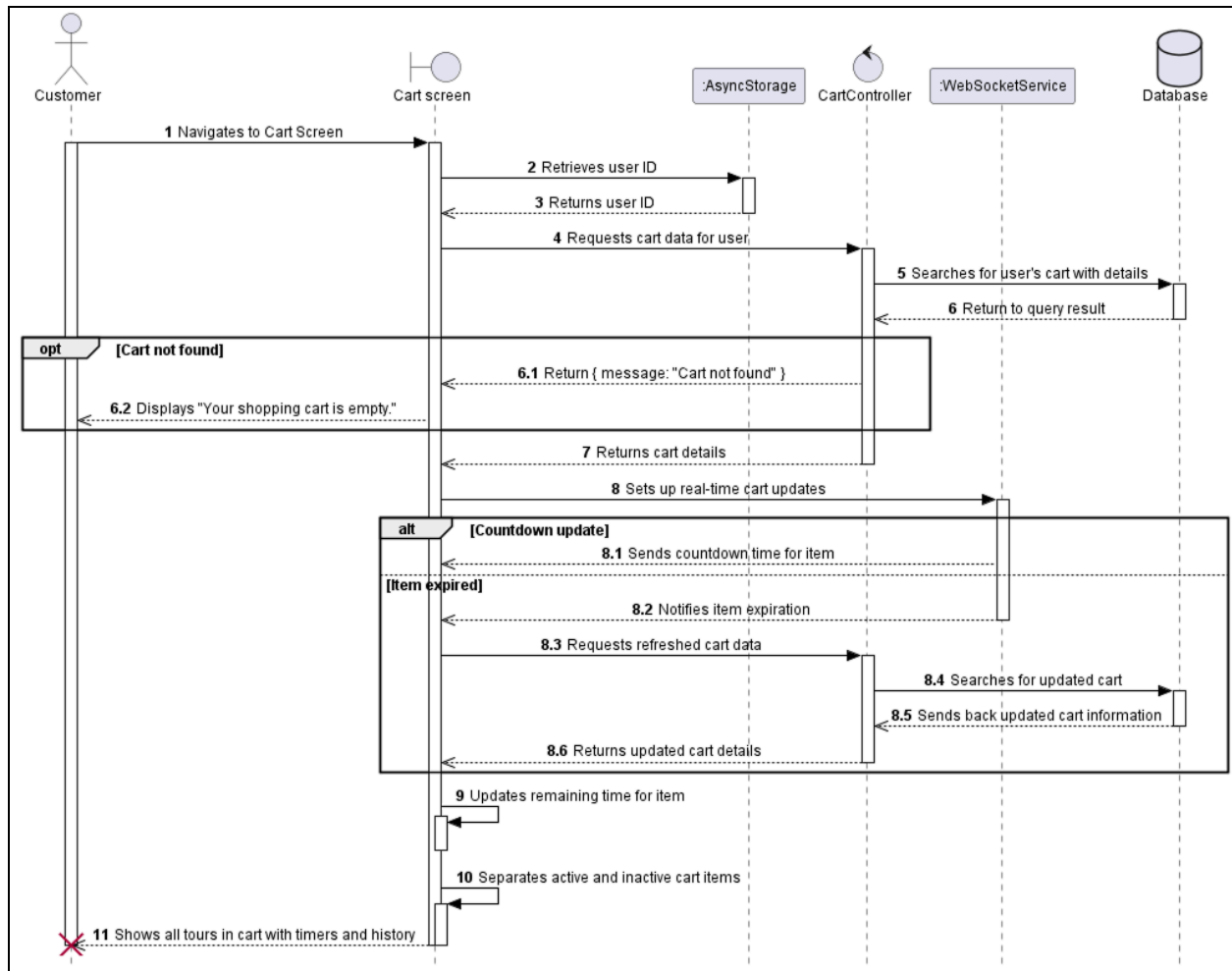


3.17 View Cart Tours

a. Class Diagram

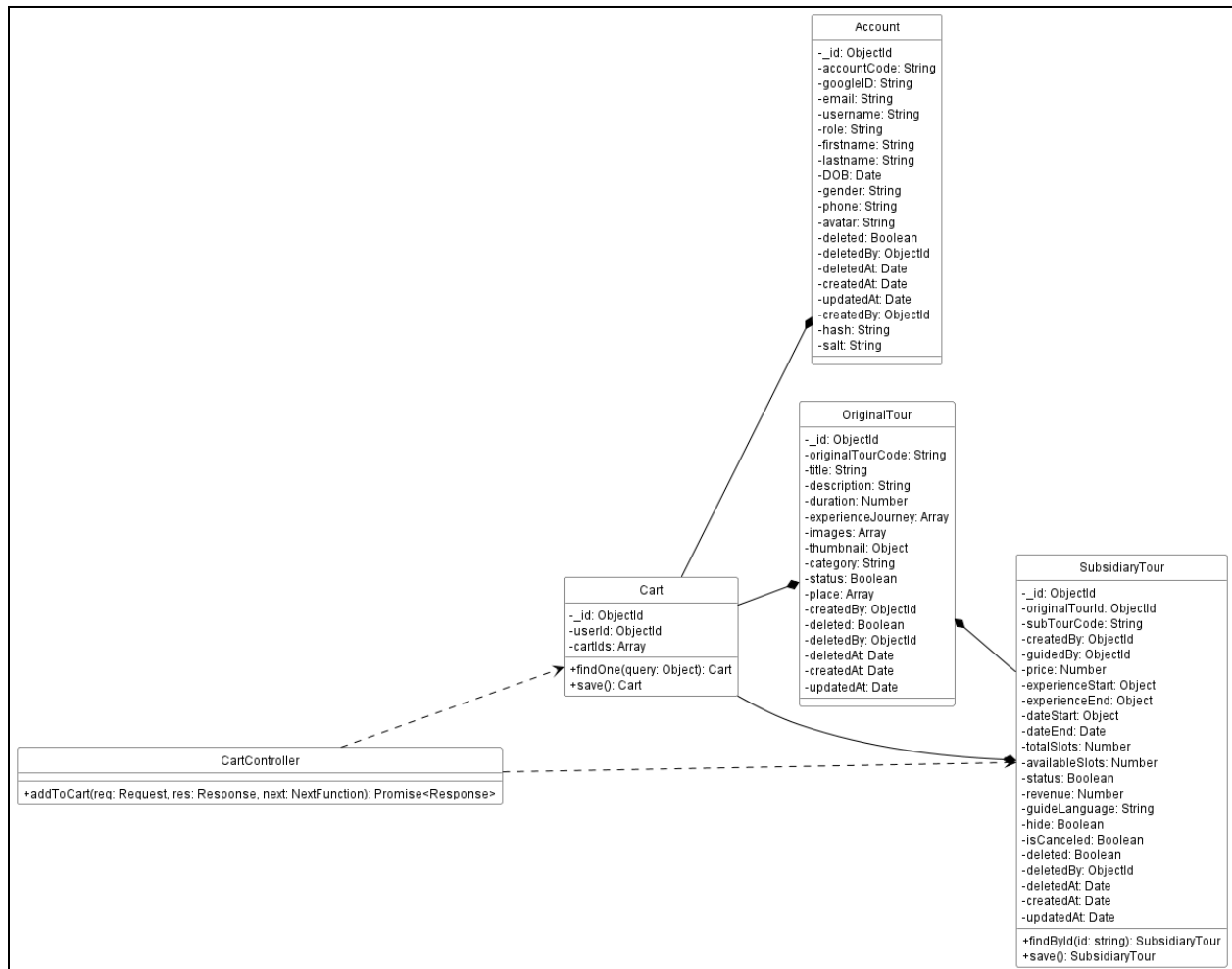


b. Sequence Diagram(s)

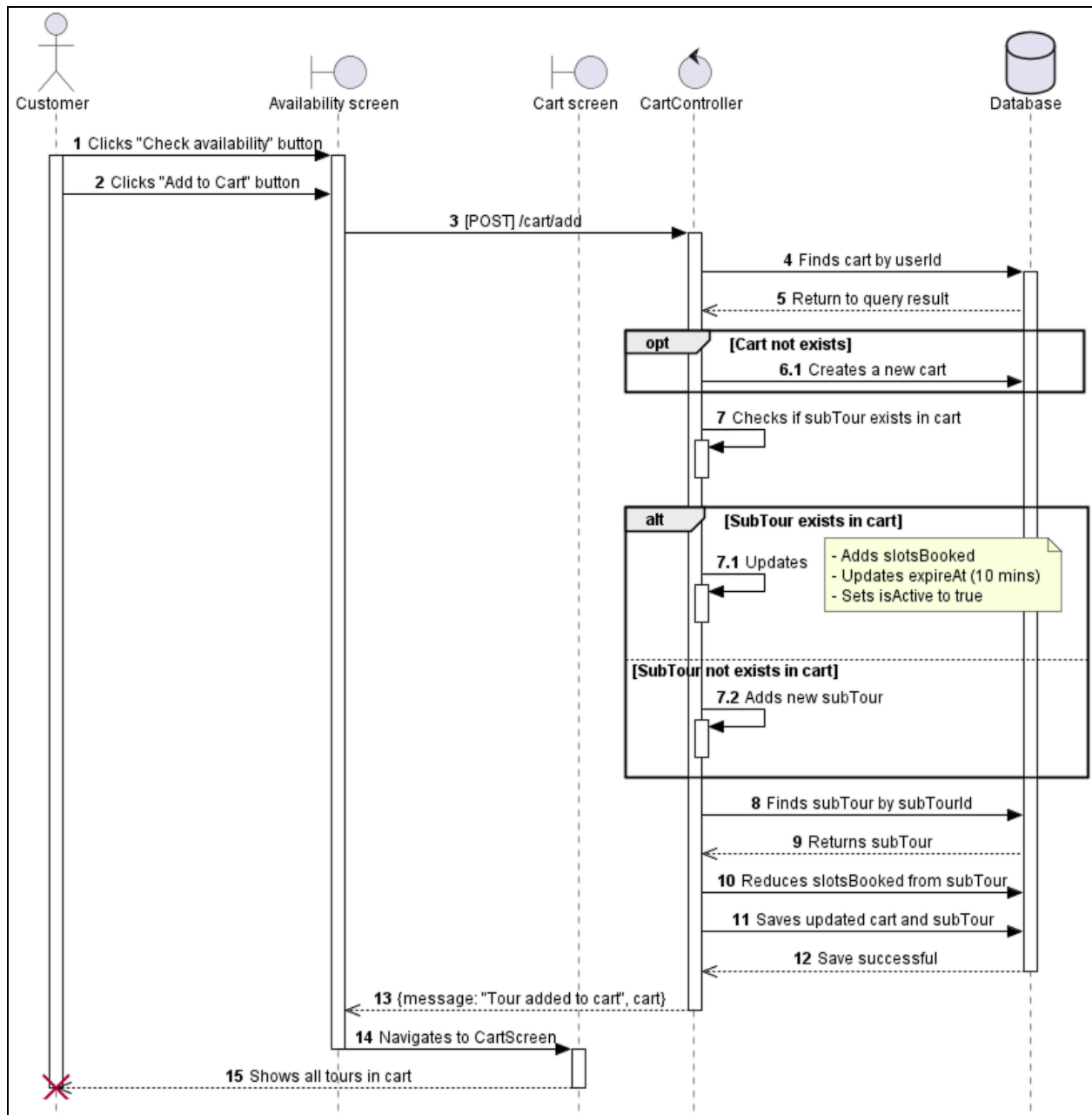


3.18 Add Tour To Cart

a. Class Diagram

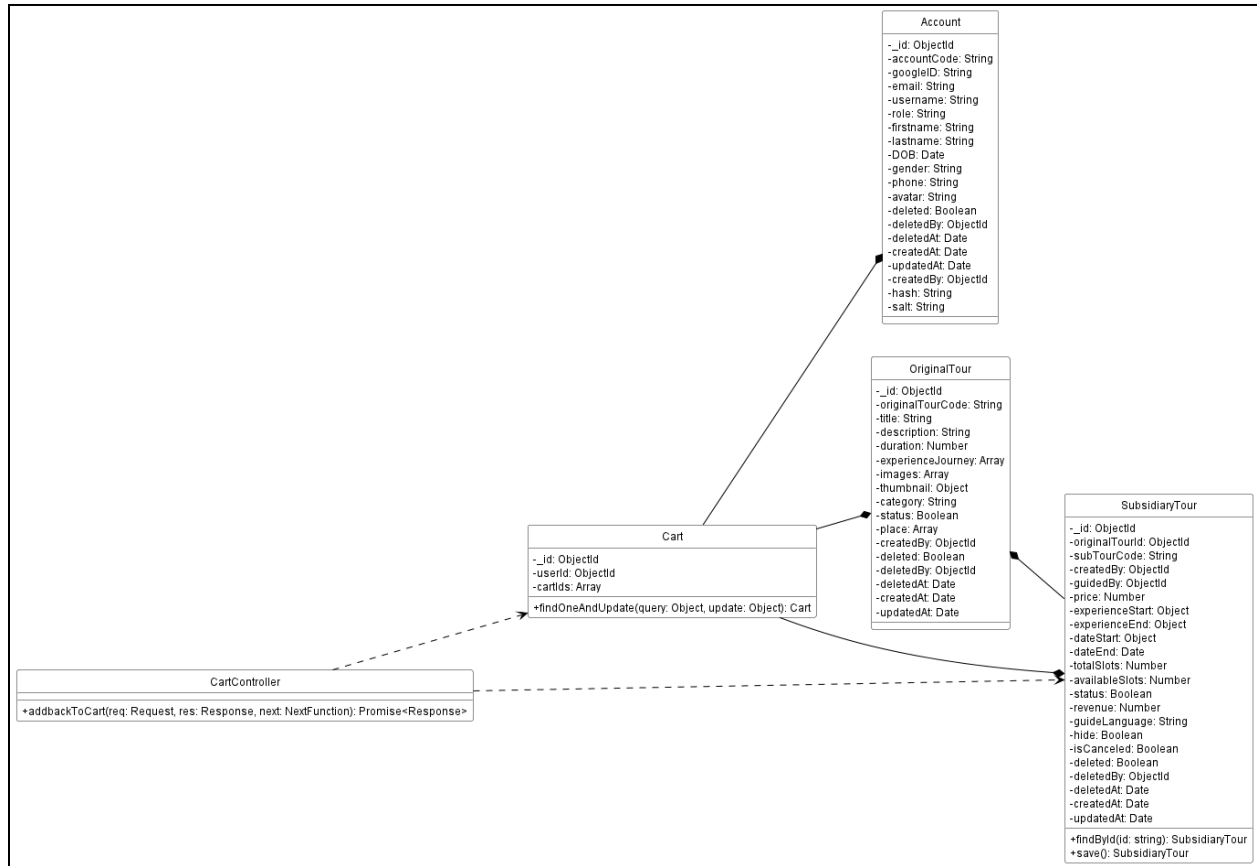


b. Sequence Diagram(s)

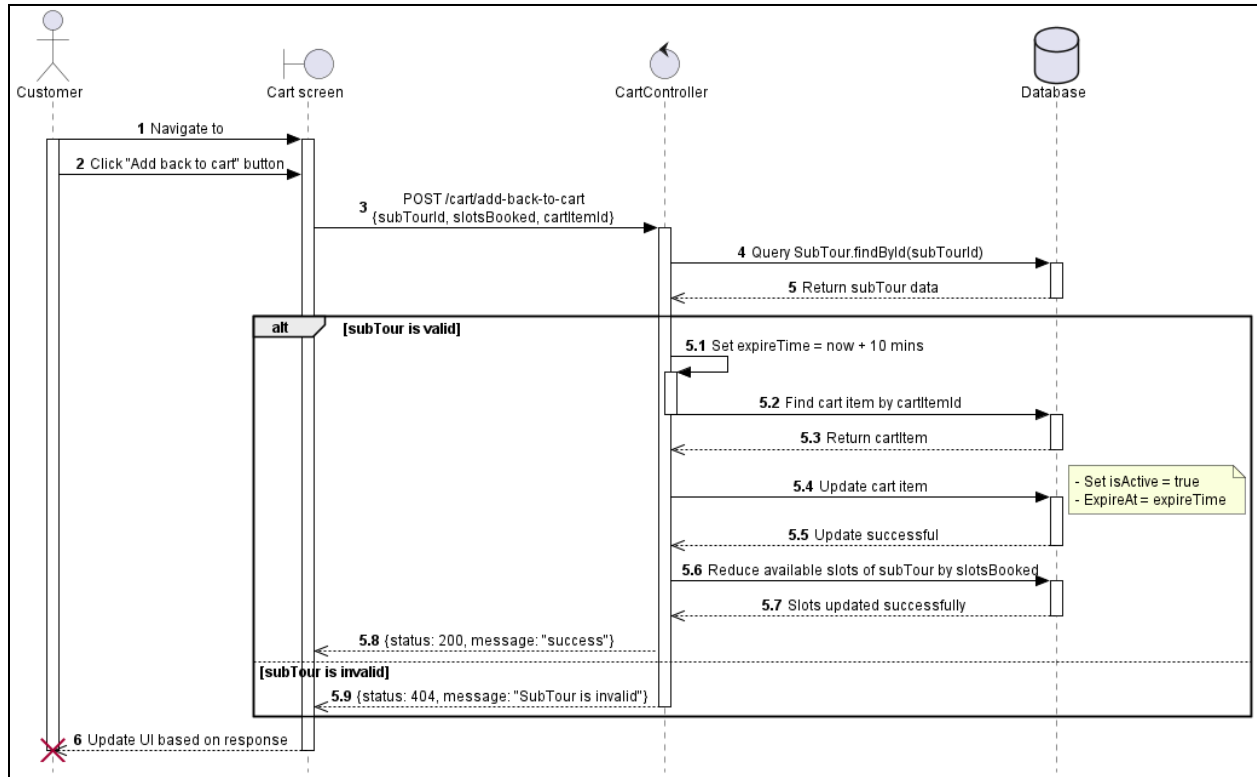


3.19 Add Back To Cart

a. Class Diagram

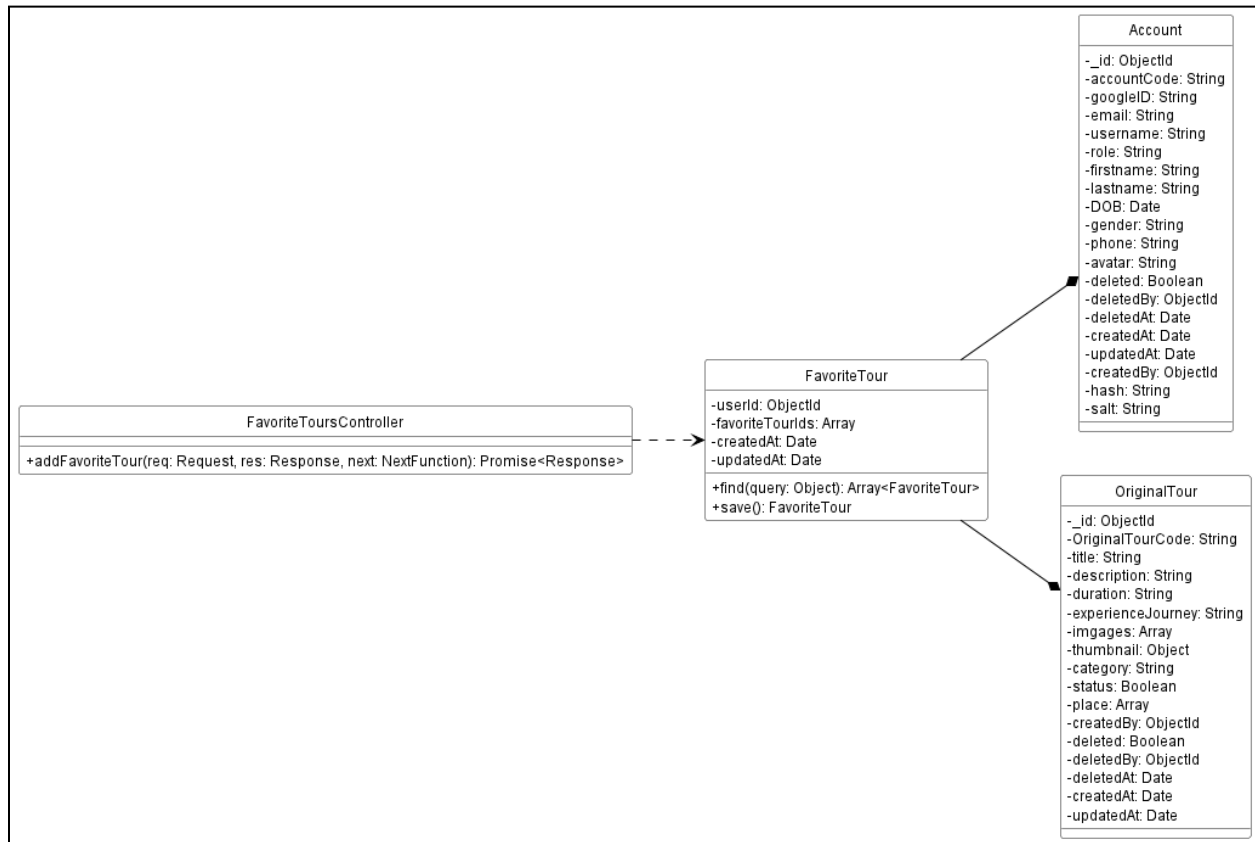


b. Sequence Diagram(s)

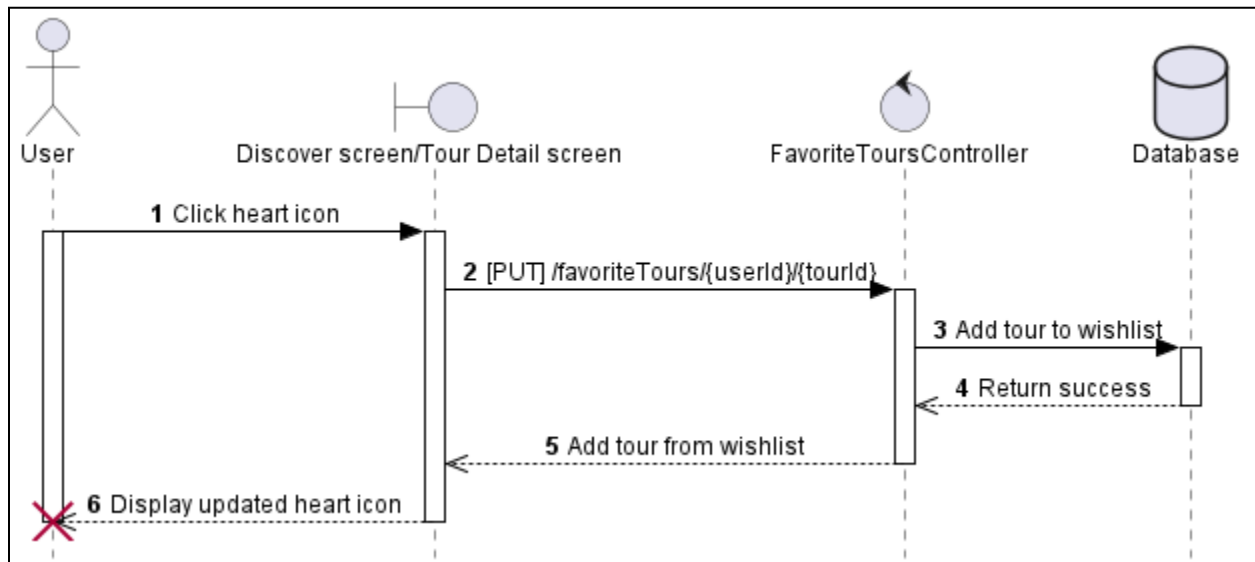


3.20 Add Tour To Wishlist

a. Class Diagram

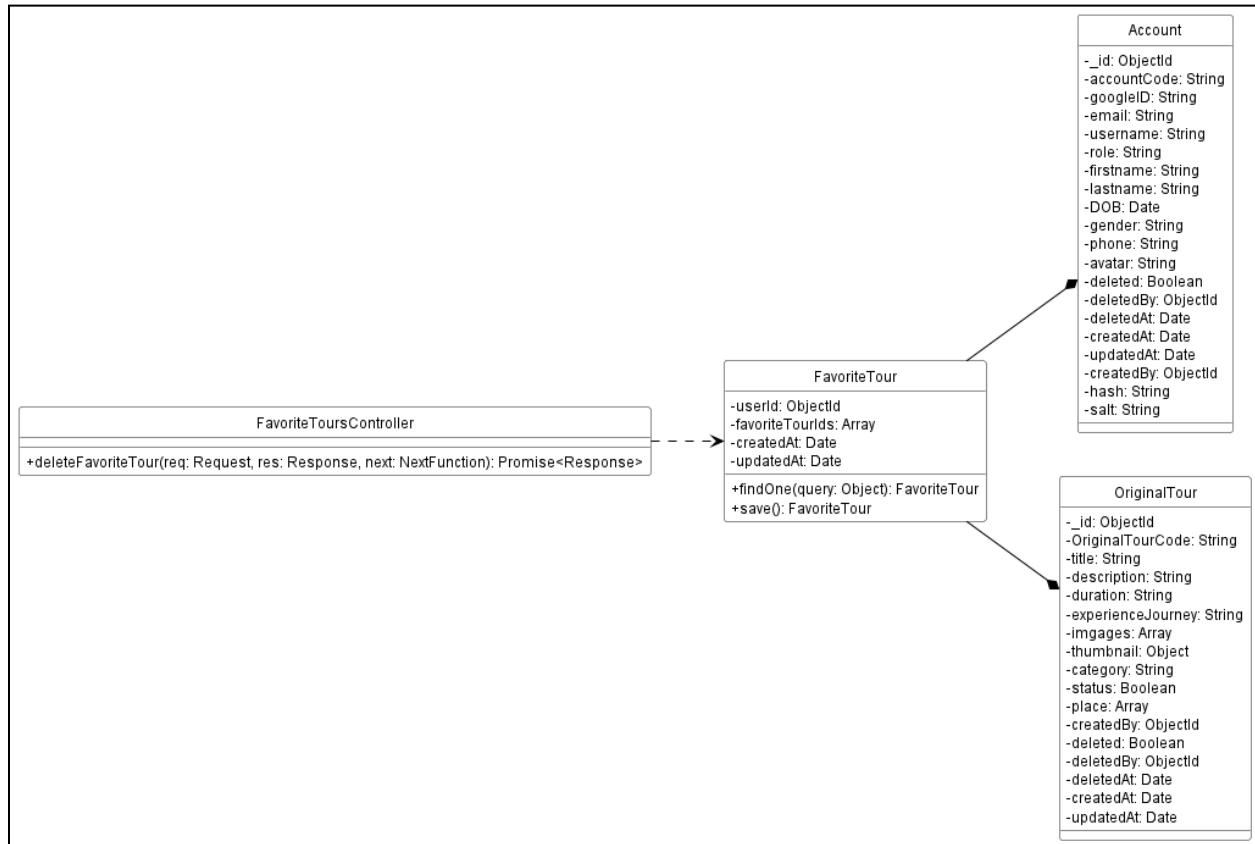


b. Sequence Diagram(s)

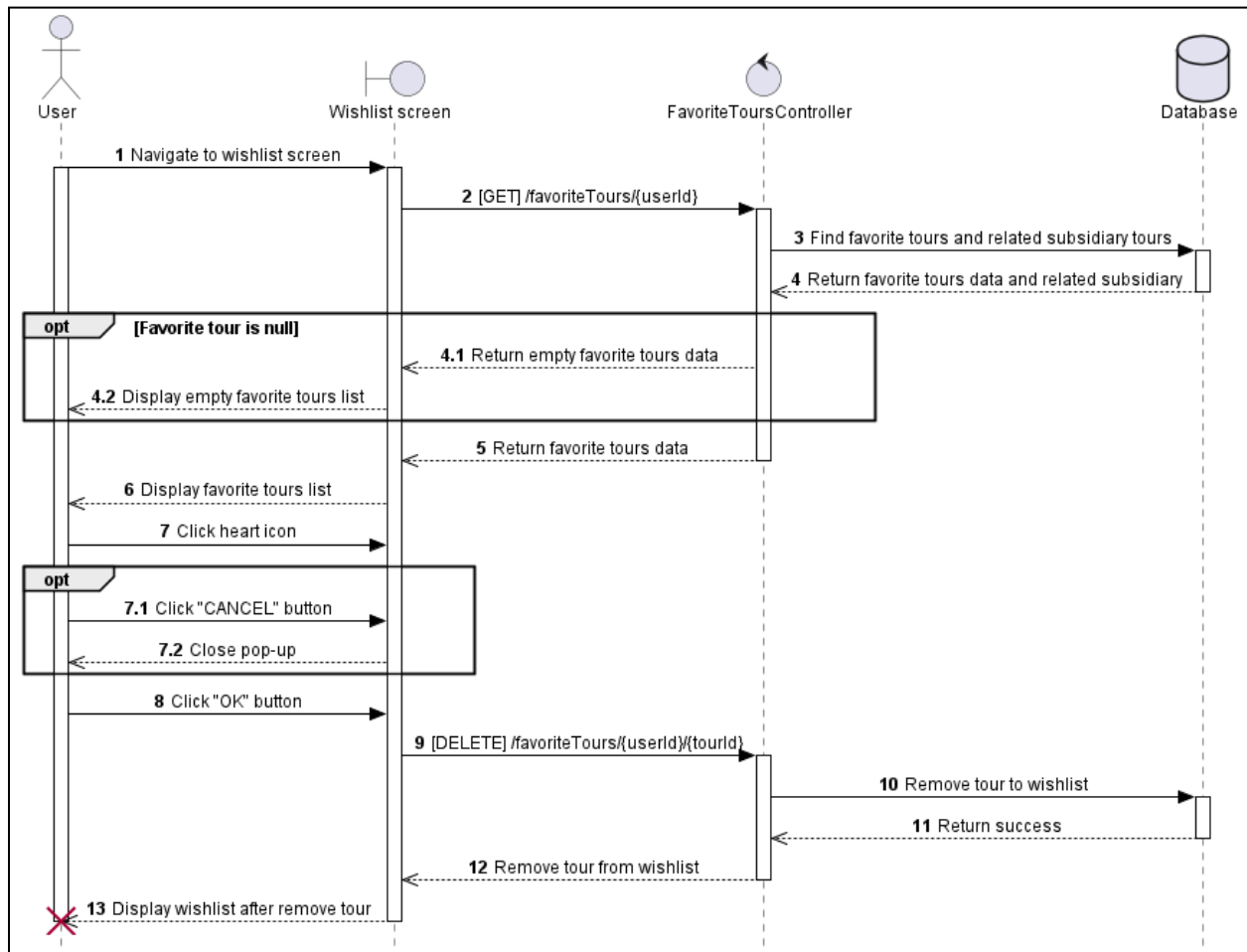


3.21 Remove Favorite Tours

a. Class Diagram

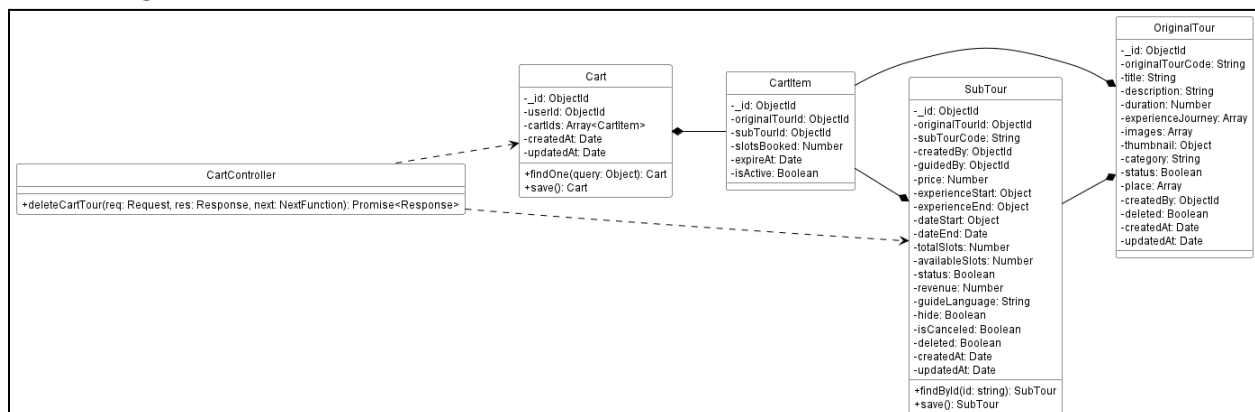


b. Sequence Diagram(s)

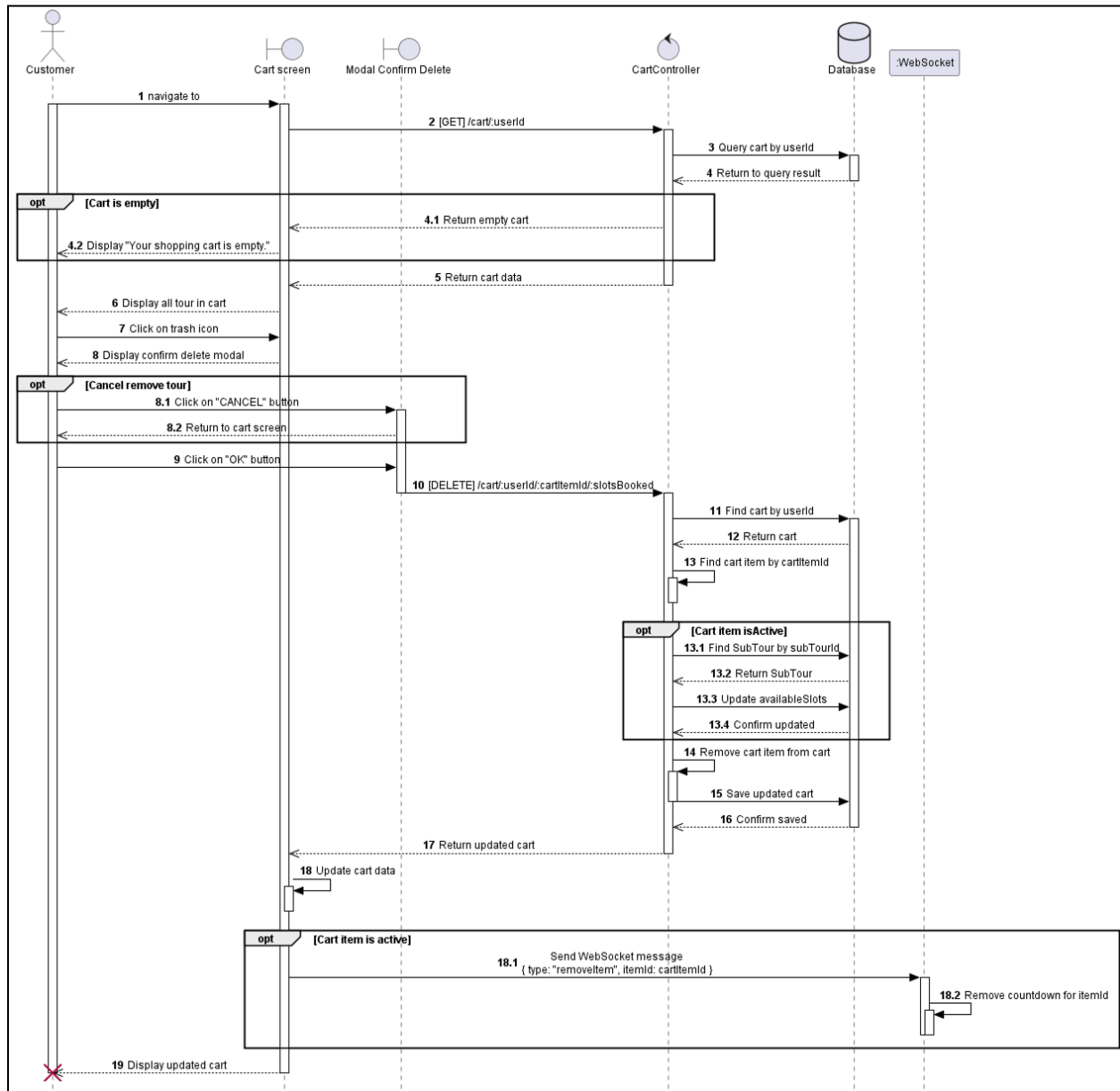


3.22 Remove Cart Tours

a. Class Diagram

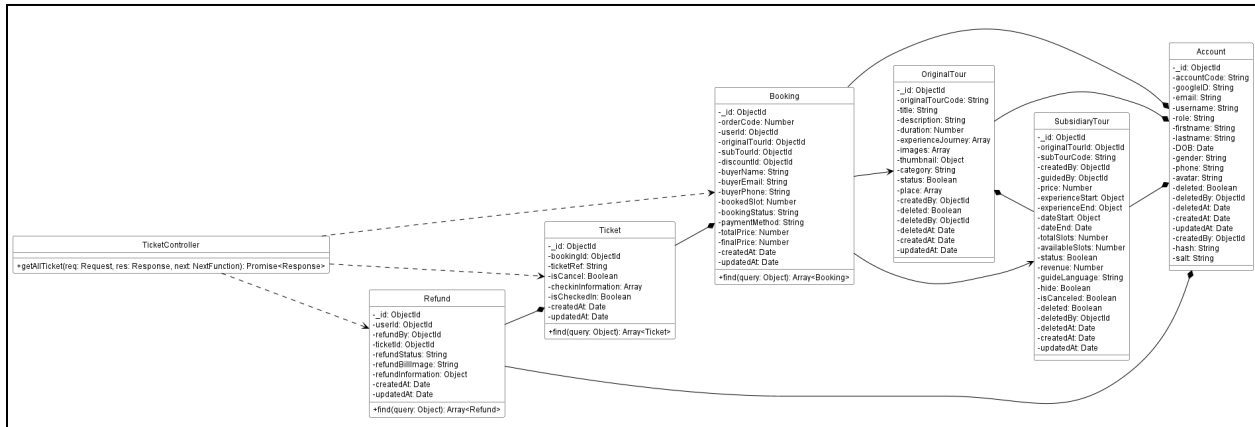


b. Sequence Diagram(s)

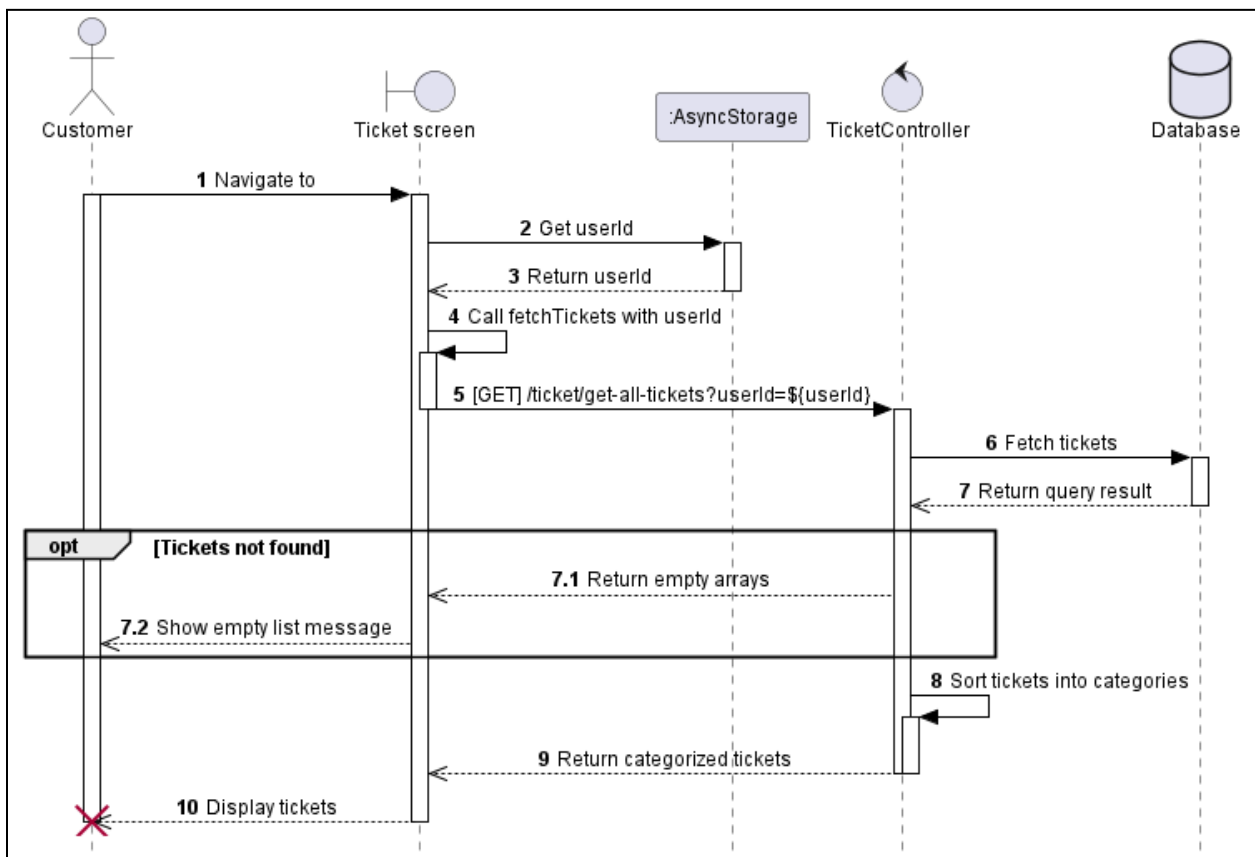


3.23 View Tickets

a. Class Diagram



b. Sequence Diagram(s)

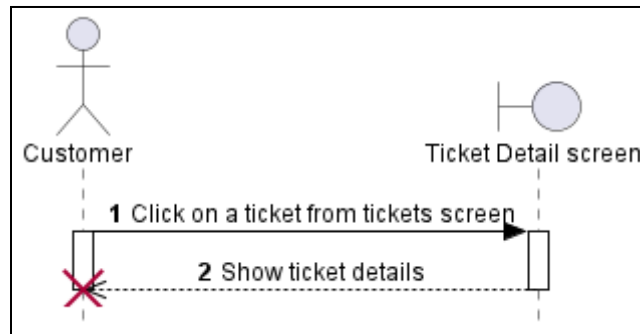


3.24 View Ticket Details

a. Class Diagram

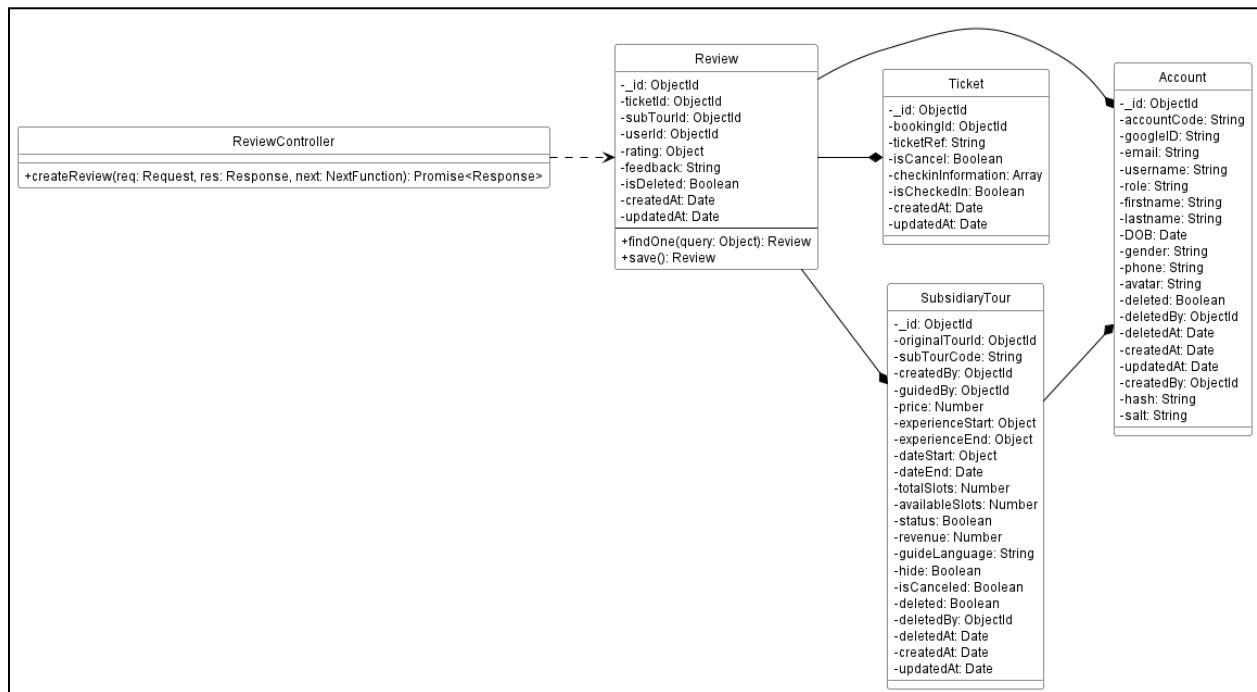
This function does not have an associated class diagram.

b. Sequence Diagram(s)

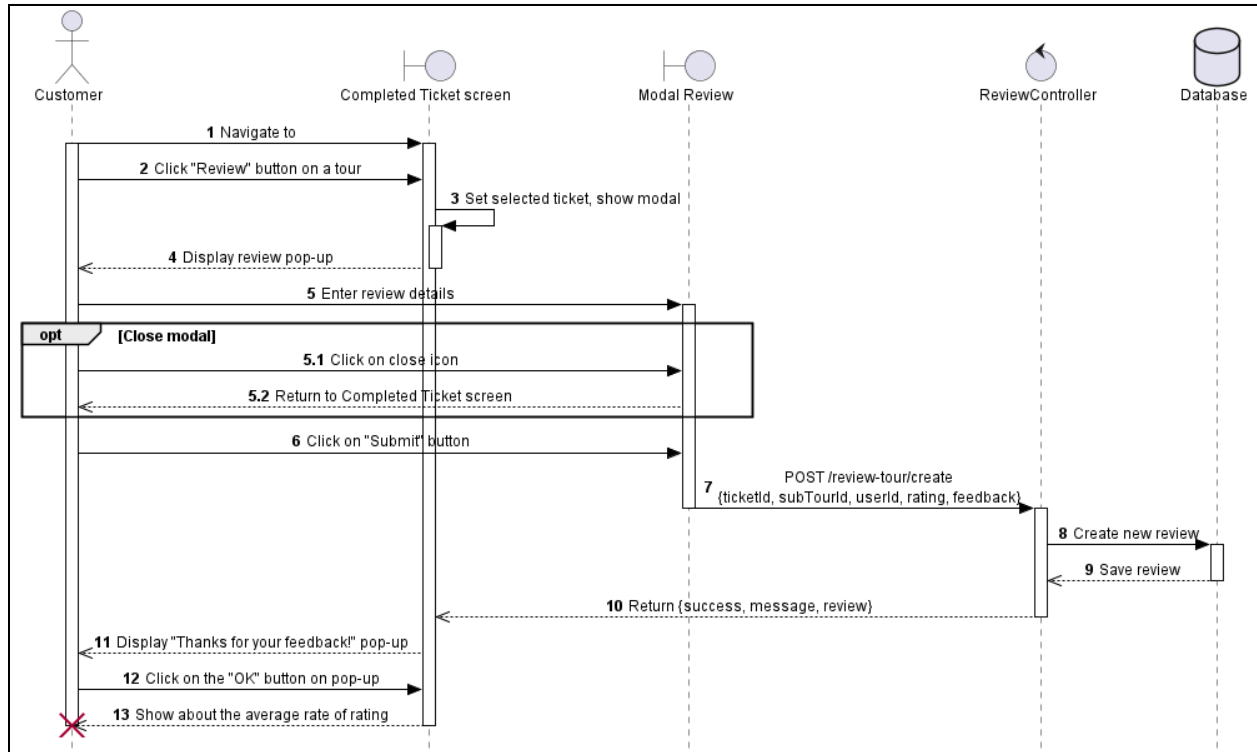


3.25 Review Tour

a. Class Diagram

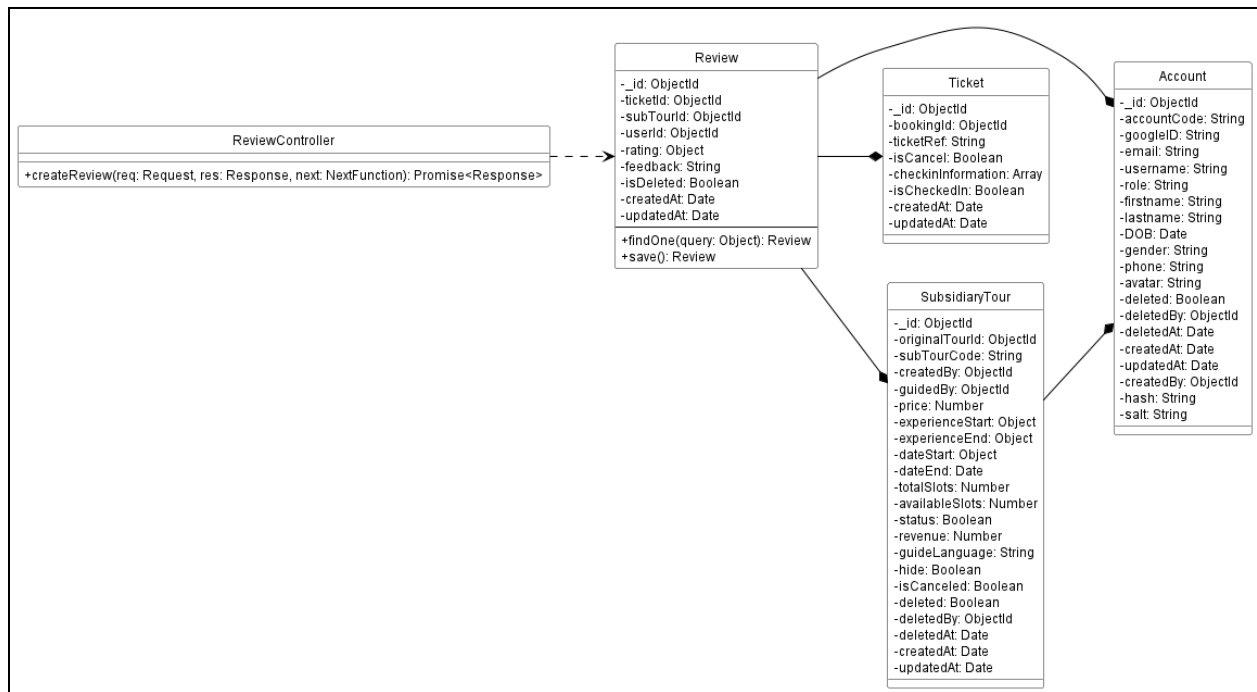


b. Sequence Diagram(s)

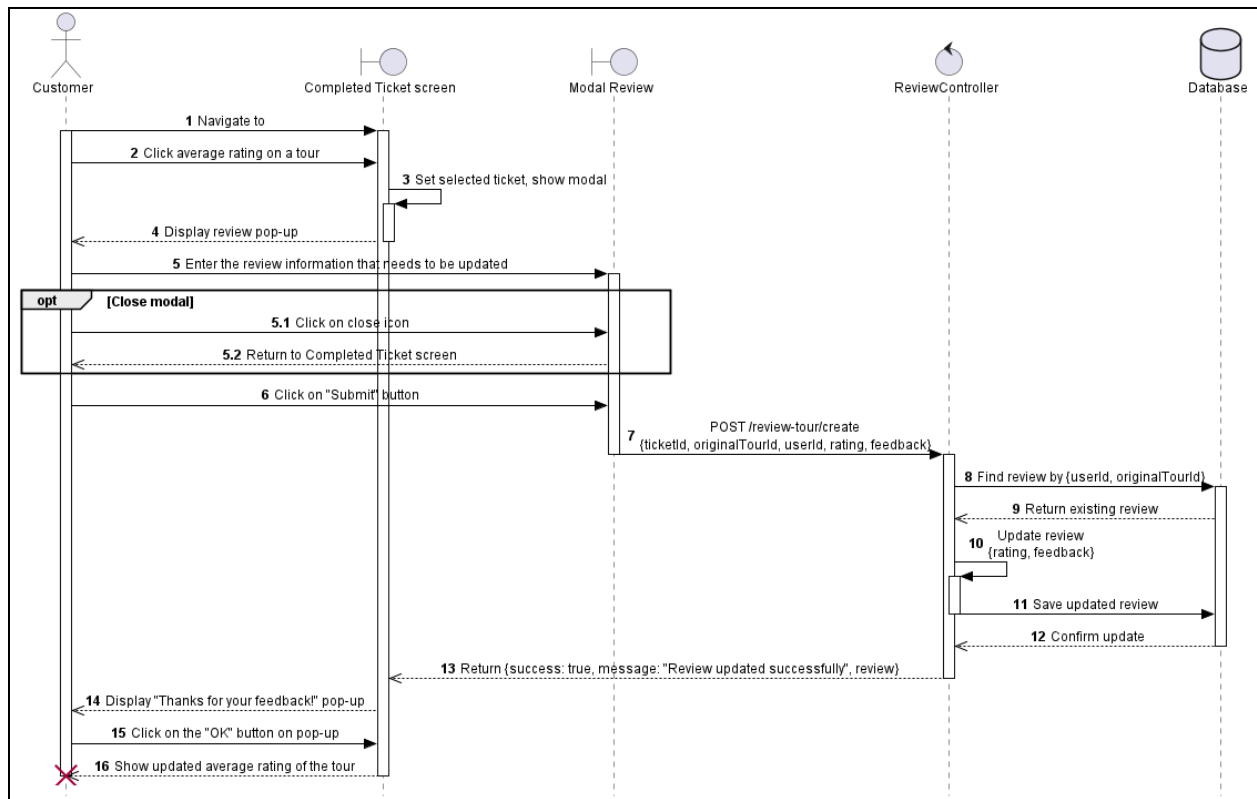


3.26 Update Review Tour

a. Class Diagram

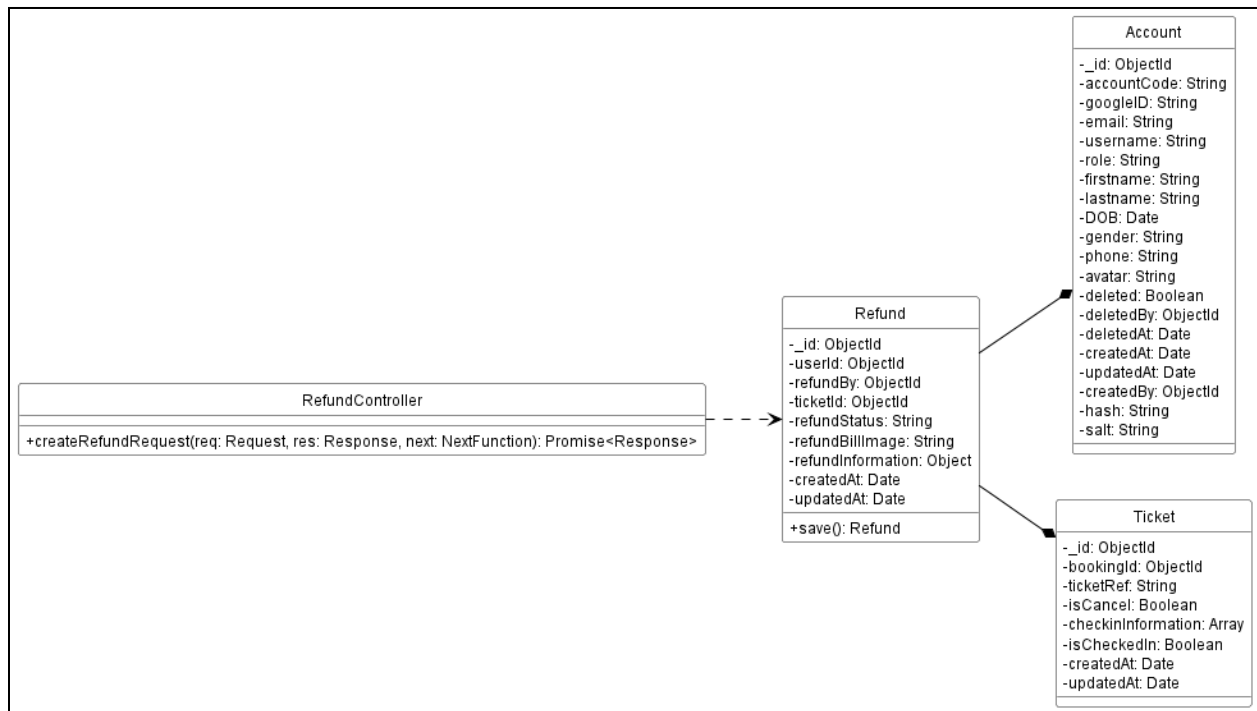


b. Sequence Diagram(s)

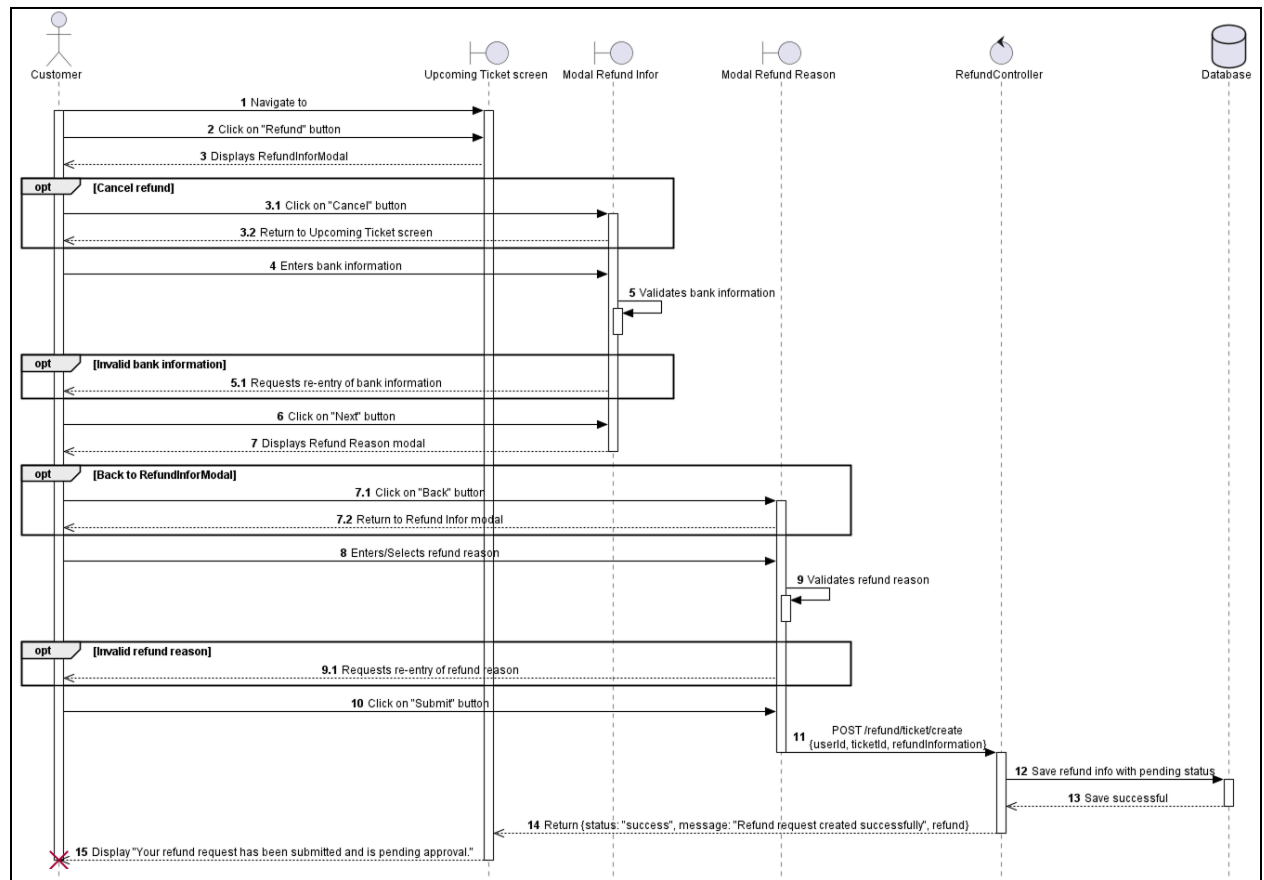


3.27 Send Refund Request

a. Class Diagram

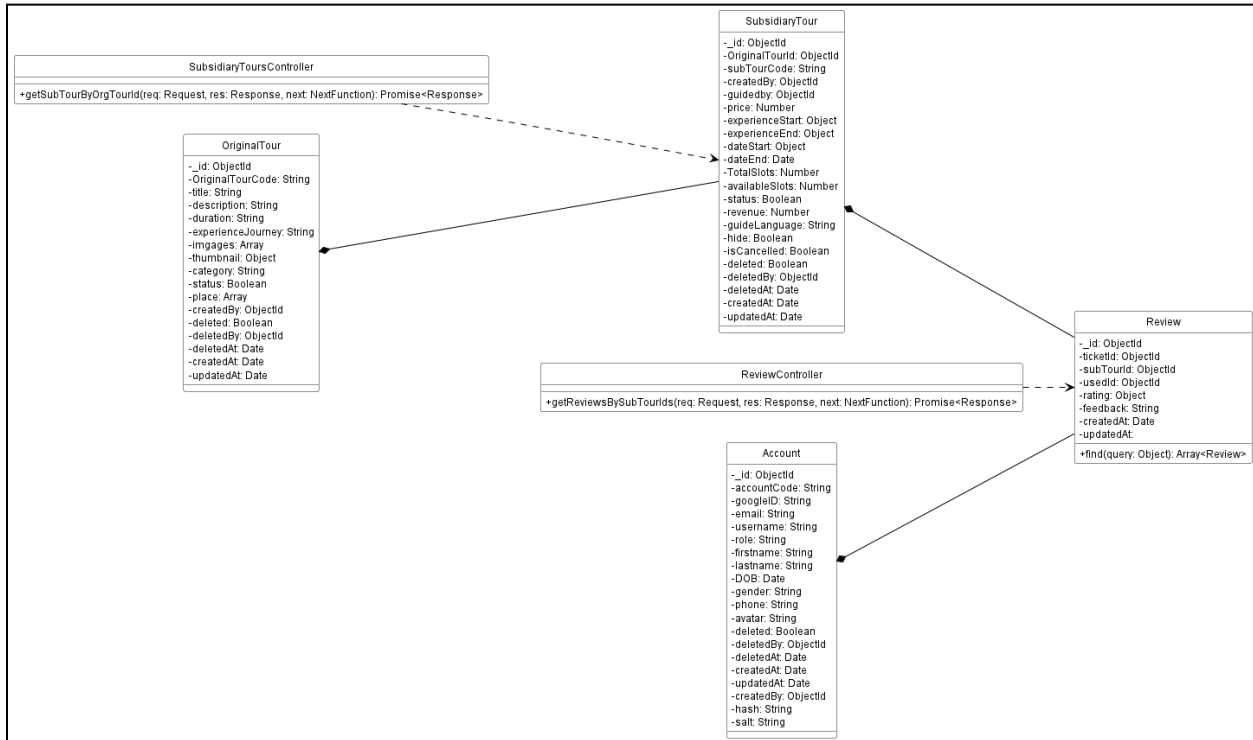


b. Sequence Diagram(s)

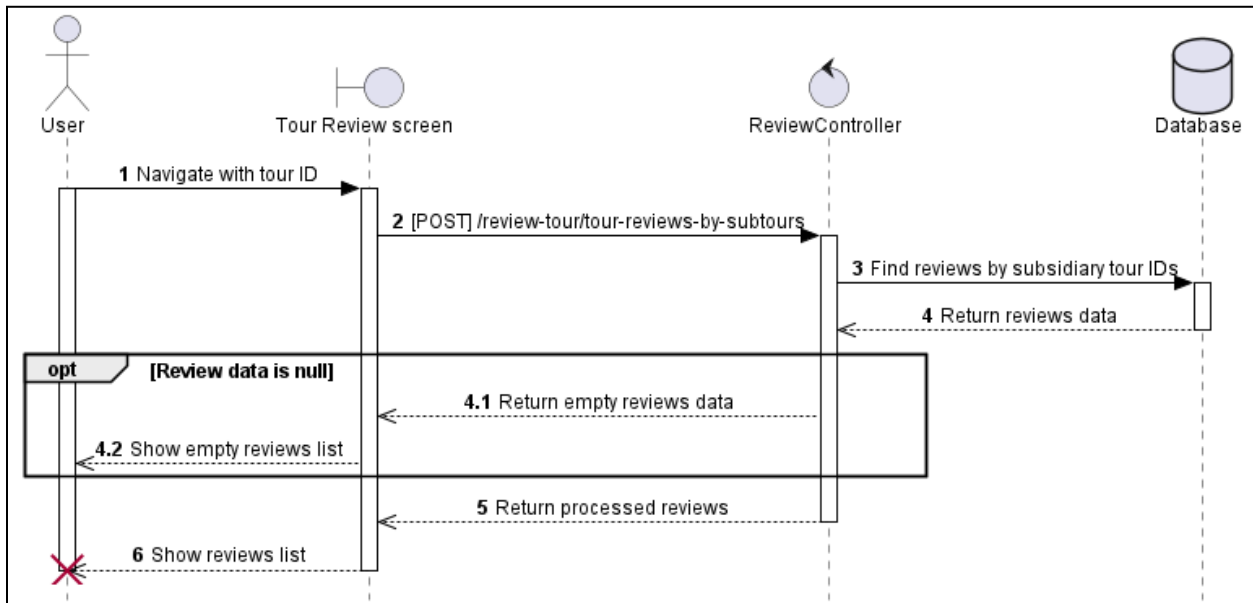


3.28 View Tour Reviews

a. Class Diagram



b. Sequence Diagram(s)

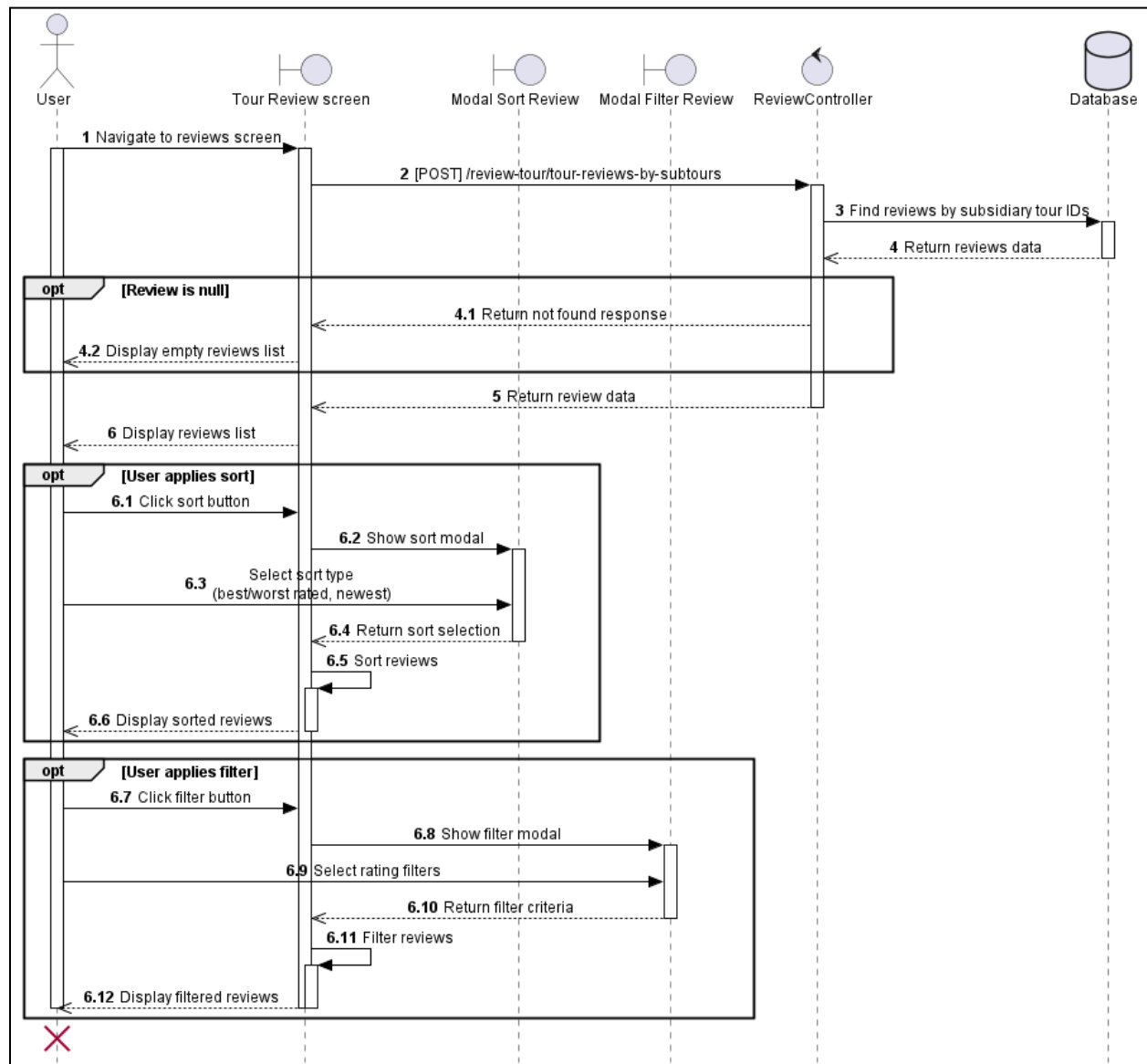


3.29 Filter Tour Reviews

a. Class Diagram

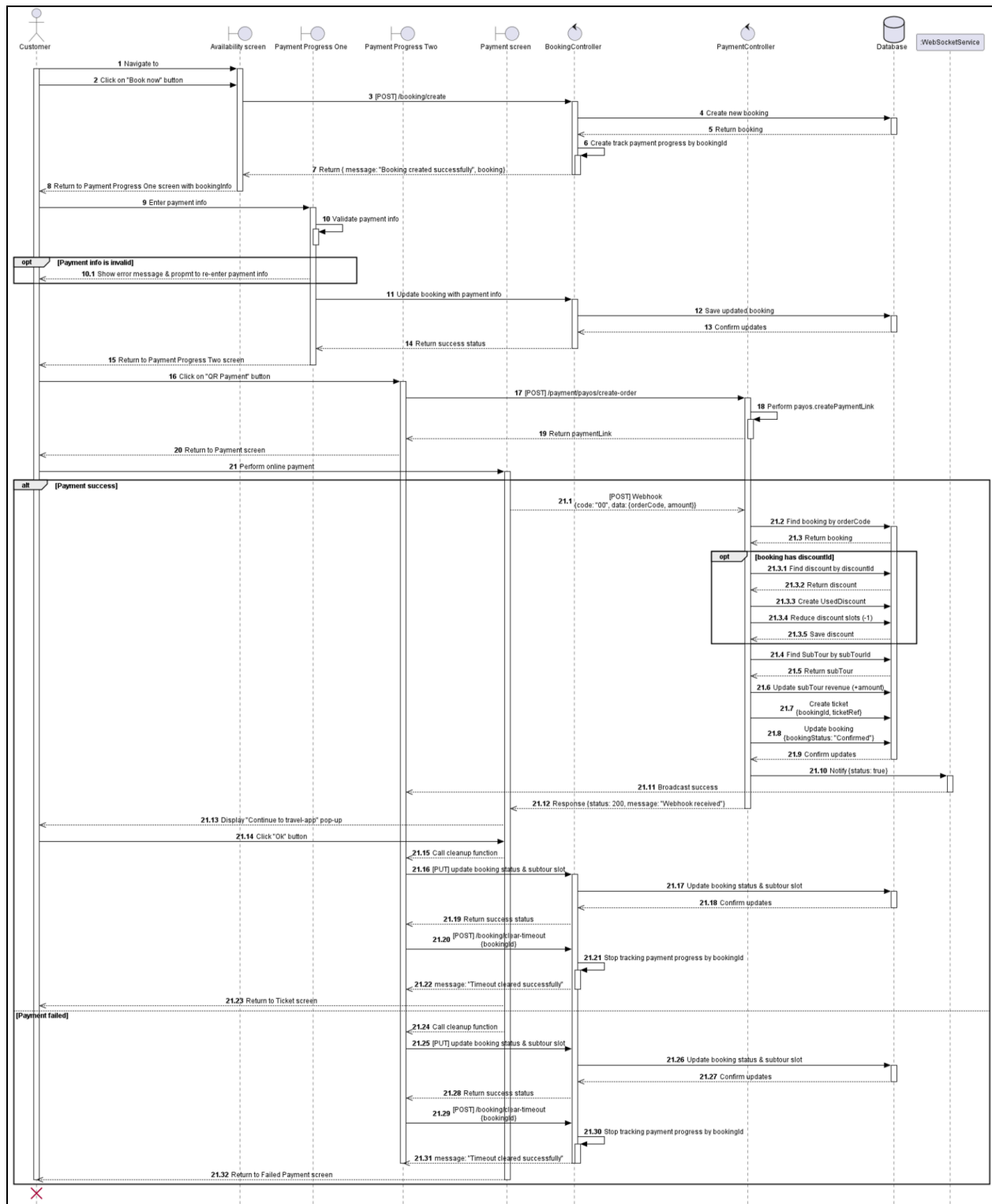
This function does not have an associated class diagram.

b. Sequence Diagram(s)



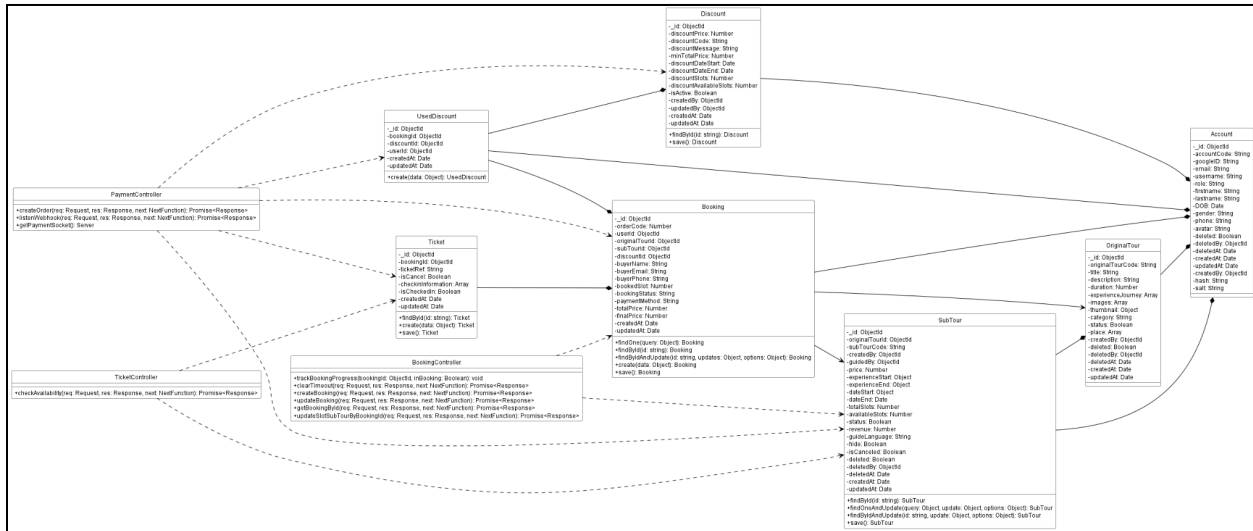
3.30 Book Tour

a. Class Diagram

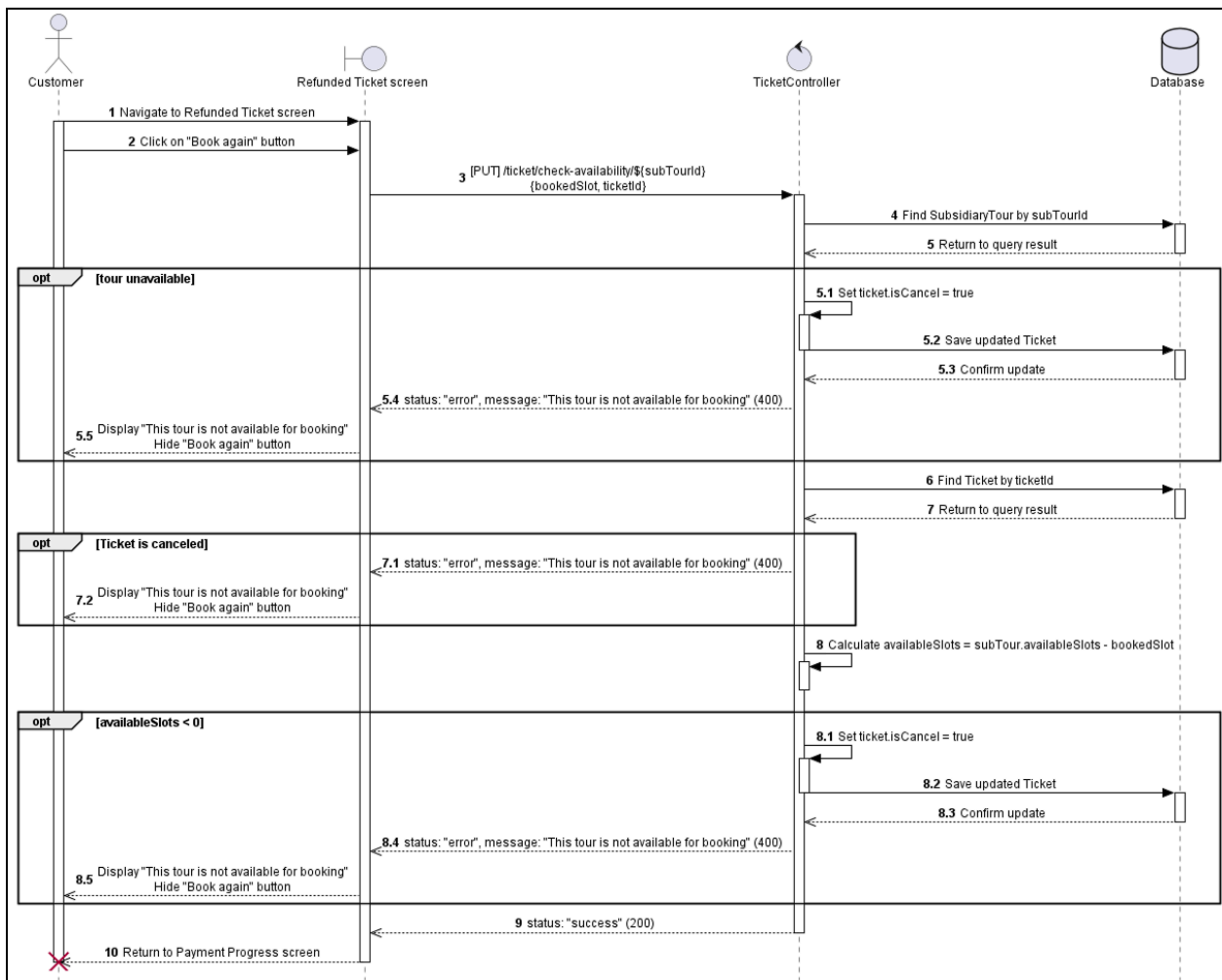


3.31 Book Again

a. Class Diagram



b. Sequence Diagram(s)

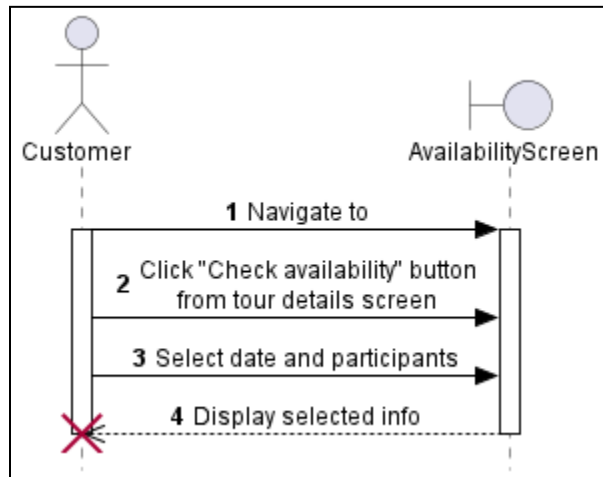


3.32 Check Availability Of A Tour

a. Class Diagram

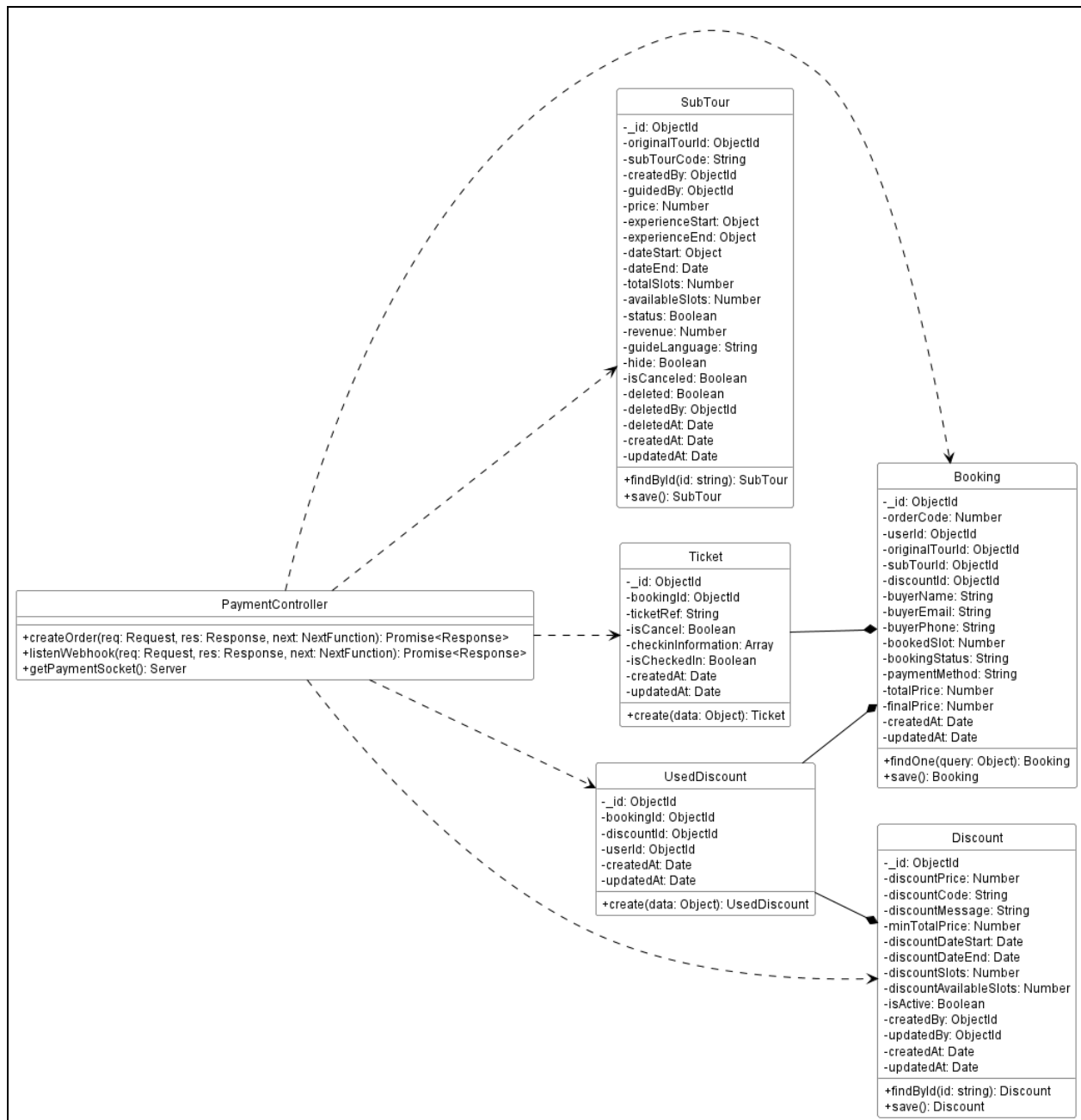
This function does not have an associated class diagram.

b. Sequence Diagram(s)

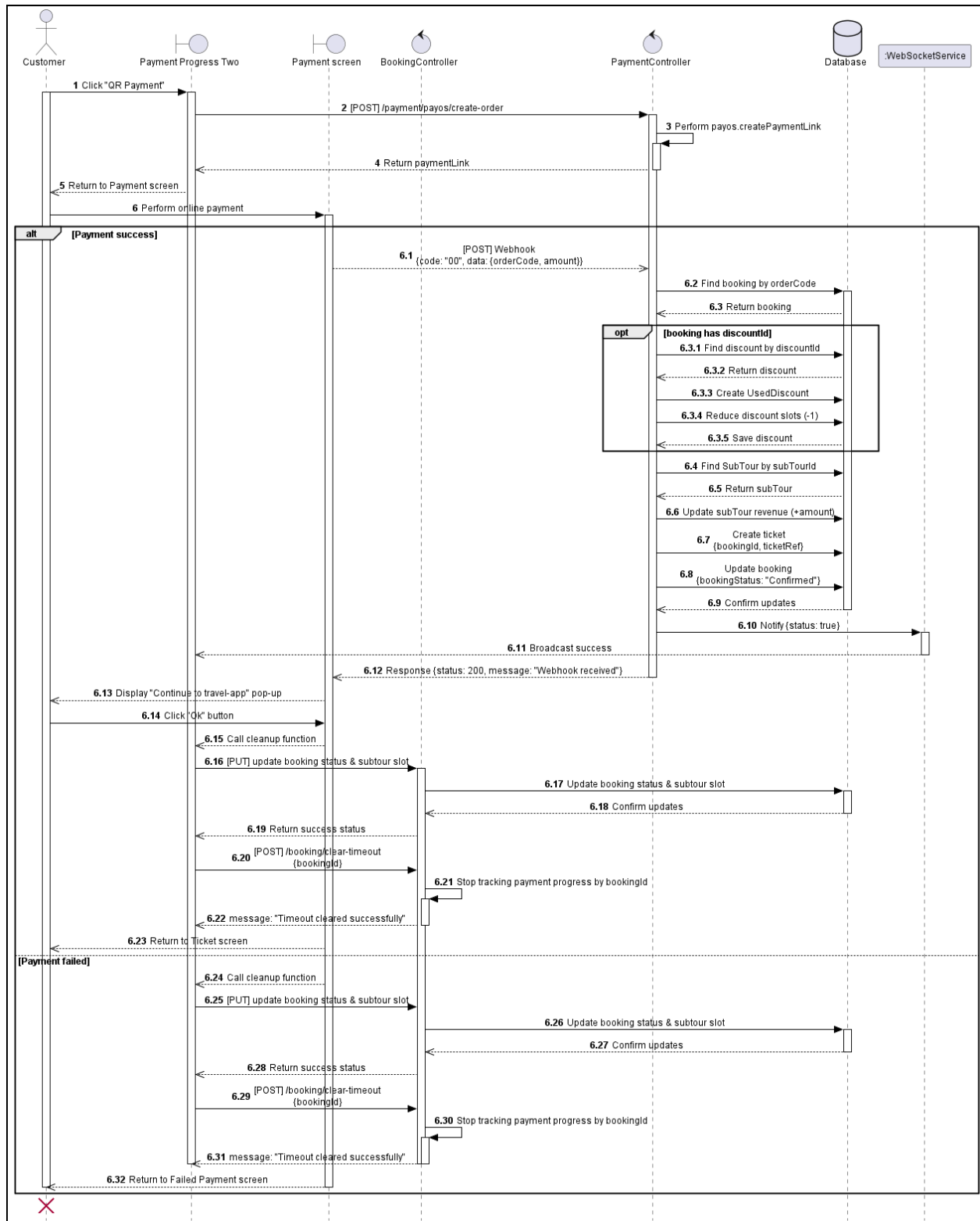


3.33 Pay Online

a. Class Diagram

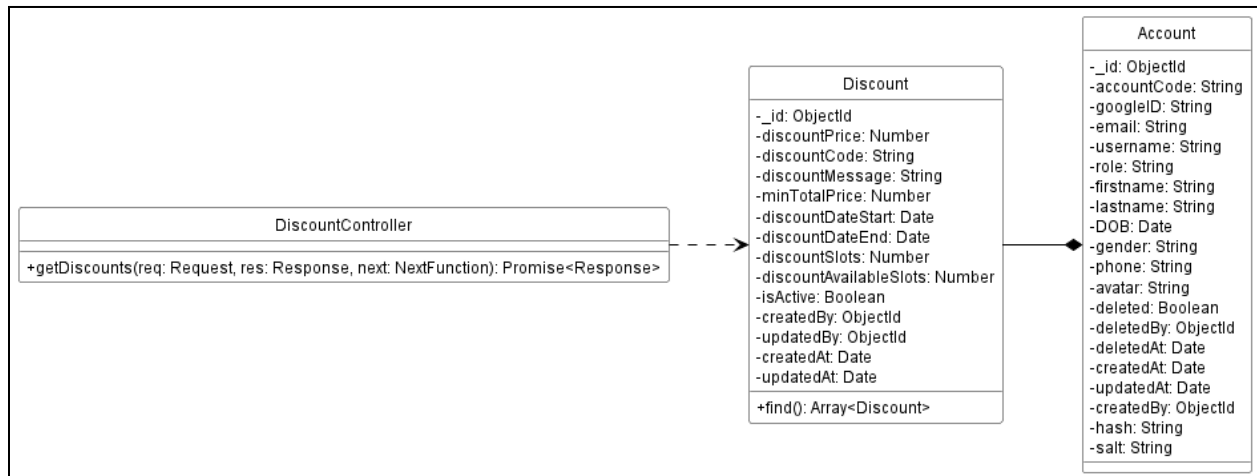


b. Sequence Diagram(s)

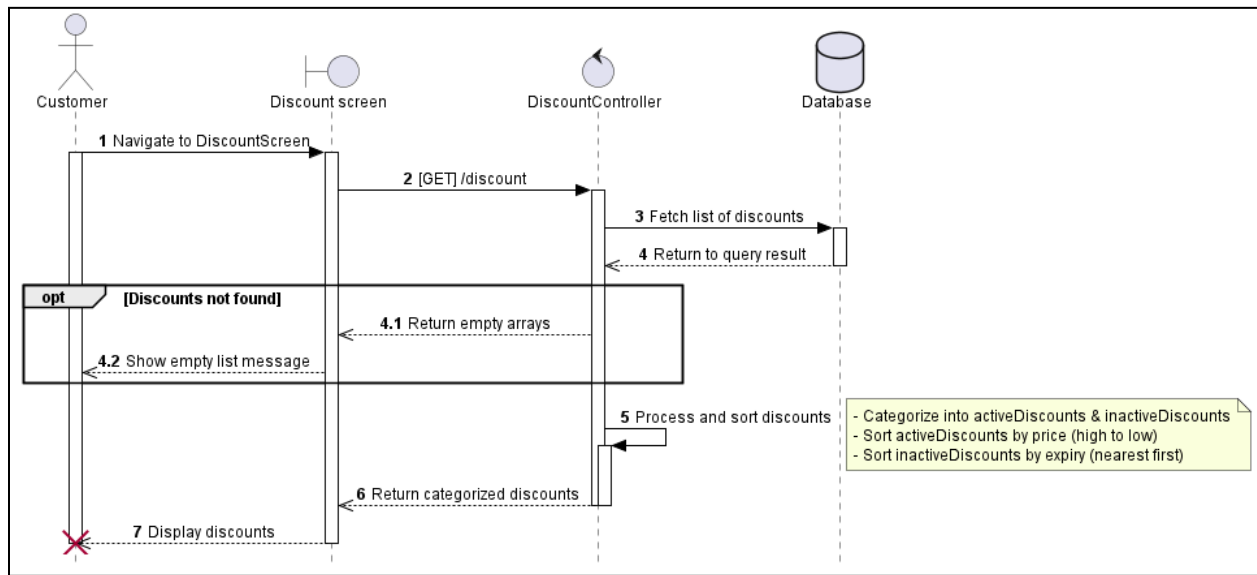


3.34 View Discounts

a. Class Diagram

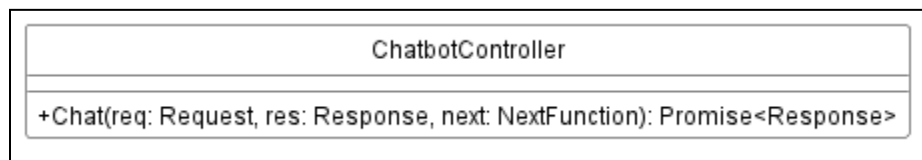


b. Sequence Diagram(s)

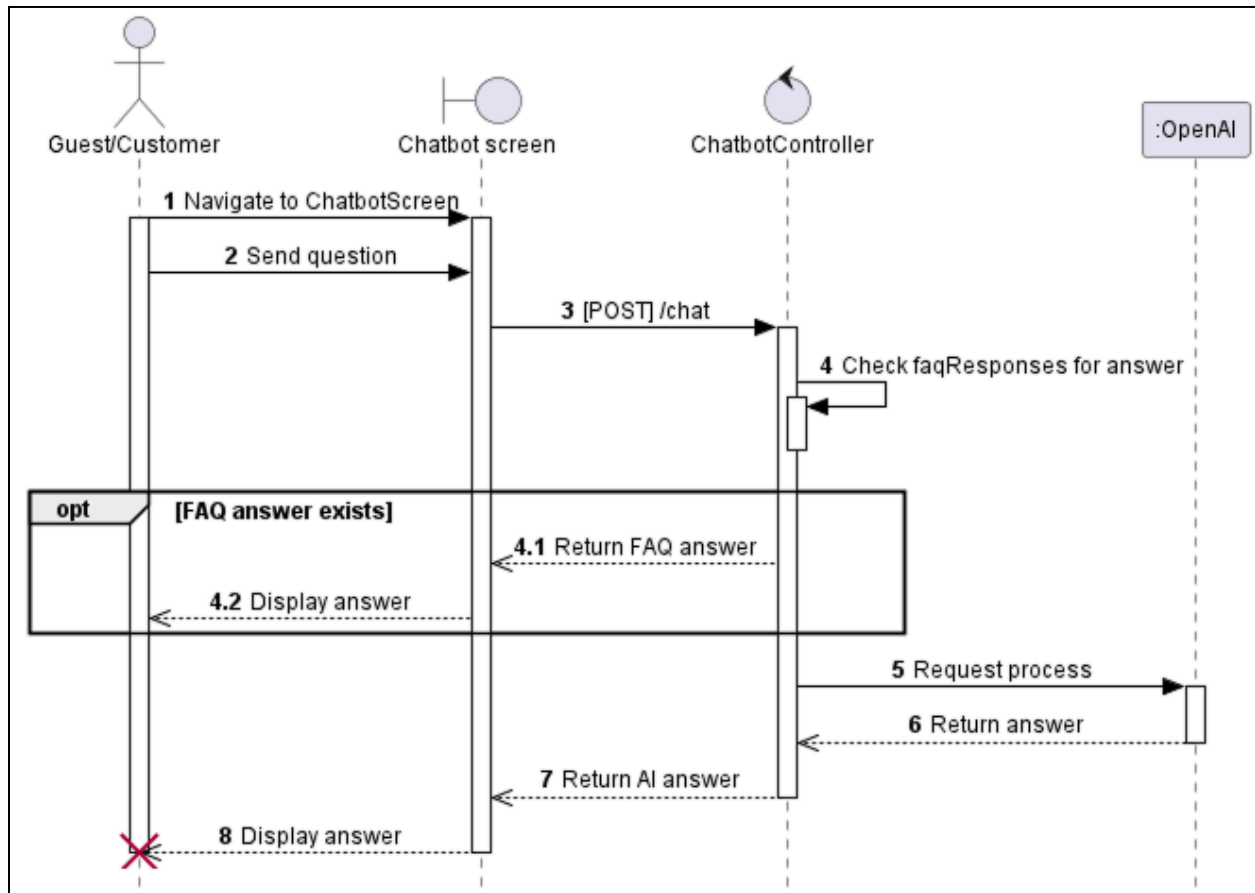


3.35 Chat With Us

a. Class Diagram

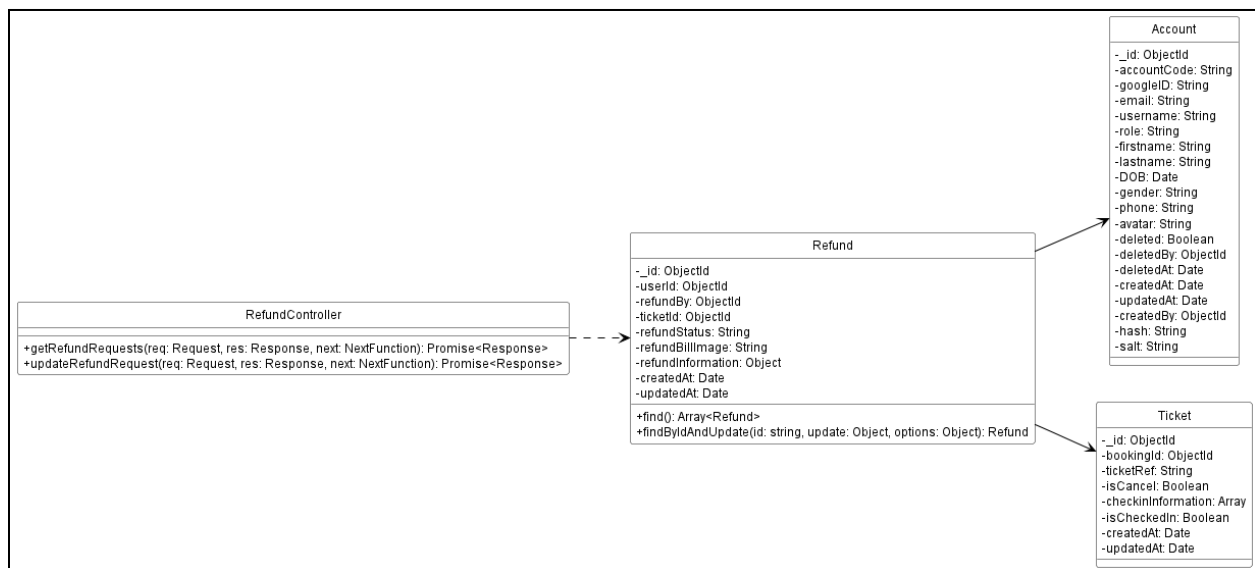


b. Sequence Diagram(s)



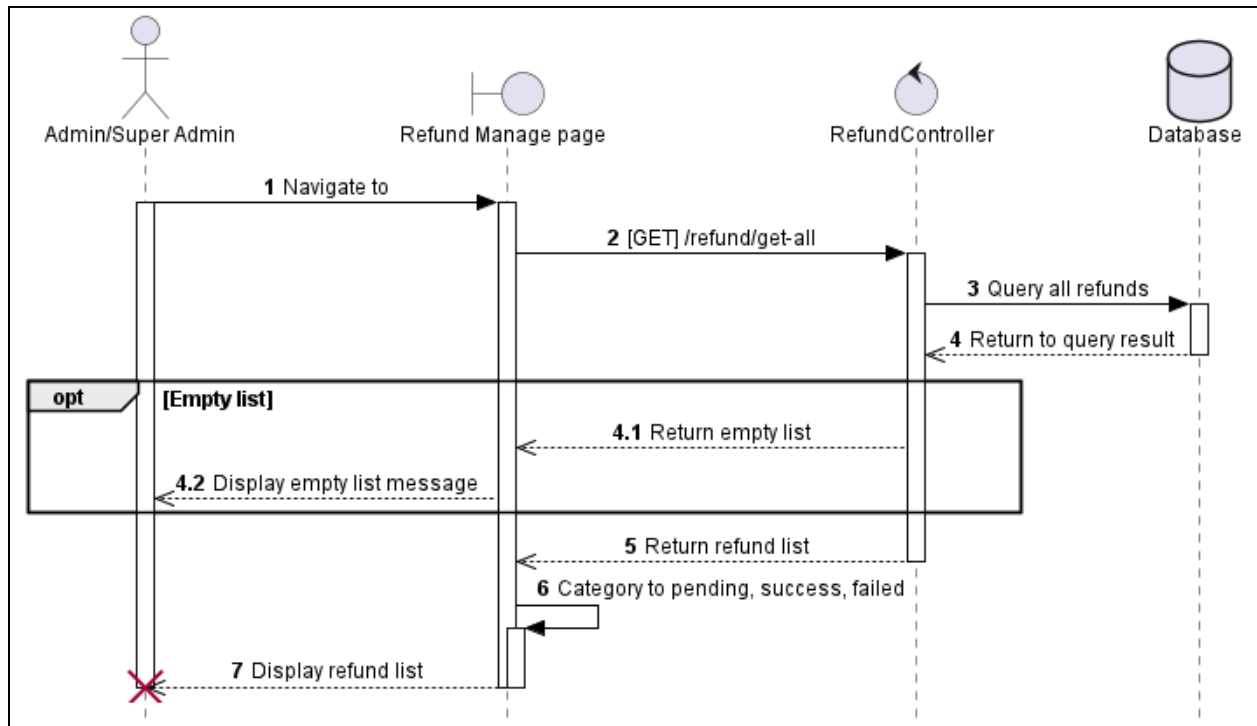
3.36 Manage Refund

a. Class Diagram

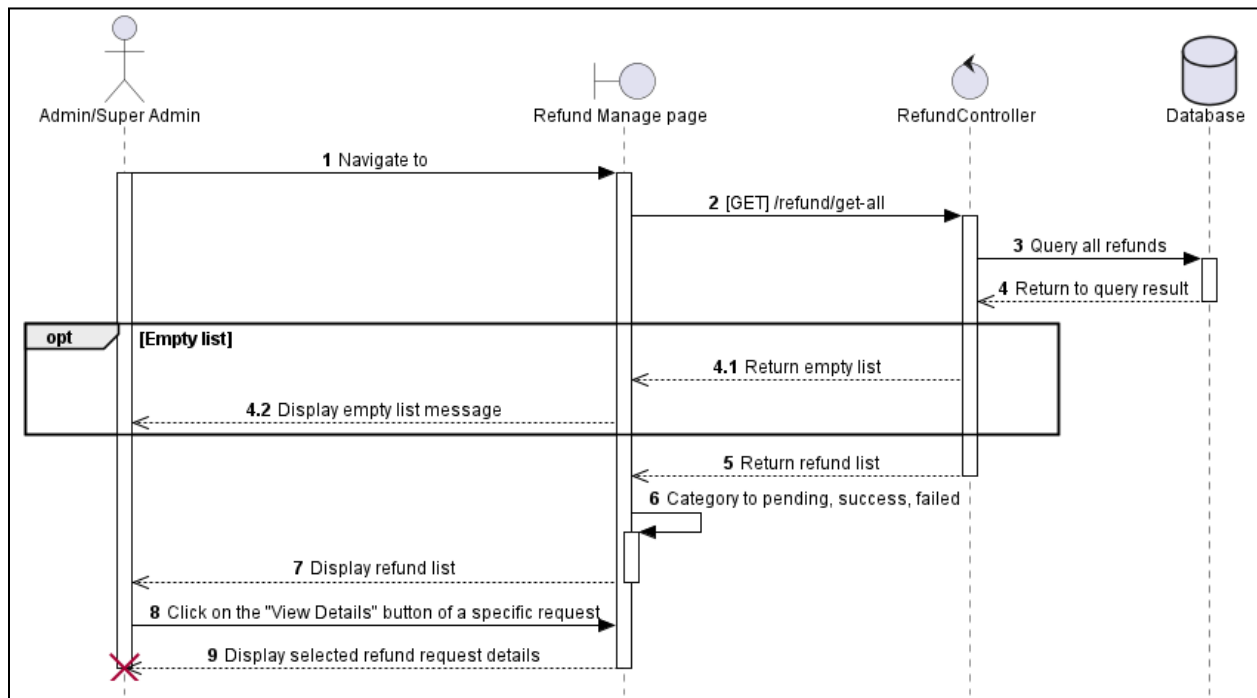


b. Sequence Diagram(s)

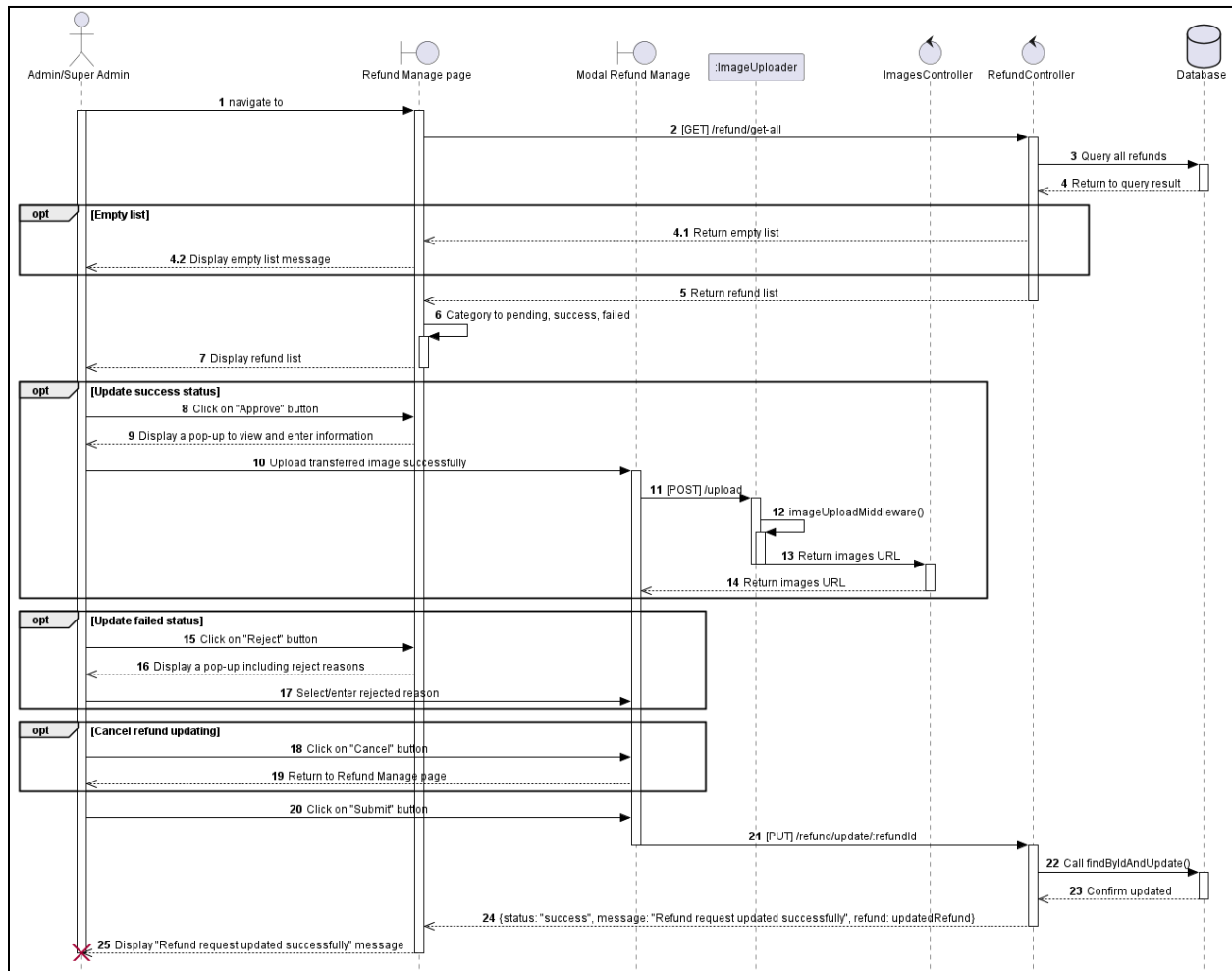
Sequence diagram of View Refund List function:



Sequence diagram of View Refund Details function:

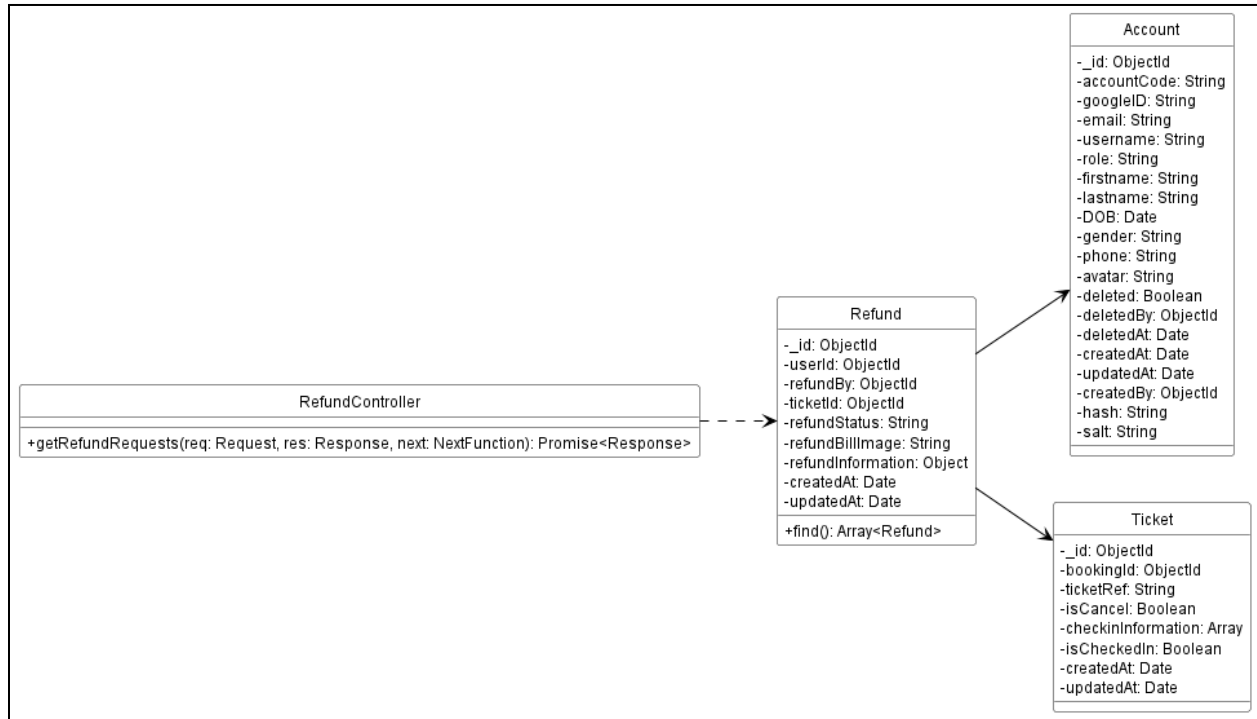


Sequence diagram of Update Refund Request function:



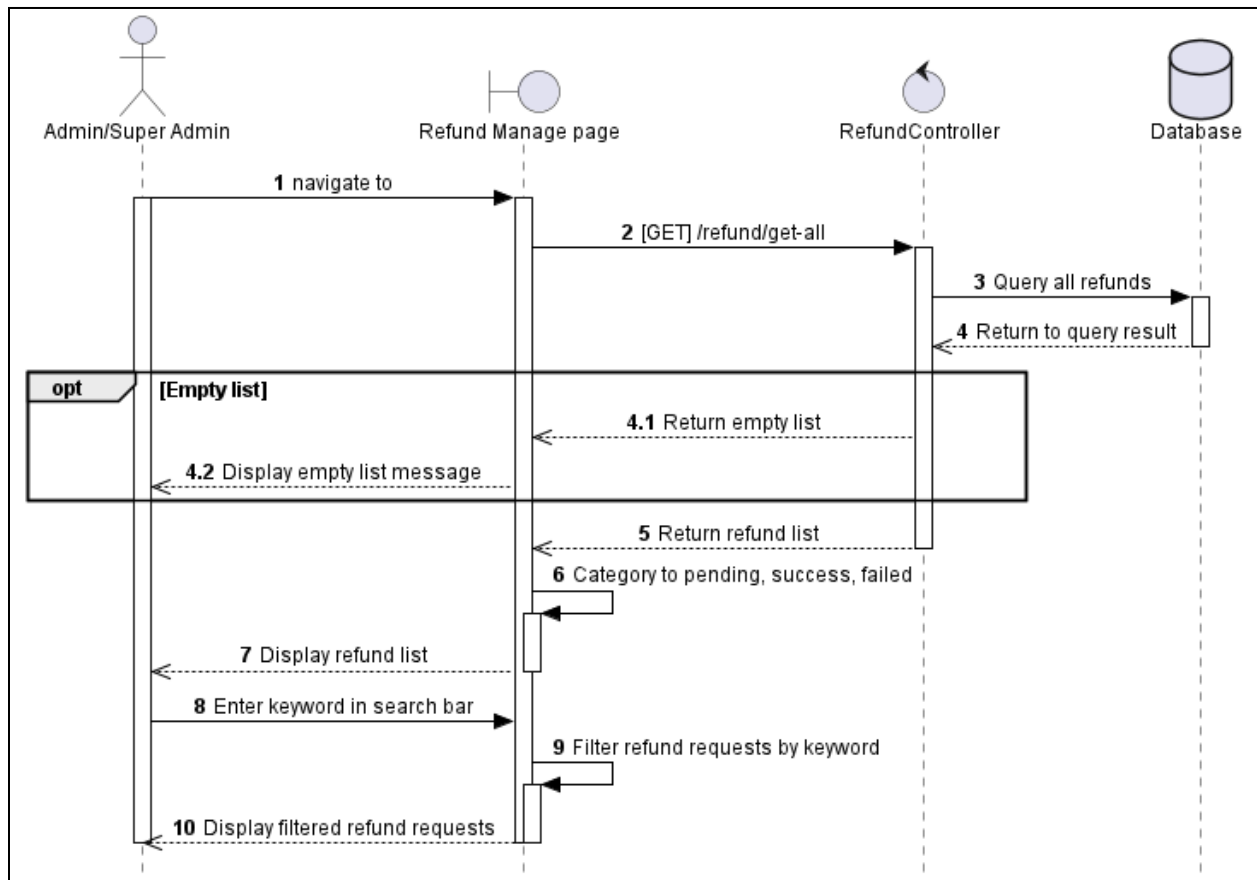
3.37 Search & Filter Refund

a. Class Diagram

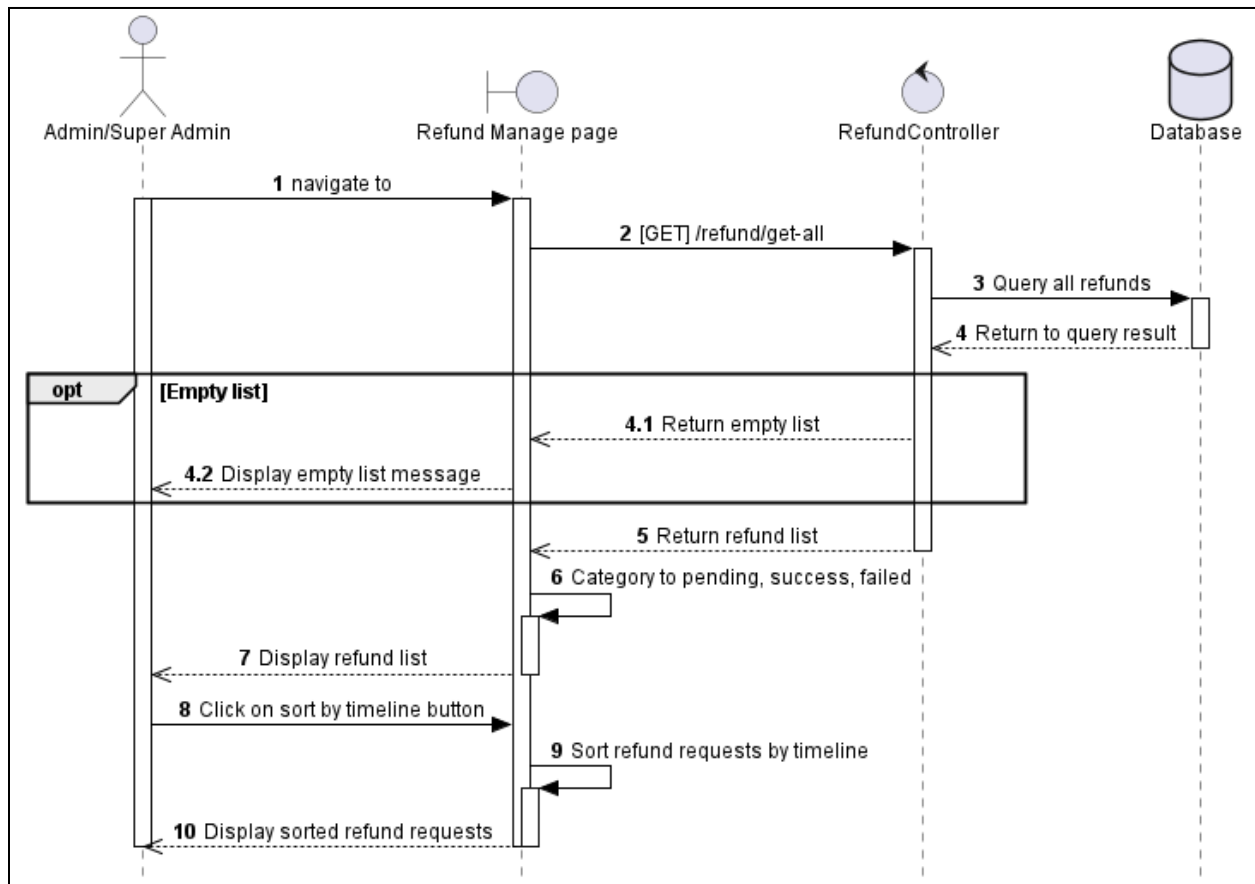


b. Sequence Diagram(s)

Sequence diagram of Search by keyword function:

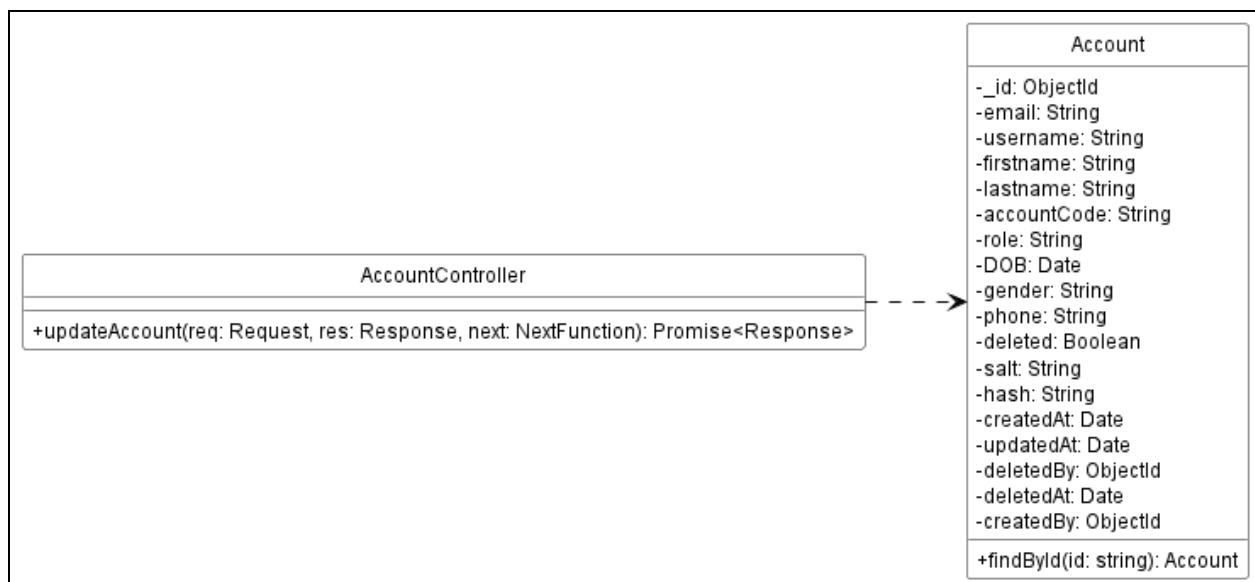


Sequence diagram of Sort by timeline function:



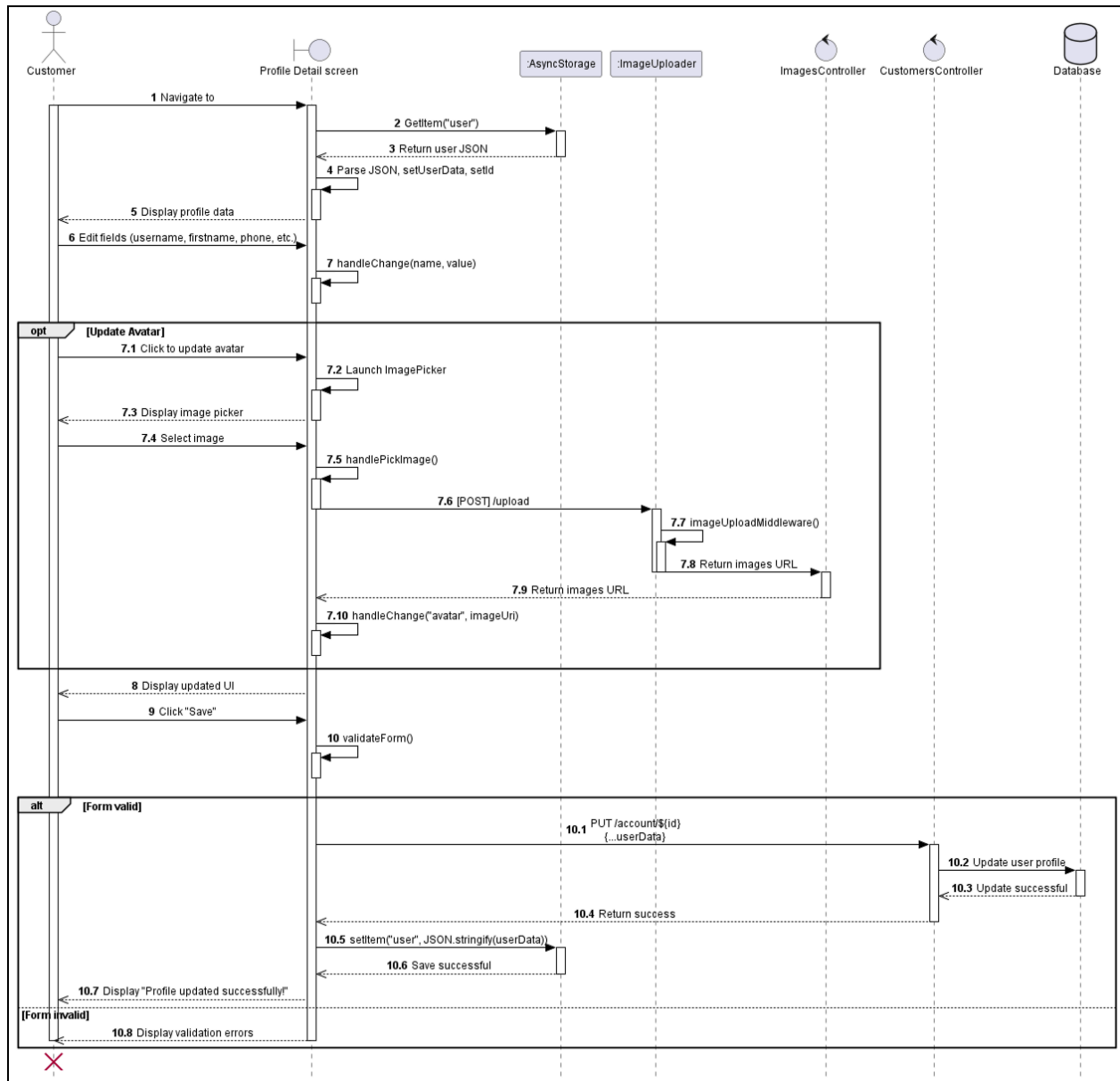
3.38 Update Profile

a. Class Diagram

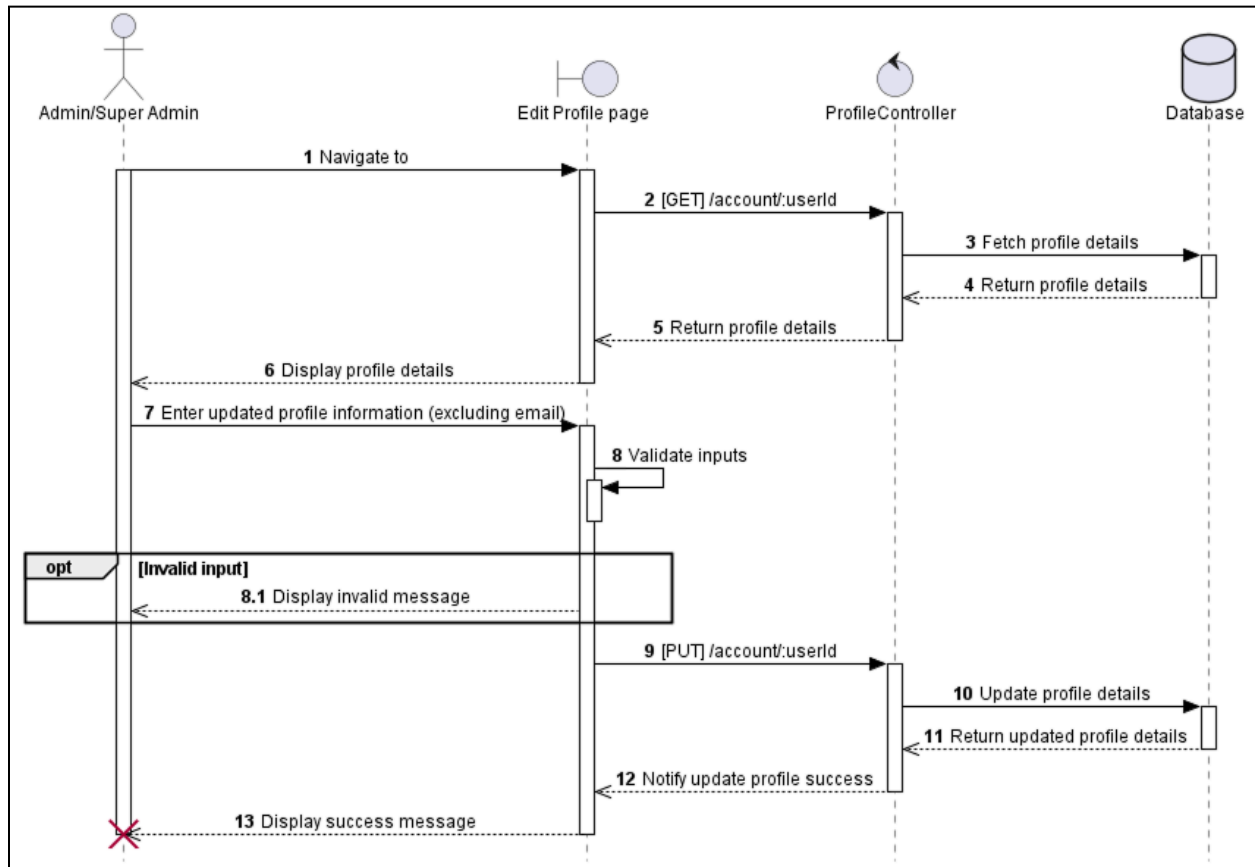


b. Sequence Diagram(s)

* Sequence diagram of Update Profile function on App:

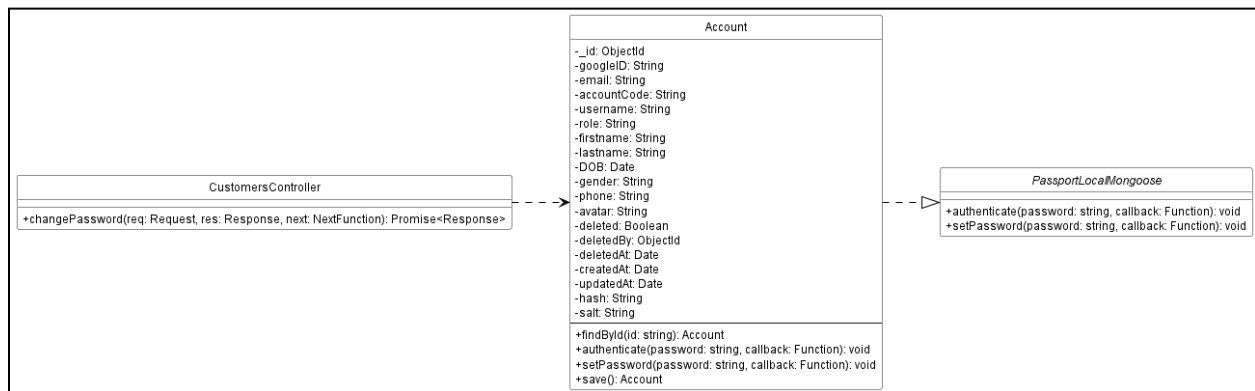


*Sequence diagram of Update Profile function on website:



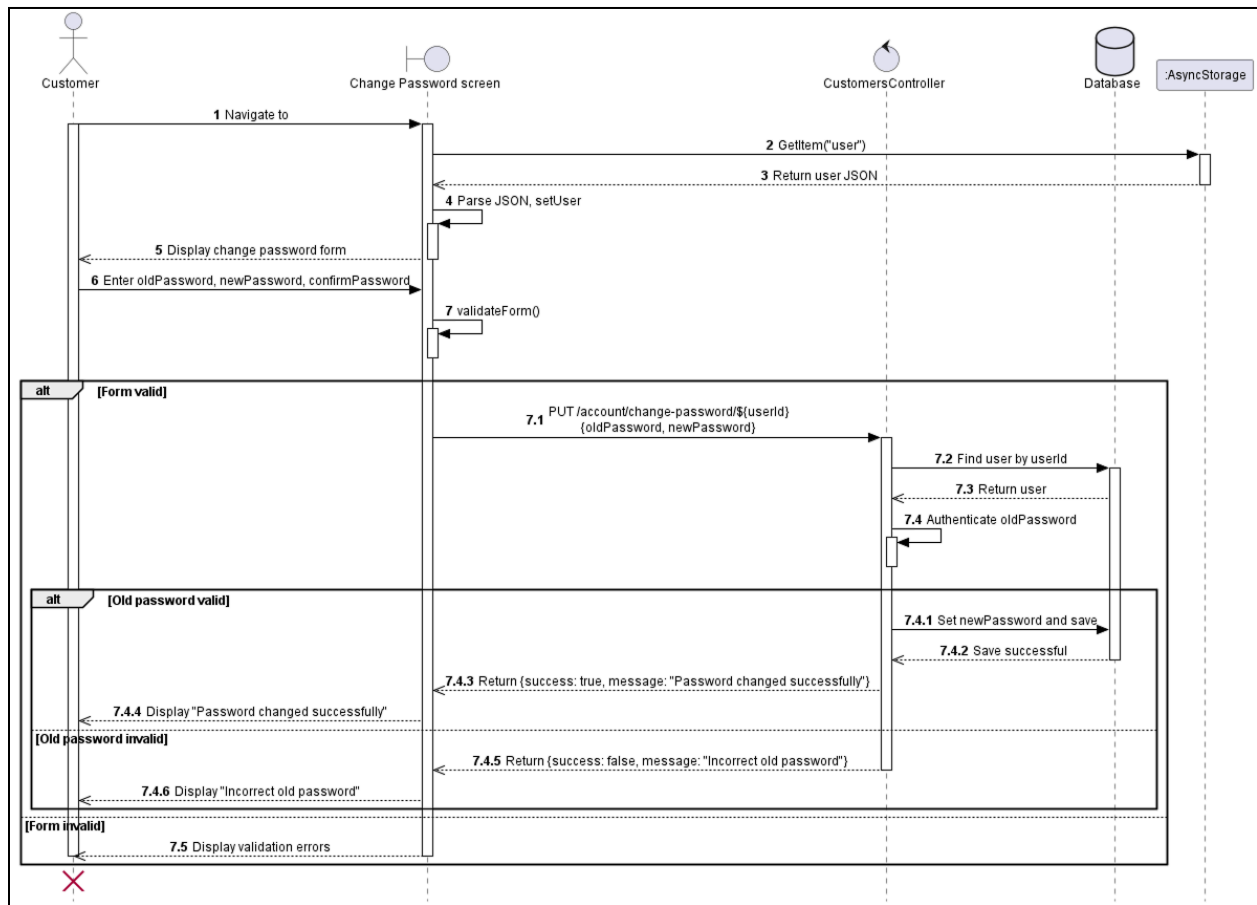
3.39 Change Password

a. Class Diagram

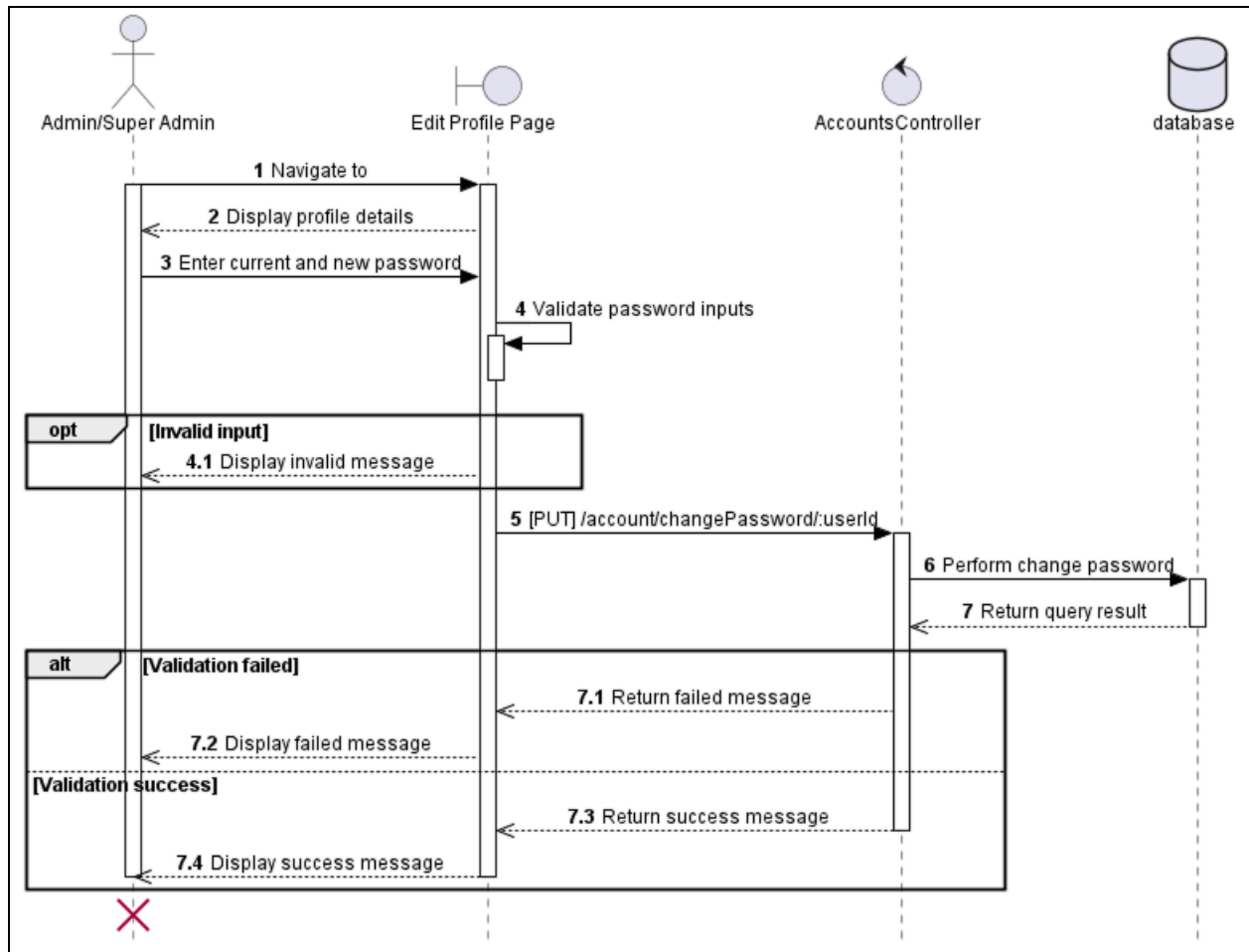


b. Sequence Diagram(s)

* Sequence diagram of Change Password function on App:

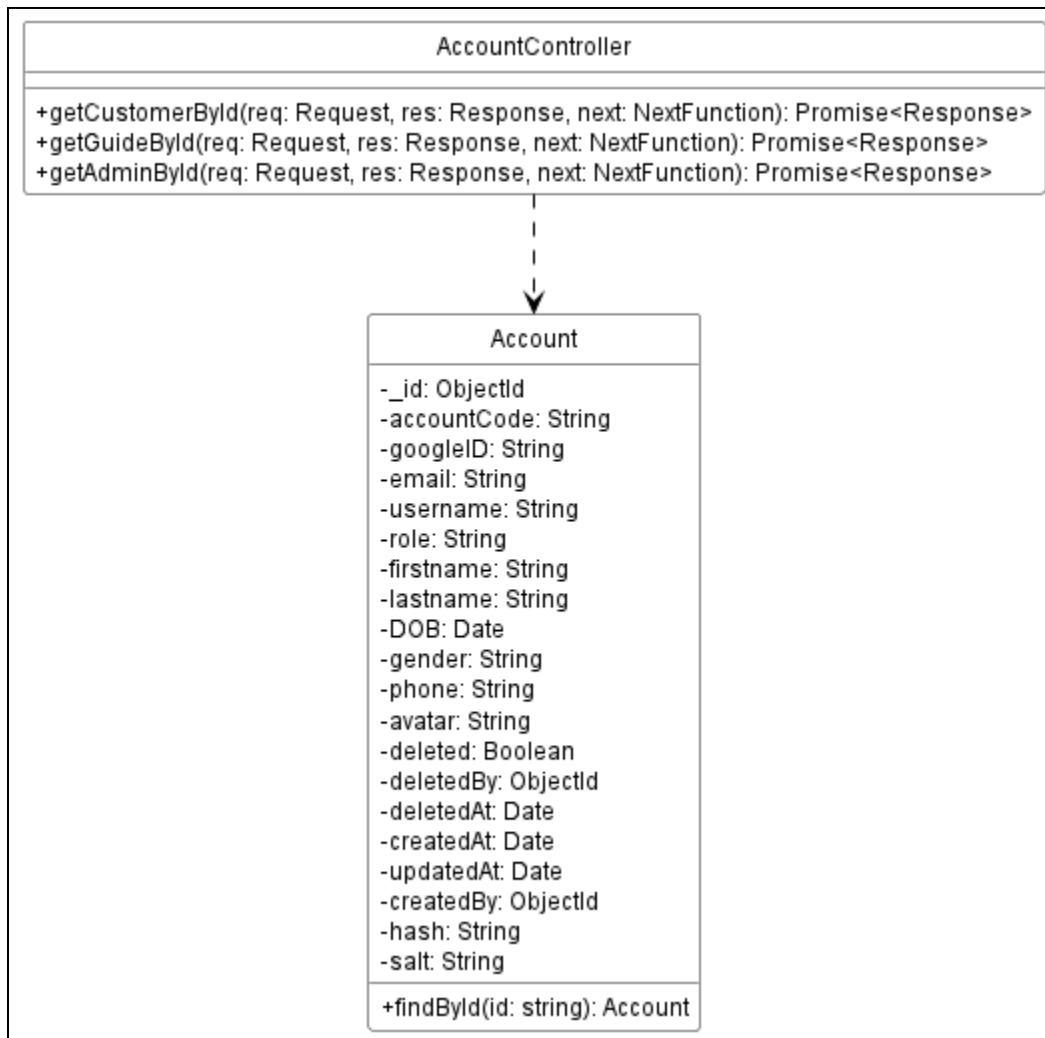


*Sequence diagram of Change Password function on website:



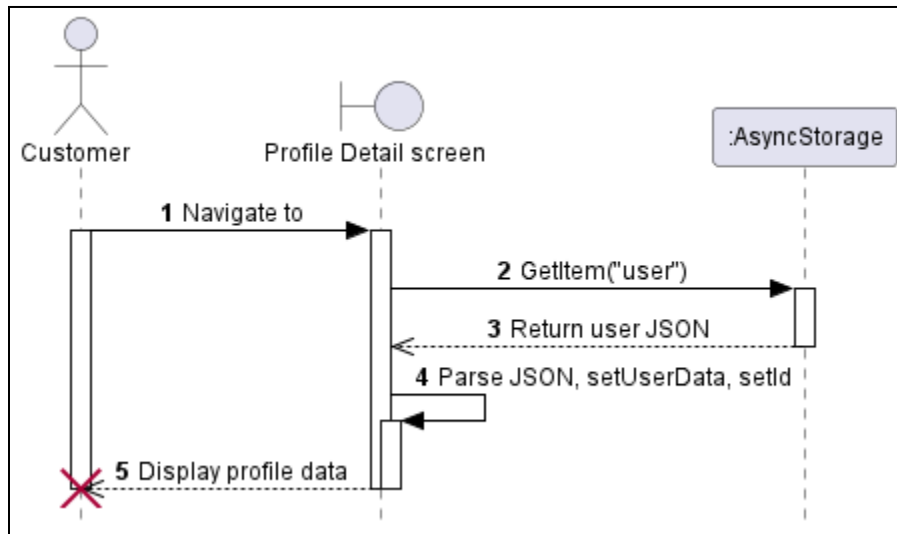
3.40 View Profile

a. Class Diagram

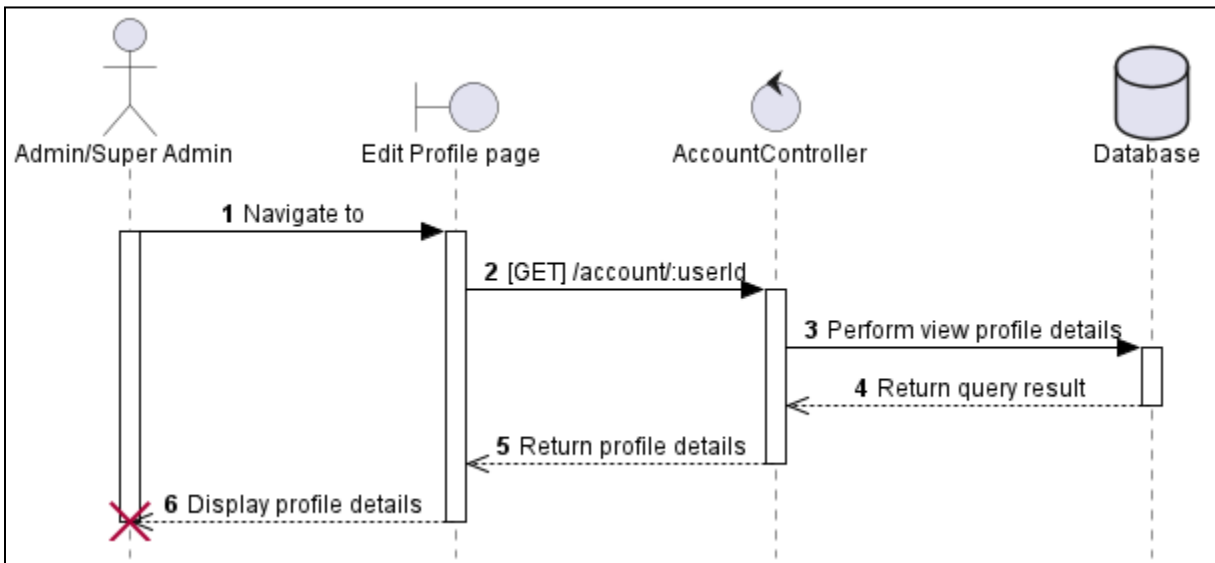


b. Sequence Diagram(s)

* Sequence diagram of View Profile function on App:

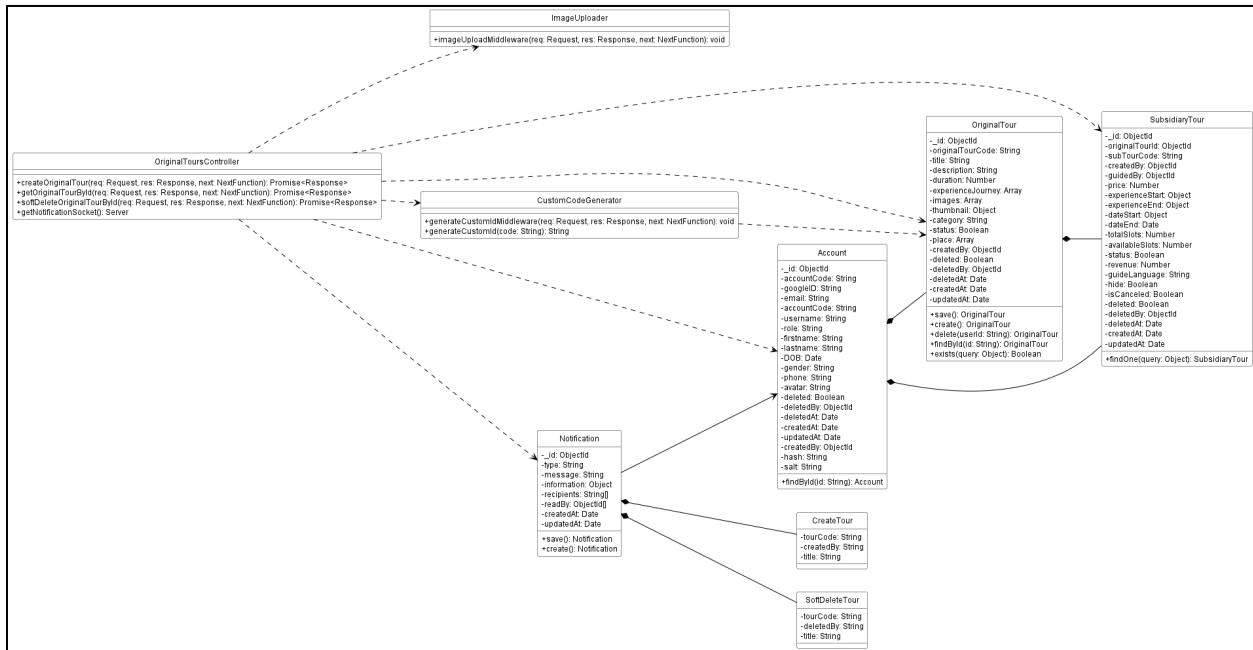


* Sequence diagram of View Profile function on Website:



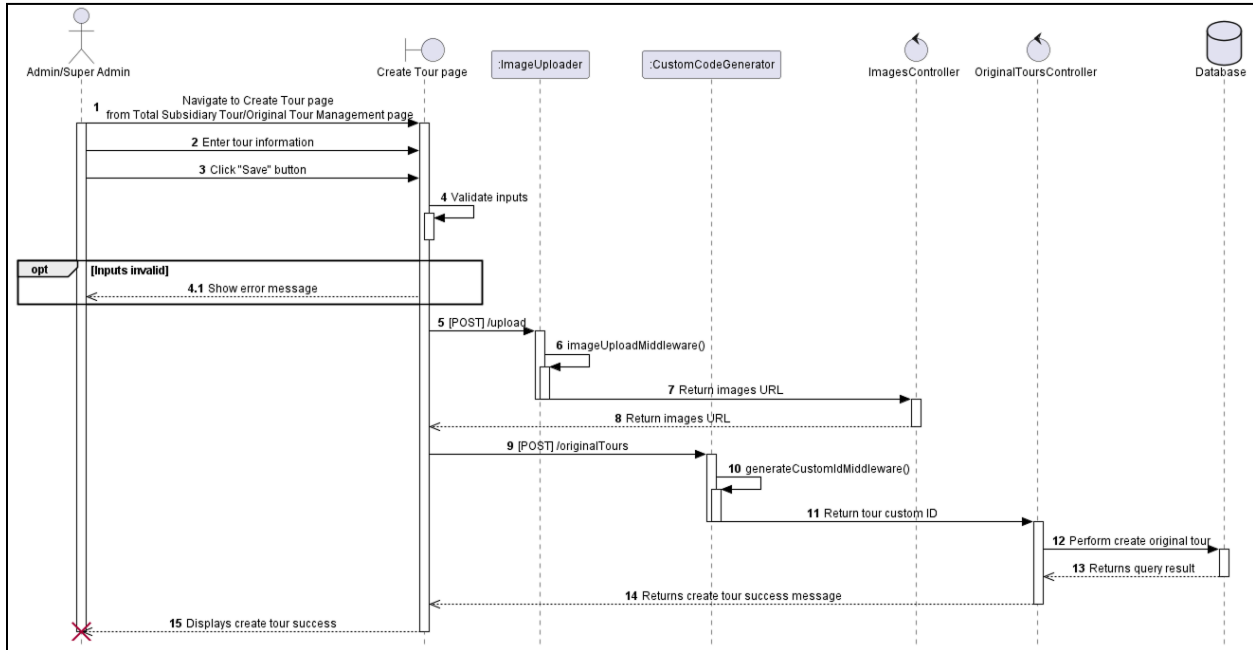
3.41 Manage Original Tour

a. Class Diagram

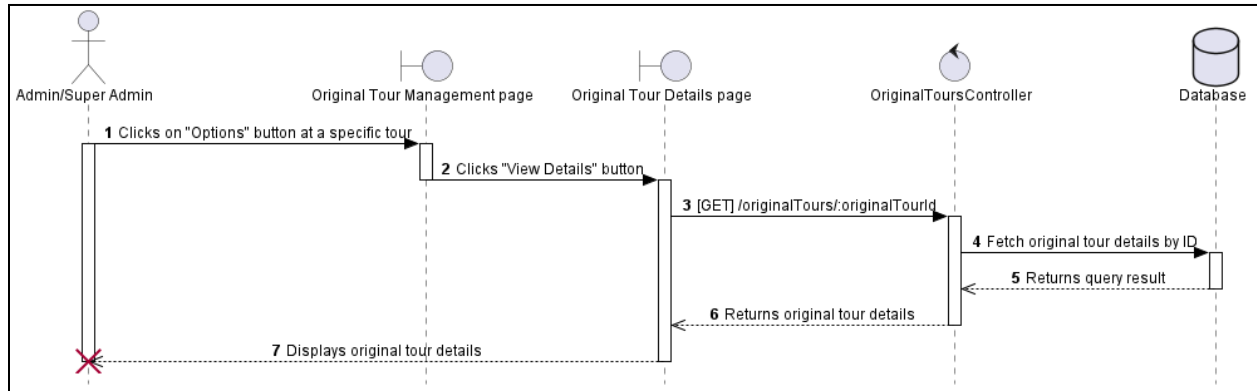


b. Sequence Diagram(s)

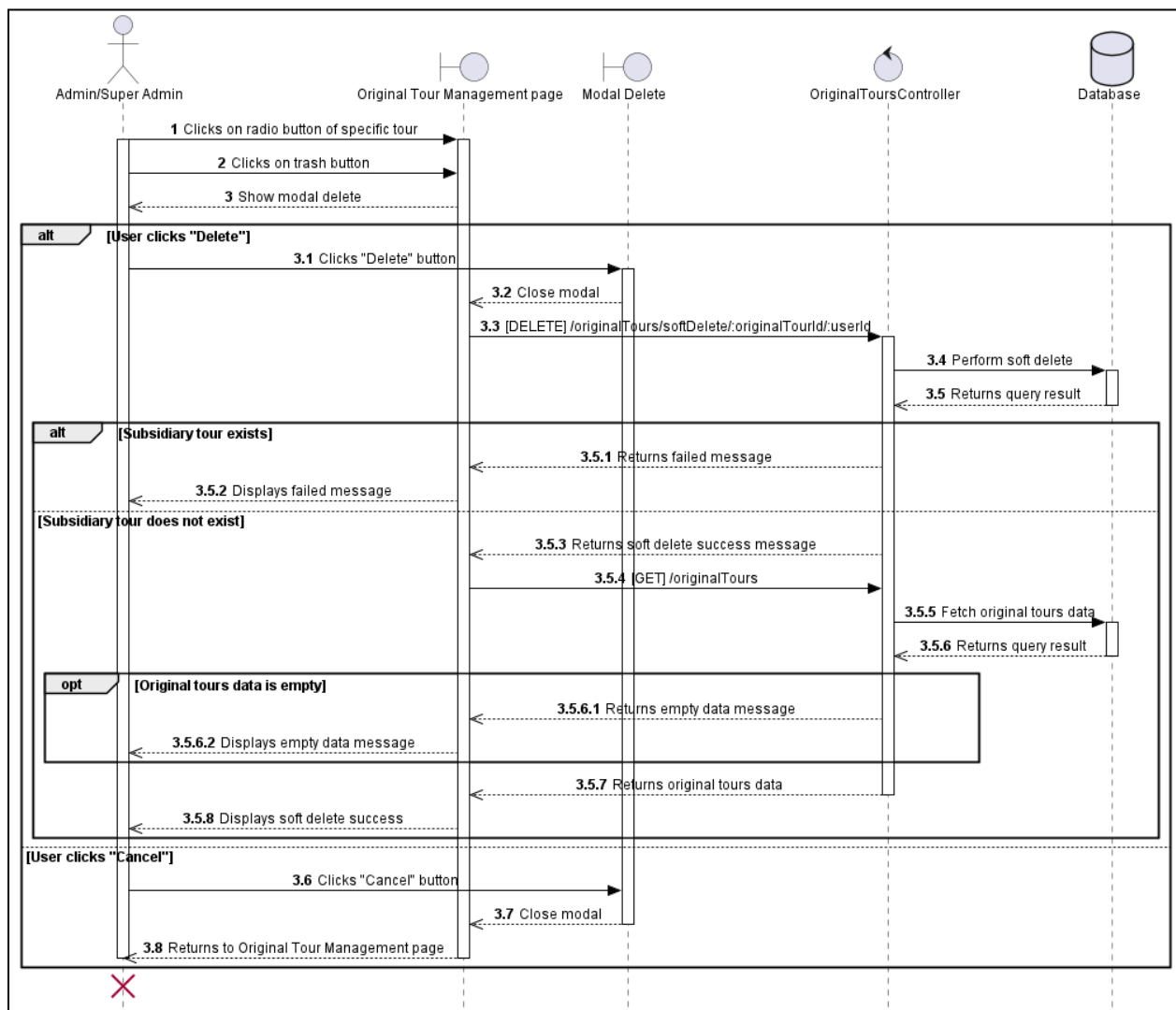
*Sequence diagram for Create Original Tour



*Sequence diagram for View Original Tour Details

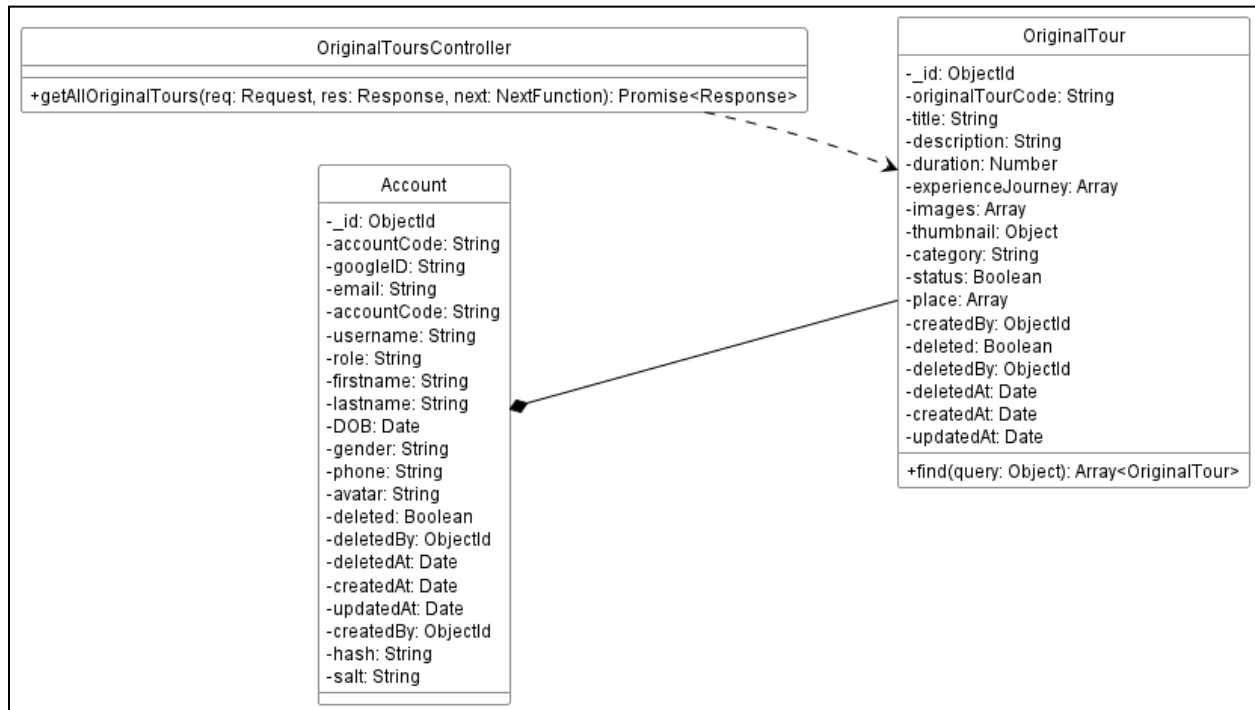


*Sequence diagram for Soft Delete Original Tour

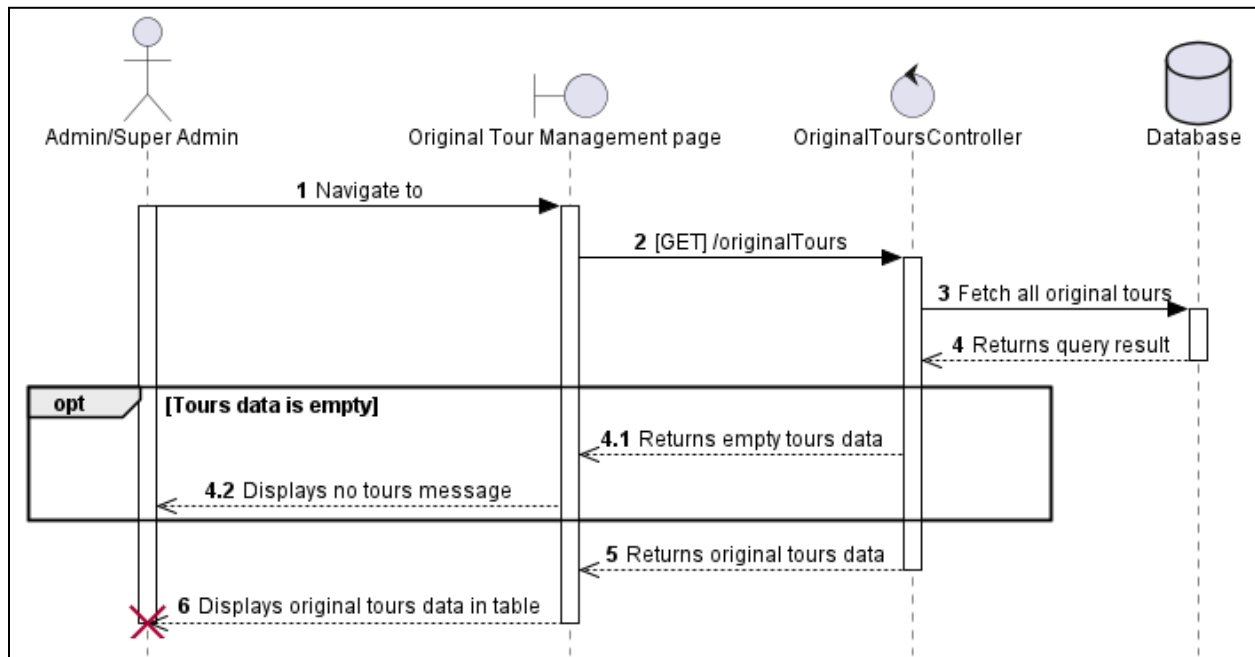


3.42 View All Original Tours

a. Class Diagram

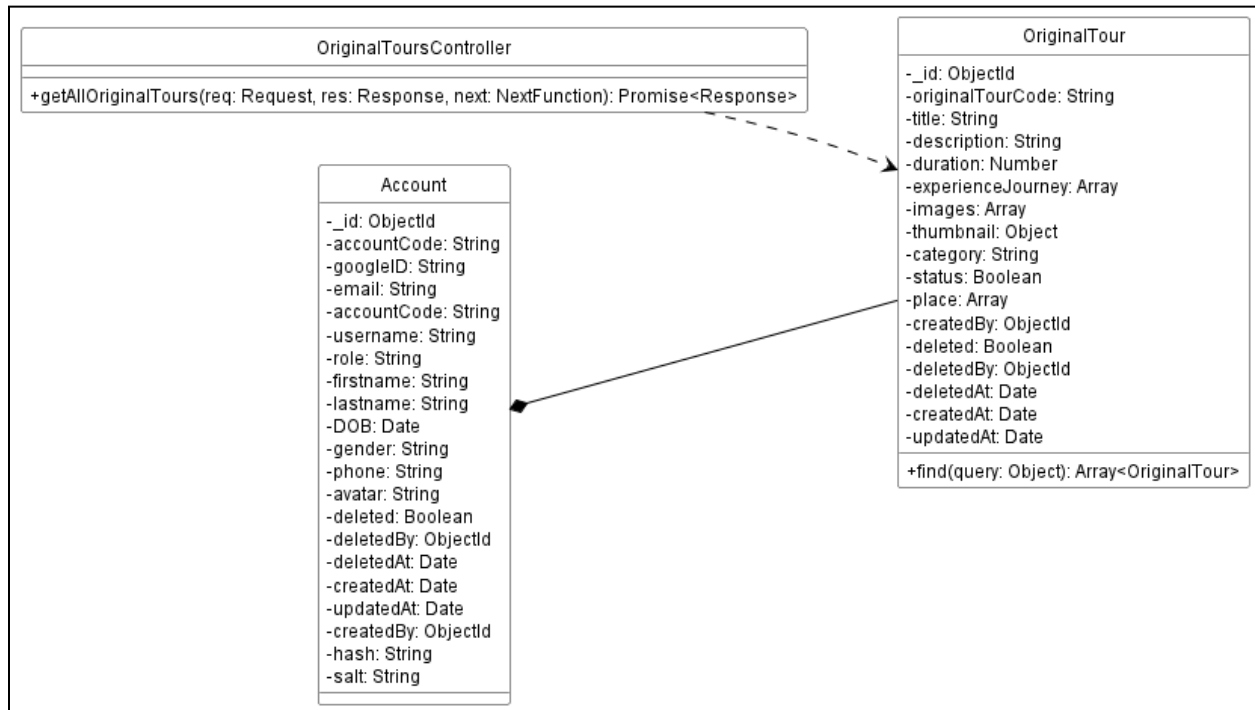


b. Sequence Diagram(s)

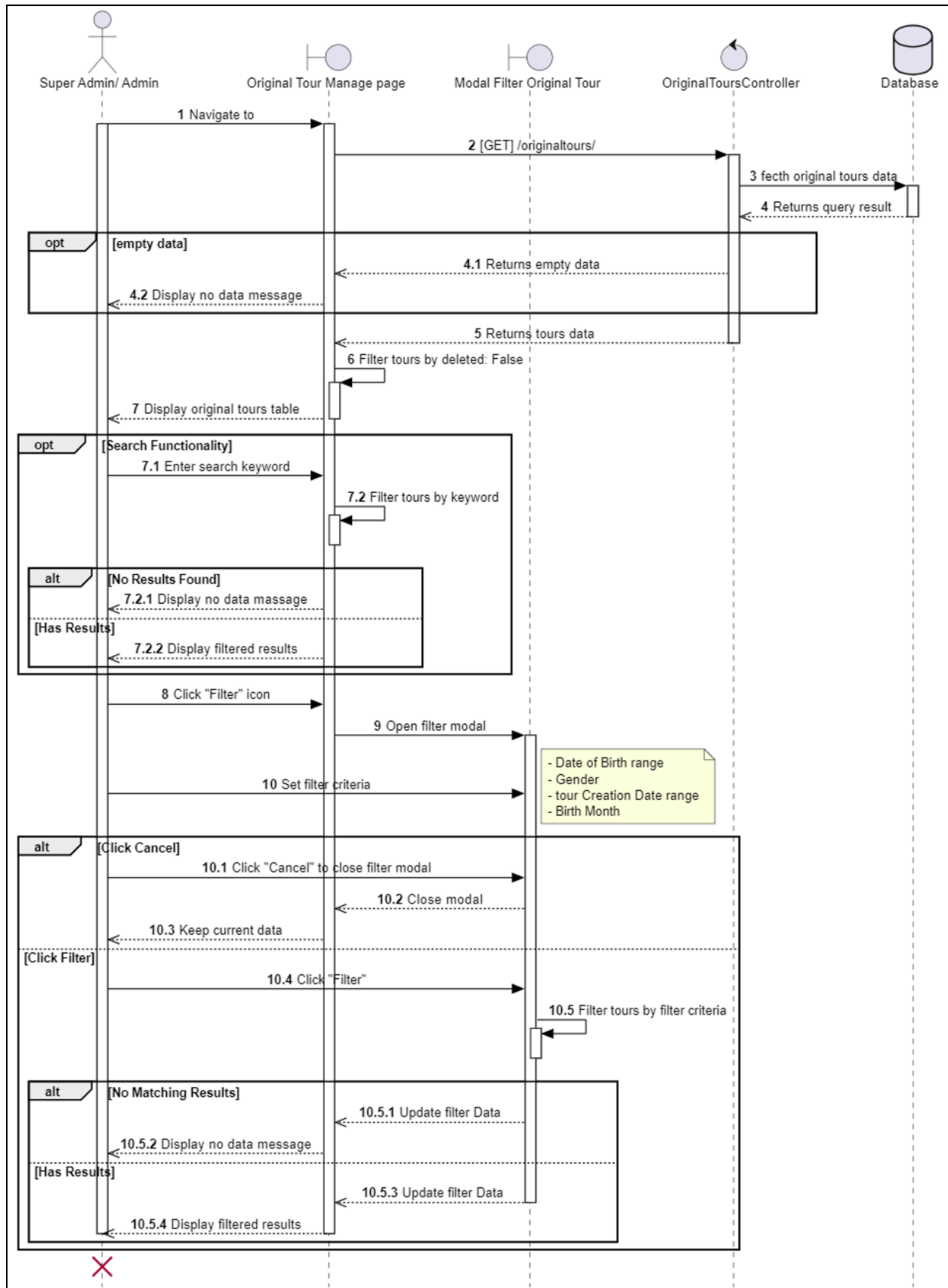


3.43 Filter & Search Original Tour

a. Class Diagram

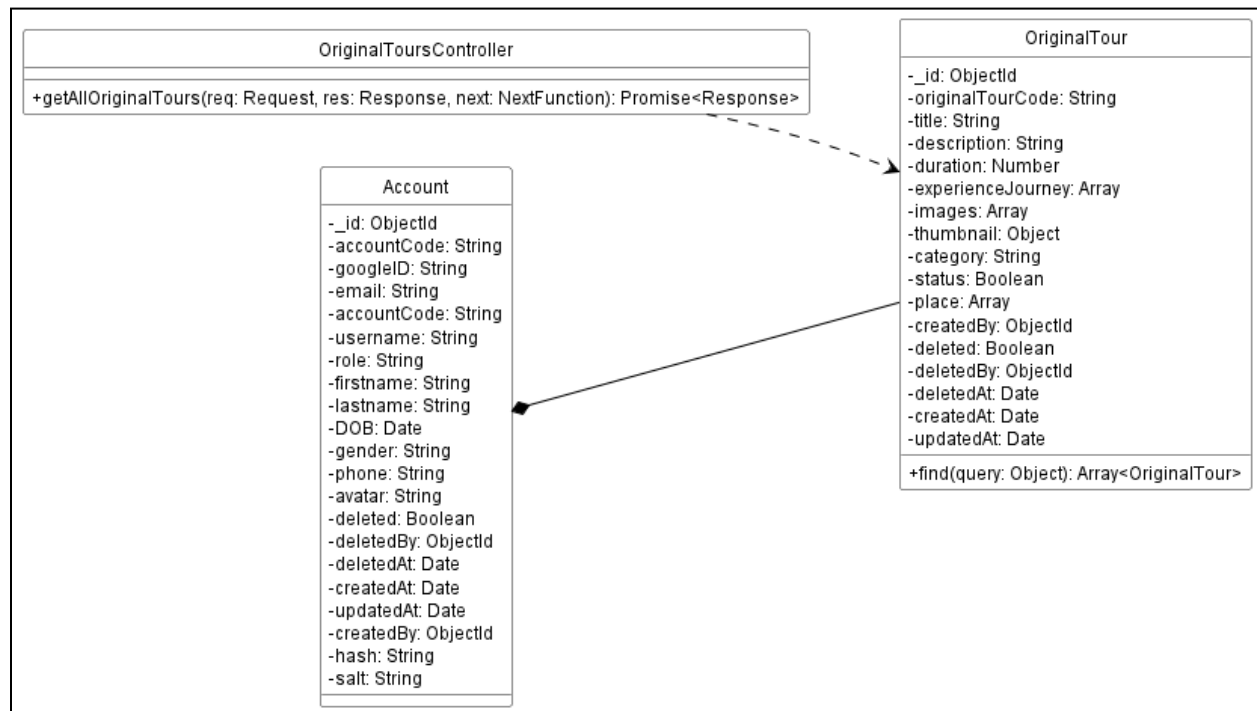


b. Sequence Diagram(s)

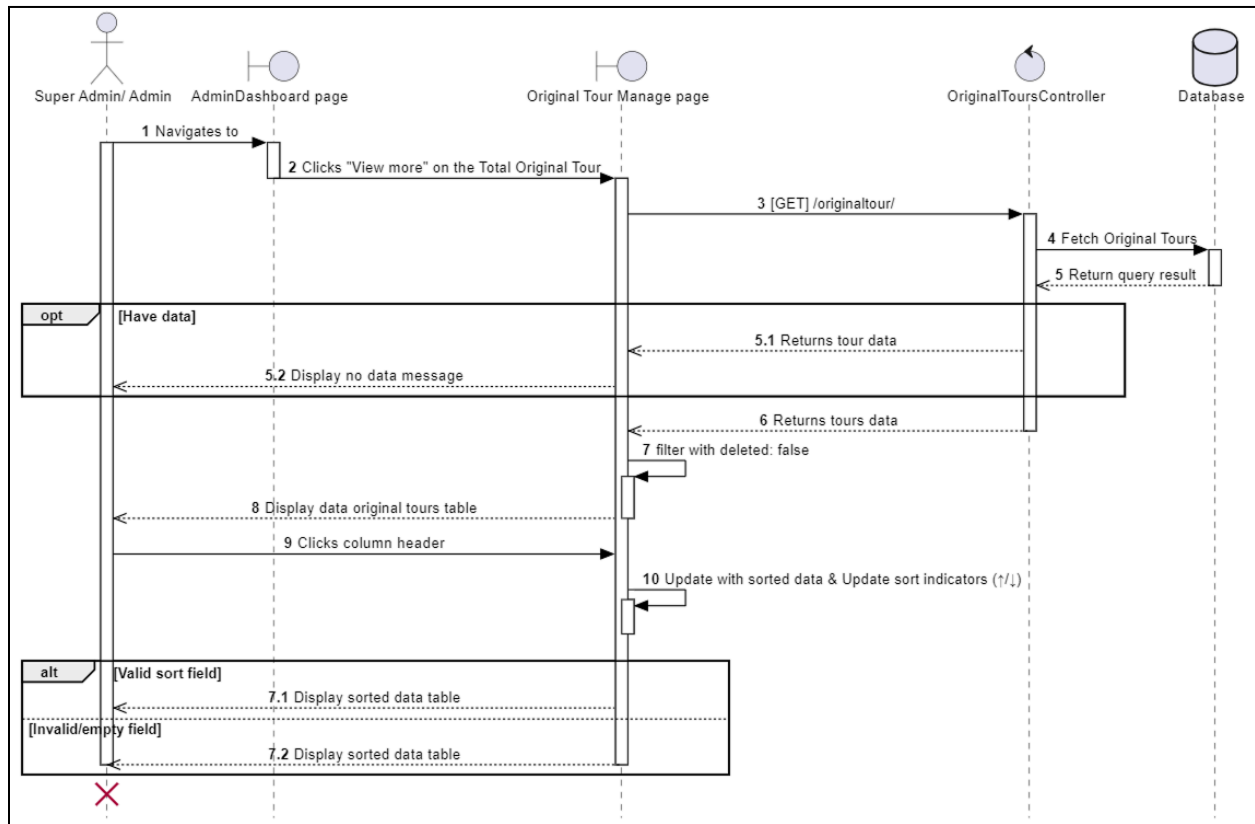


3.44 Sort Original Tours

a. Class Diagram

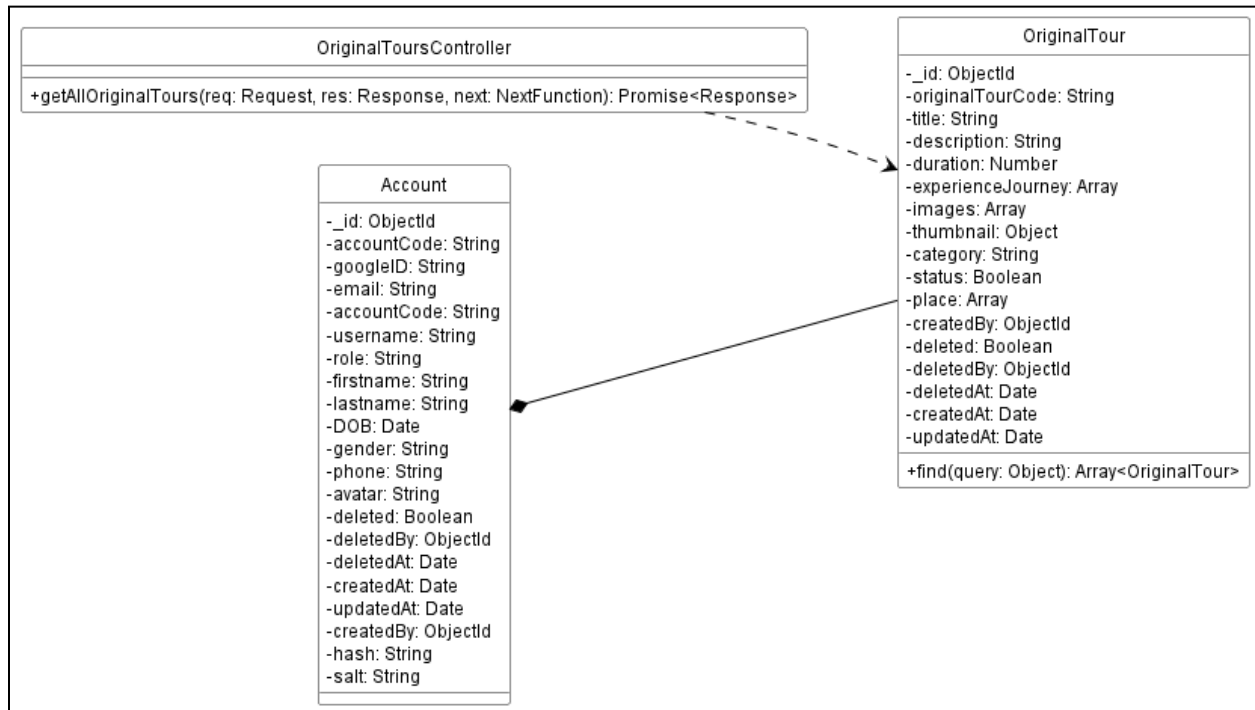


b. Sequence Diagram(s)

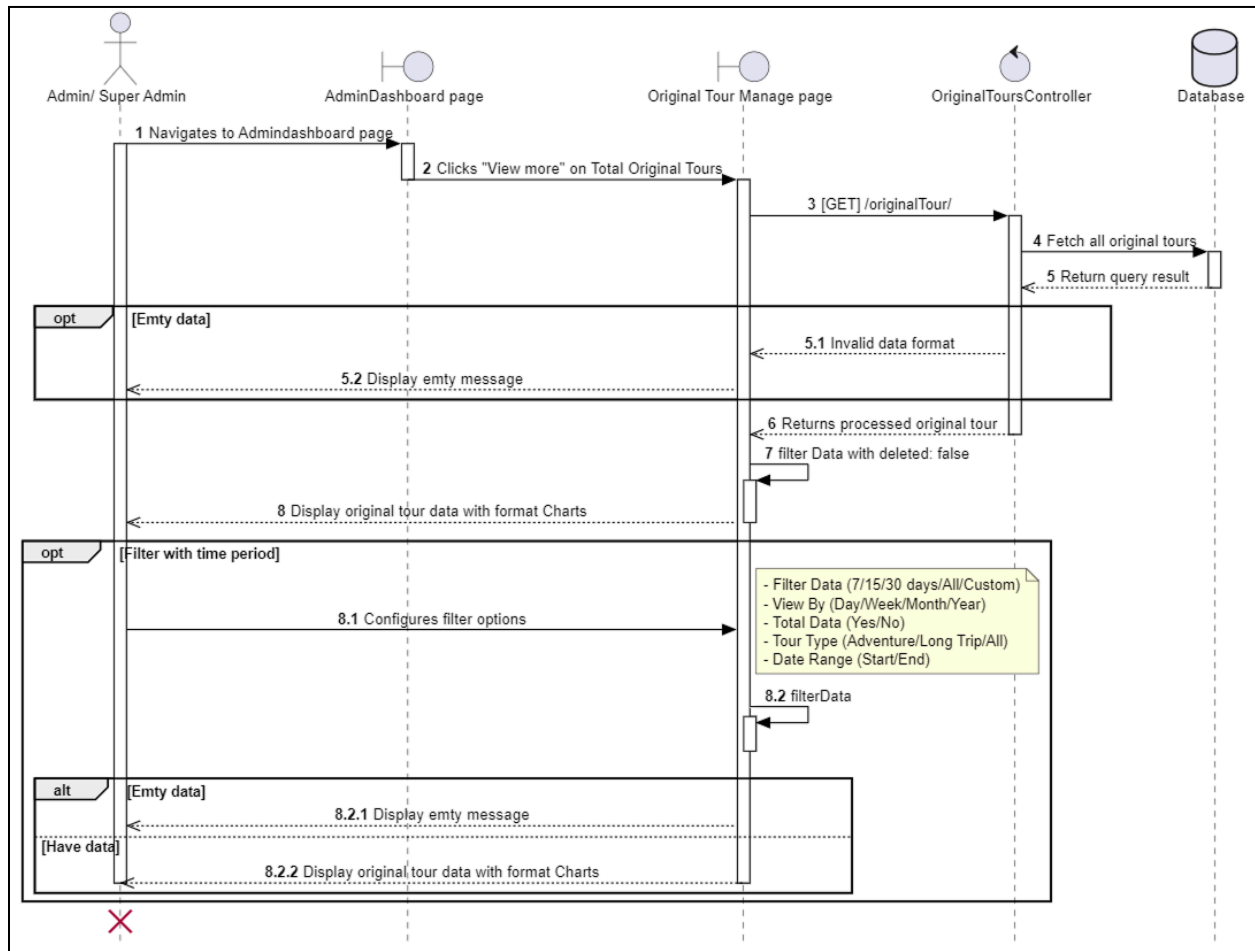


3.45 View Original Tour Statistic

a. Class Diagram

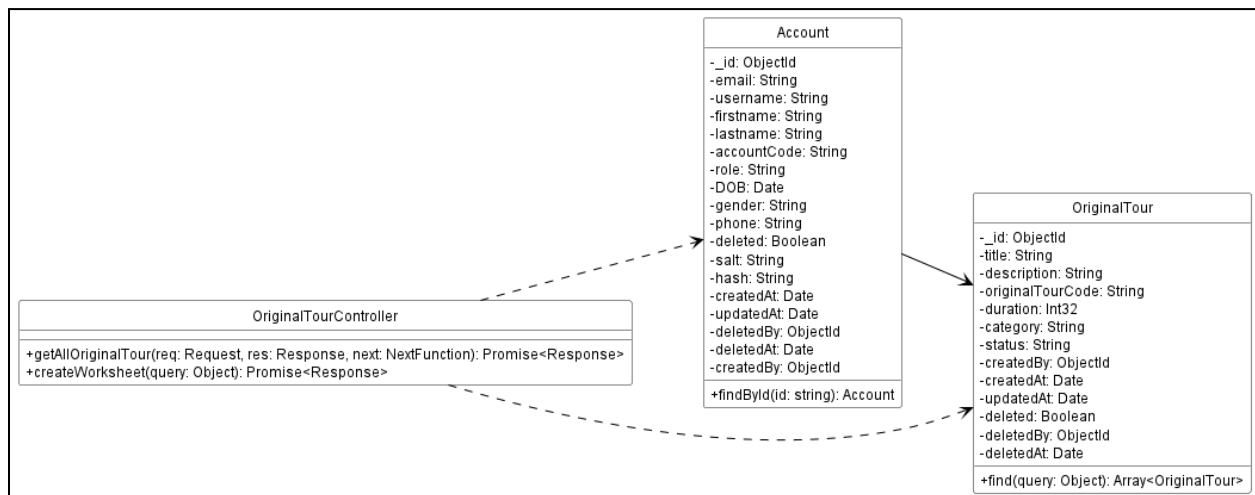


b. Sequence Diagram(s)

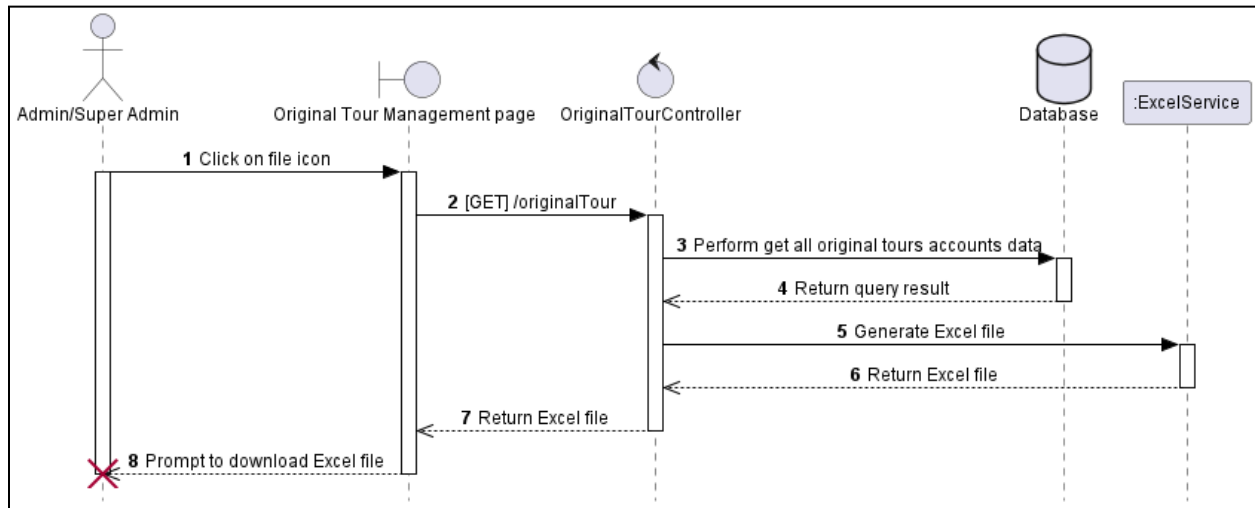


3.46 Export Original Tour Data To Excel

a. Class diagram

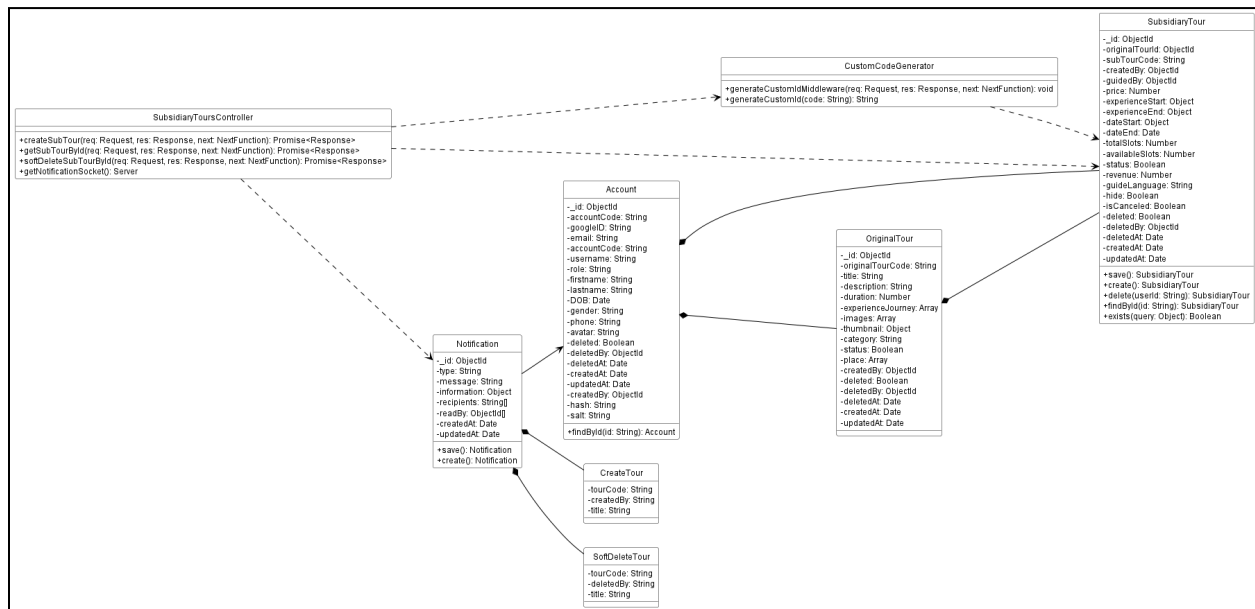


b. Sequence diagram



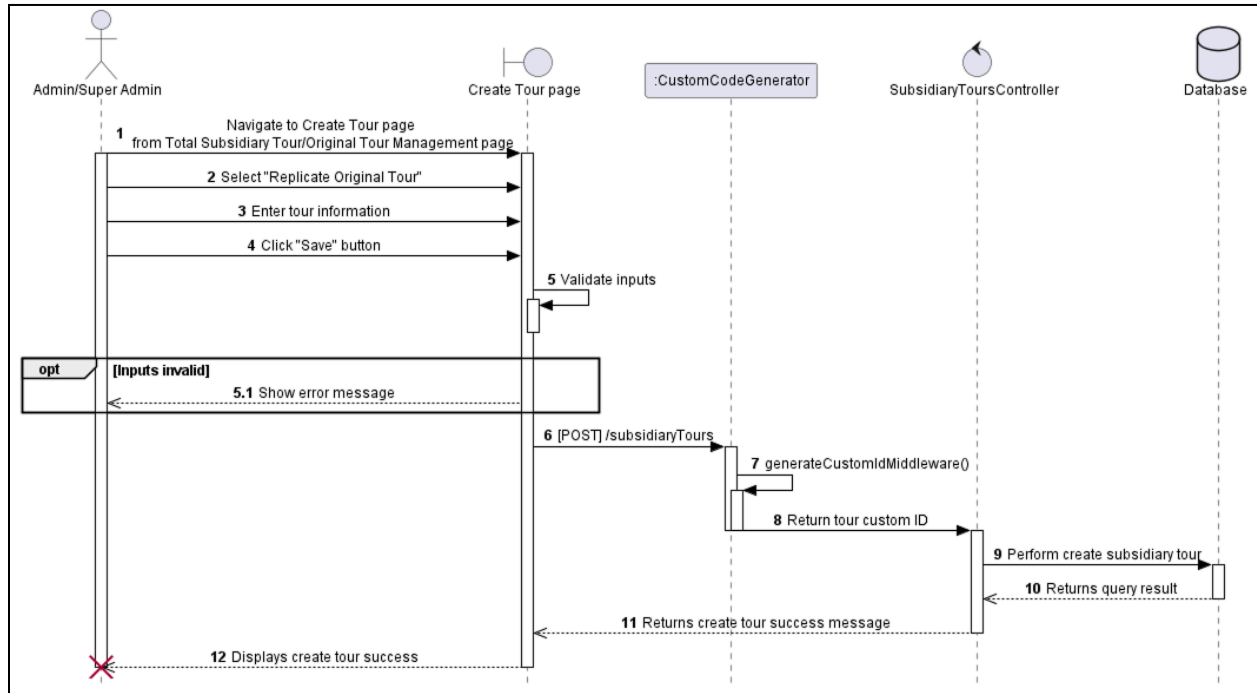
3.47 Manage Subsidiary Tour

a. Class Diagram

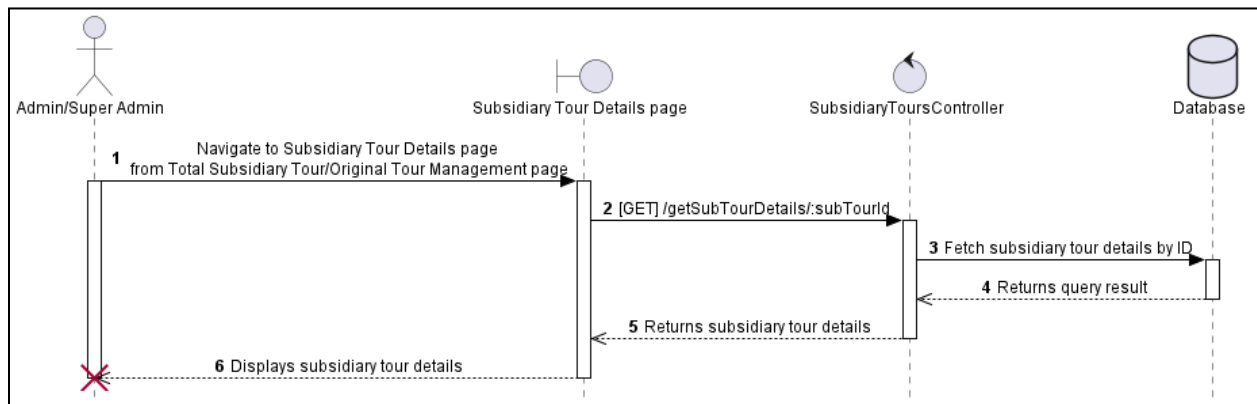


b. Sequence Diagram(s)

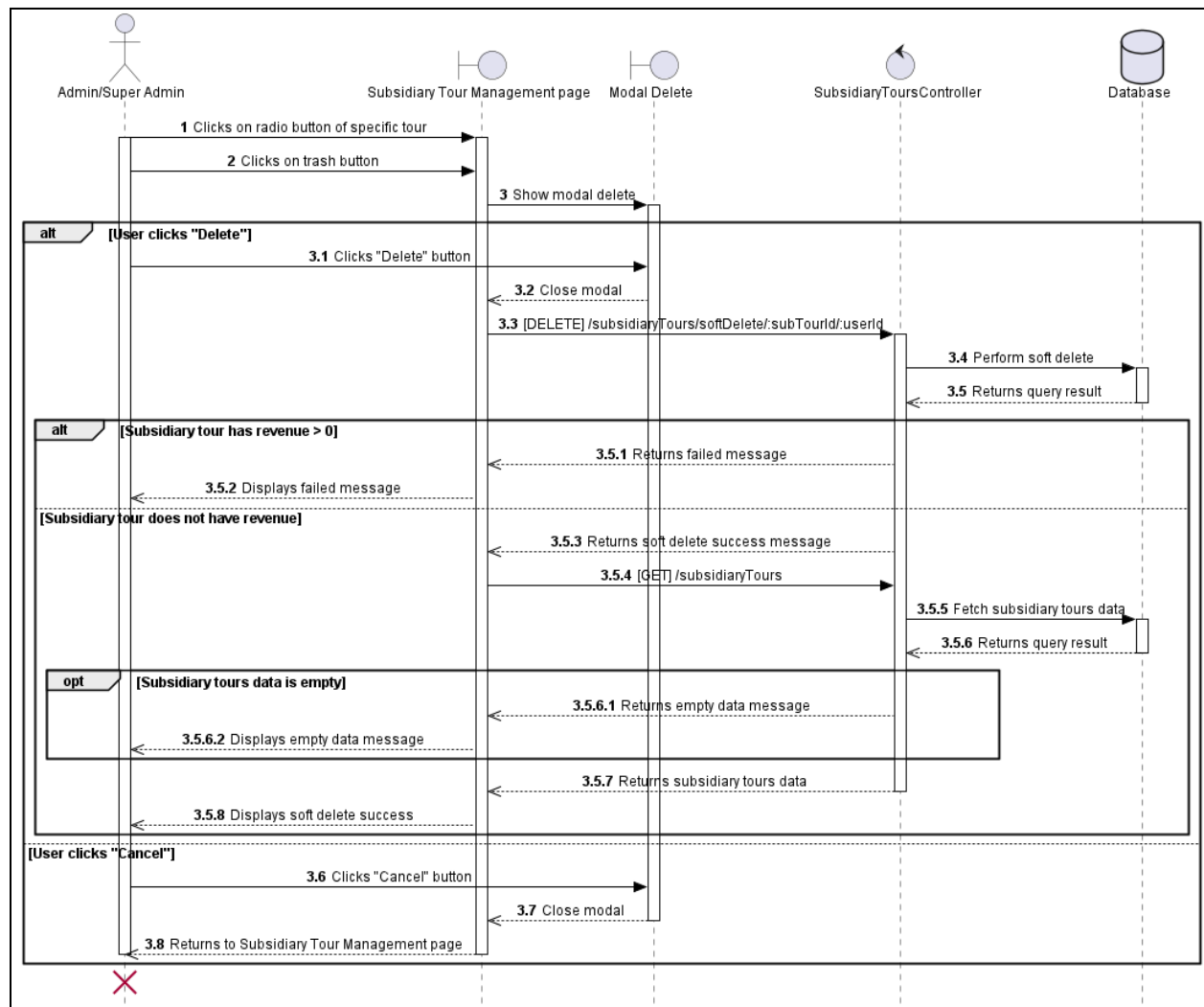
*Sequence diagram for Create Subsidiary Tour



*Sequence diagram for View Subsidiary Tour Details

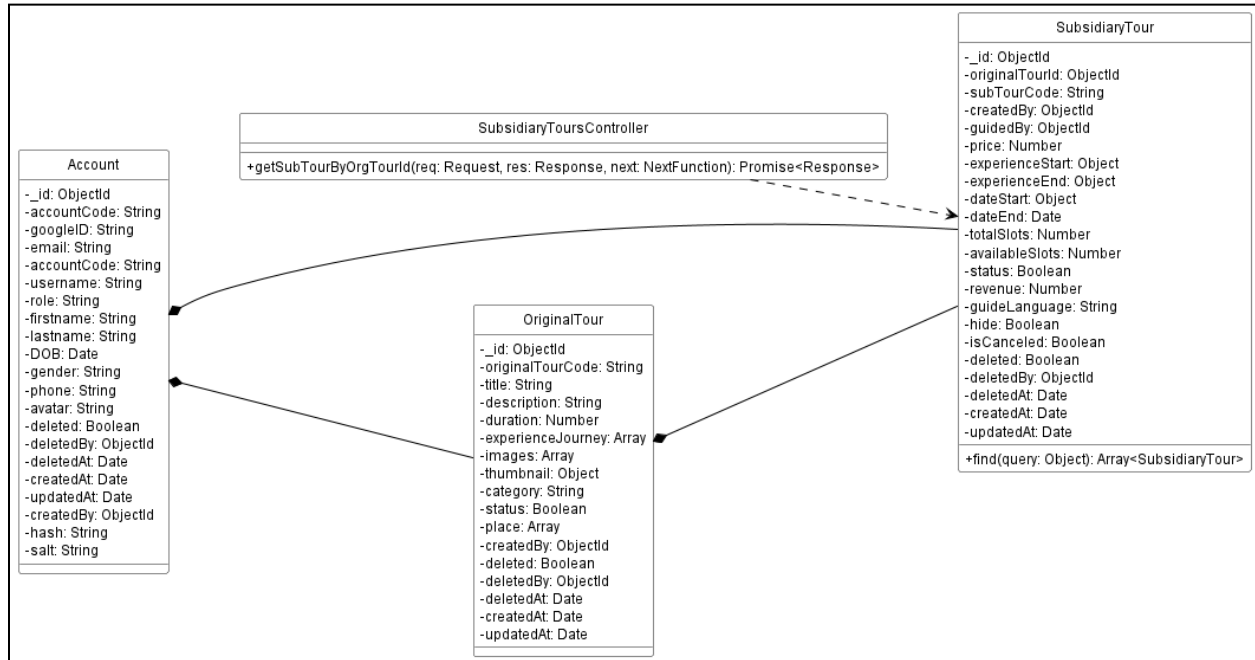


*Sequence diagram for Soft Delete Subsidiary Tour

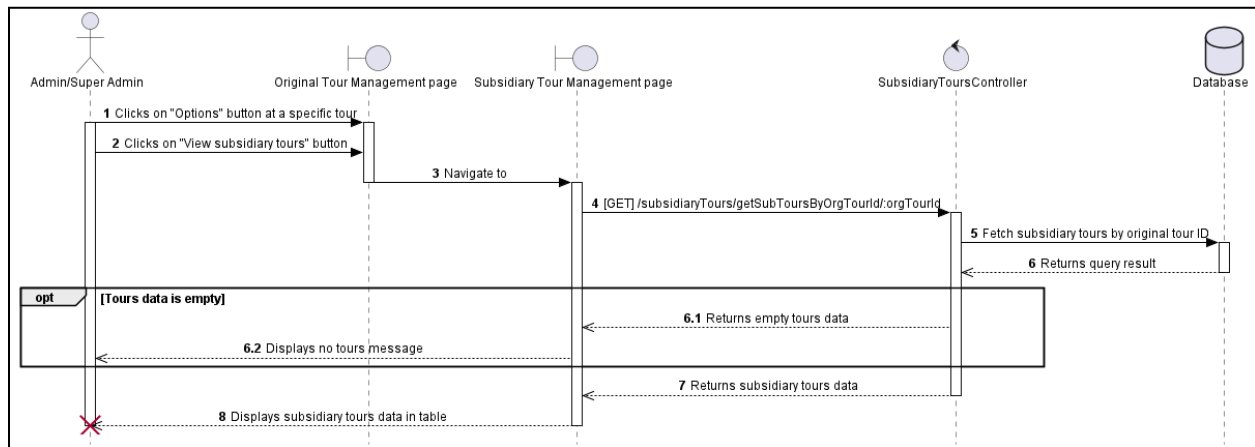


3.48 View All Subsidiary Tours Of Original Tour

a. Class Diagram

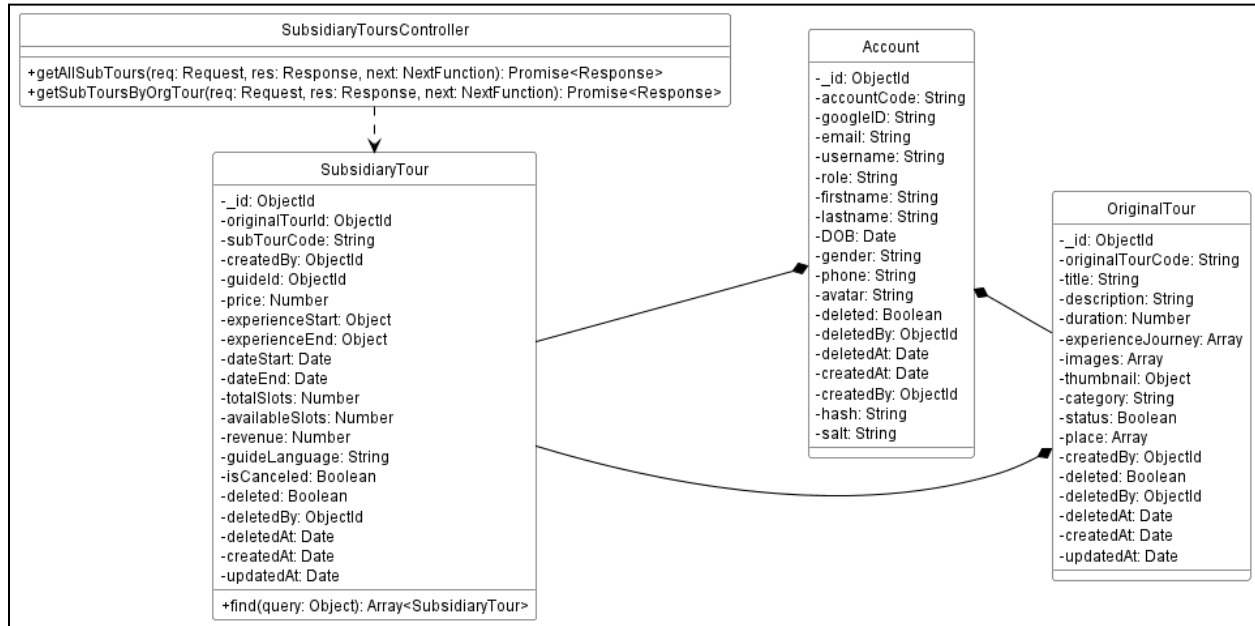


b. Sequence Diagram(s)



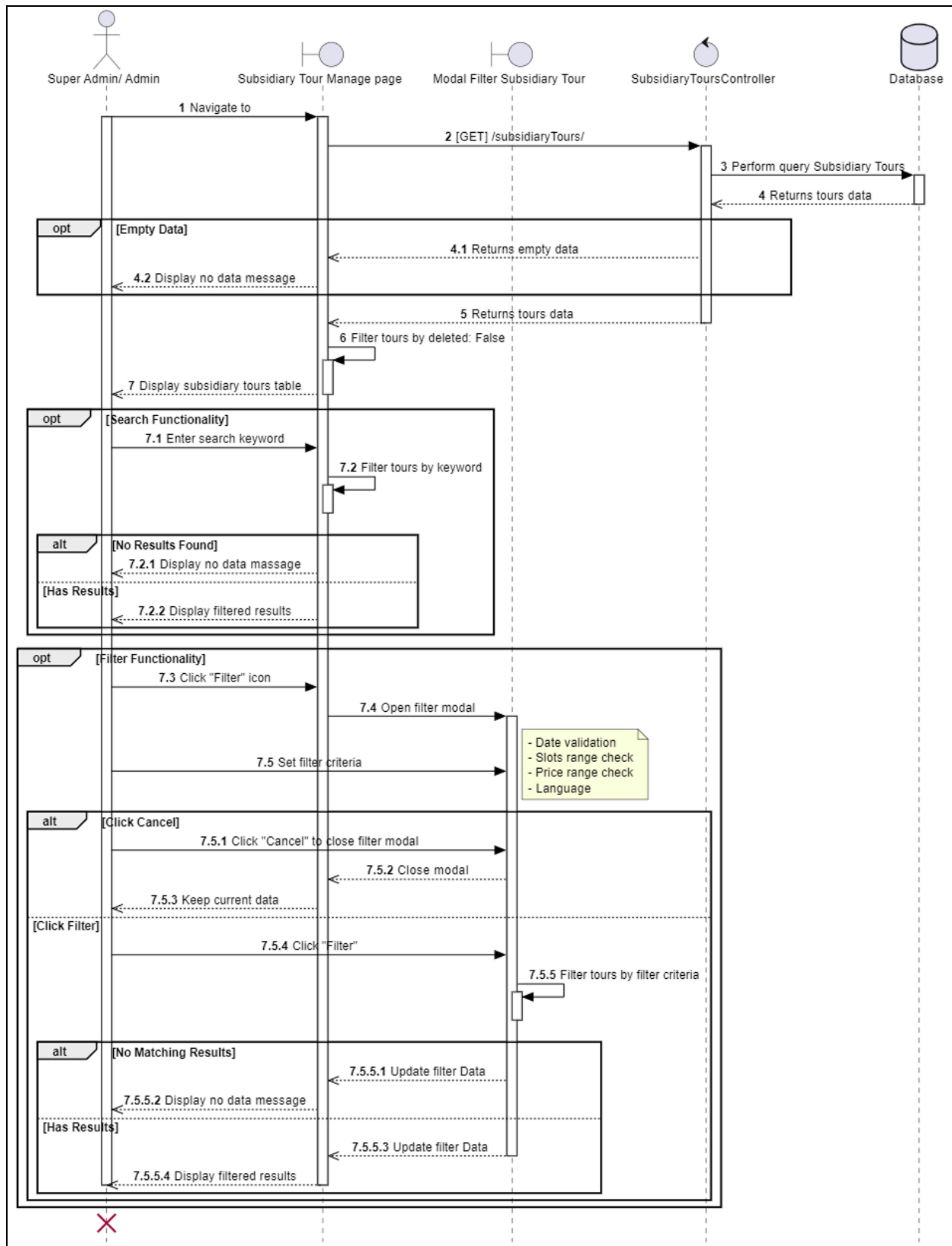
3.49 Filter & Search Subsidiary Tour

a. Class Diagram

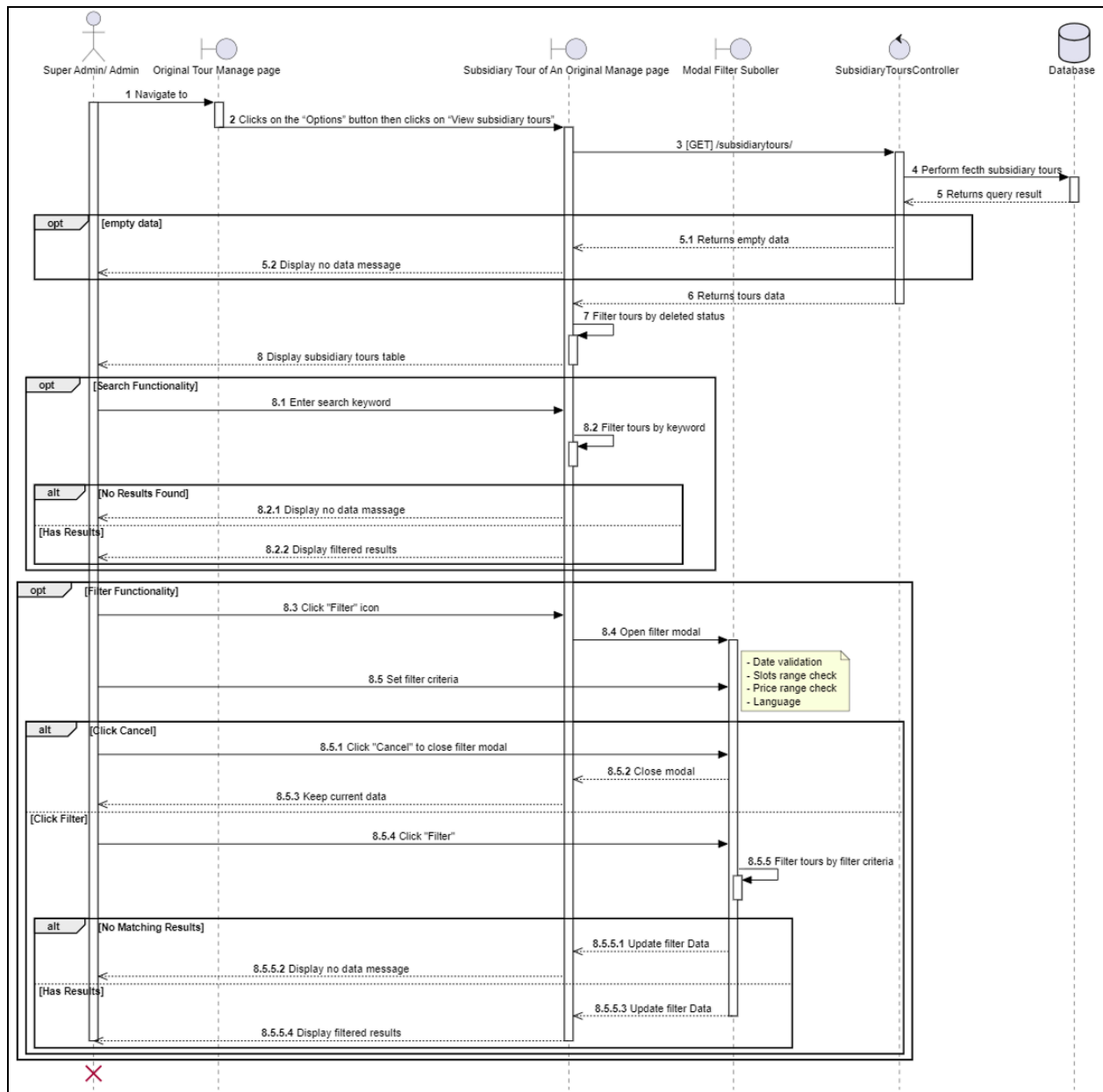


b. Sequence Diagram(s)

*Sequence diagram for filter & search subsidiary tour

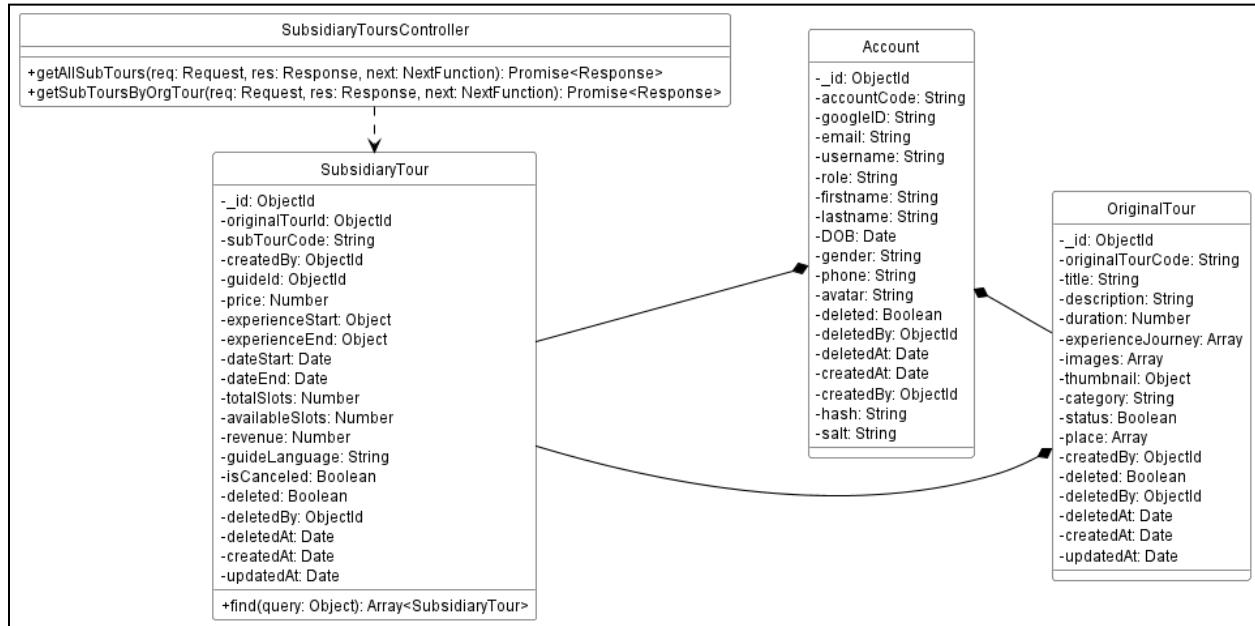


*Sequence diagram for filter & search subsidiary tour of an original tour



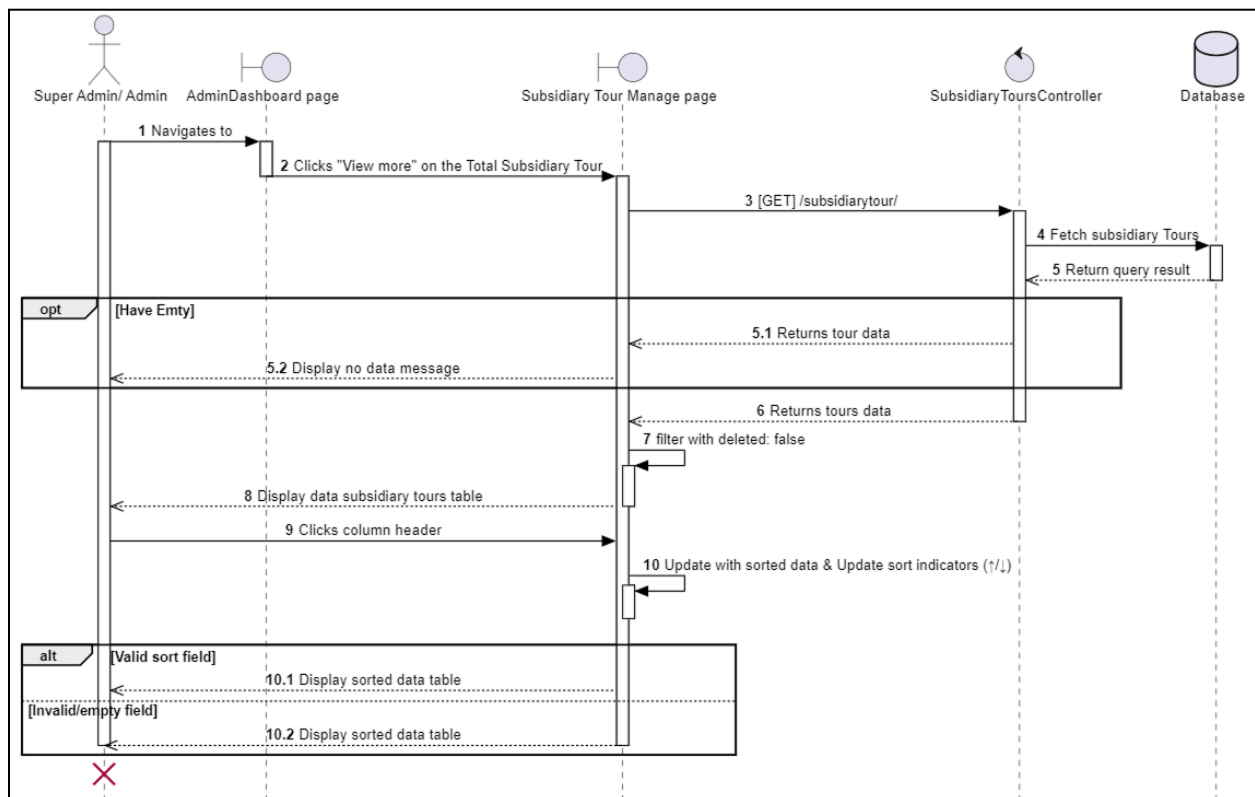
3.50 Sort Subsidiary Tour

a. Class Diagram

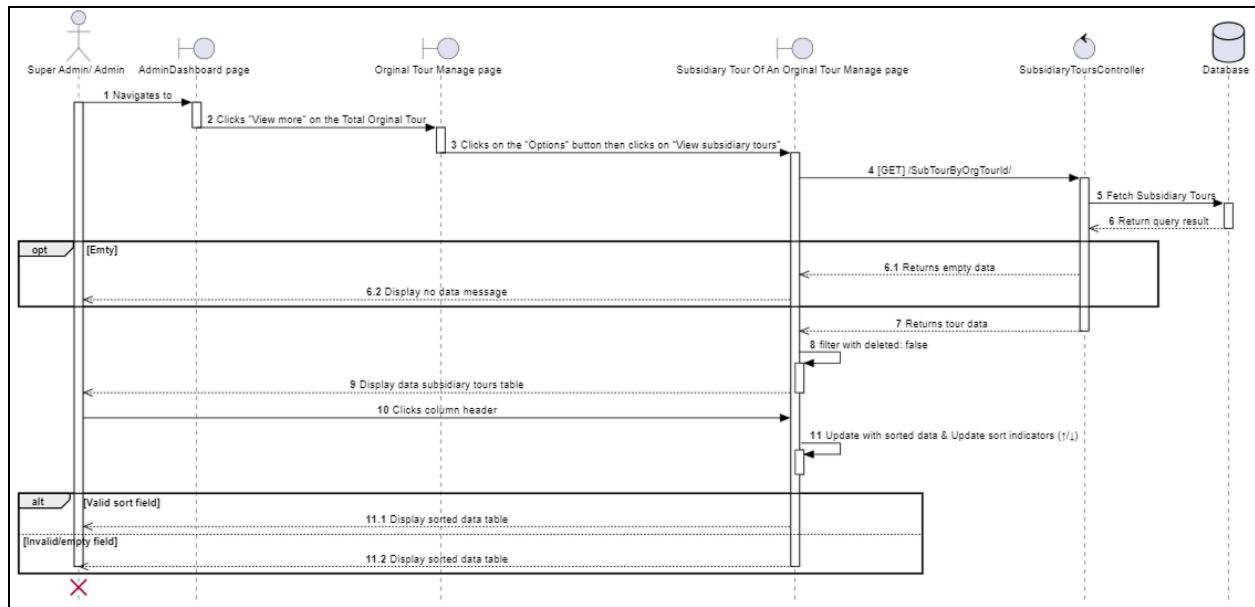


b. Sequence Diagram(s)

*Sequence diagram for sort subsidiary tour.

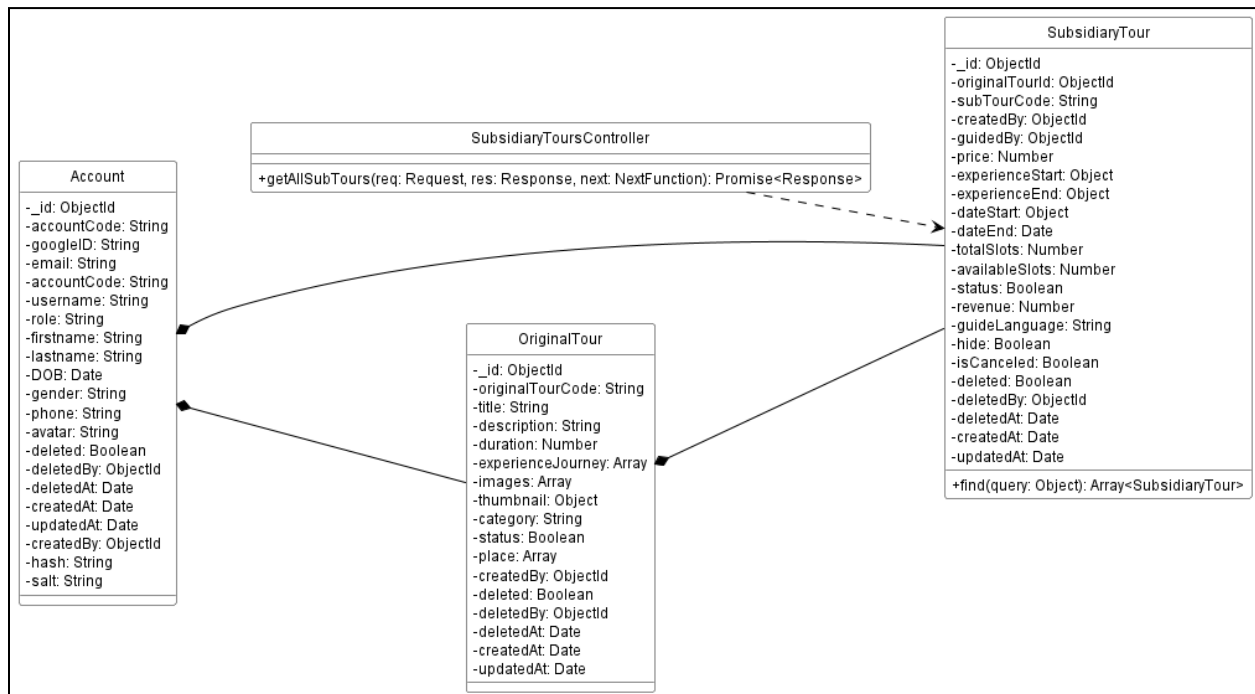


*Sequence diagram for sort subsidiary tour of an original tour.

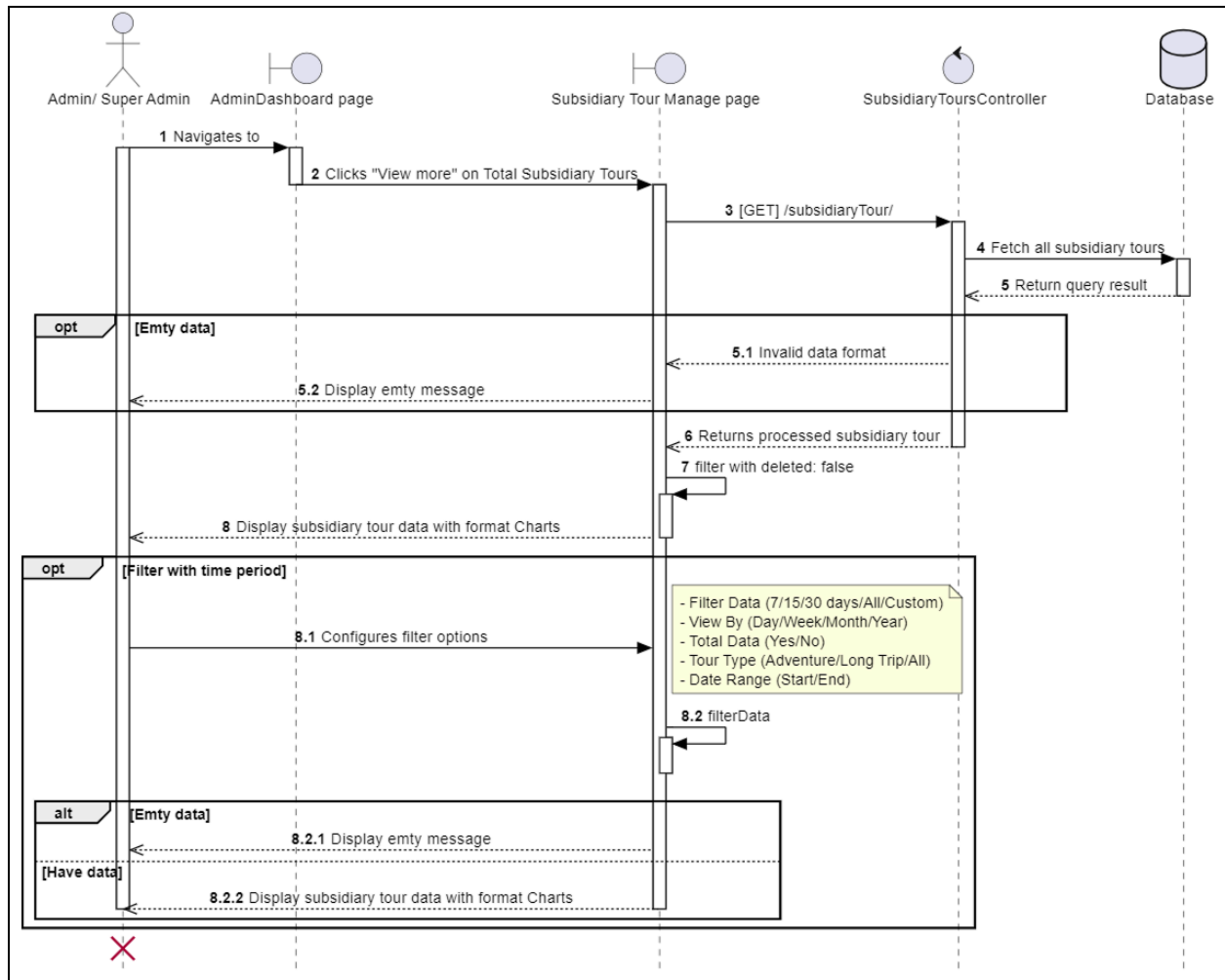


3.51 View Subsidiary Tour Statistic

a. Class Diagram

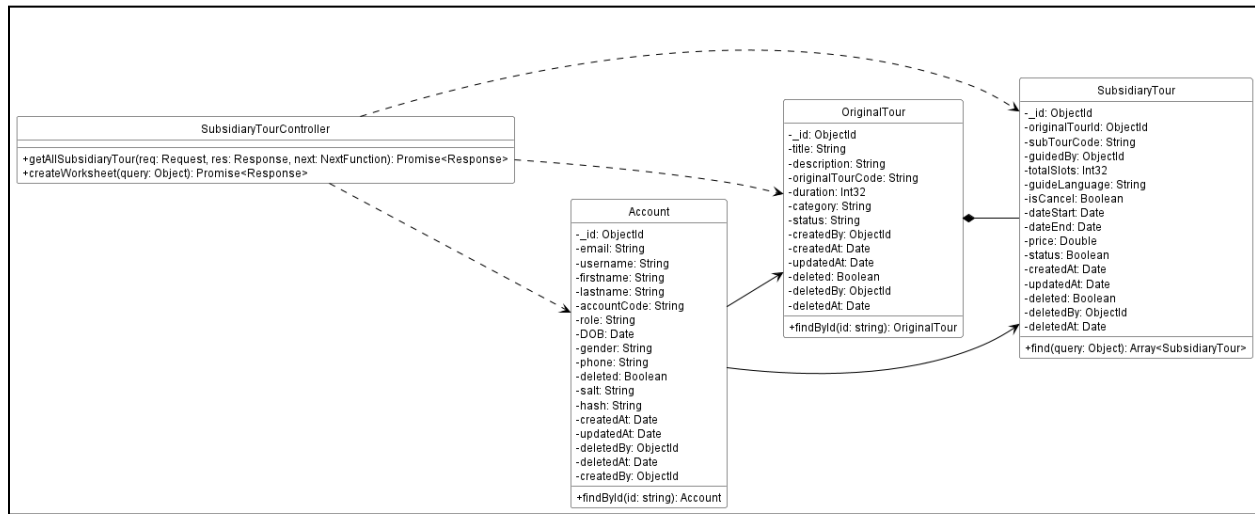


b. Sequence Diagram(s)

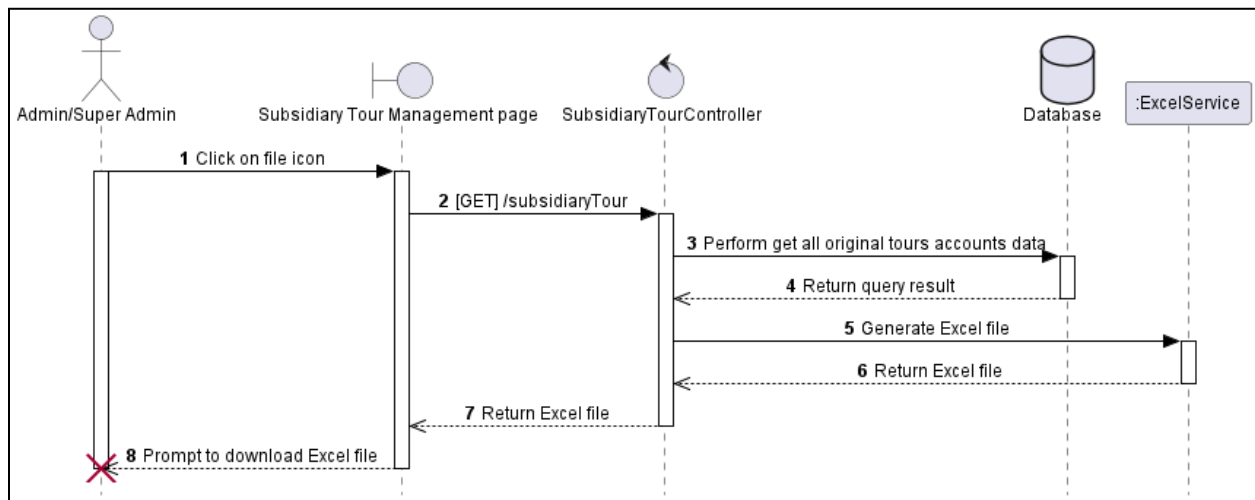


3.52 Export Subsidiary Tour Data To Excel File

a. Class Diagram

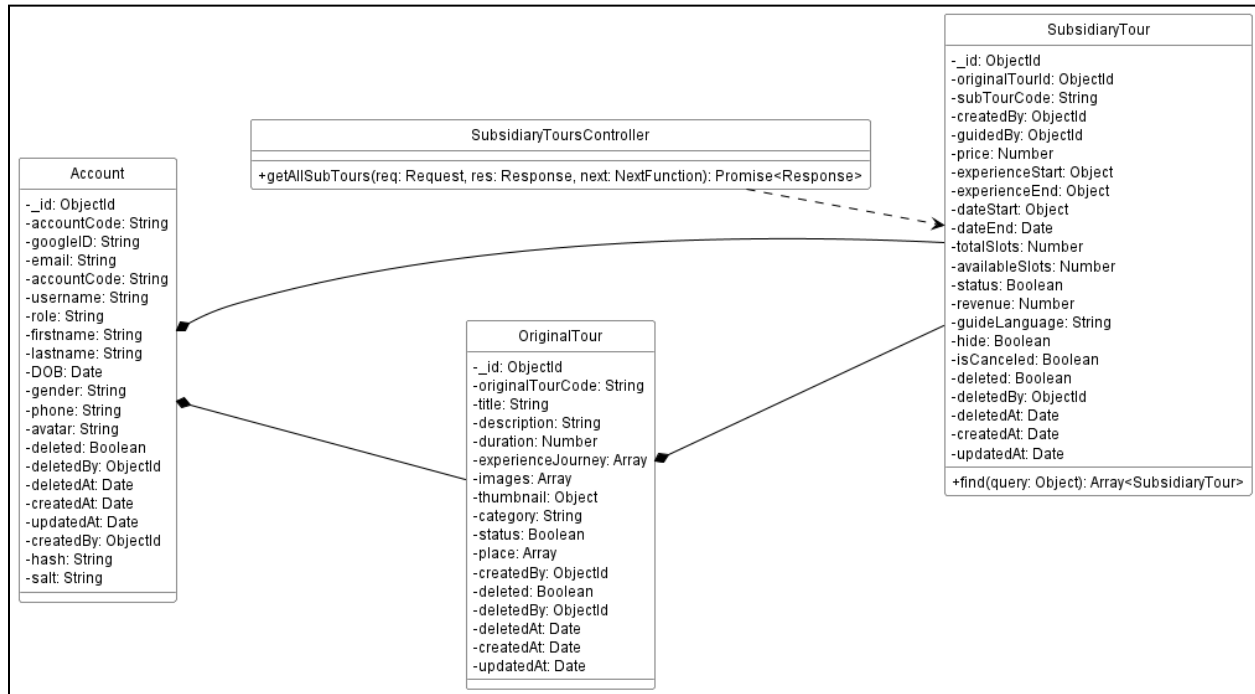


b. Sequence Diagram(s)

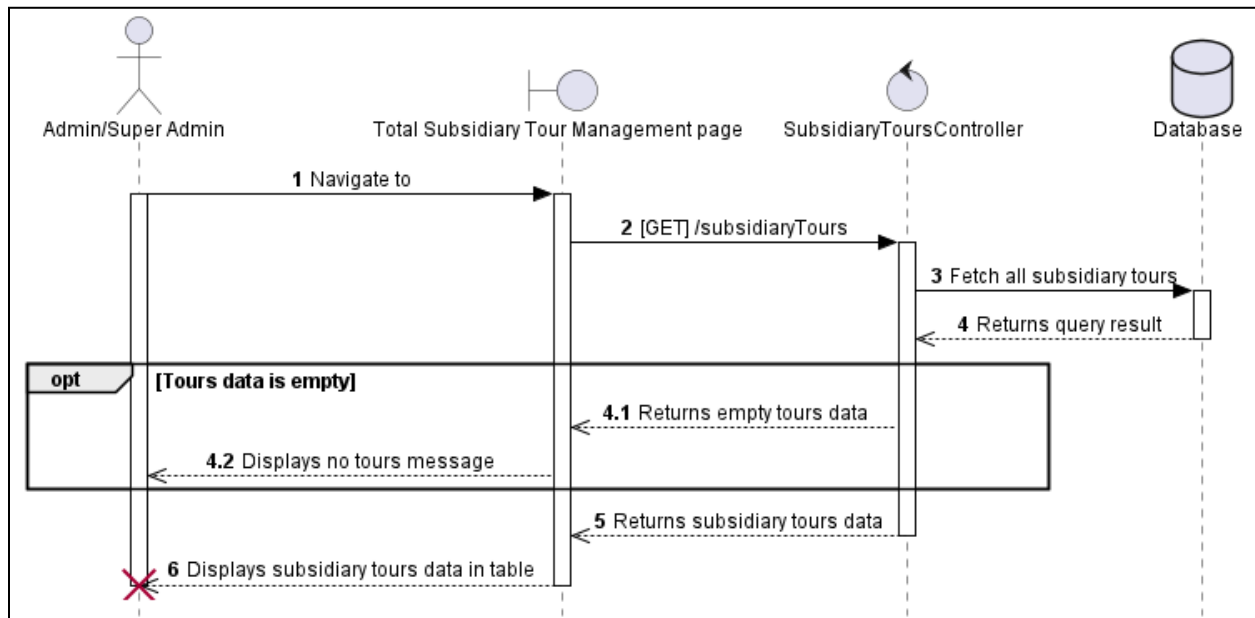


3.53 View All Subsidiary Tours

a. Class Diagram

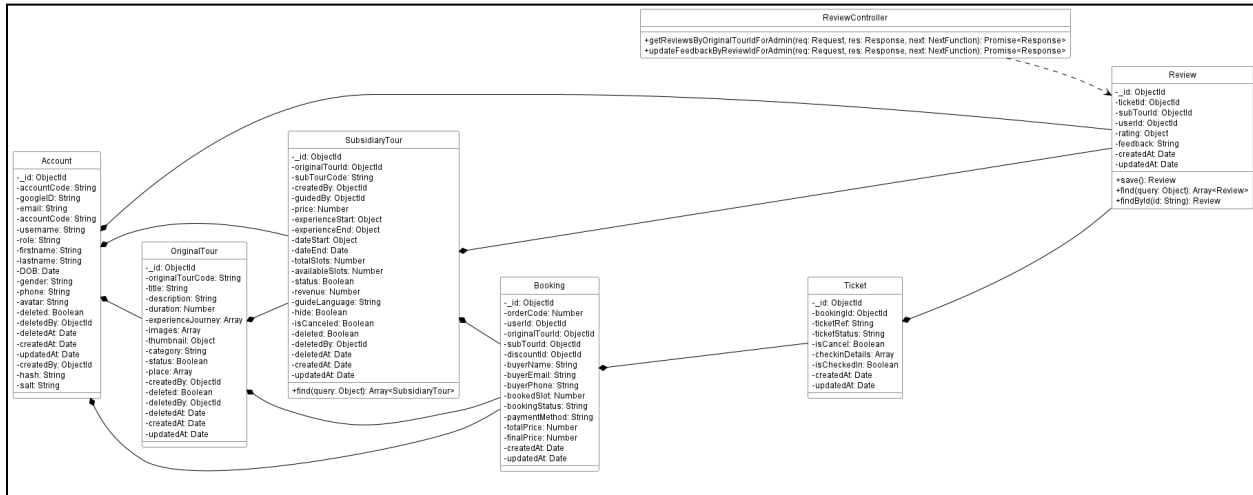


b. Sequence Diagram(s)



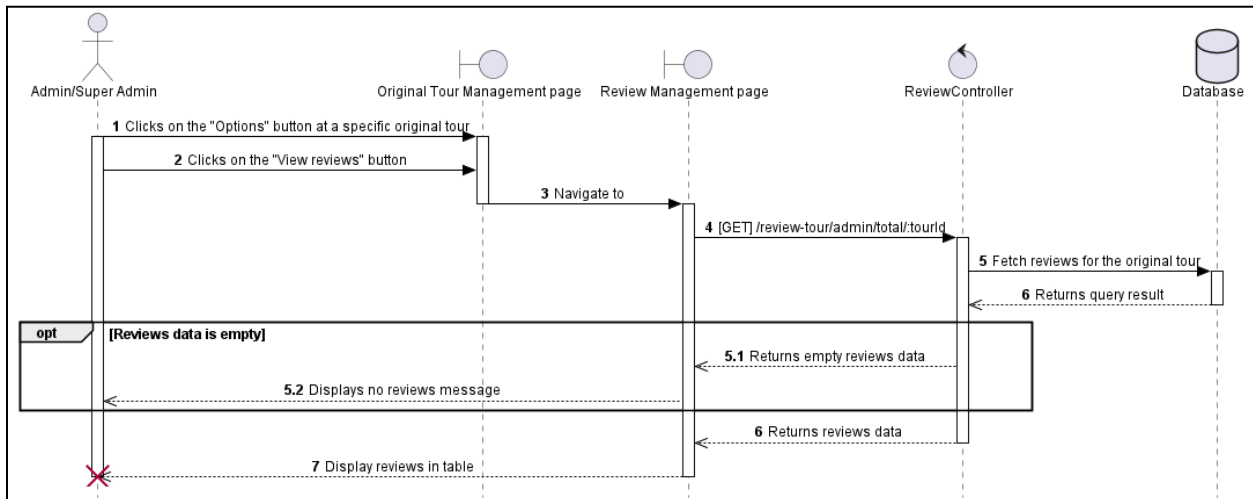
3.54 Manage Original Tour Reviews

a. Class Diagram

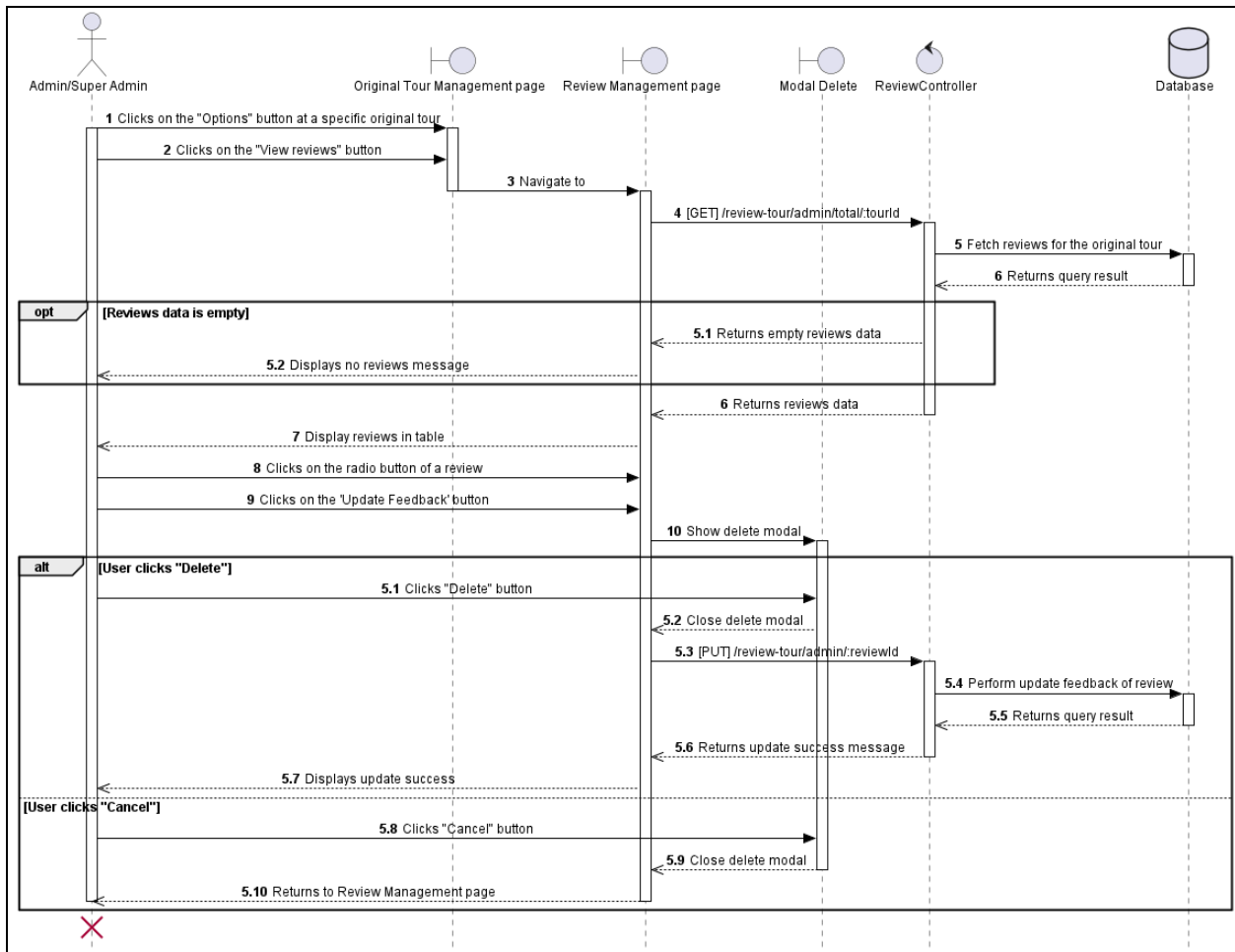


b. Sequence diagram

*Sequence diagram for View All Reviews

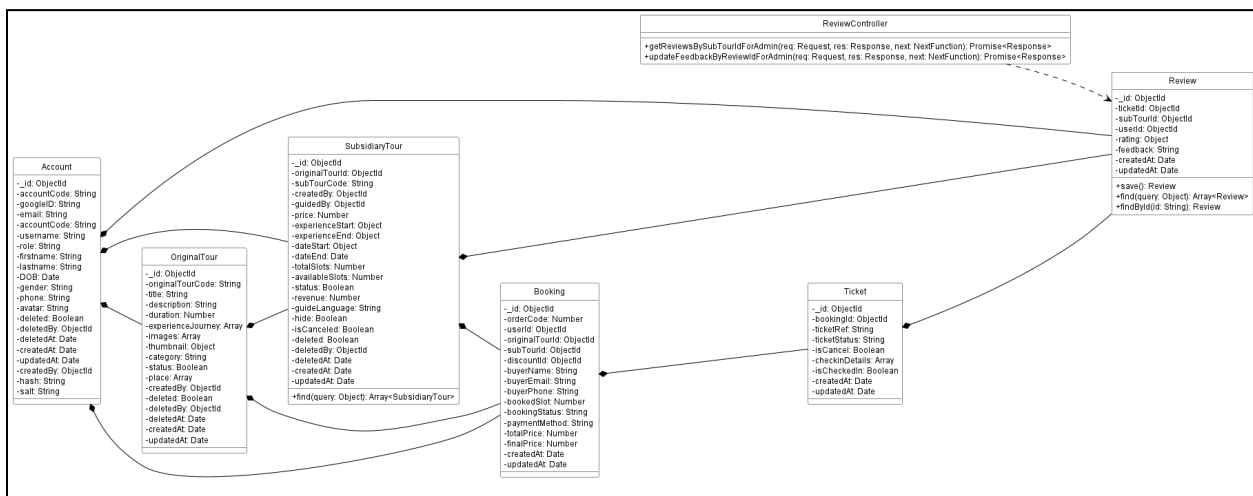


*Sequence diagram for Update Feedback



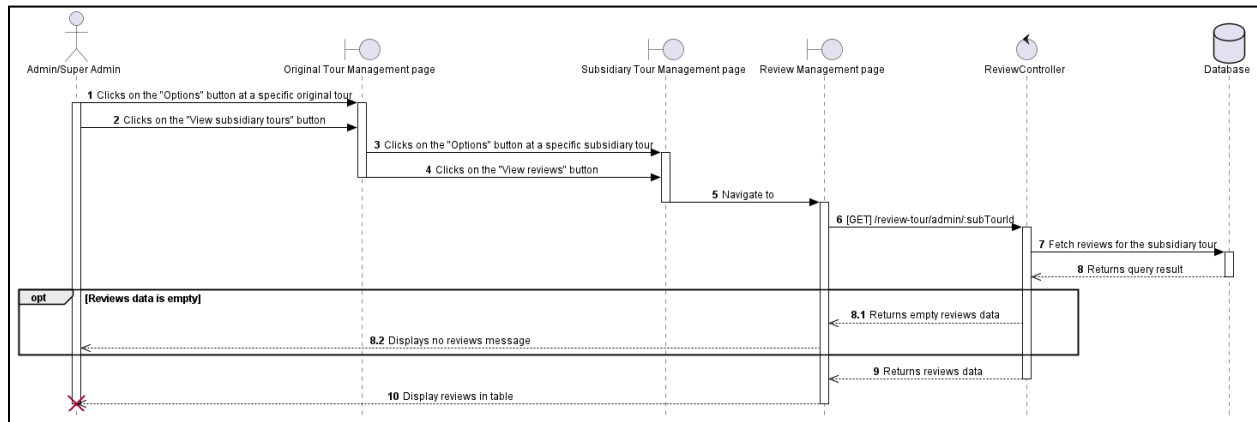
3.55 Manage Subsidiary Tour Reviews

a. Class Diagram

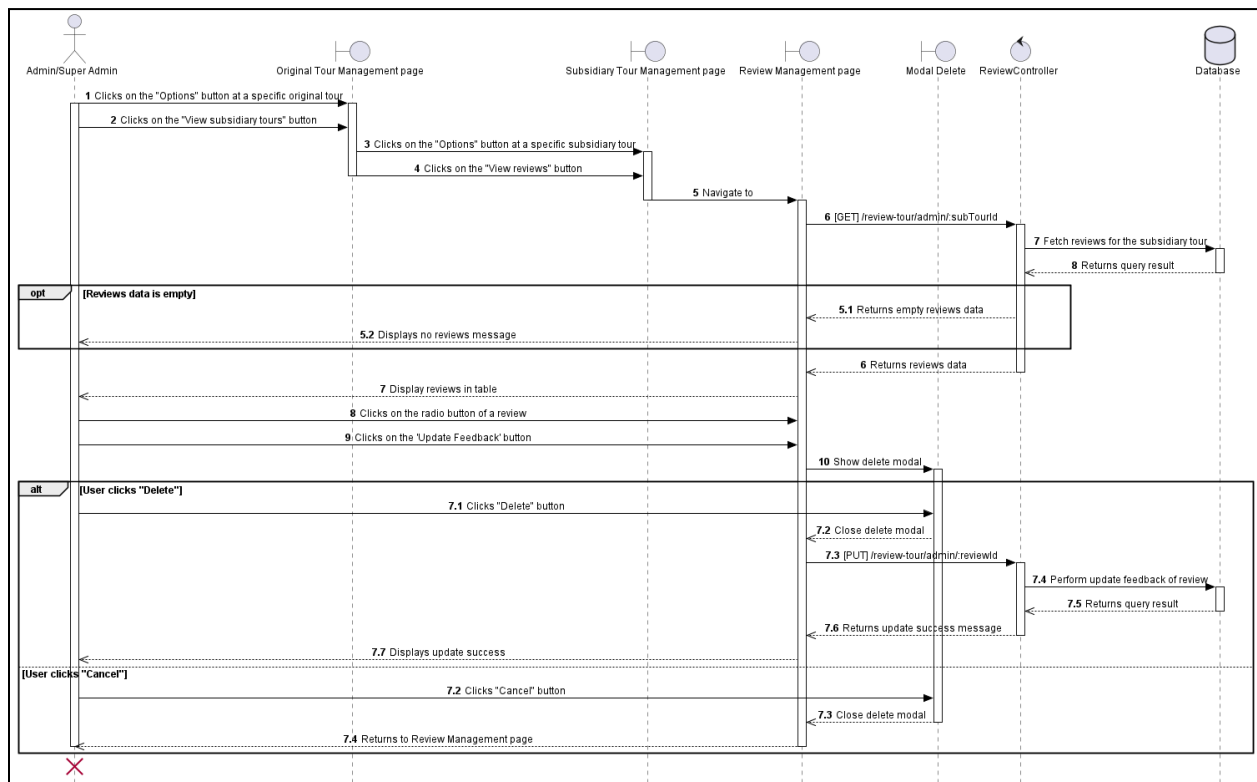


b. Sequence diagram

*Sequence diagram for View All Reviews

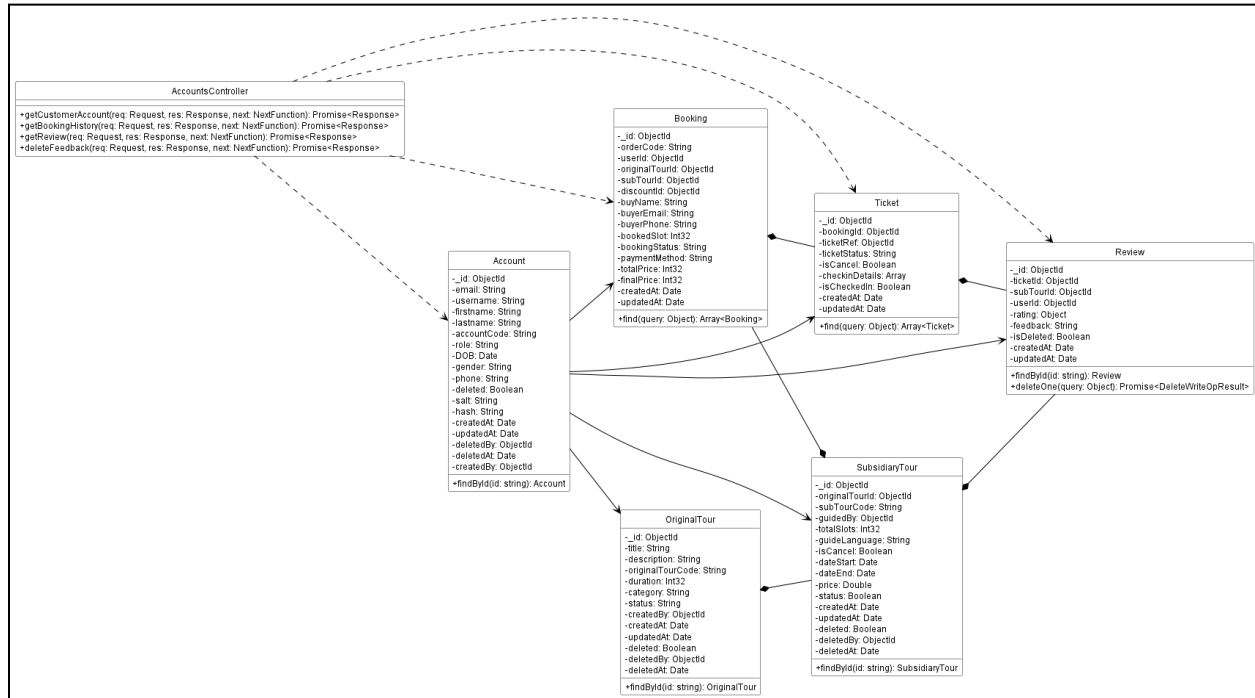


*Sequence diagram for Update Feedback



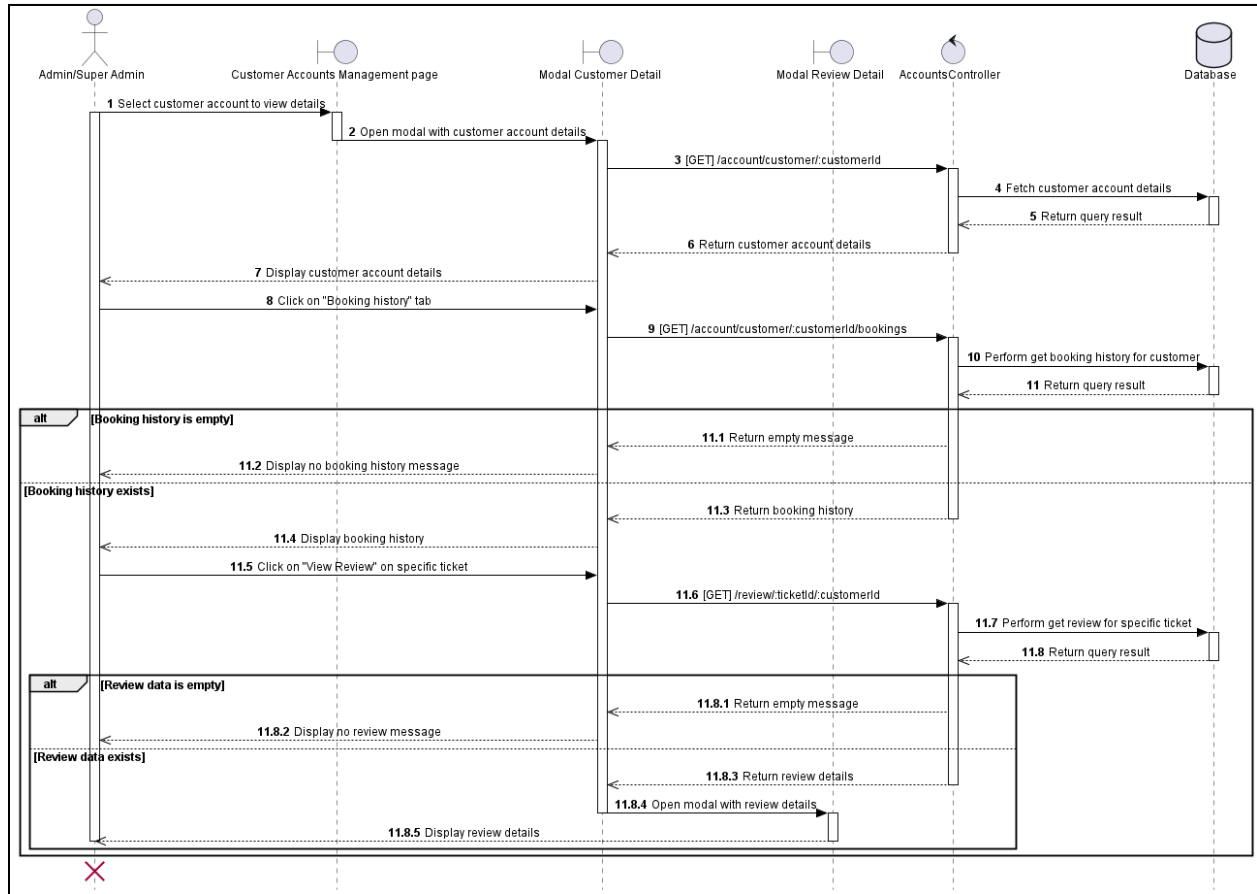
3.56 Manage Customer's Ticket Review

a. Class Diagram

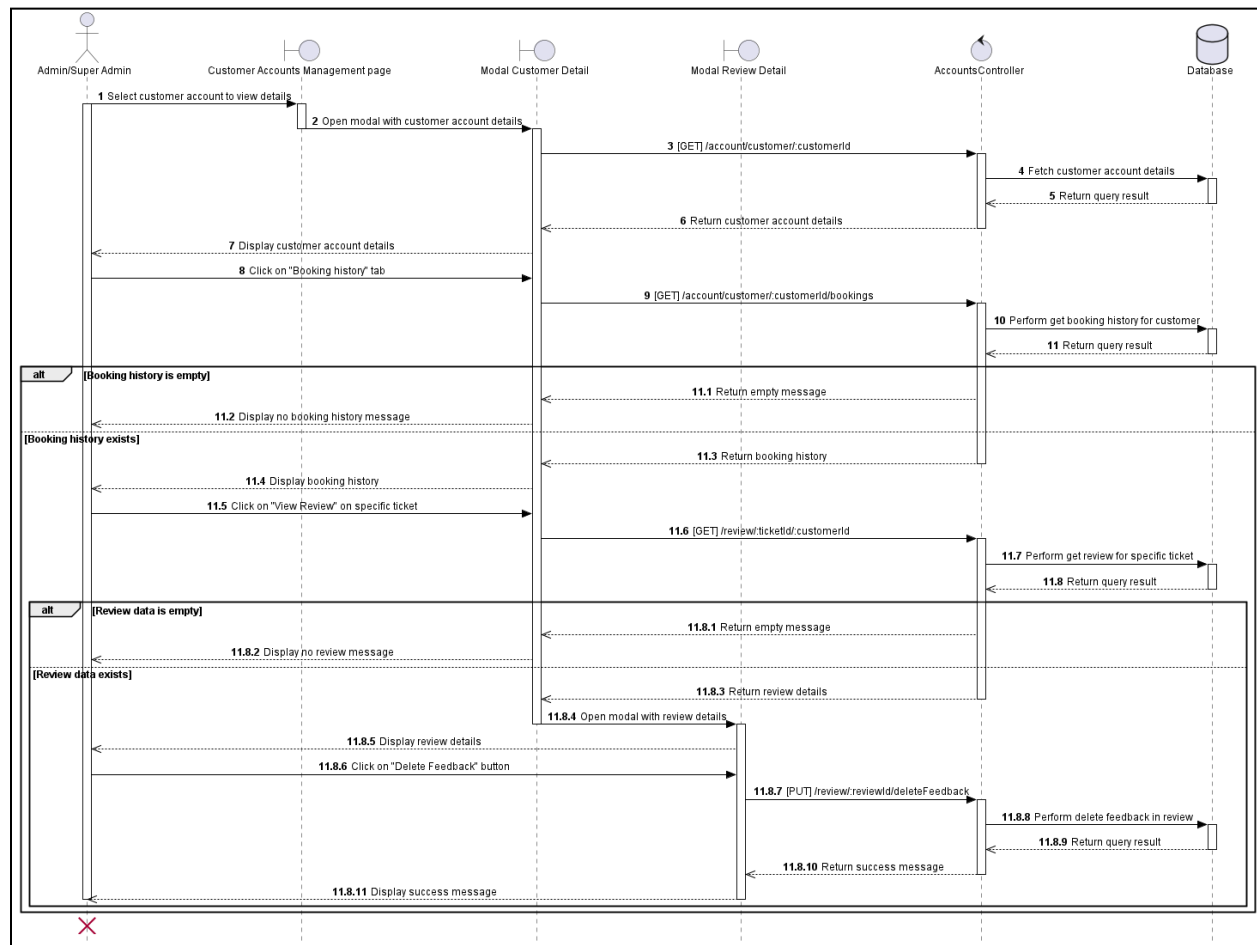


b. Sequence diagram

* Sequence diagram of View Review:

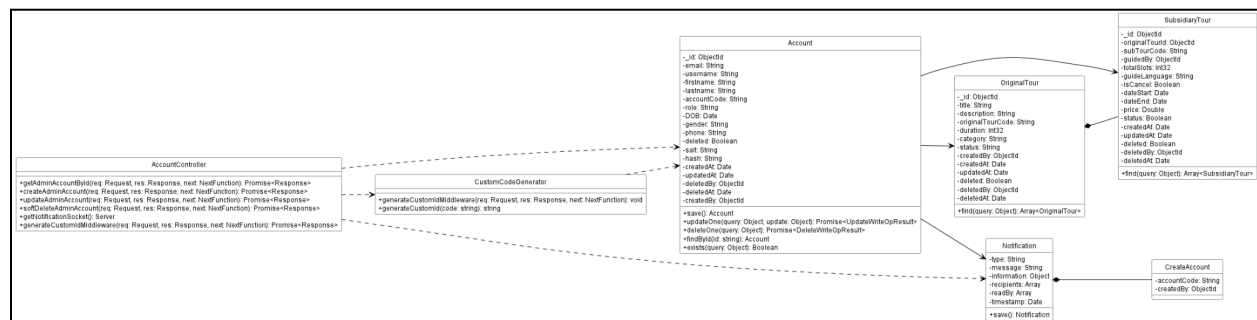


* Sequence diagram of Delete Feedback



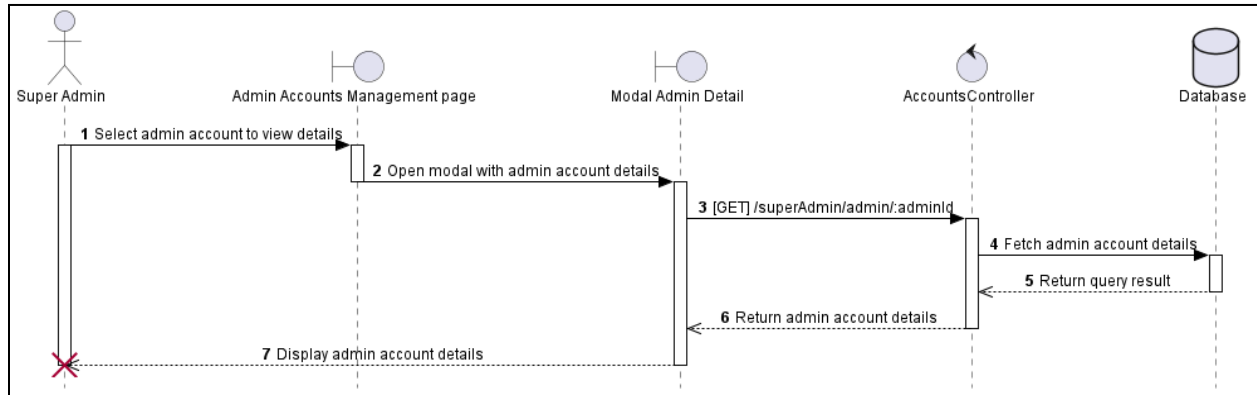
3.57 Manage Admin

a. Class diagram

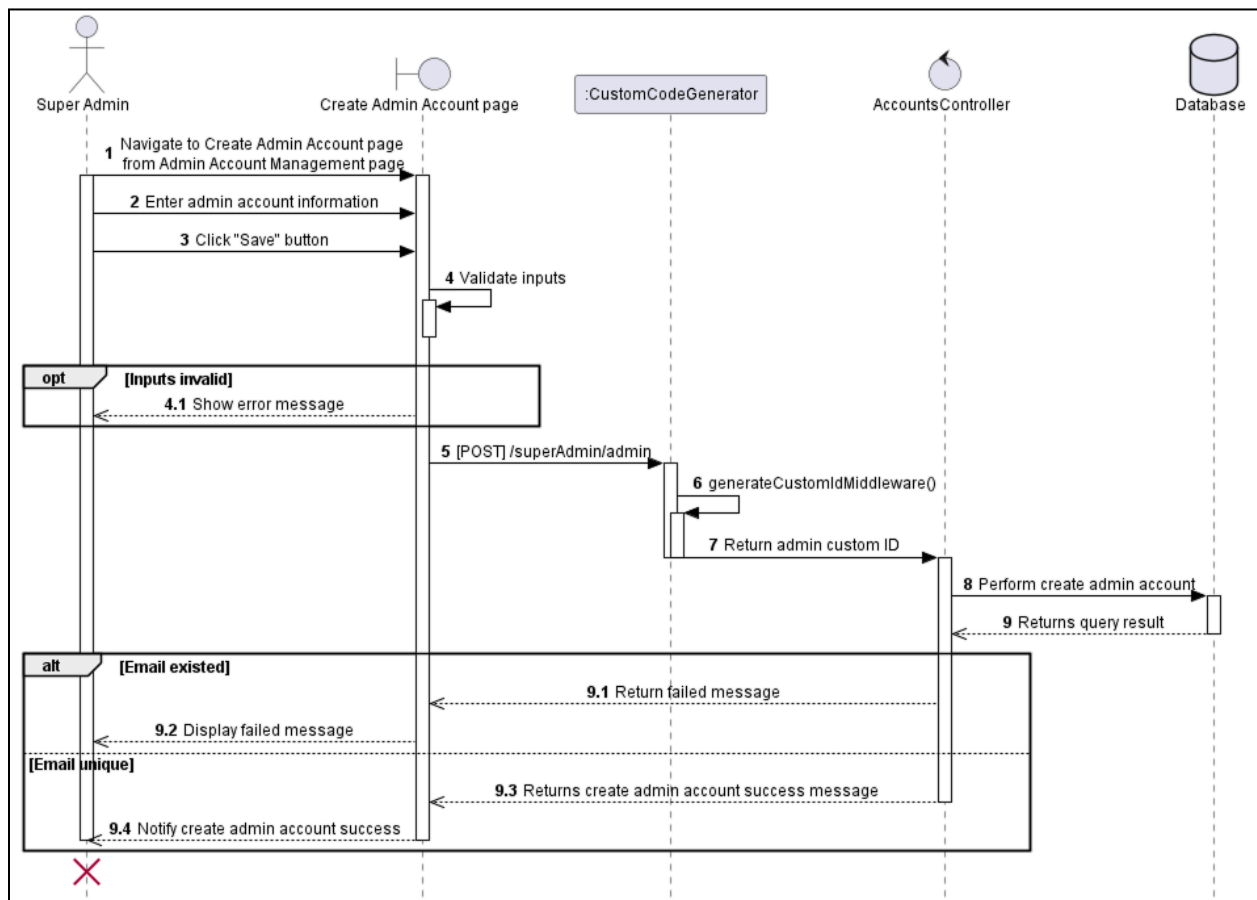


b. Sequence diagram

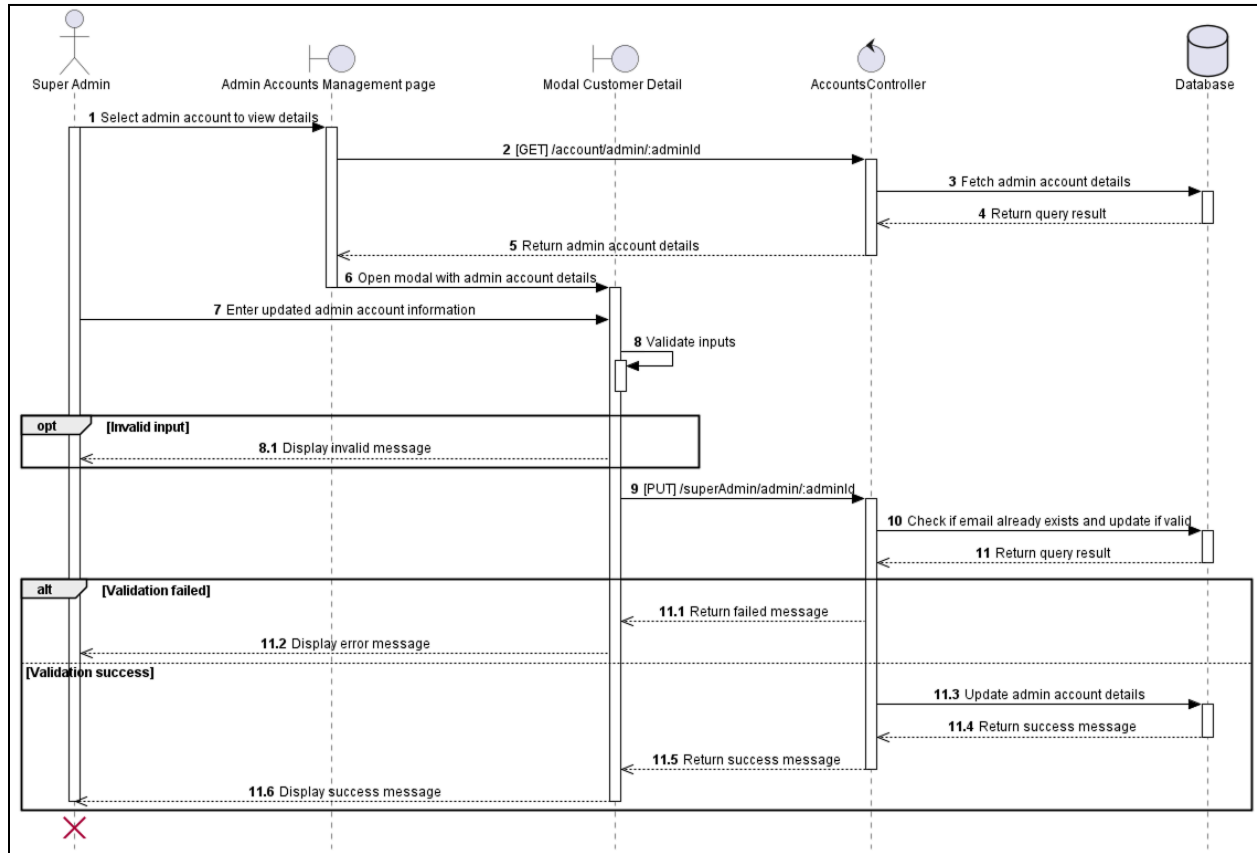
- Sequence diagram for View Admin Account:



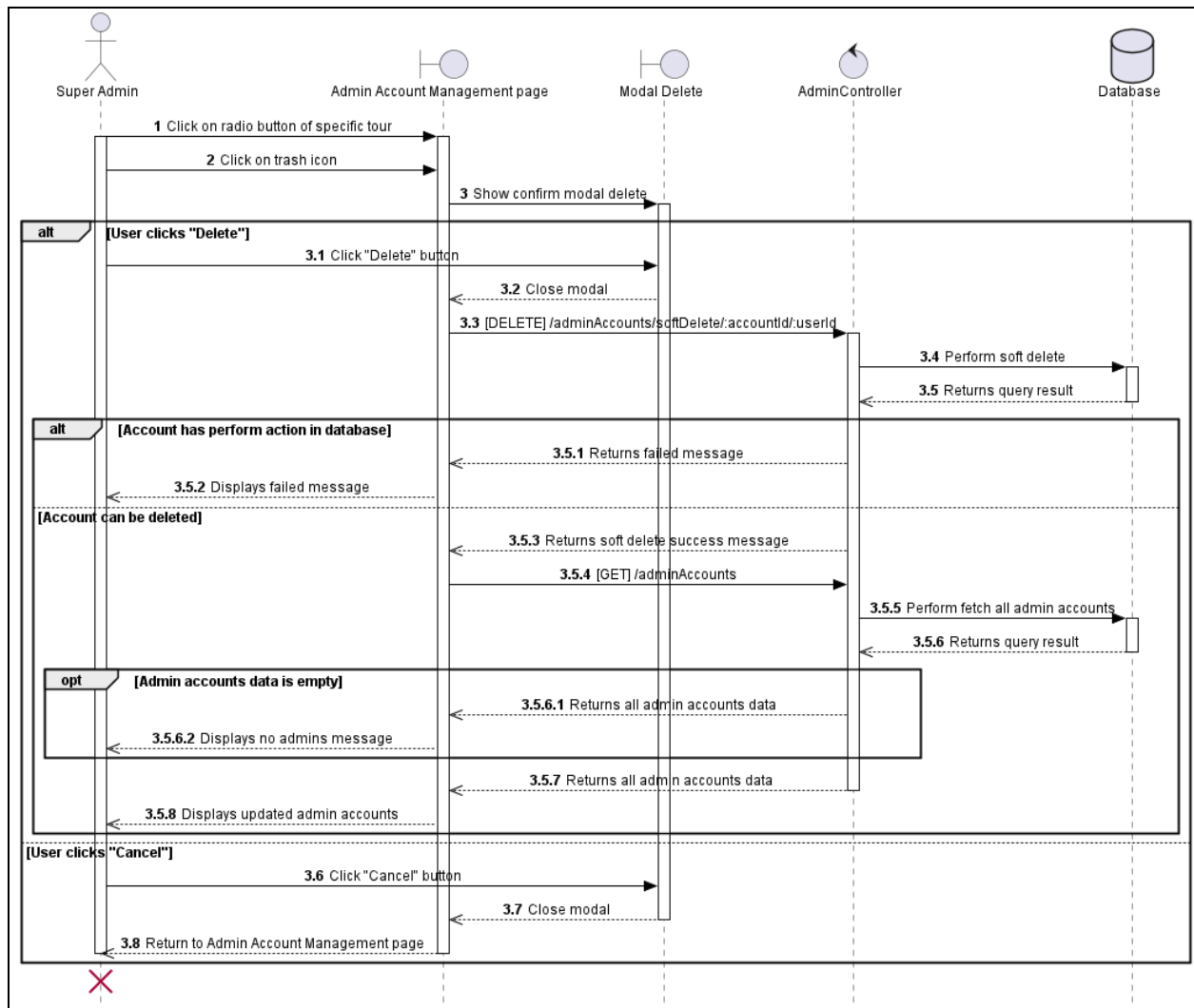
- Sequence diagram for Create Admin Account:



- Sequence diagram for Update Admin Account:

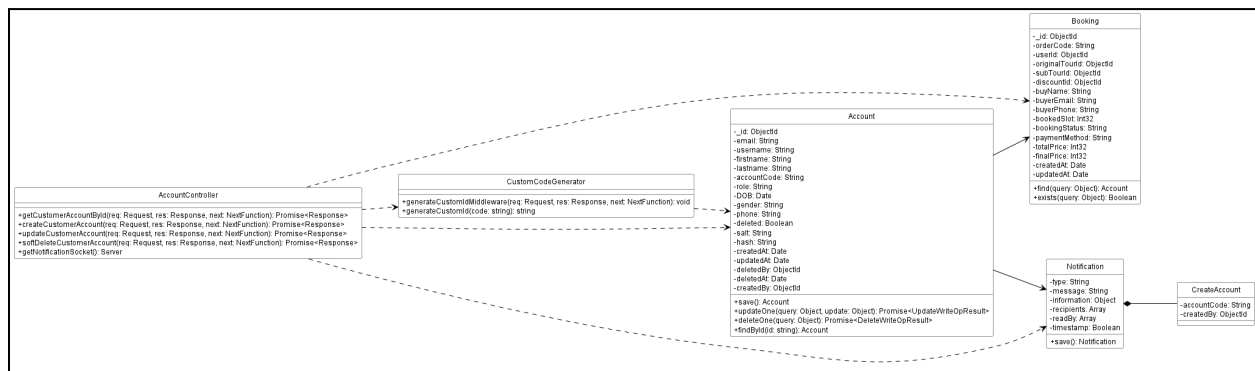


- Sequence diagram for Soft Delete Admin Account:



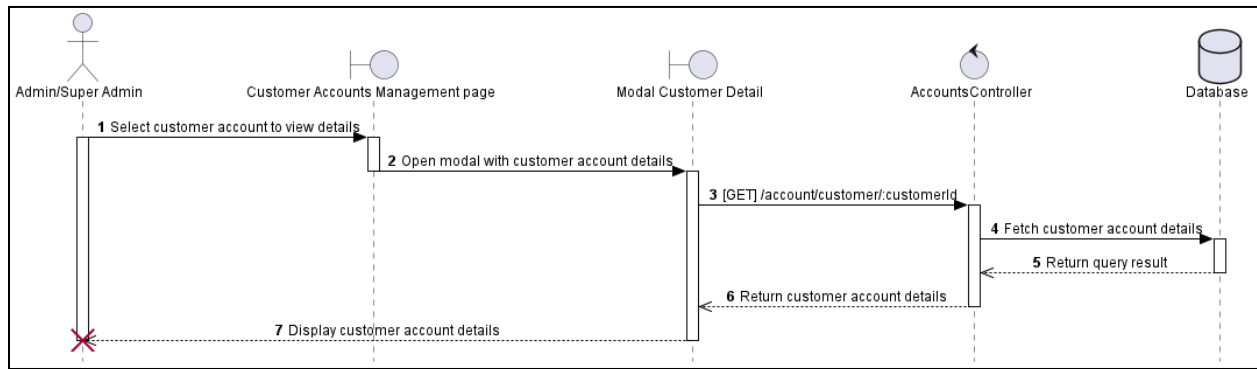
3.58 Manage Customer

a. Class diagram

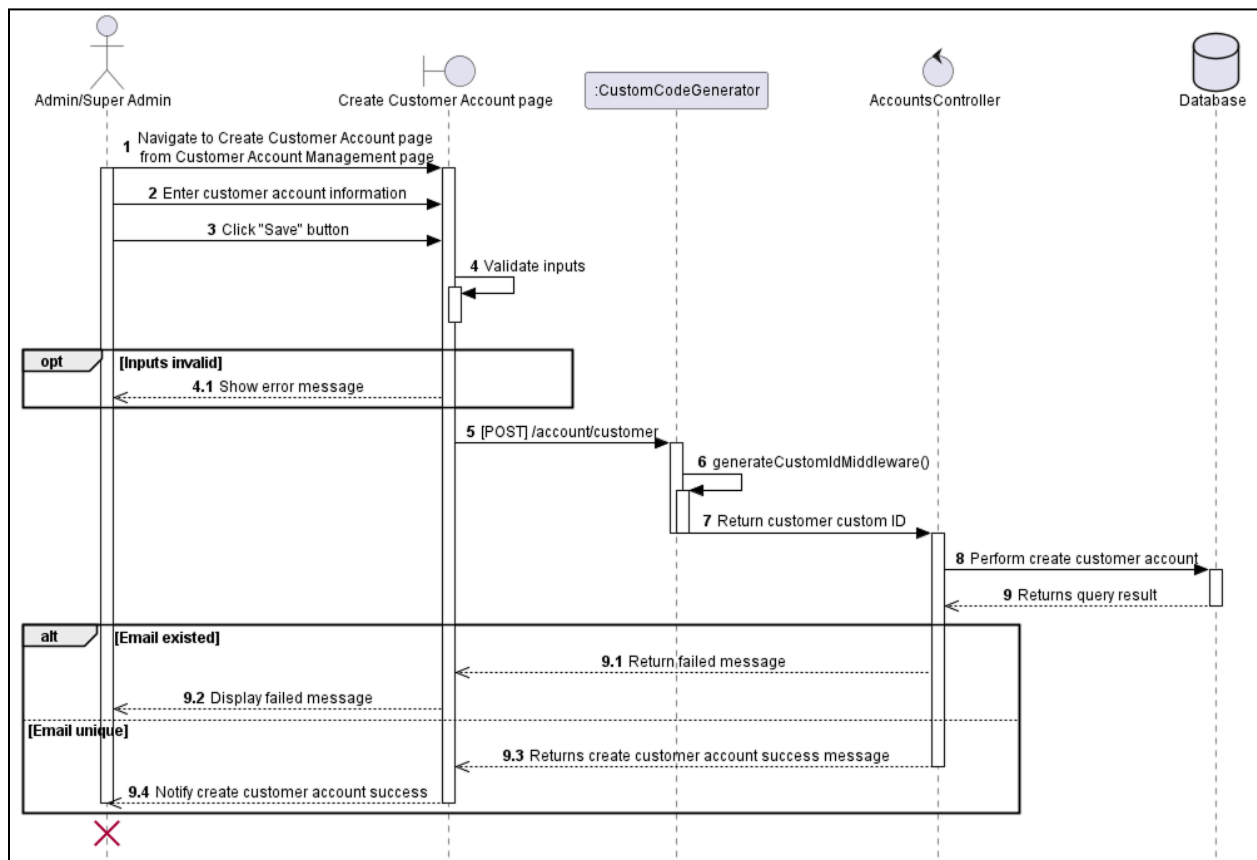


b. Sequence diagram

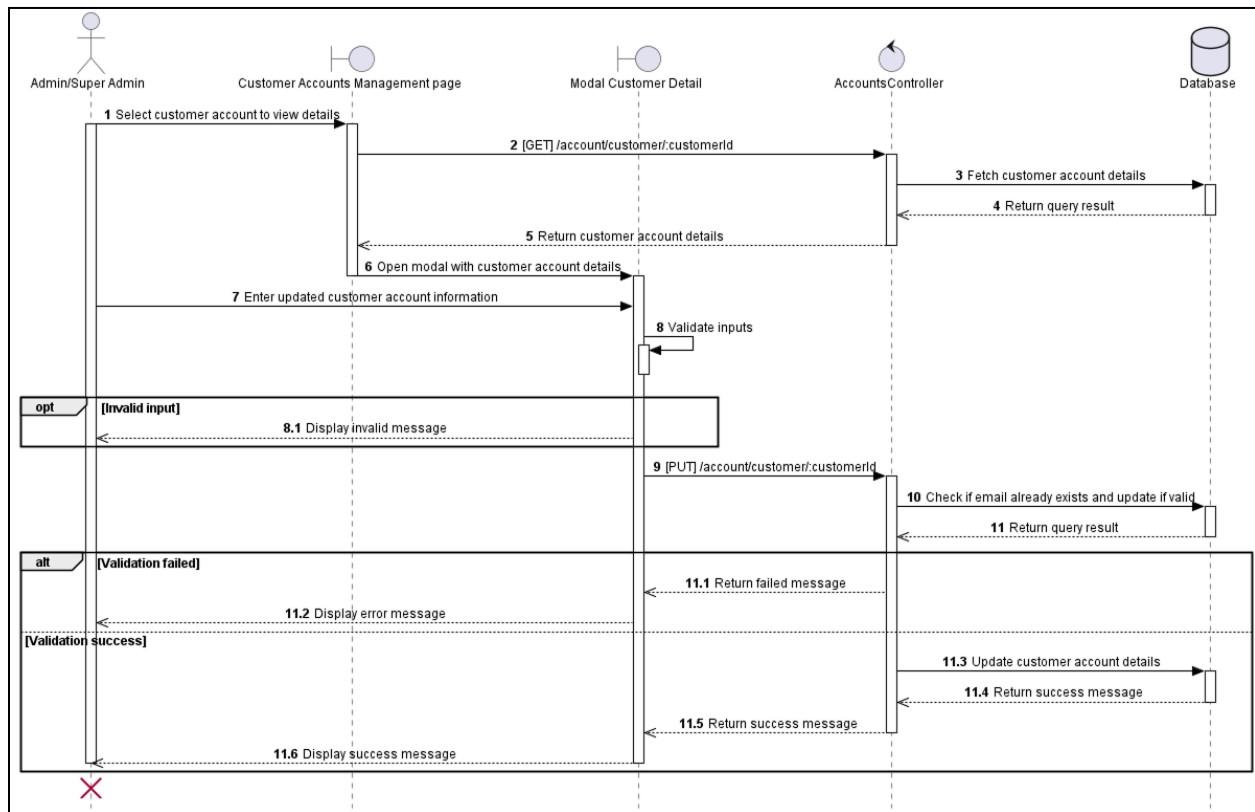
- Sequence for view customer details:



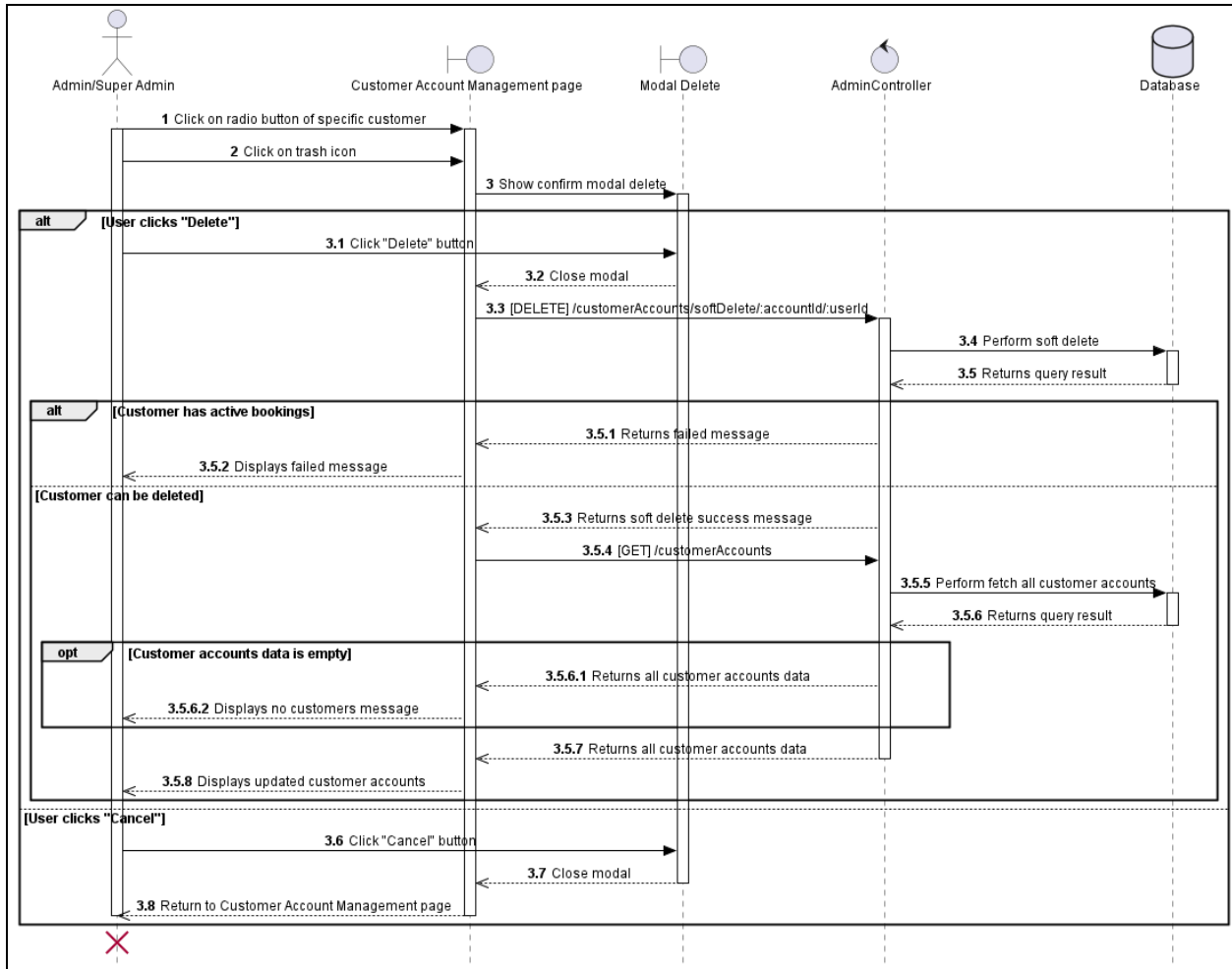
- Sequence diagram for create customer:



- Sequence diagram for update customer account:

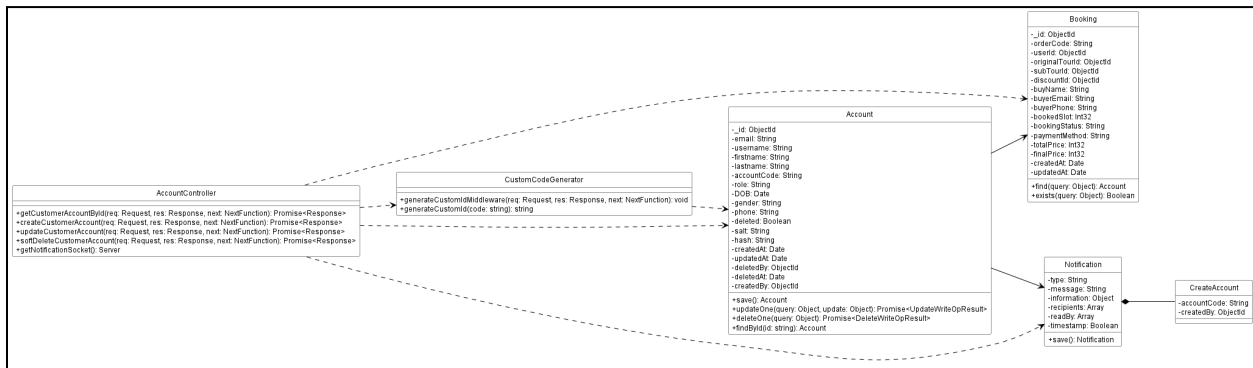


- Sequence diagram for soft delete customer:



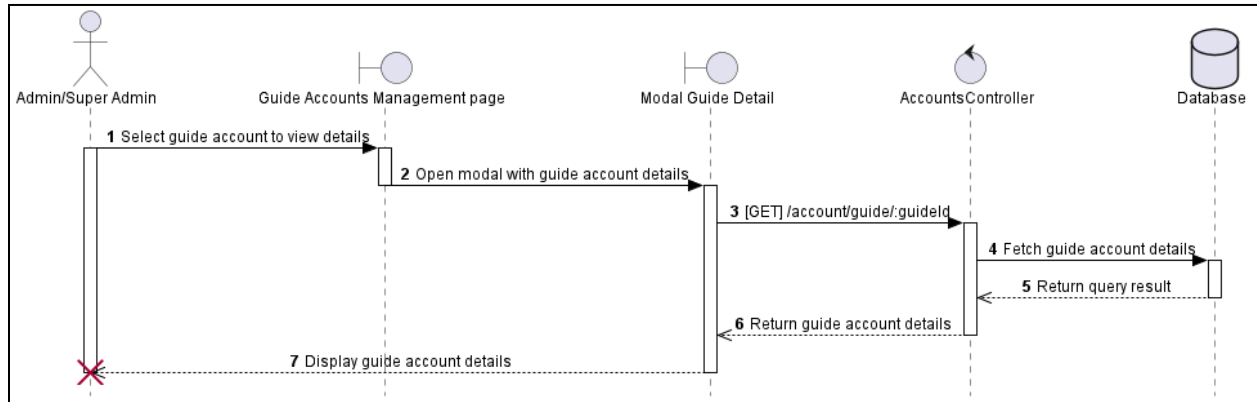
3.59 Manage Guide

a. Class diagram

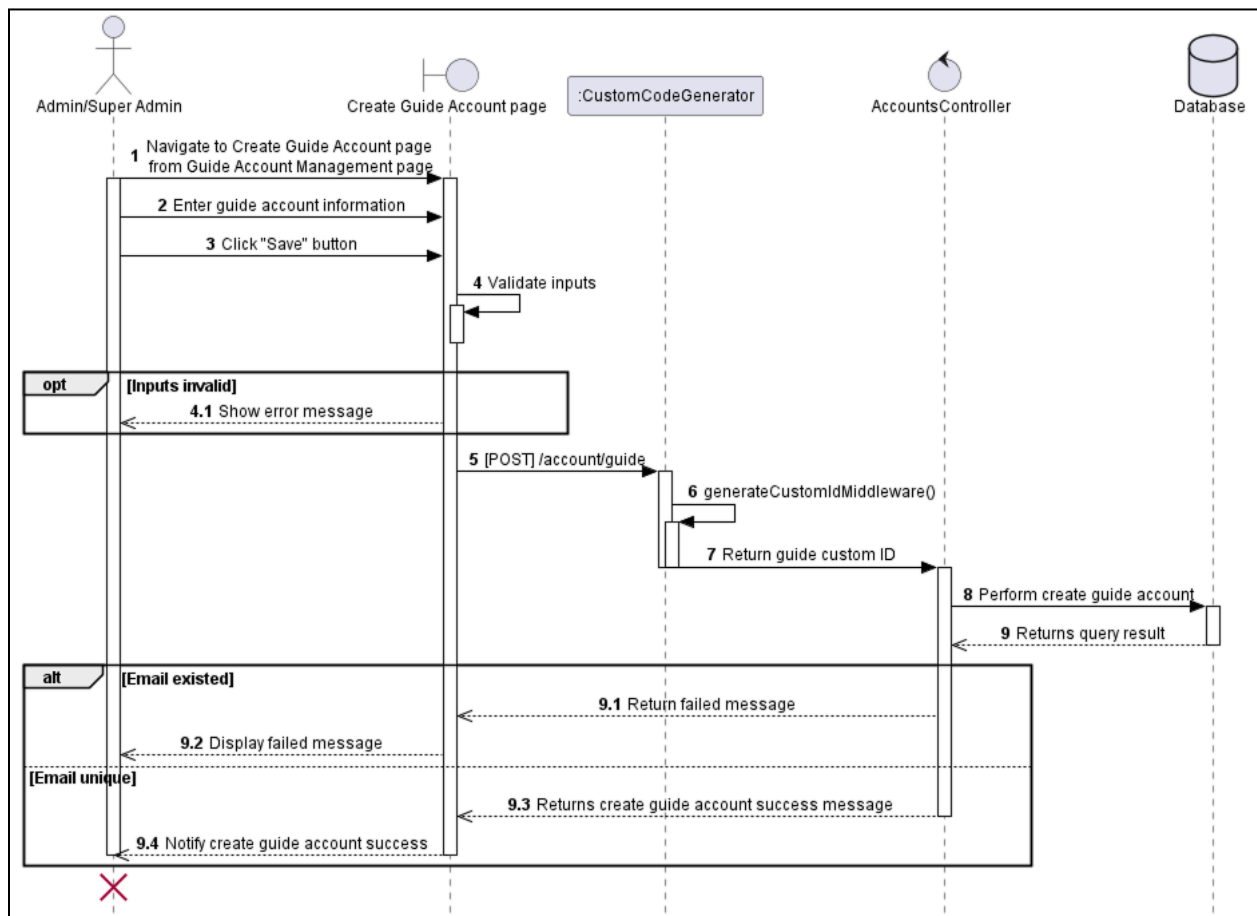


b. Sequence diagram

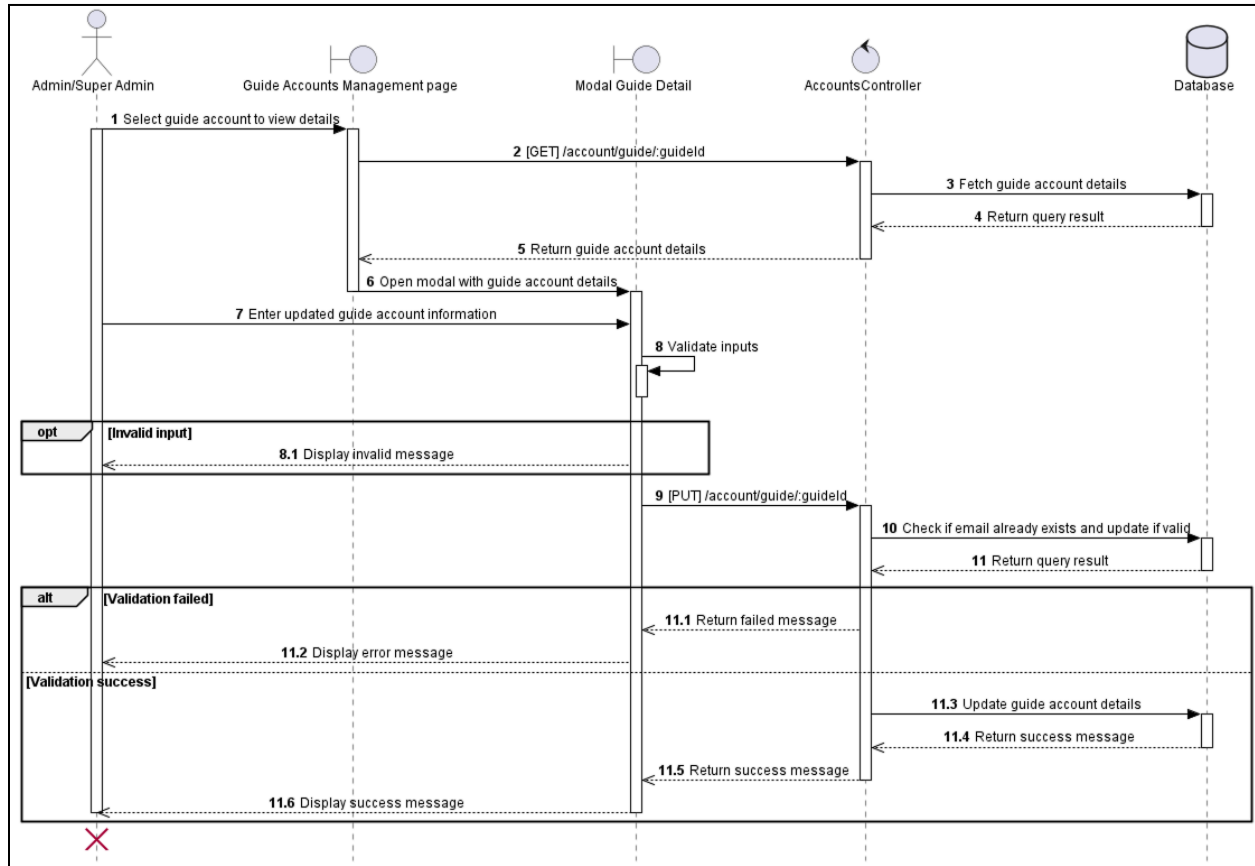
* Sequence diagram for View Guide Account:



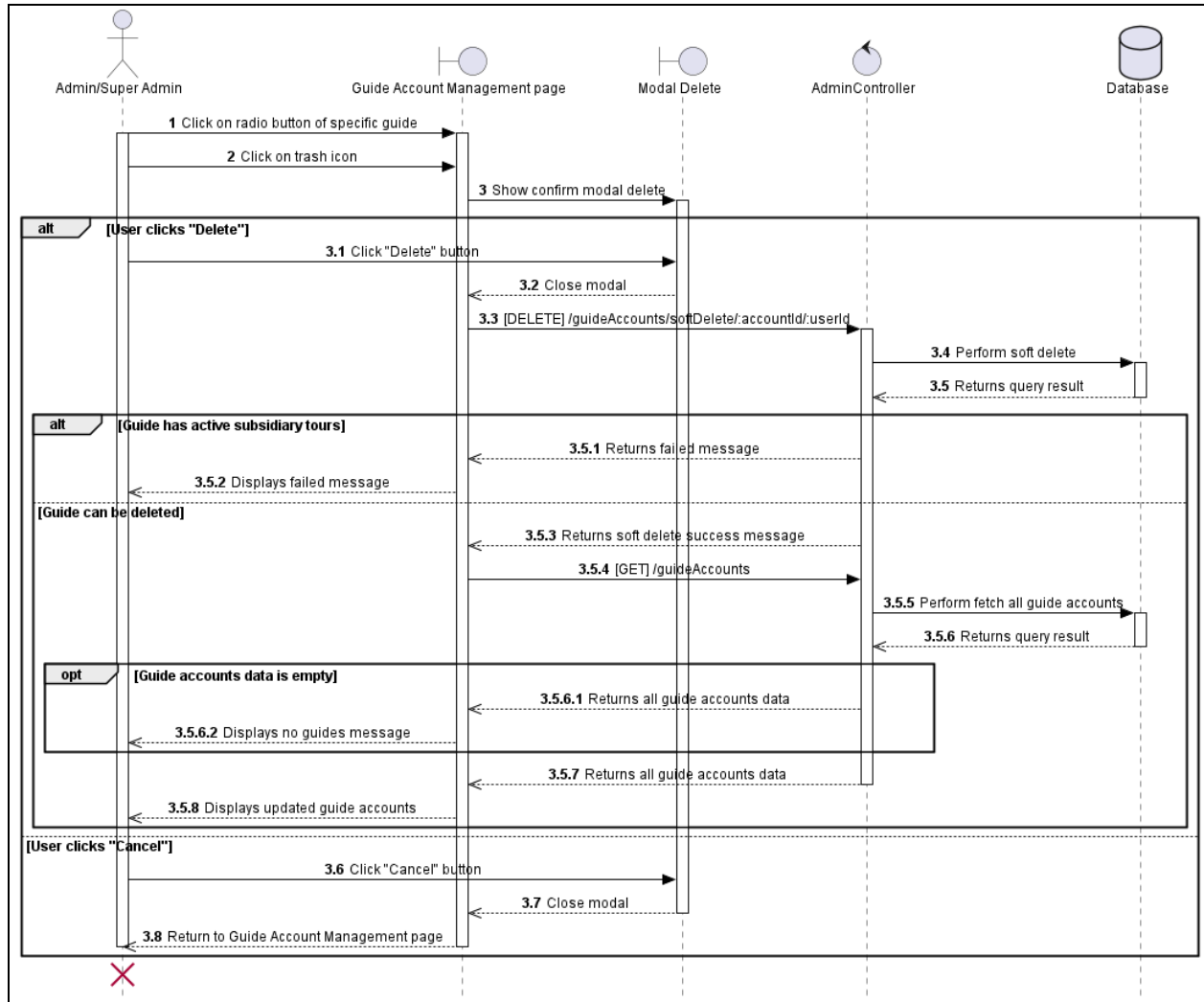
* Sequence diagram for Create Guide Account:



* Sequence diagram for Update Guide Account:

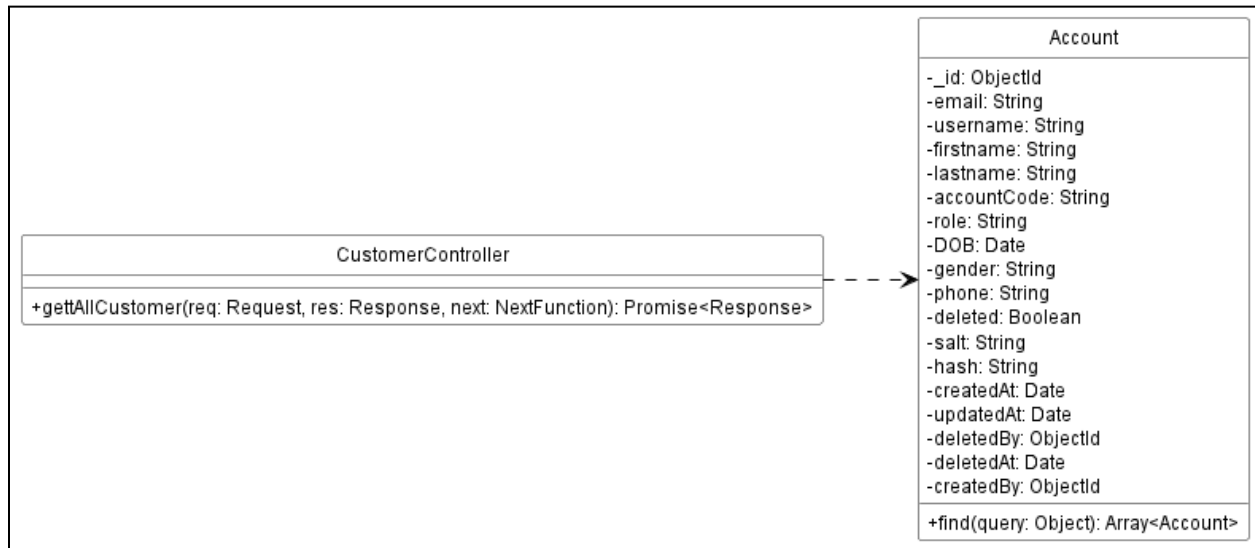


* Sequence diagram for Soft Delete Guide Account:

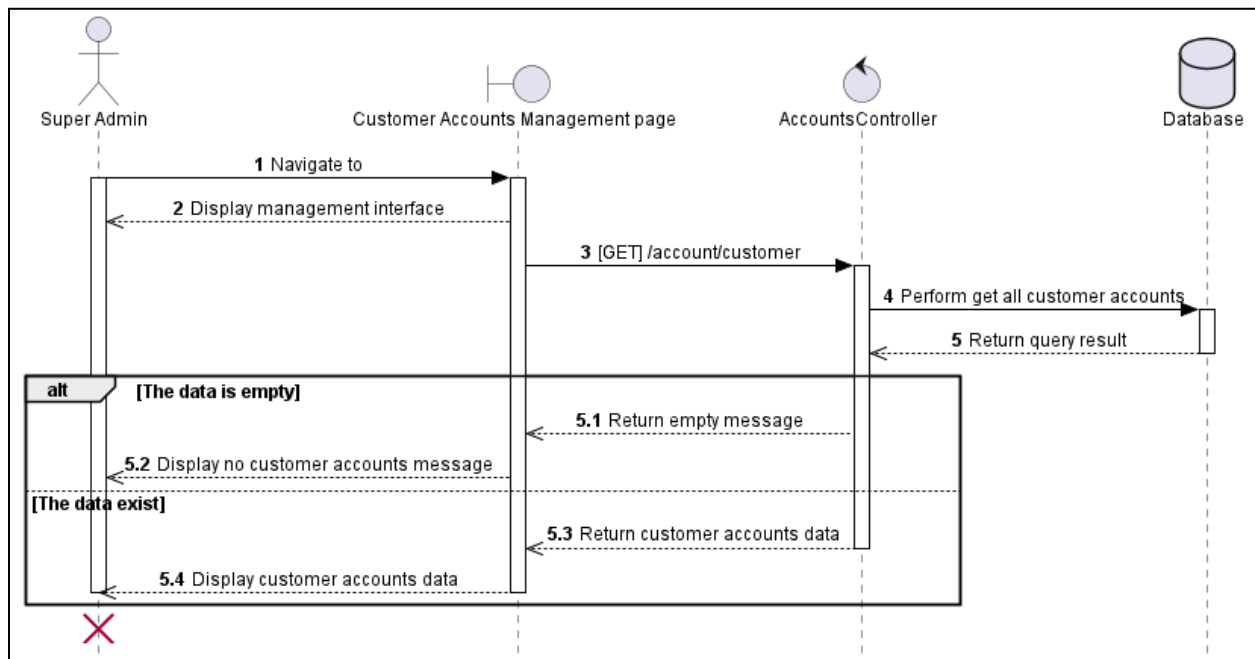


3.60 View All Customer

a. Class diagram

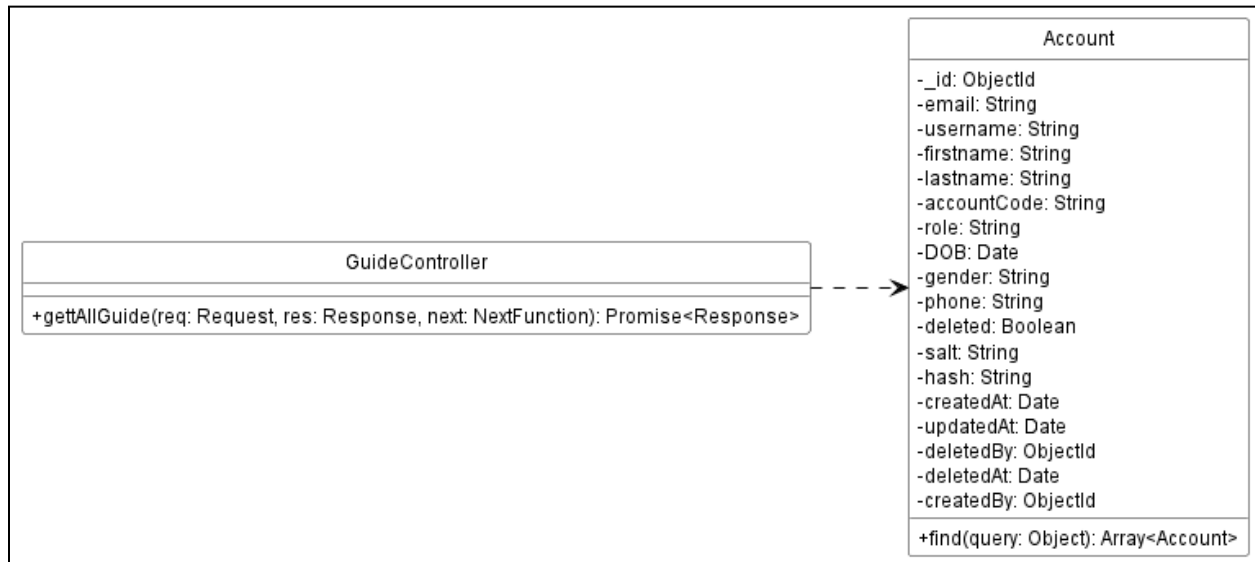


b. Sequence diagram

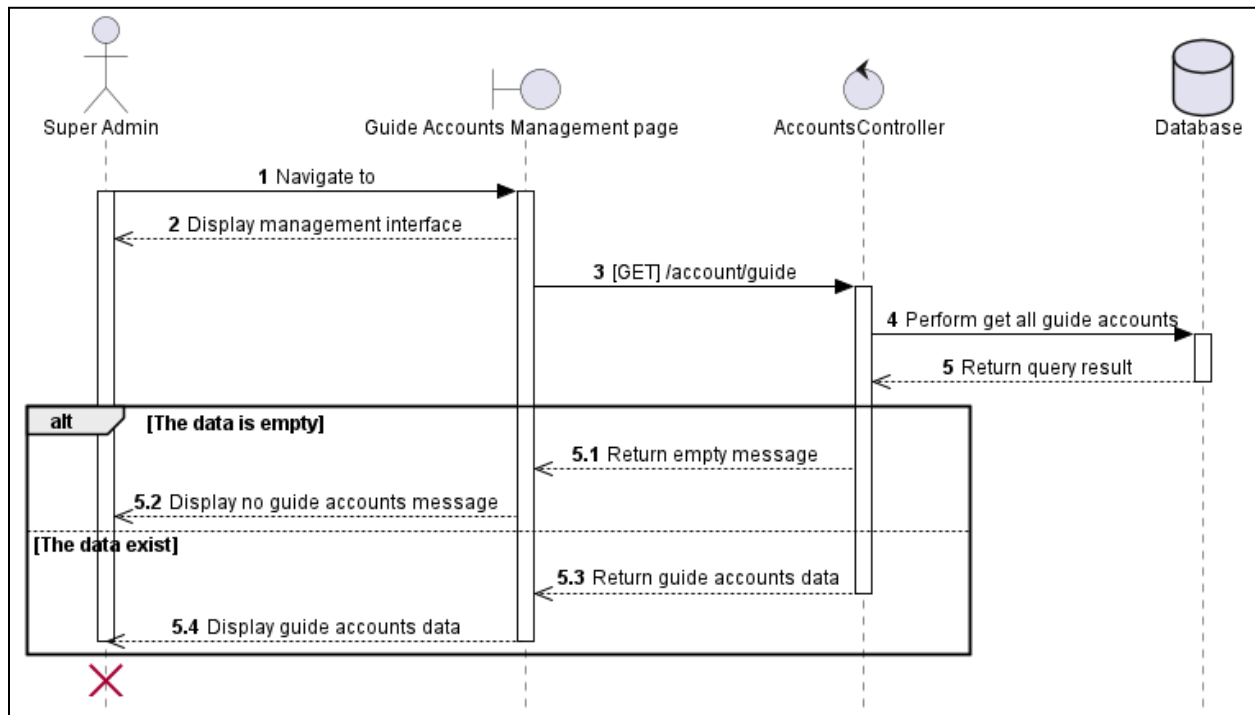


3.61 View All Guide

a. Class diagram

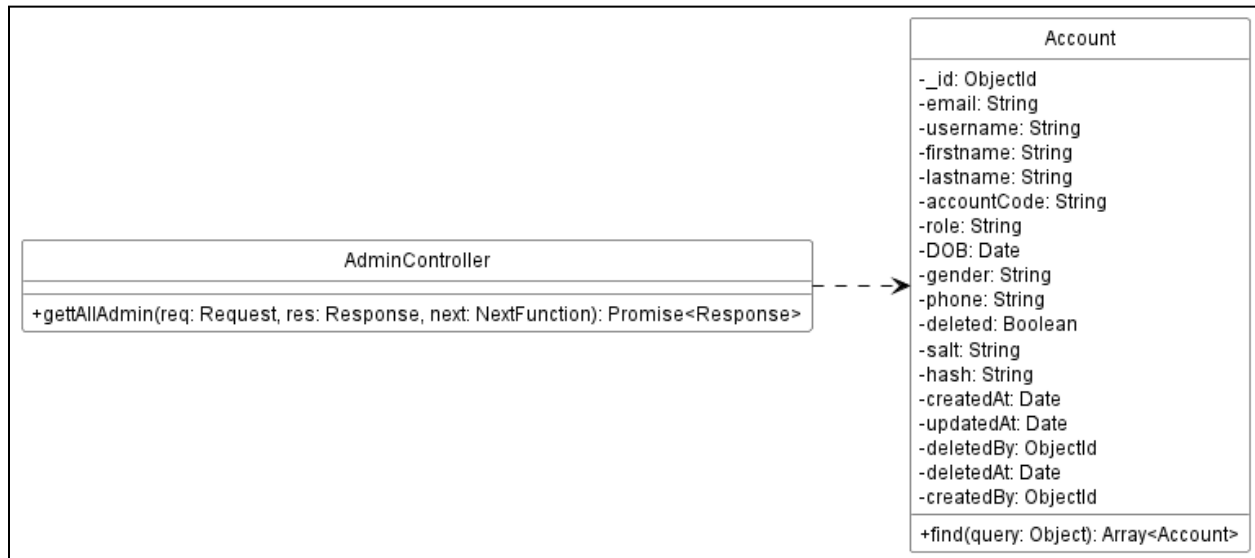


b. Sequence diagram

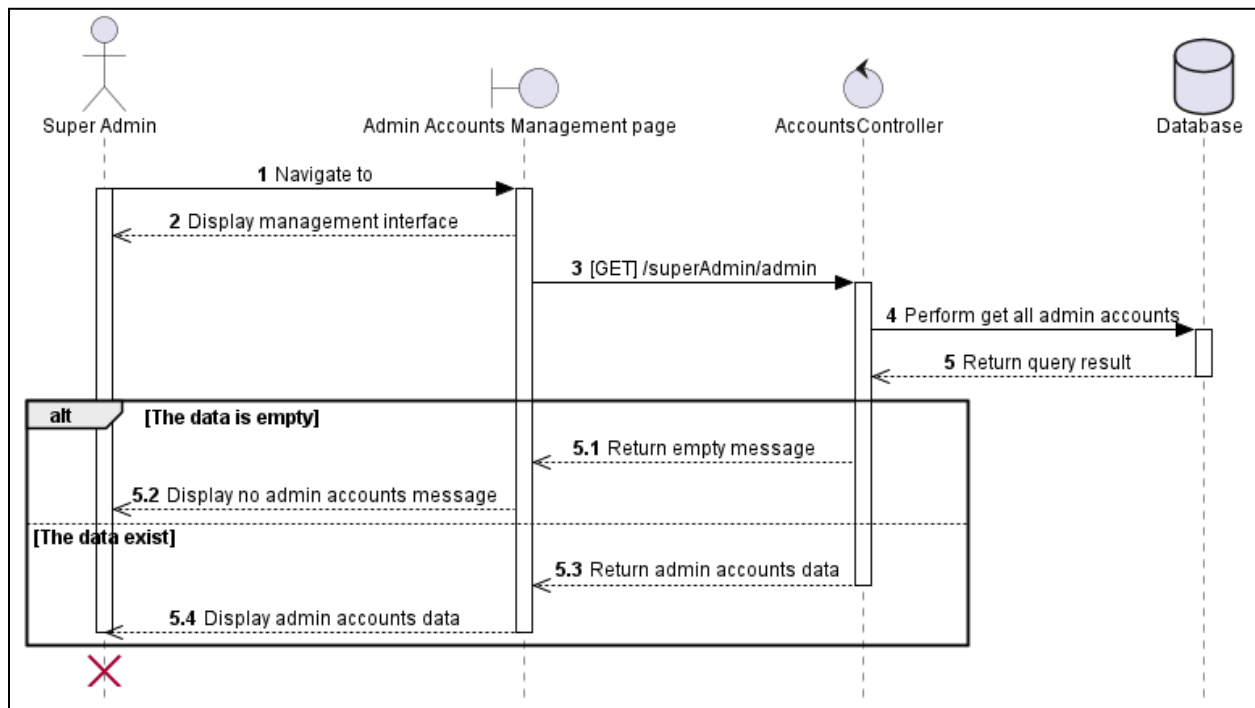


3.62 View All Admin

c. Class diagram

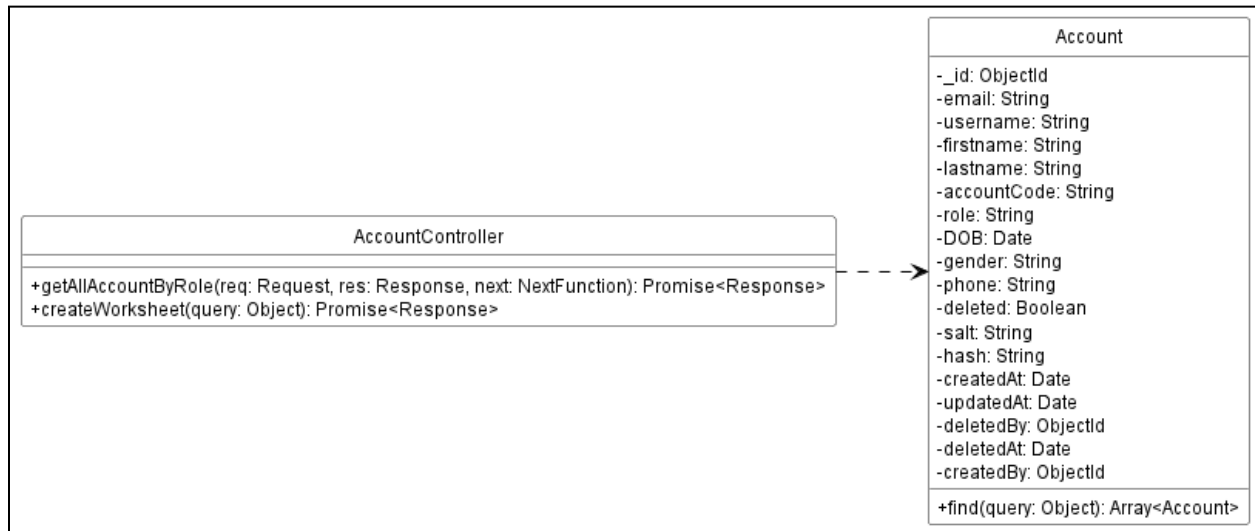


d. Sequence diagram



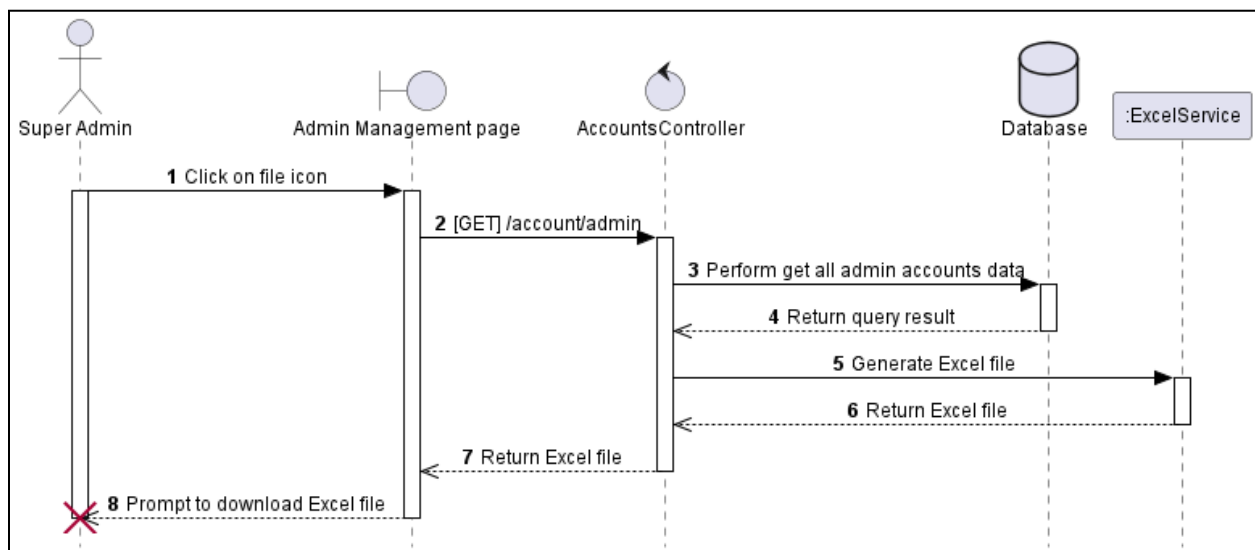
3.63 Export Account Data To Excel File

a. Class diagram

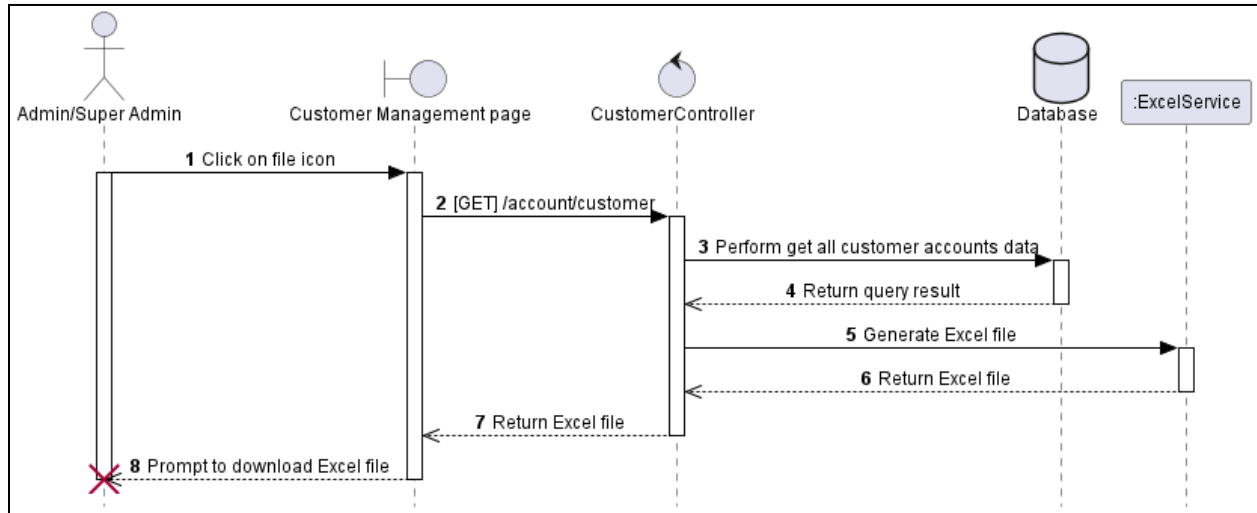


b. Sequence diagram

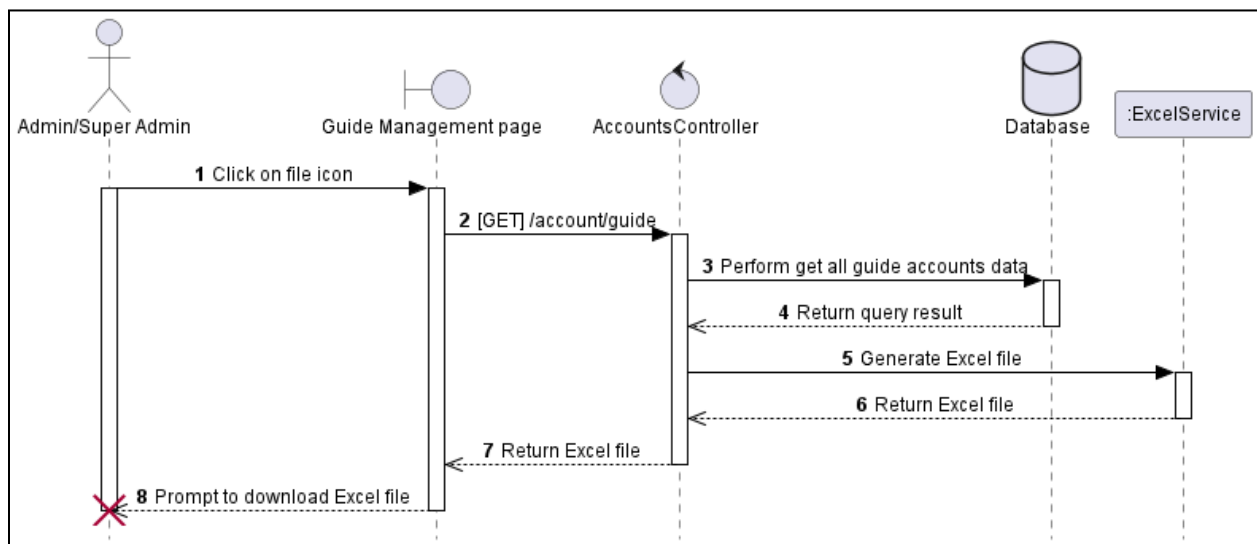
- Sequence diagram for Export Admin Account to excel file:



- Sequence diagram for Export Customer Account to excel file:

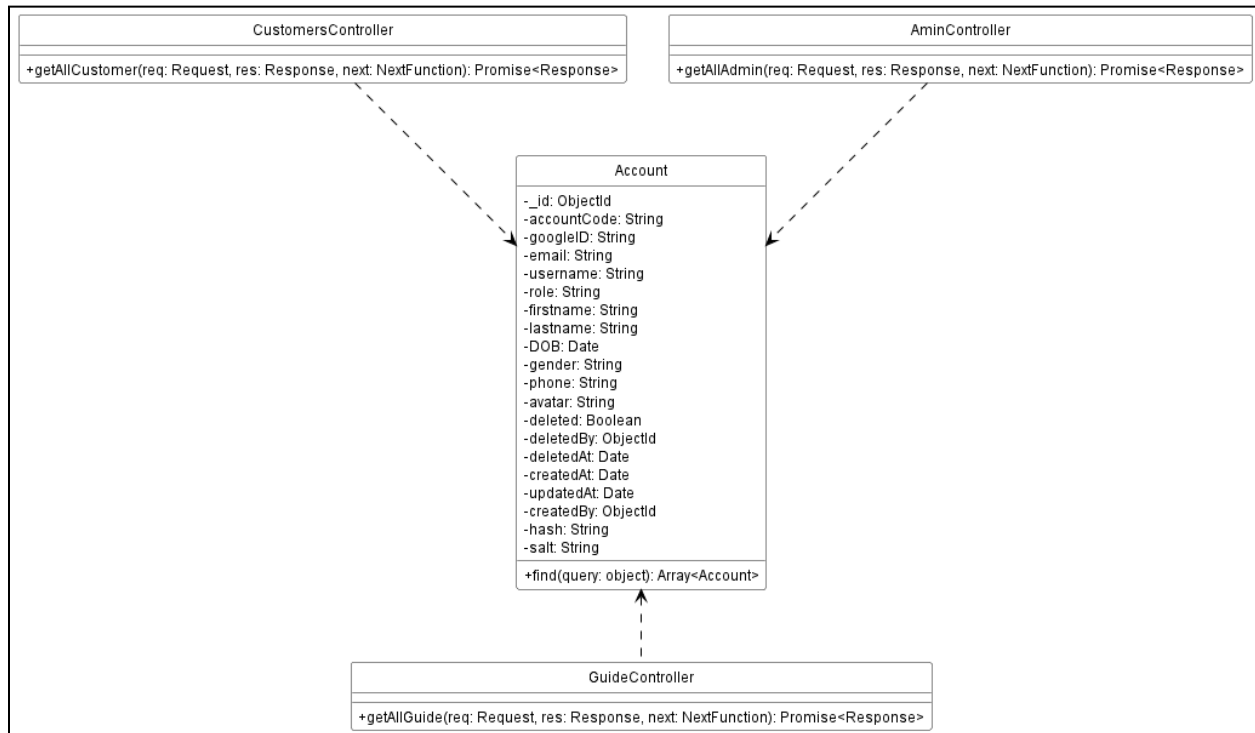


- Sequence diagram for Export Guide Account to excel file:



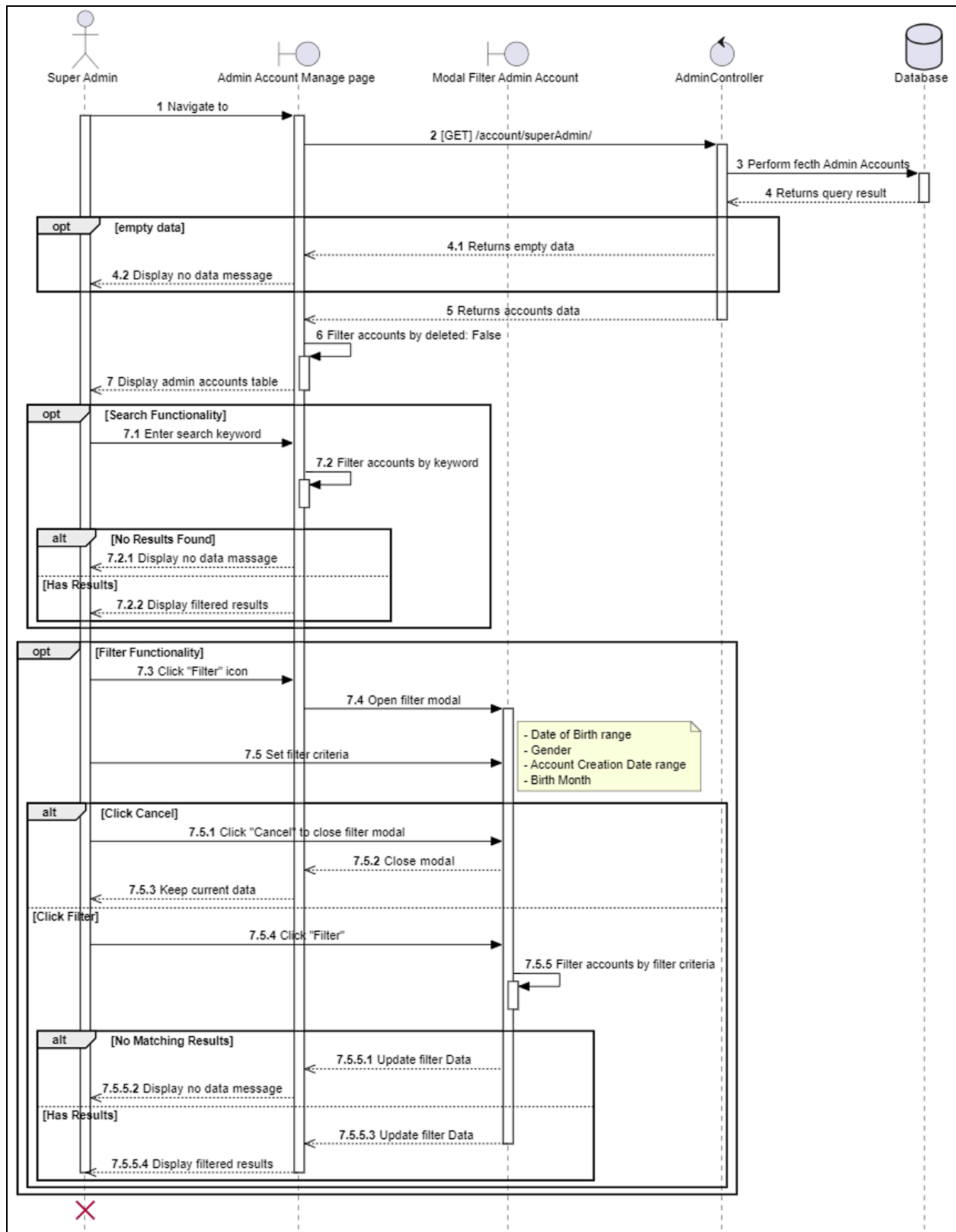
3.64 Filter & Search Account

a. Class Diagram

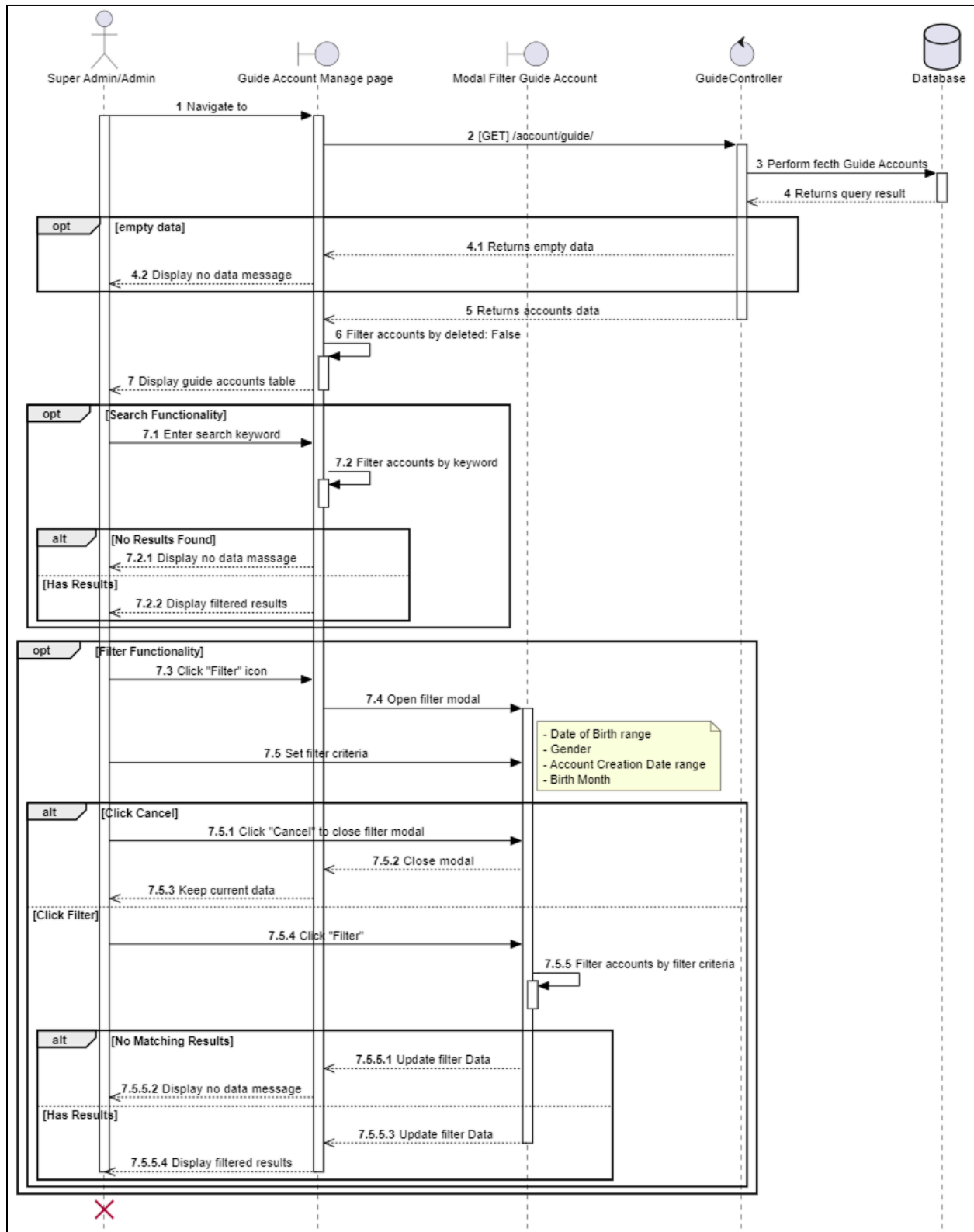


b. Sequence Diagram(s)

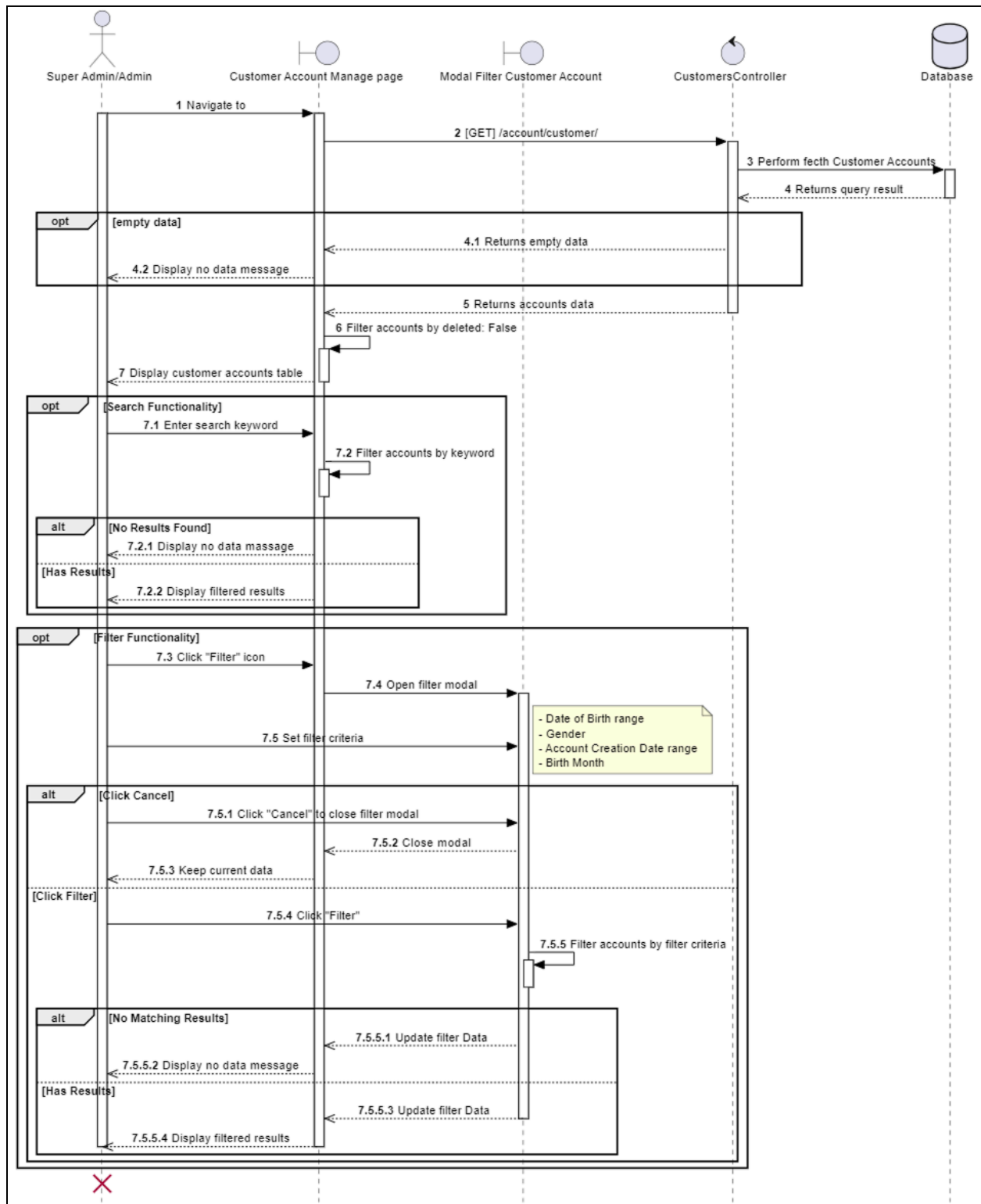
*Sequence diagram for Filter & Search Admin Account.



*Sequence diagram for Filter & Guide Admin Account.

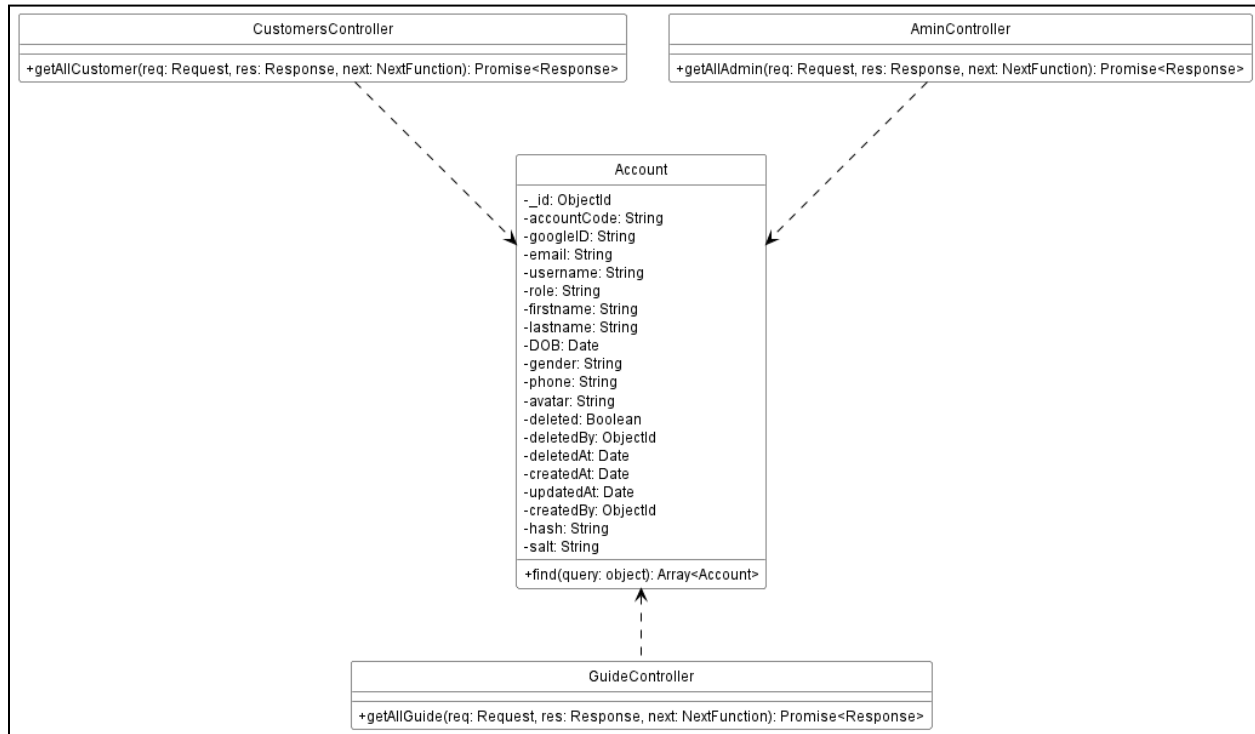


*Sequence diagram for Filter & Search Customer Account.



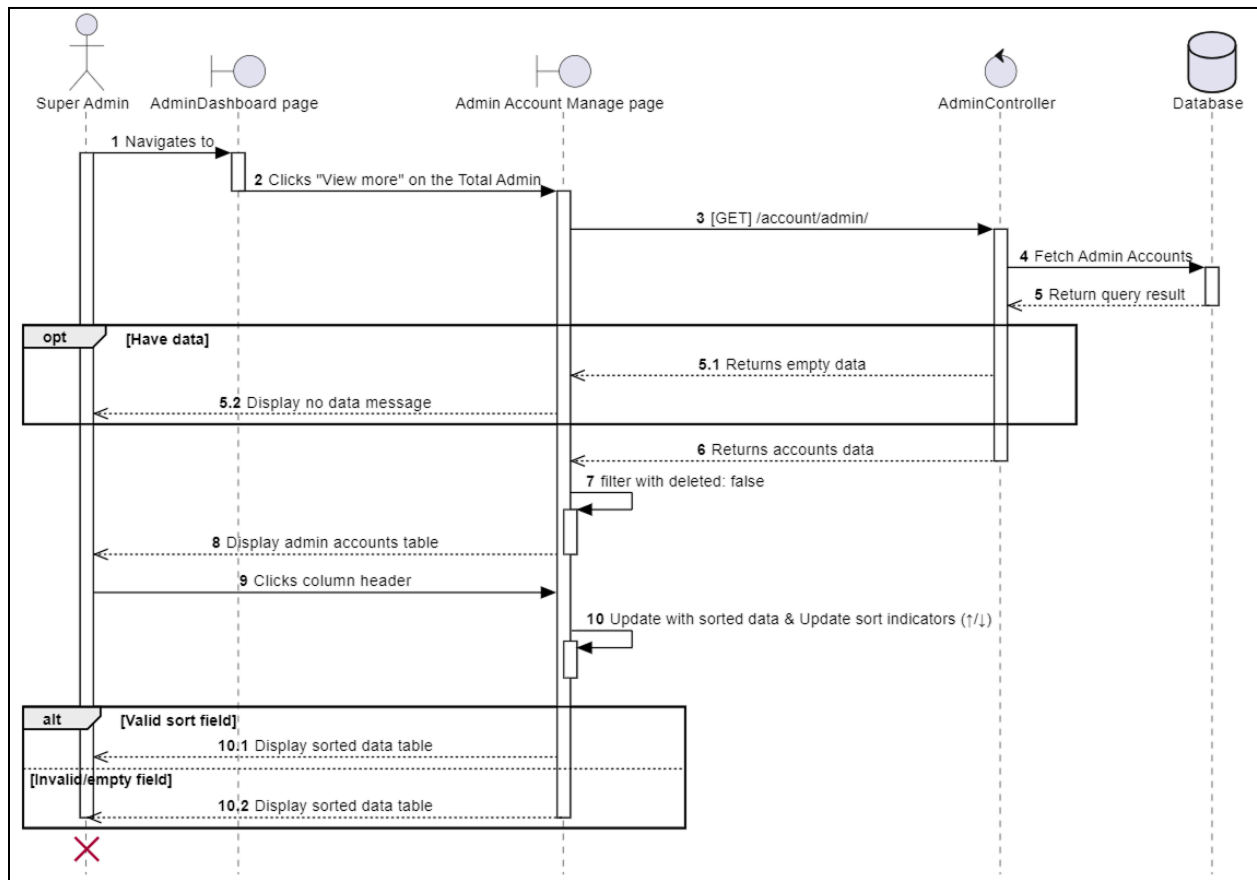
3.65 Sort Account

a. Class Diagram

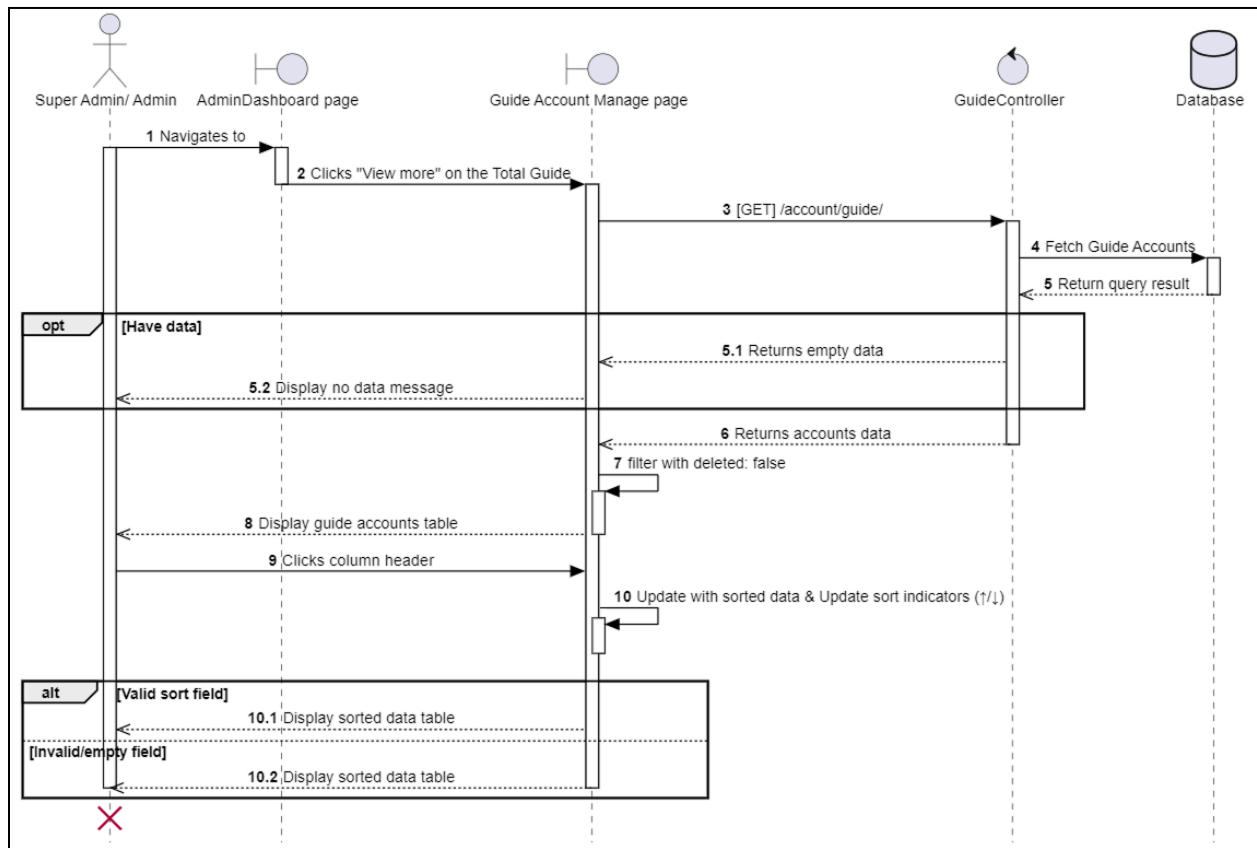


b. Sequence Diagram(s)

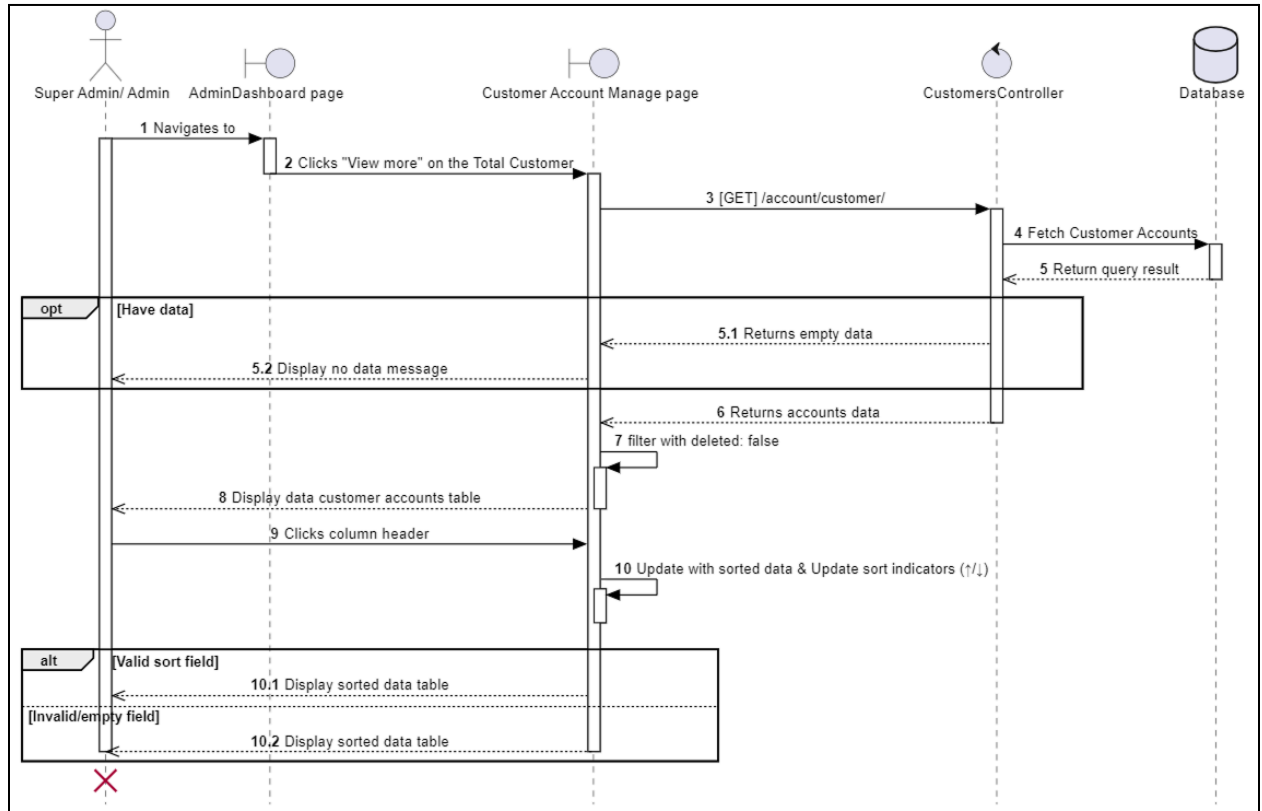
*Sequence diagram for Sort Admin Account.



*Sequence diagram for Sort Guide Account.

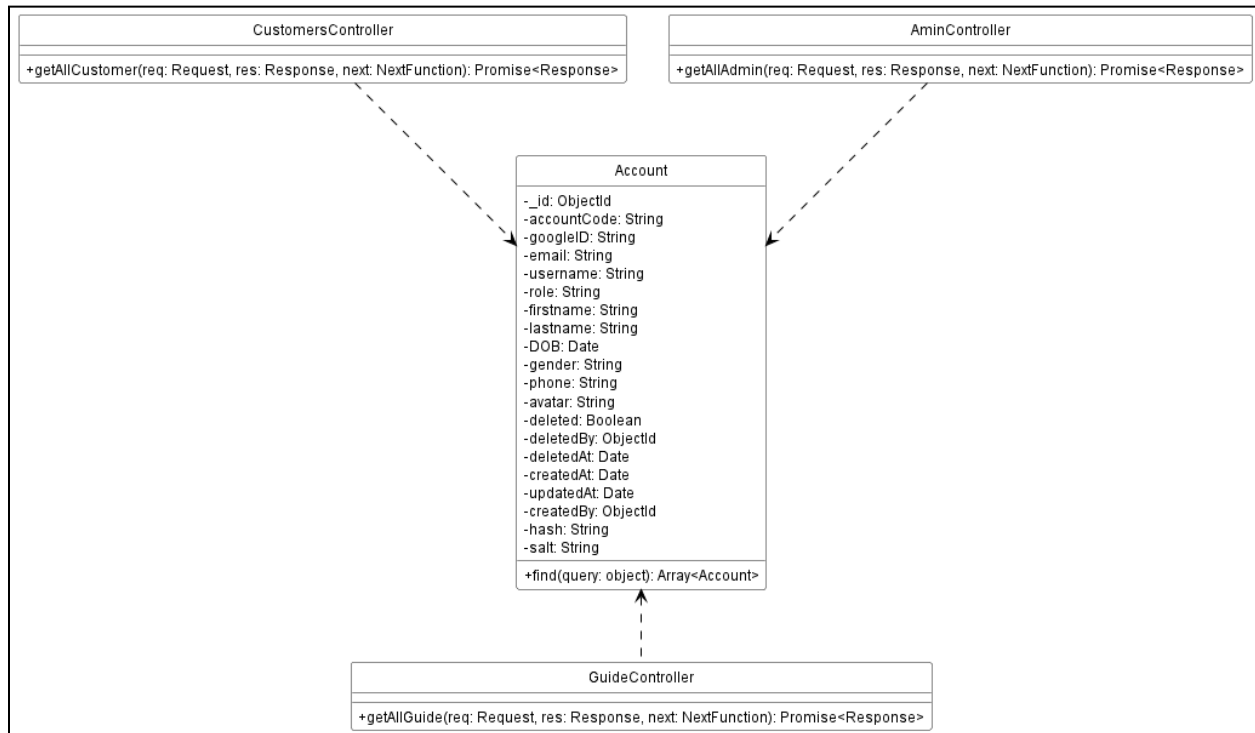


*Sequence diagram for Sort Customer Account.



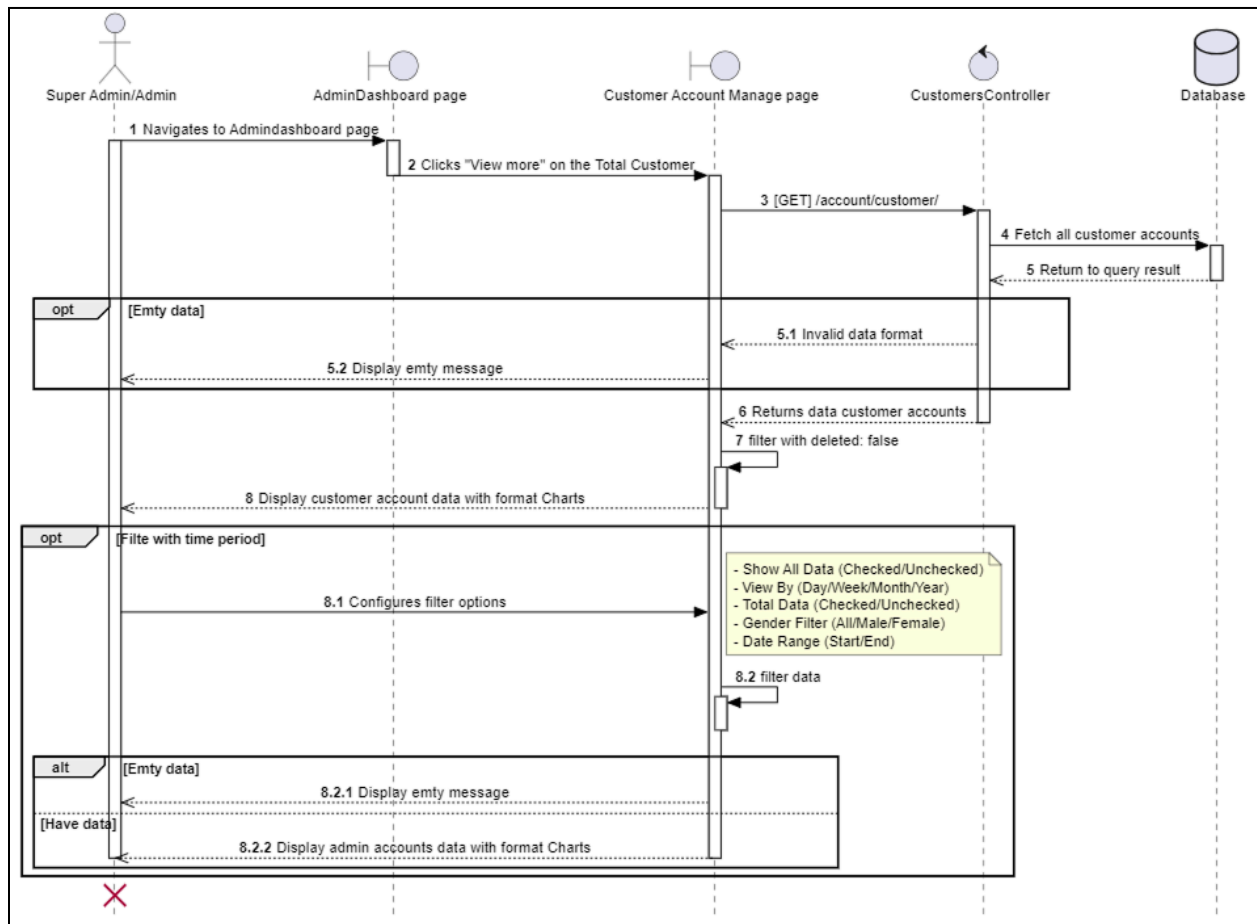
3.66 View Account Statistic

a. Class Diagram

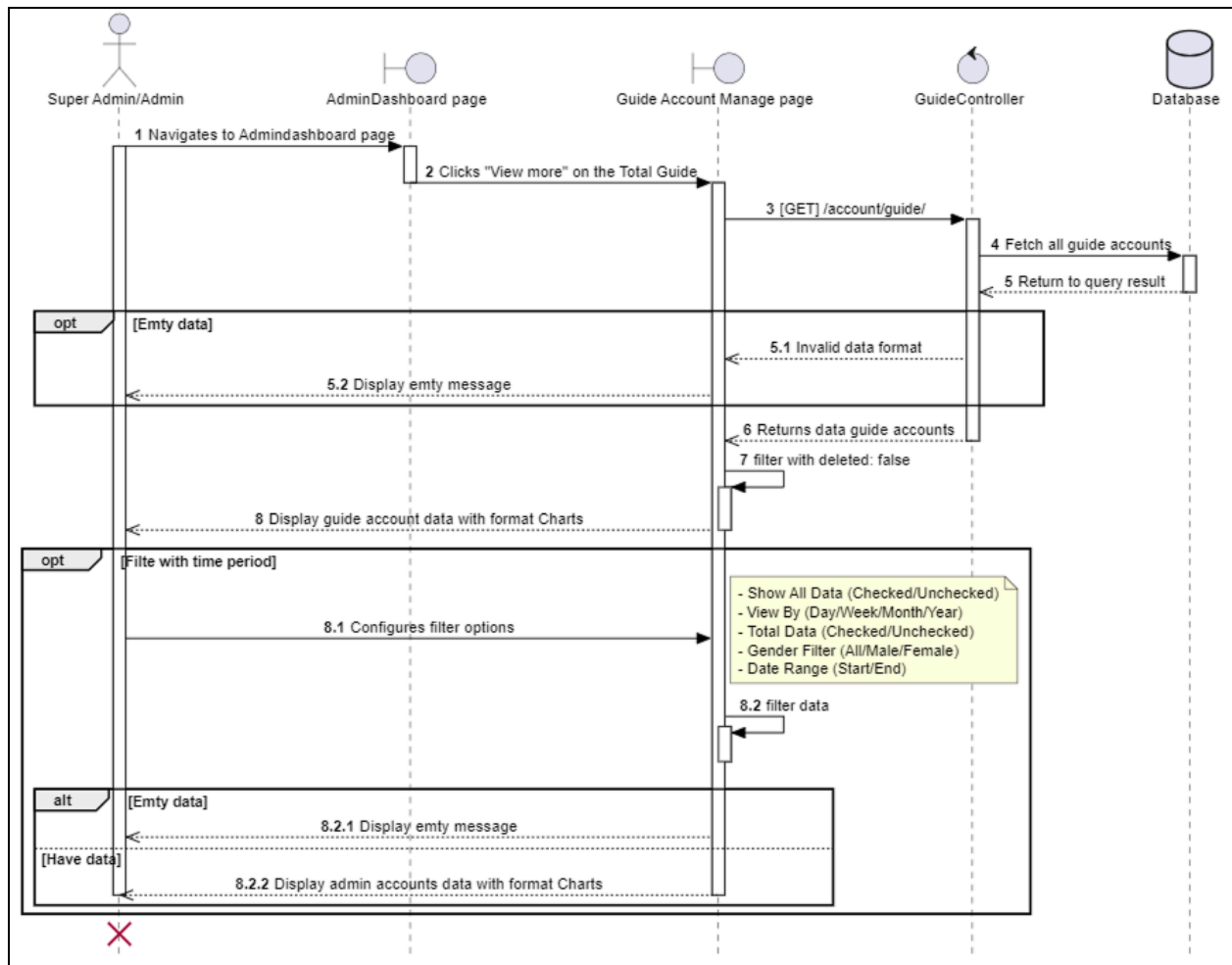


b. Sequence Diagram(s)

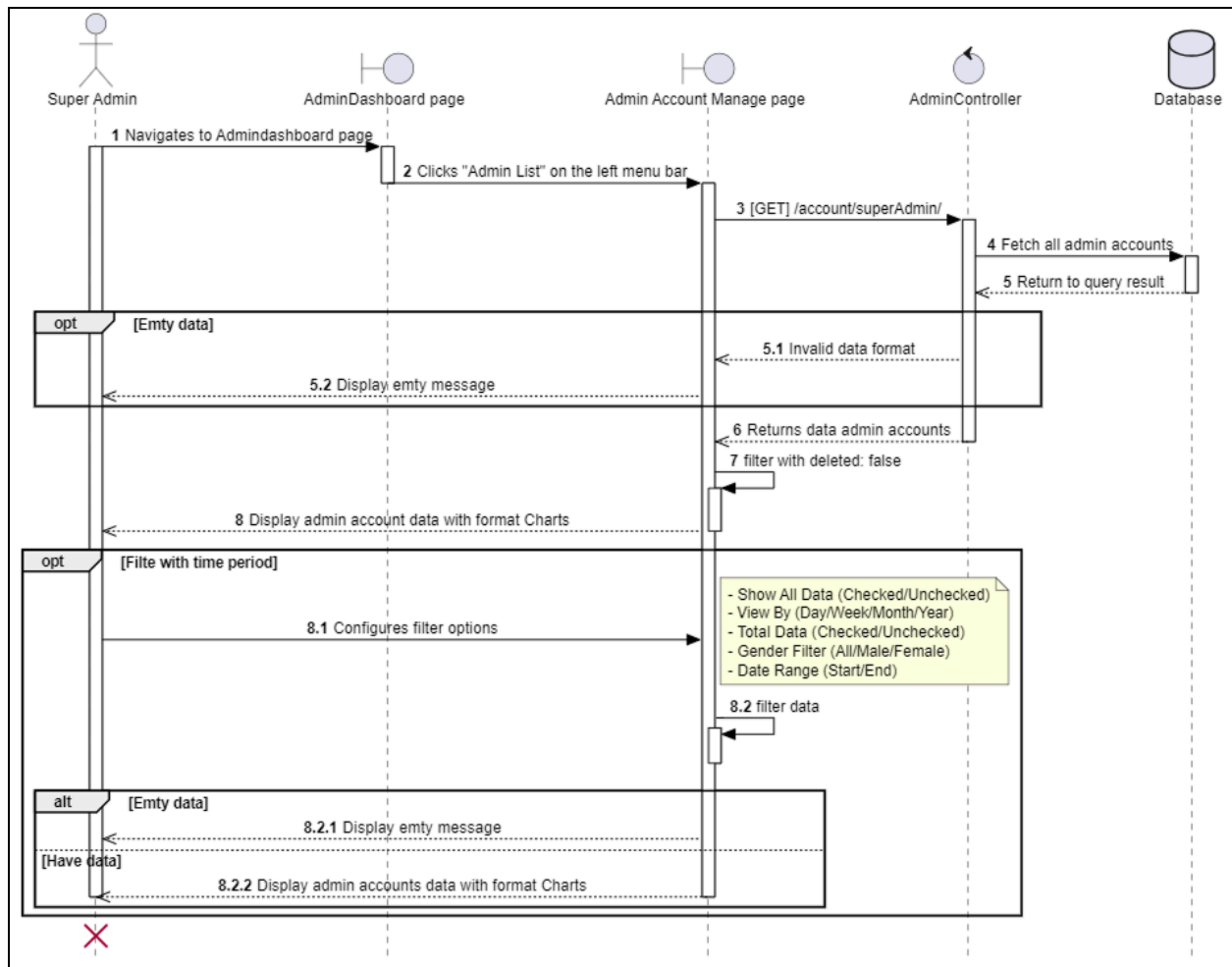
*Sequence diagram for View Customer Statistics.



*Sequence diagram for View Guide Statistics.

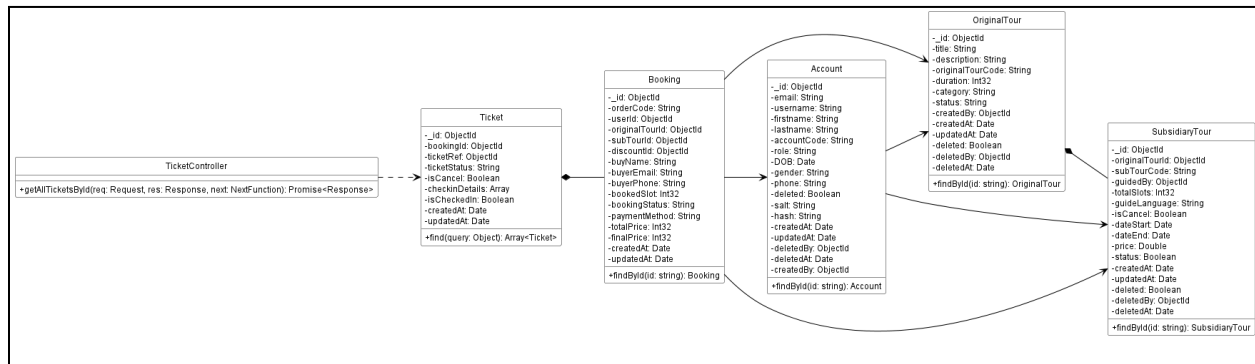


*Sequence diagram for View Admin Statistics.

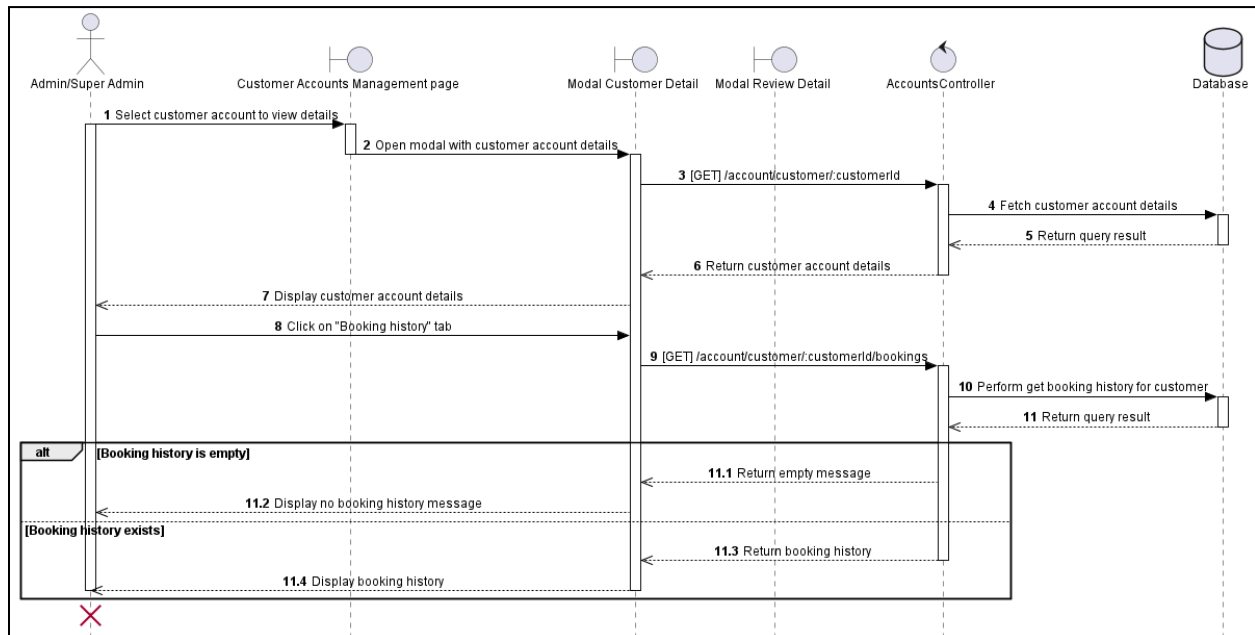


3.67 View All Tickets Of Customer

a. Class Diagram

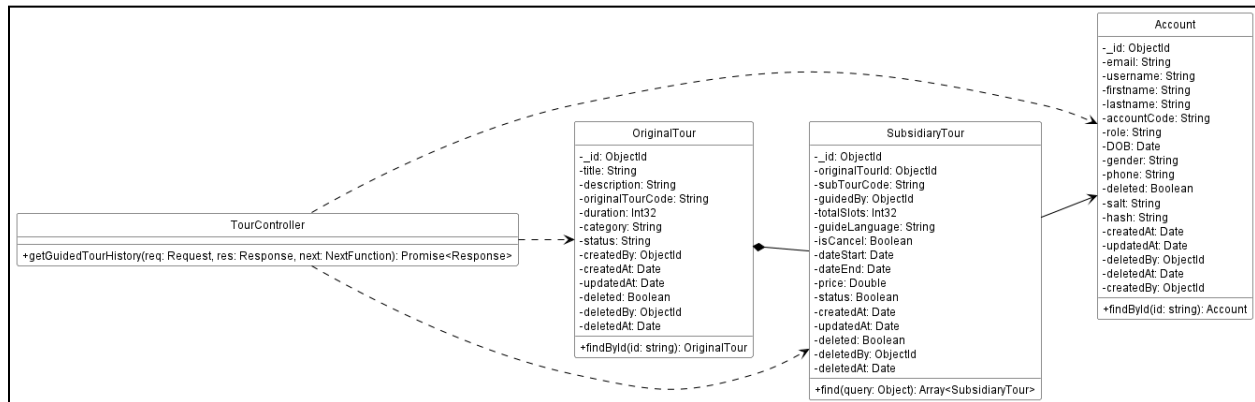


b. Sequence Diagram

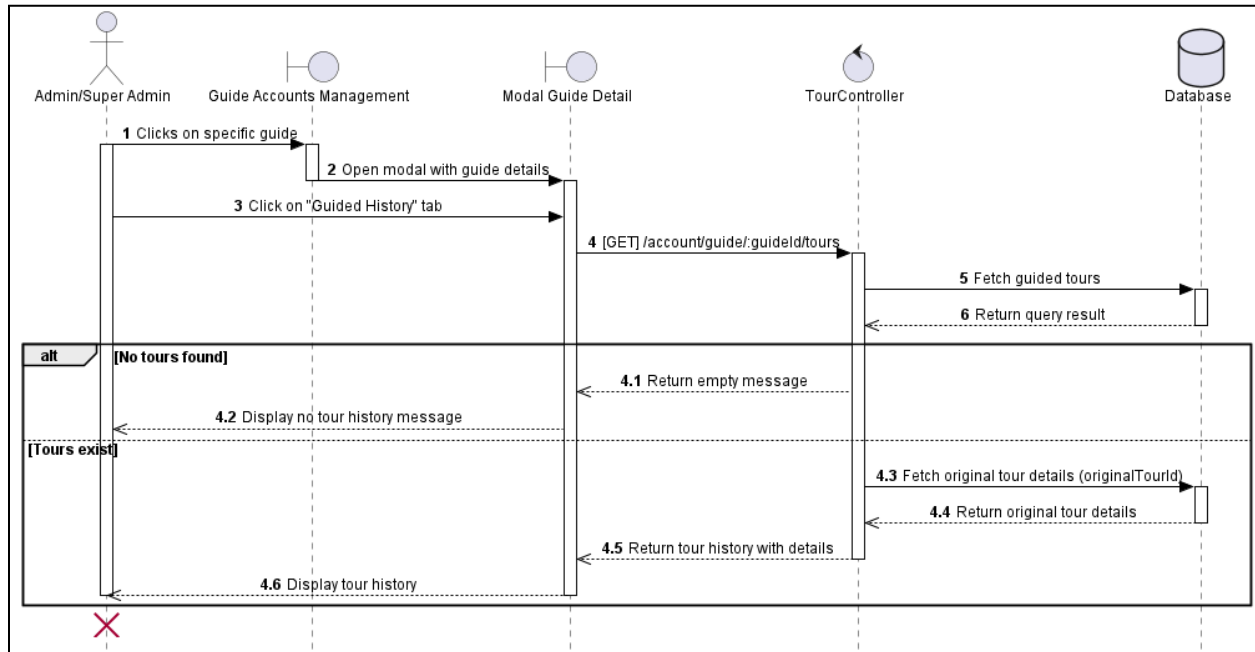


3.68 View Guided Tour History

a. Class Diagram

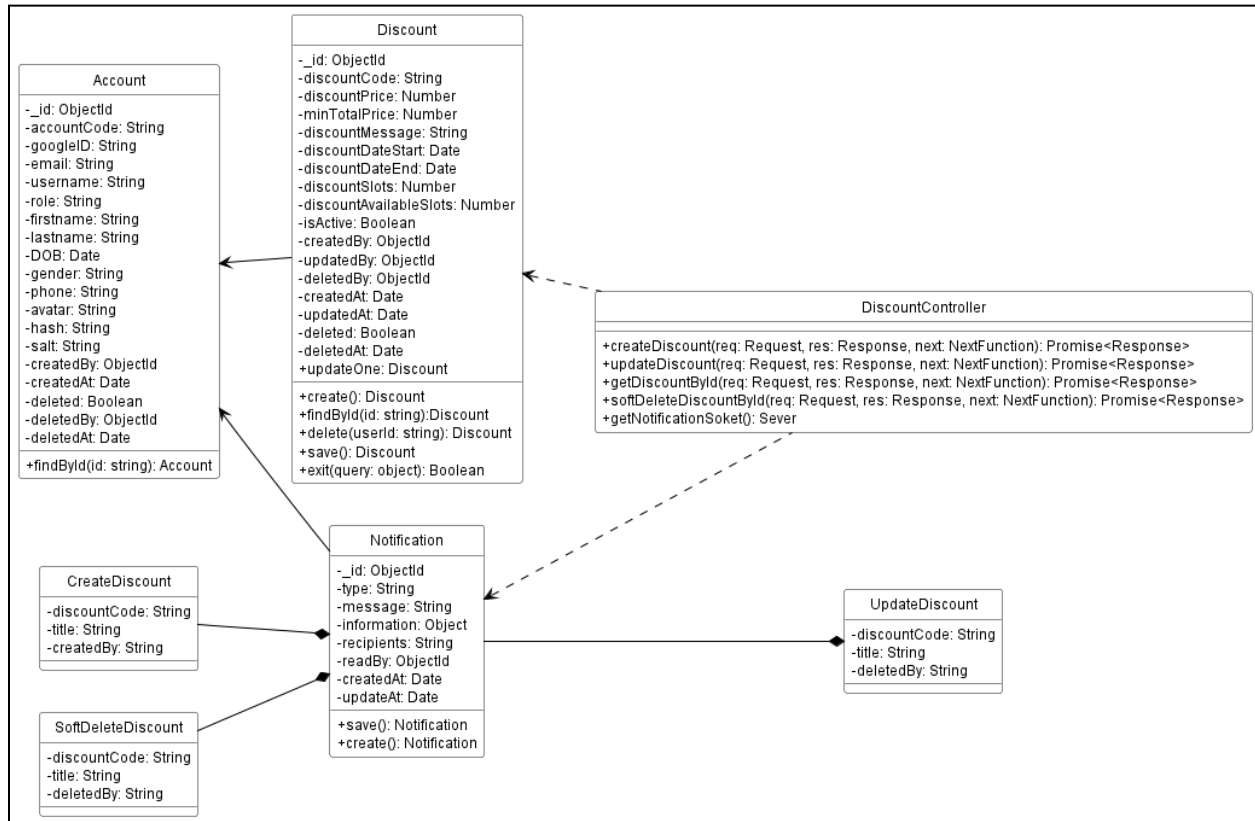


b. Sequence Diagram(s)



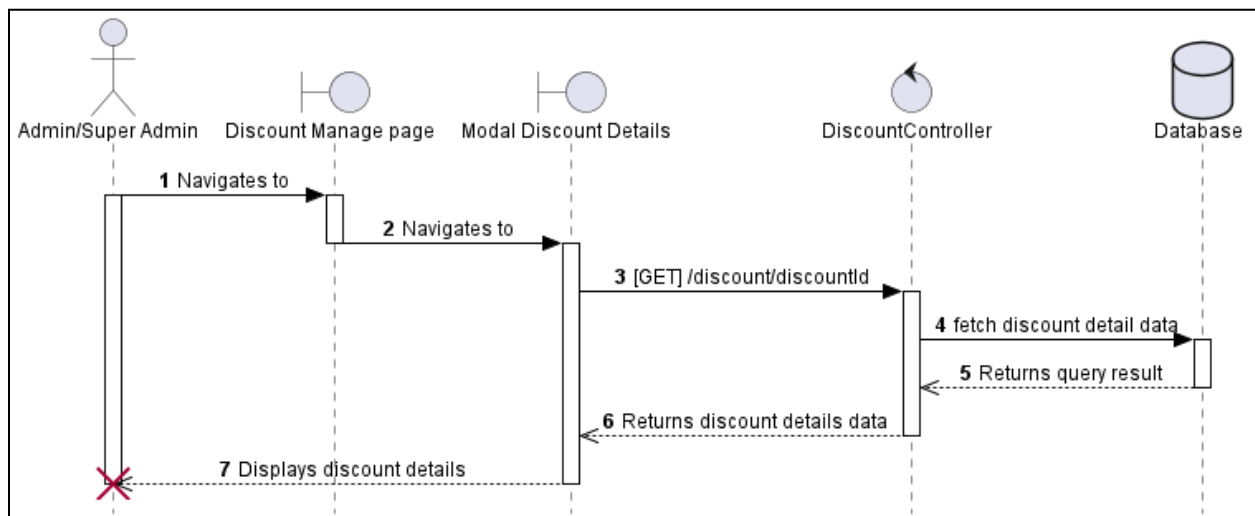
3.69 Manage Discount

a. Class Diagram

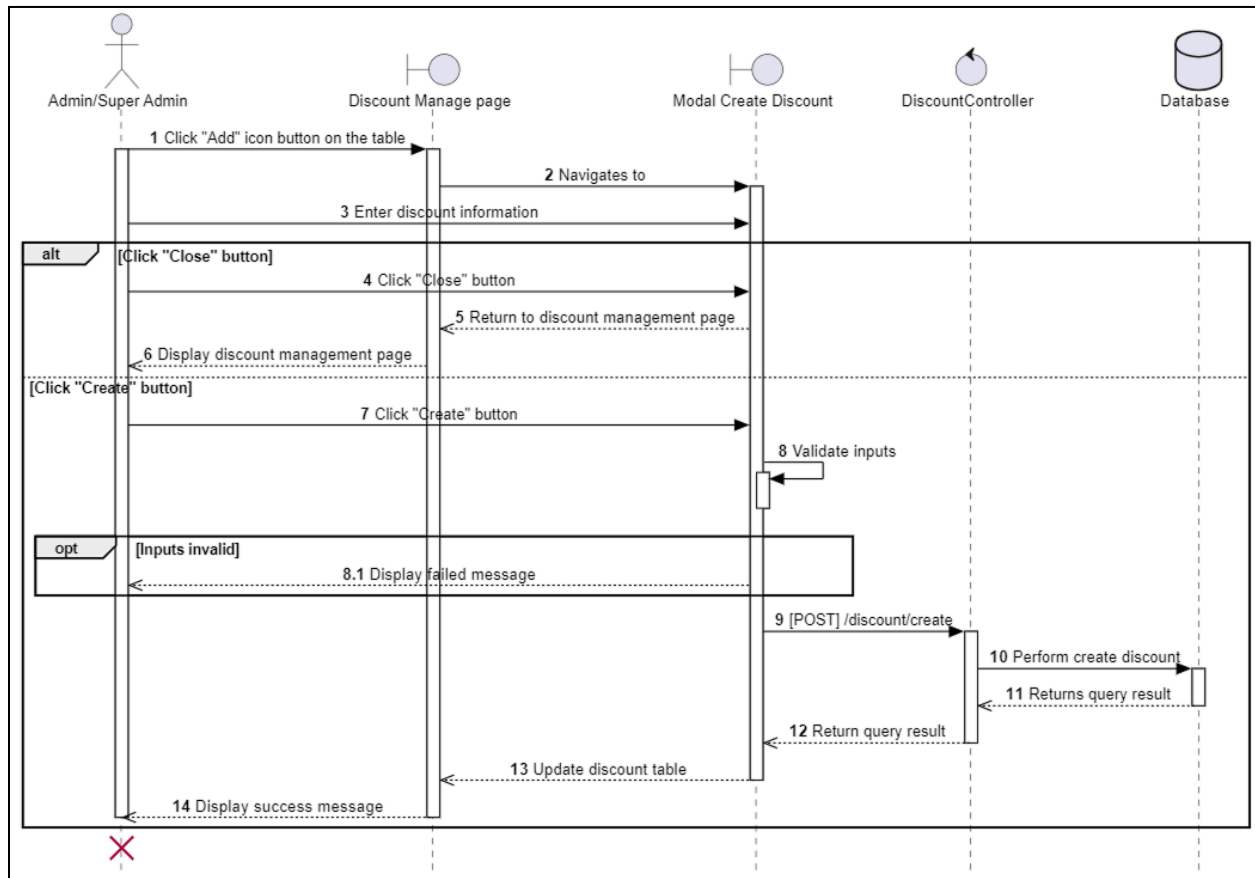


b. Sequence Diagram(s)

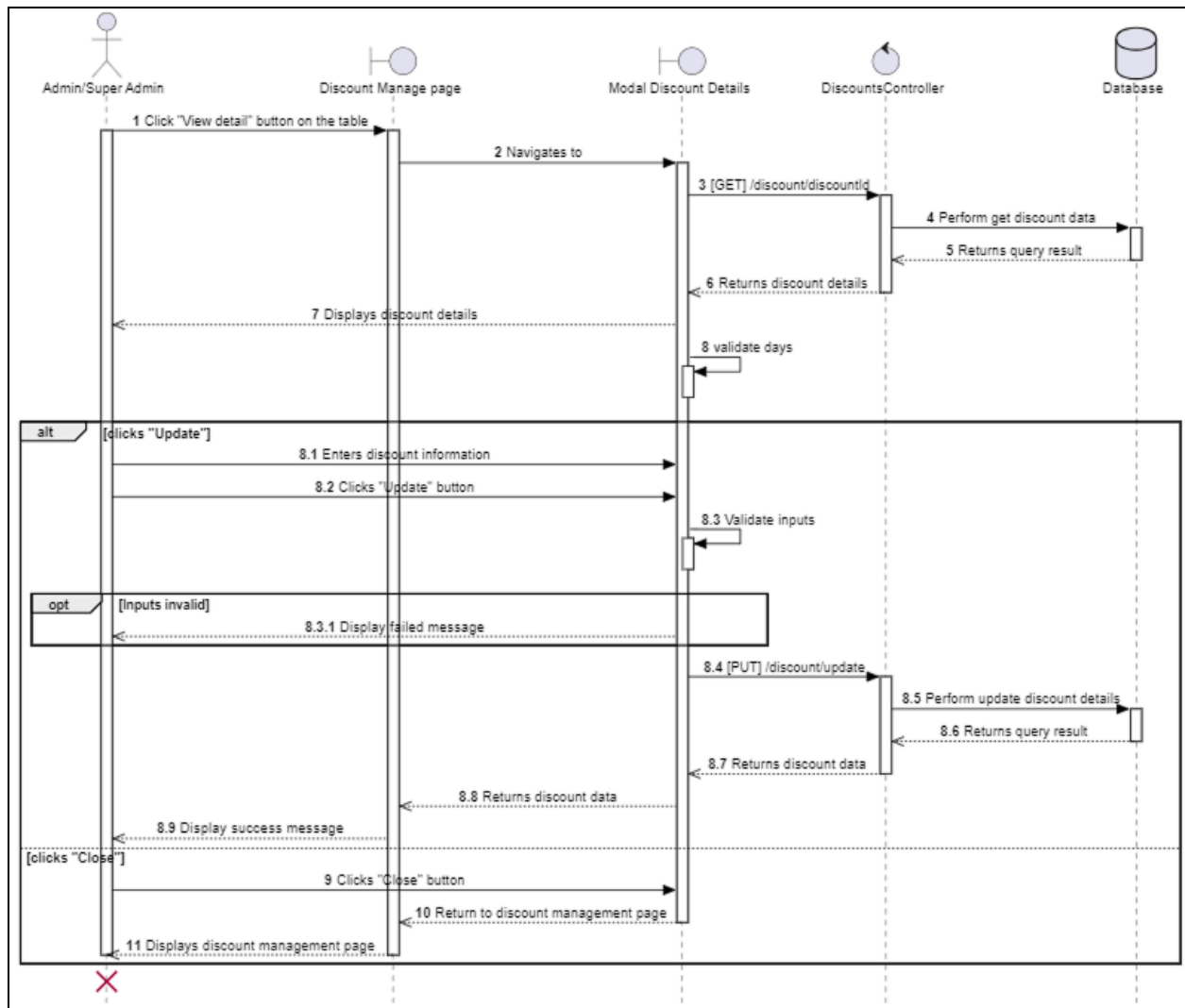
*Sequence diagram for View Discount Details.



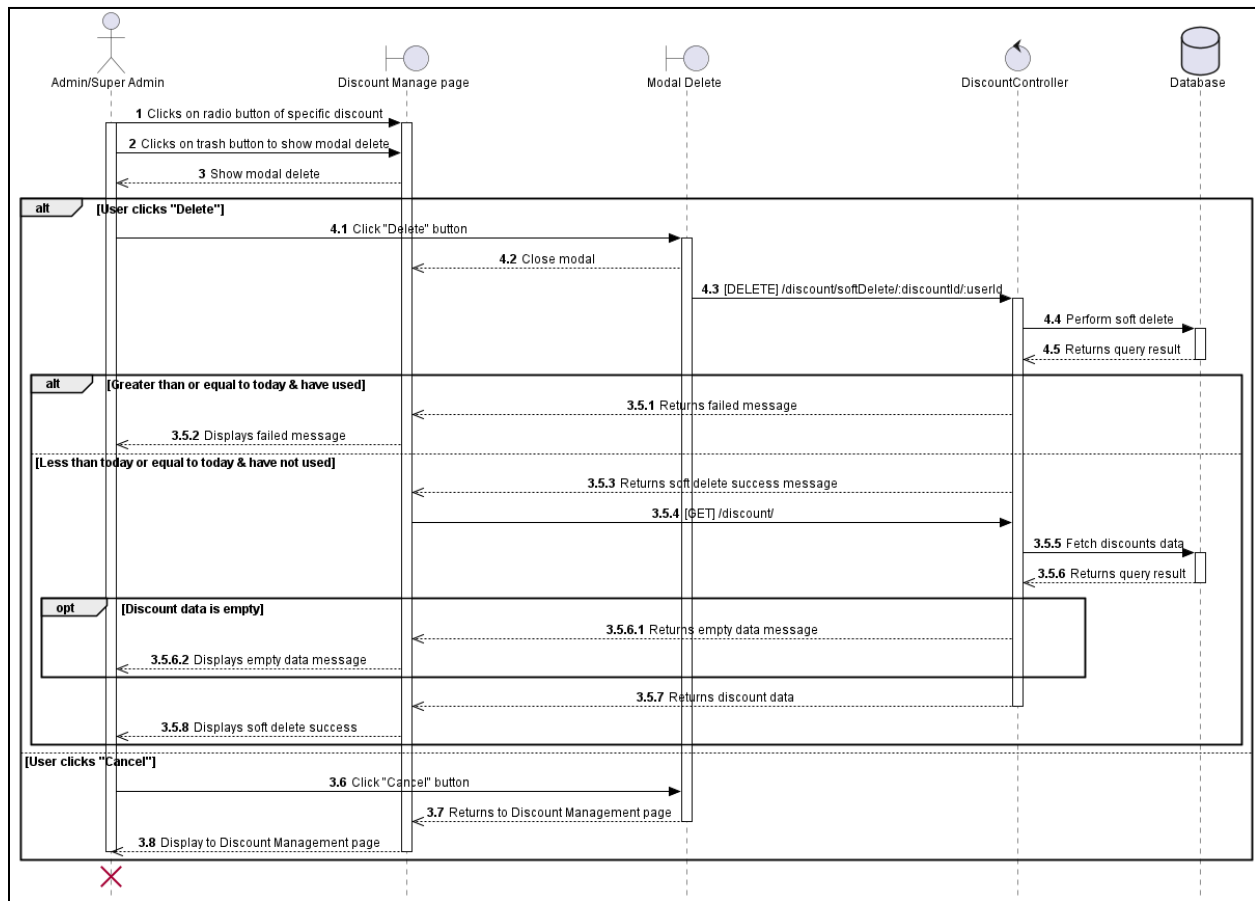
*Sequence diagram for Create Discount.



*Sequence diagram for Update Discount.

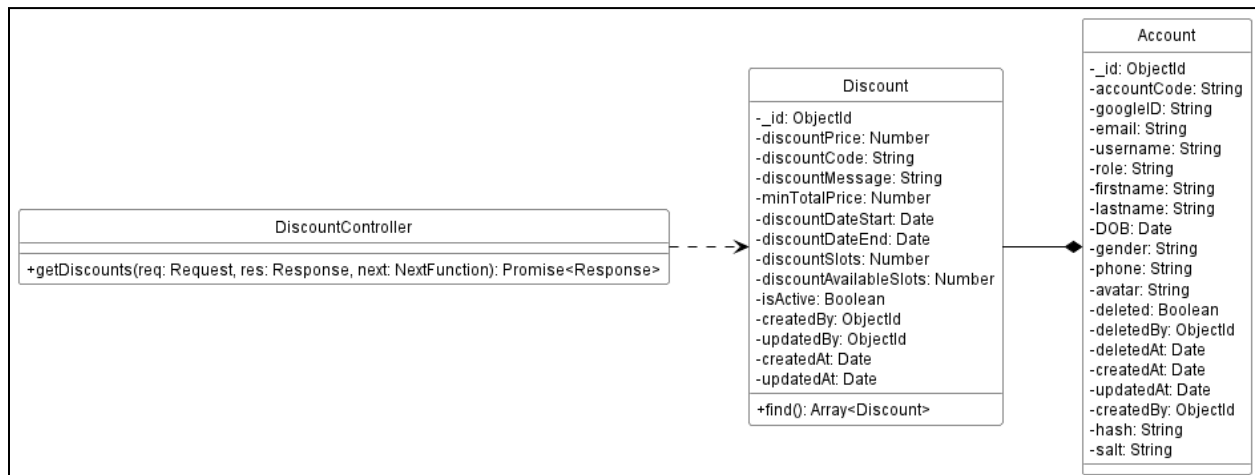


*Sequence diagram for Soft Delete Discount.

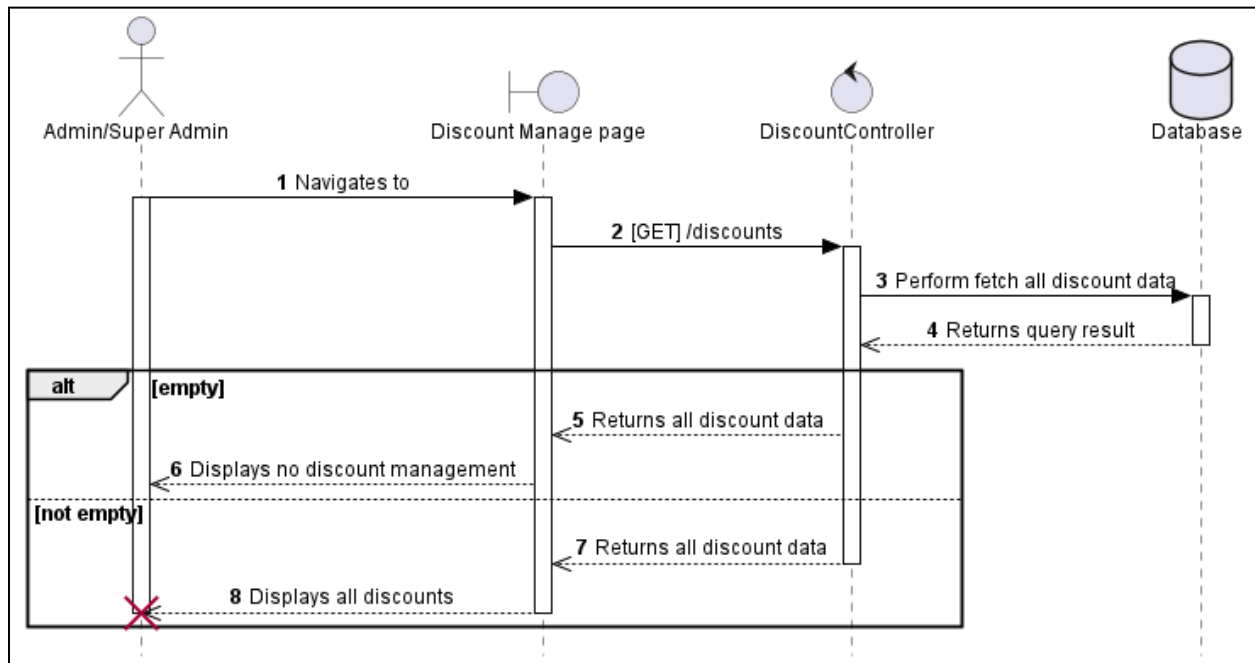


3.70 View All Discounts

a. Class Diagram

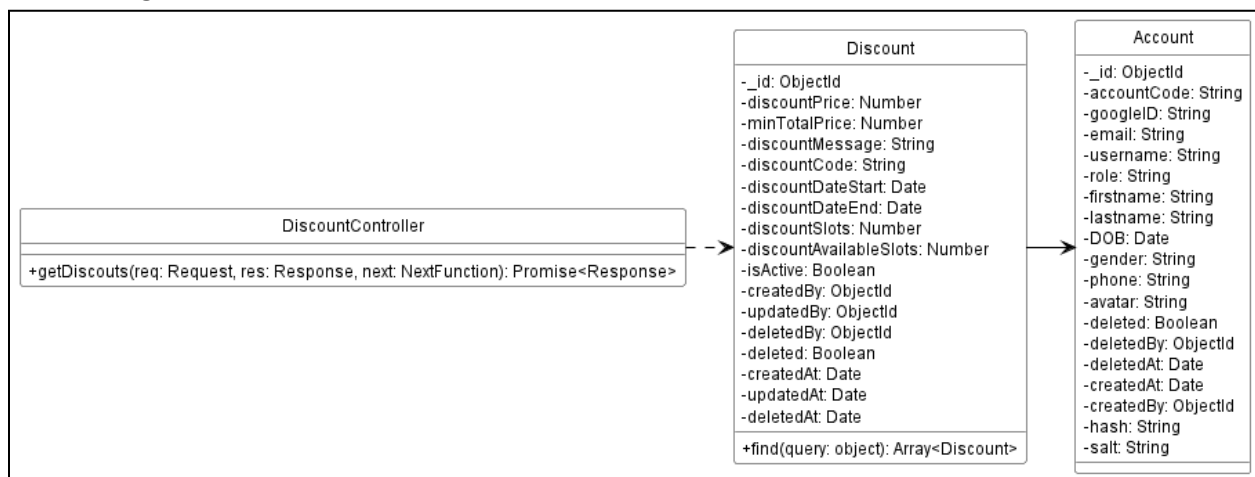


b. Sequence Diagram(s)



3.71 Filter & Search Discount

a. Class Diagram



```
sequenceDiagram
    actor Admin as Super Admin/ Admin
    participant DMP as Discount Manage page
    participant MFD as Modal Filter Discount
    participant DC as DiscountController
    participant DB as Database

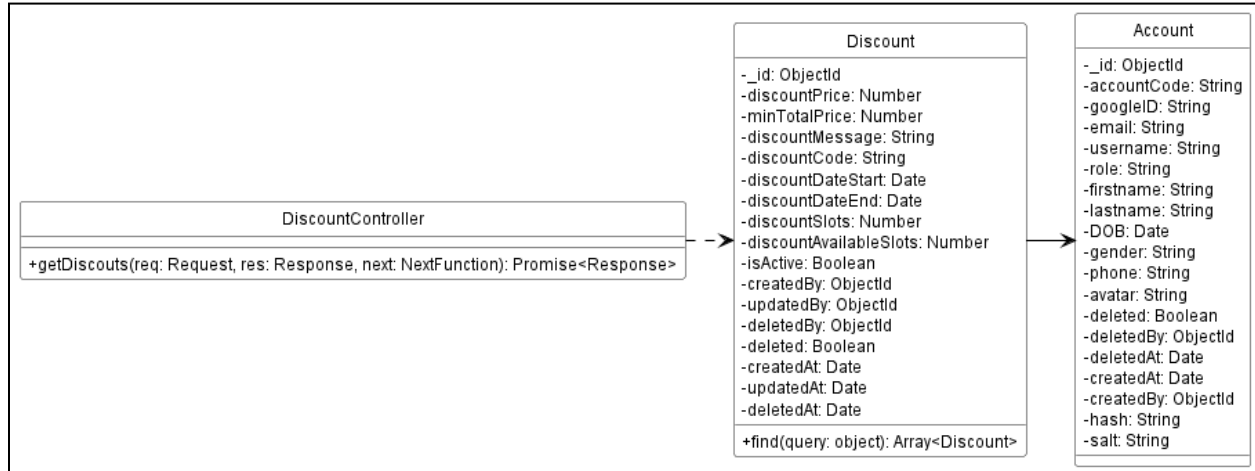
    Admin->>DMP: 1 Navigate to
    DMP->>DC: 2 [GET] /discount/
    DC->>DB: 3 Perform get all discounts data
    DB-->>DC: 4 Returns query result
    DC-->>DMP: 4.1 Returns empty data
    alt empty data
        DMP-->>Admin: 4.2 Display no data message
    else
        DC-->>DMP: 5 Returns discounts data
        DMP->>MFD: 6 Filter discounts by deleted: false
        MFD-->>DMP: 
        DMP-->>Admin: 7 Display discounts table
    end

    alt Search Functionality
        Admin->>DMP: 7.1 Enter search keyword
        DMP->>MFD: 7.2 Filter discounts by keyword
        MFD-->>DMP: 
        alt No Results Found
            DMP-->>Admin: 7.2.1 Display no data message
        else Has Results
            DMP-->>Admin: 7.2.2 Display filtered results
        end
    end

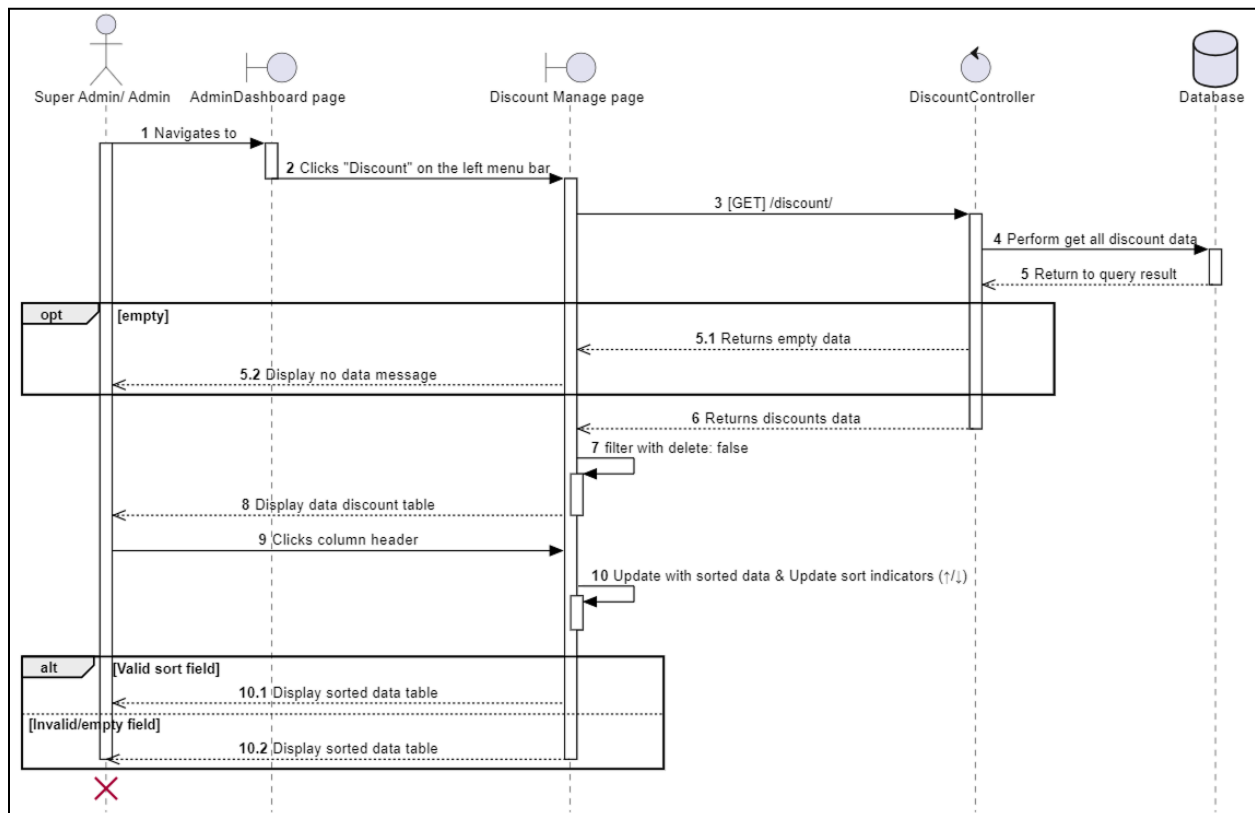
    alt Filter Functionality
        Admin->>DMP: 7.3 Click "Filter" icon
        DMP->>MFD: 7.4 Open filter modal
        MFD-->>DMP: 
        DMP->>MFD: 7.5 Set filter criteria
        Note right of MFD: Slot range  
Status Type  
Min Total price range  
Price range  
Date range
        alt Click Cancel
            Admin->>DMP: 7.5.1 Click "Cancel" to close filter modal
            DMP->>MFD: 7.5.2 Close modal
            MFD-->>DMP: 
            DMP-->>Admin: 7.5.3 Keep current data
        else Click Filter
            Admin->>DMP: 7.5.4 Click "Filter"
            DMP->>MFD: 7.5.5 Filter discounts by filter criteria
            MFD-->>DMP: 
            alt No Matching Results
                DMP-->>Admin: 7.5.5.1 Update filter Data
            else Has Results
                DMP-->>Admin: 7.5.5.3 Update filter Data
            end
            DMP-->>Admin: 7.5.5.4 Display filtered results
        end
    end
```

3.72 Sort Discount

a. Class Diagram

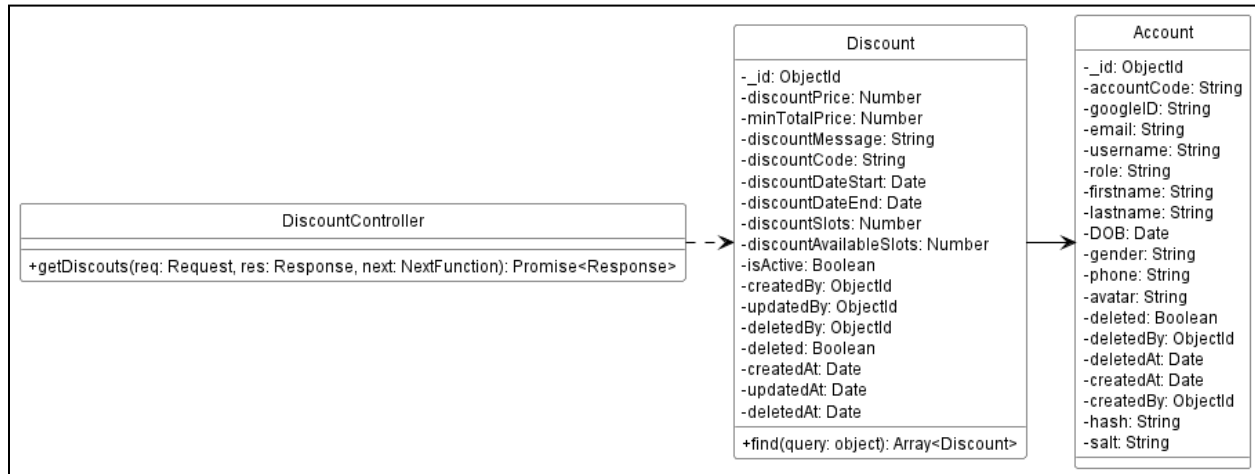


b. Sequence Diagram(s)



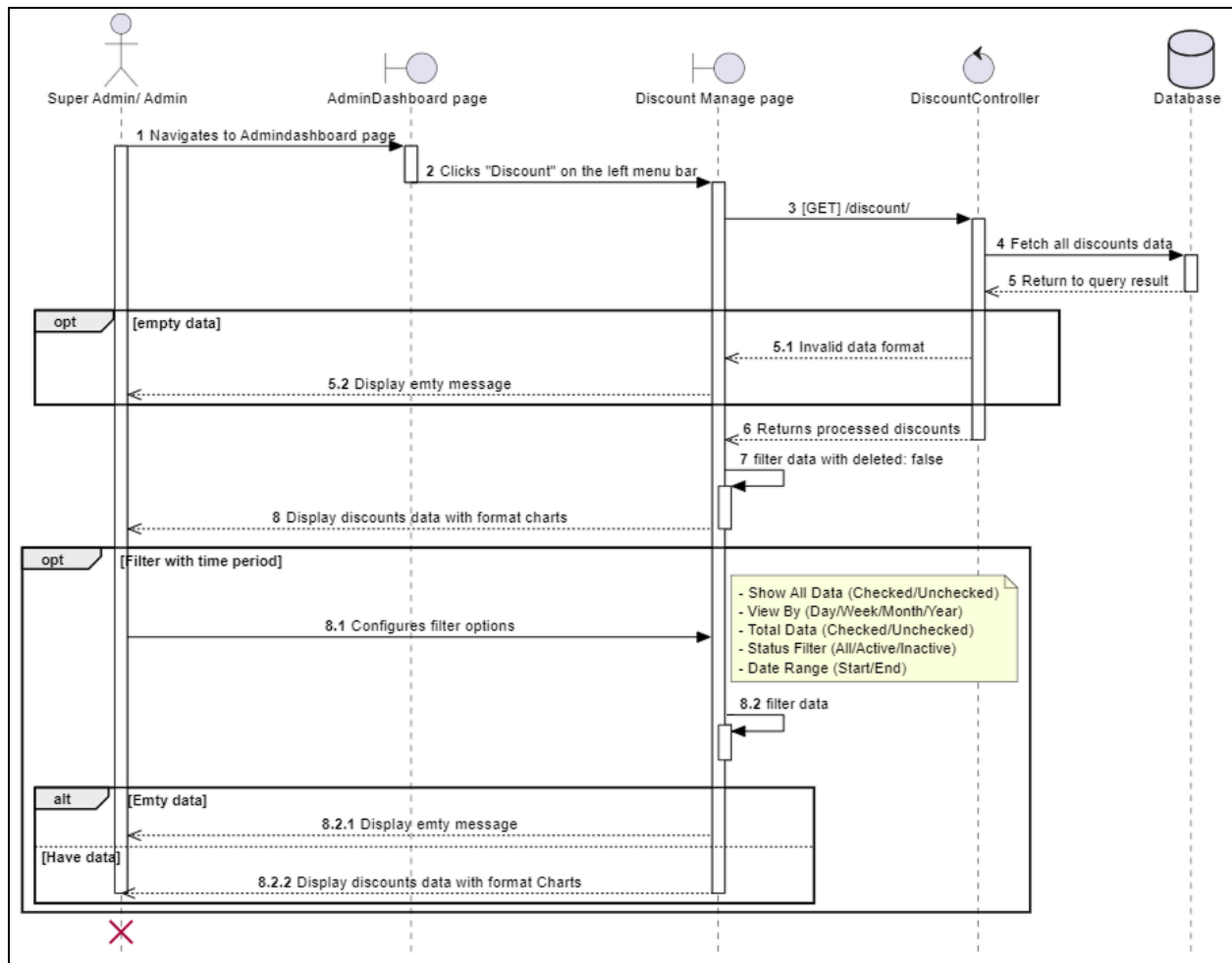
3.73 View Discount Statistic

a. Class Diagram

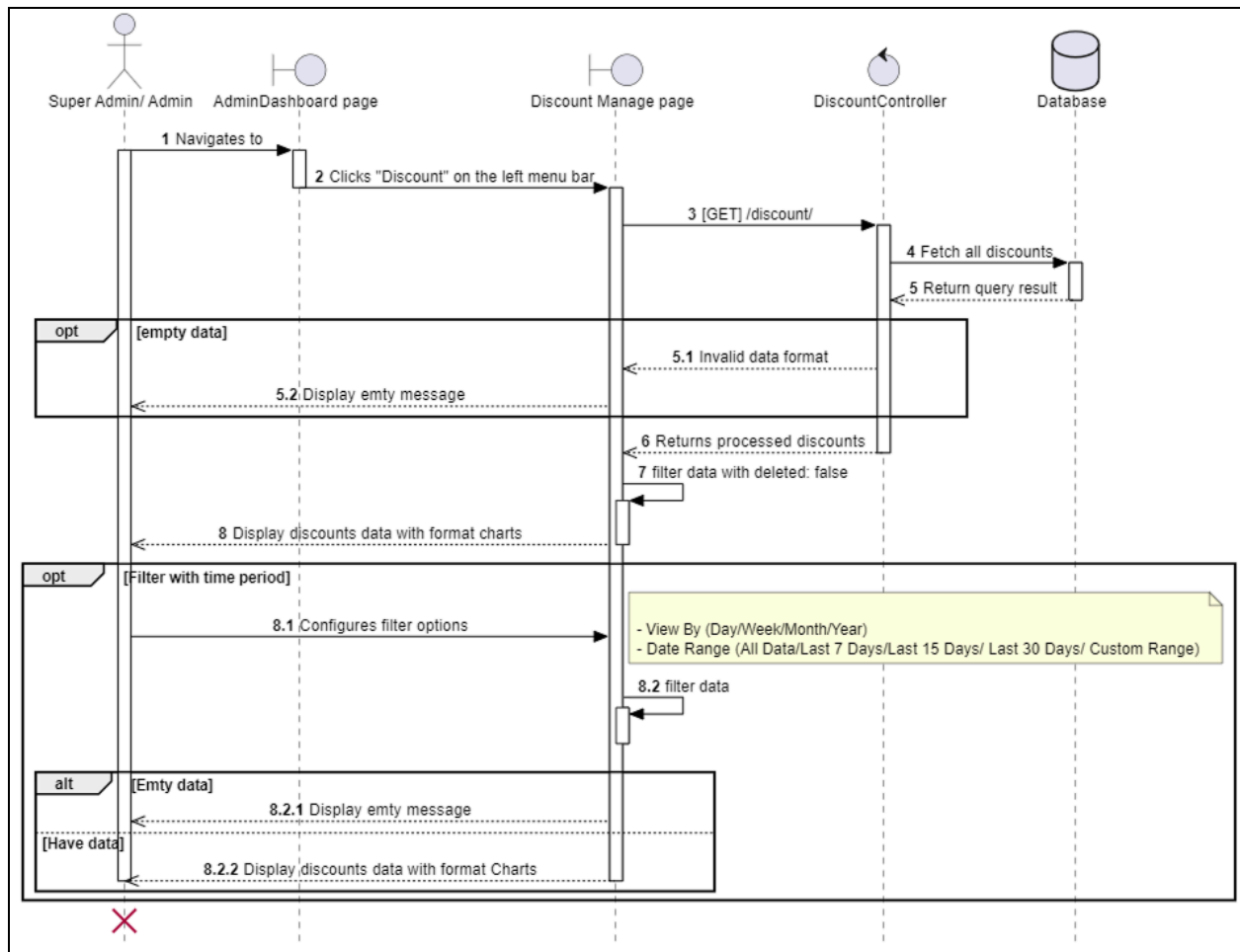


b. Sequence Diagram(s)

*Sequence diagram for View Discount Statistics (cost).

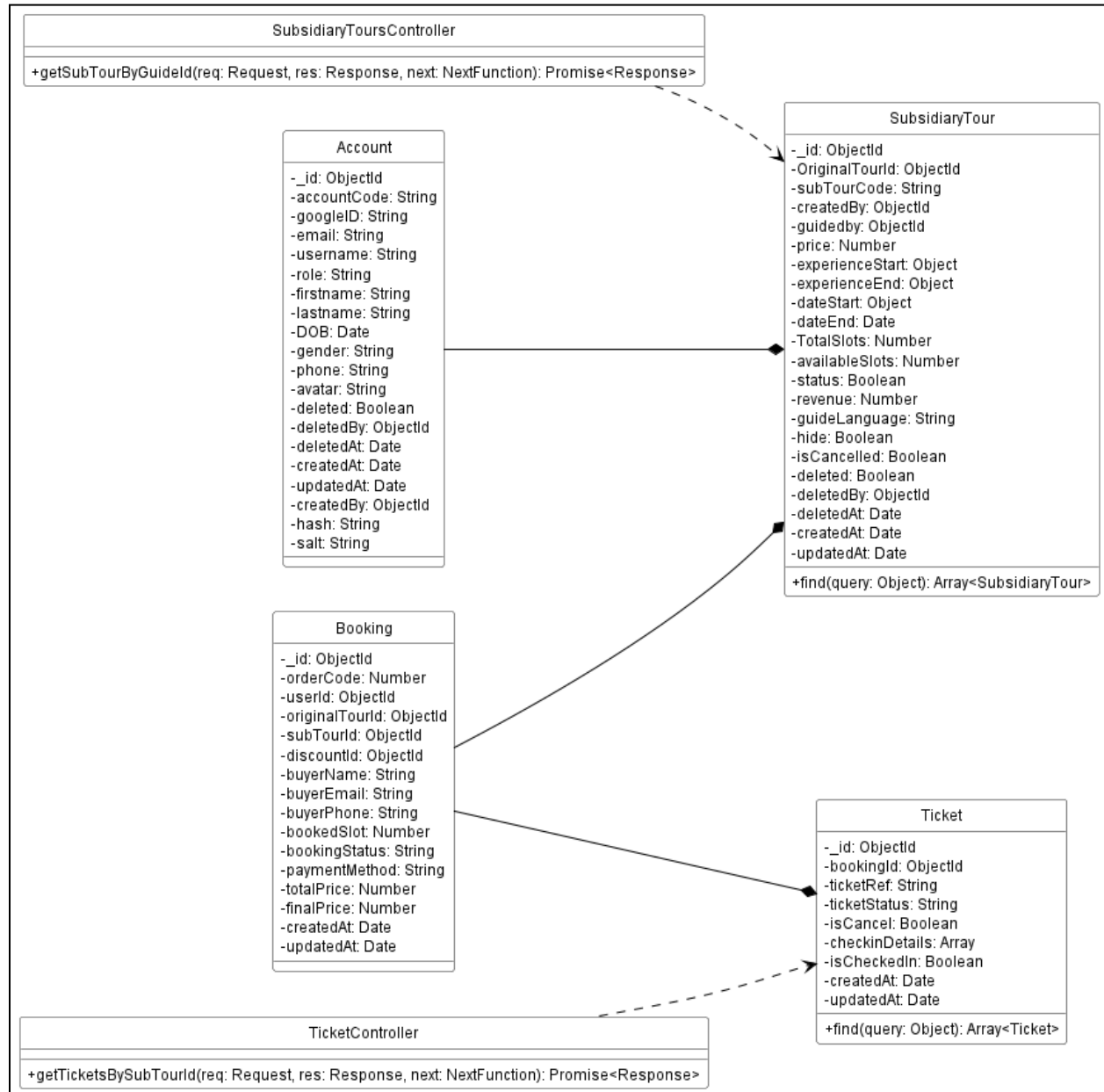


*Sequence diagram for View Discount Statistics (code).

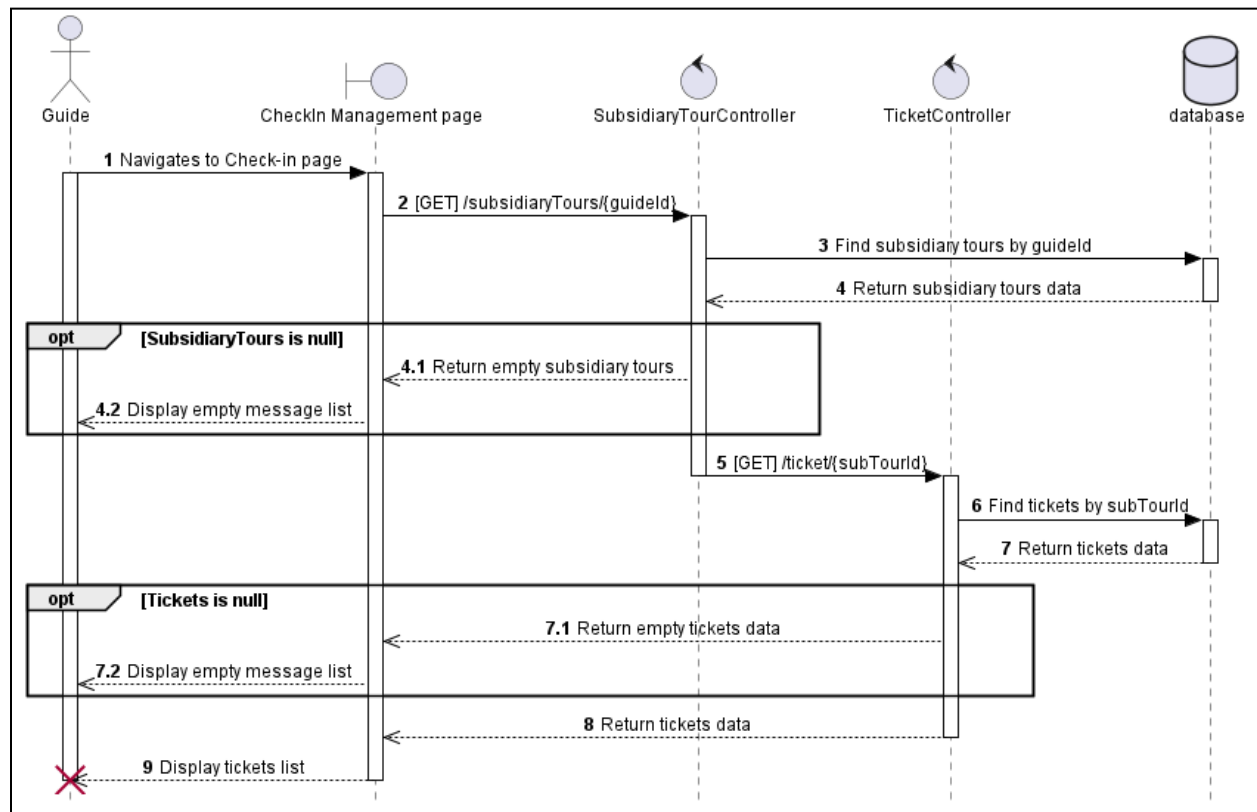


3.74 View Guide Tours

a. Class Diagram



b. Sequence Diagram(s)

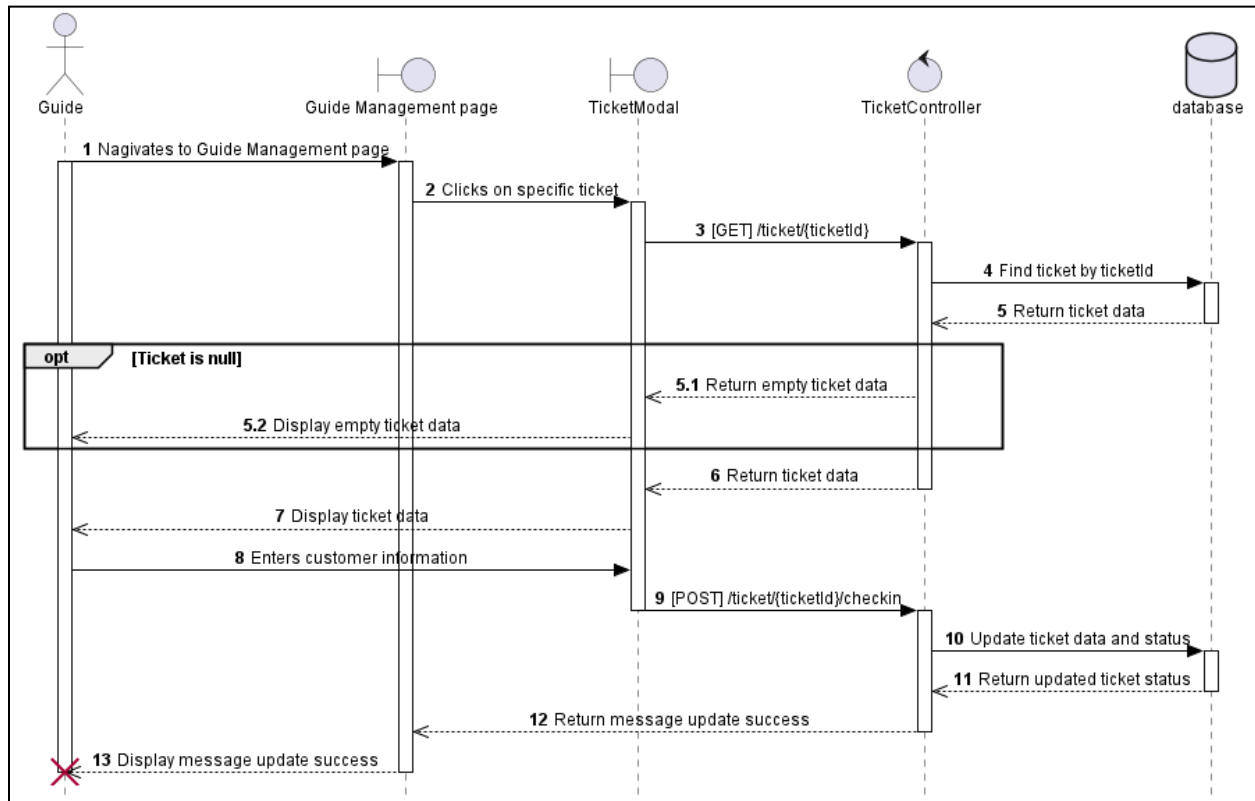


3.75 Manage Tours Attendance

a. Class Diagram

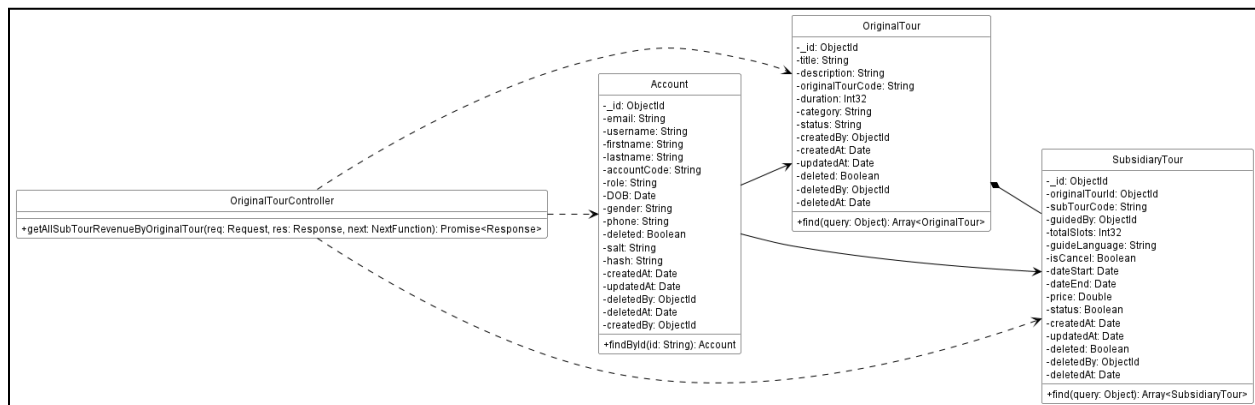


b. Sequence Diagram(s)

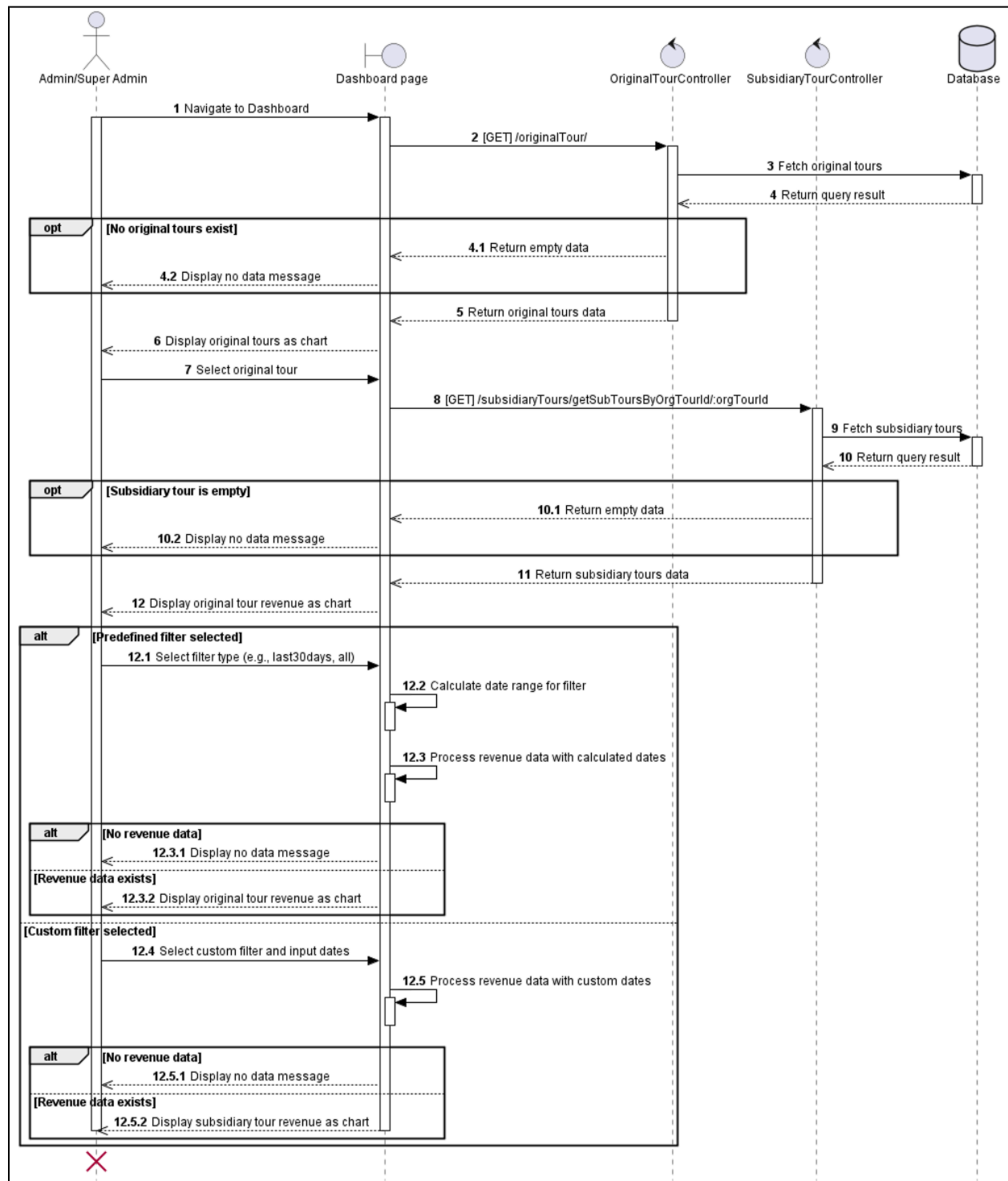


3.76 View Original Tour Revenue

a. Class Diagram

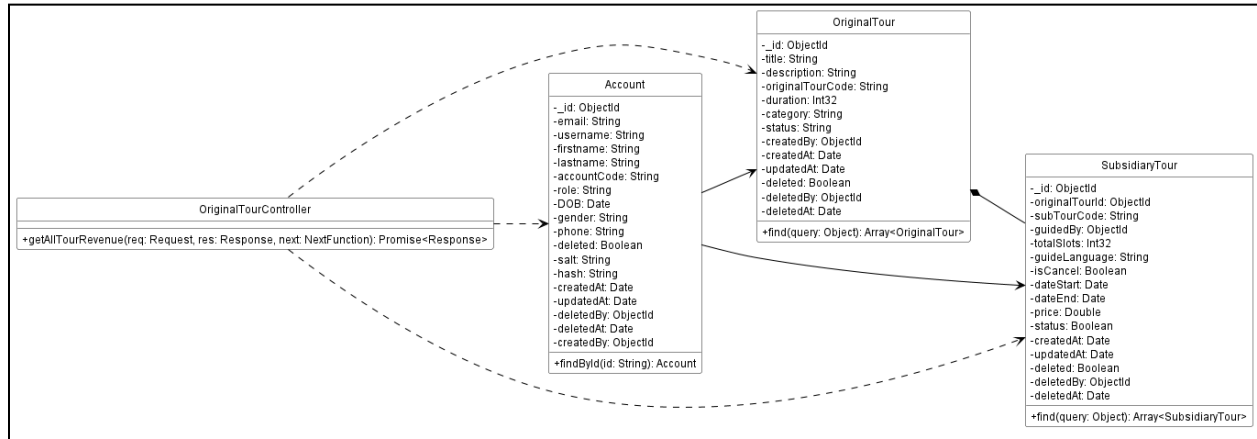


b. Sequence Diagram(s)

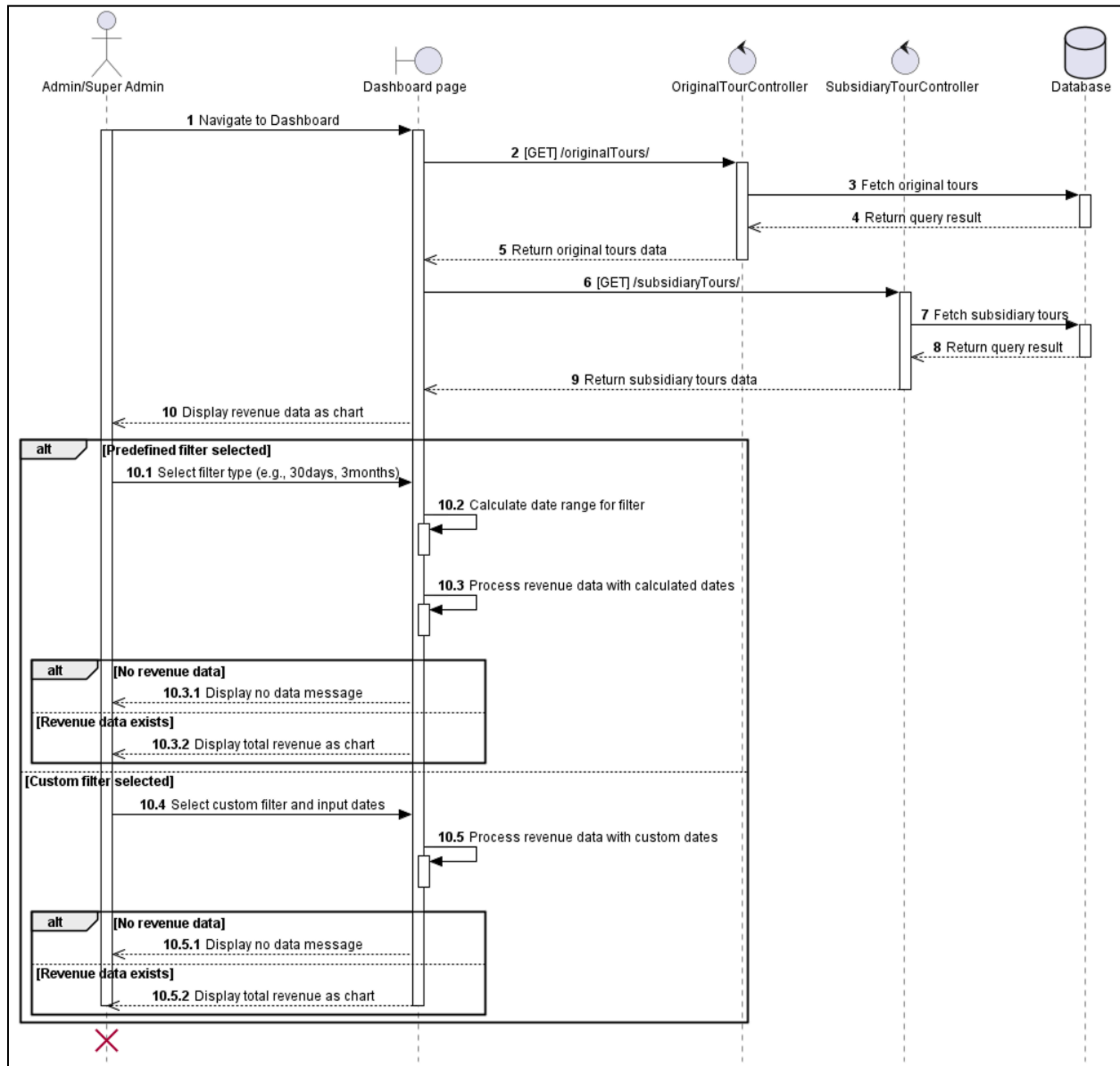


3.77 View Total Revenue

a. Class Diagram

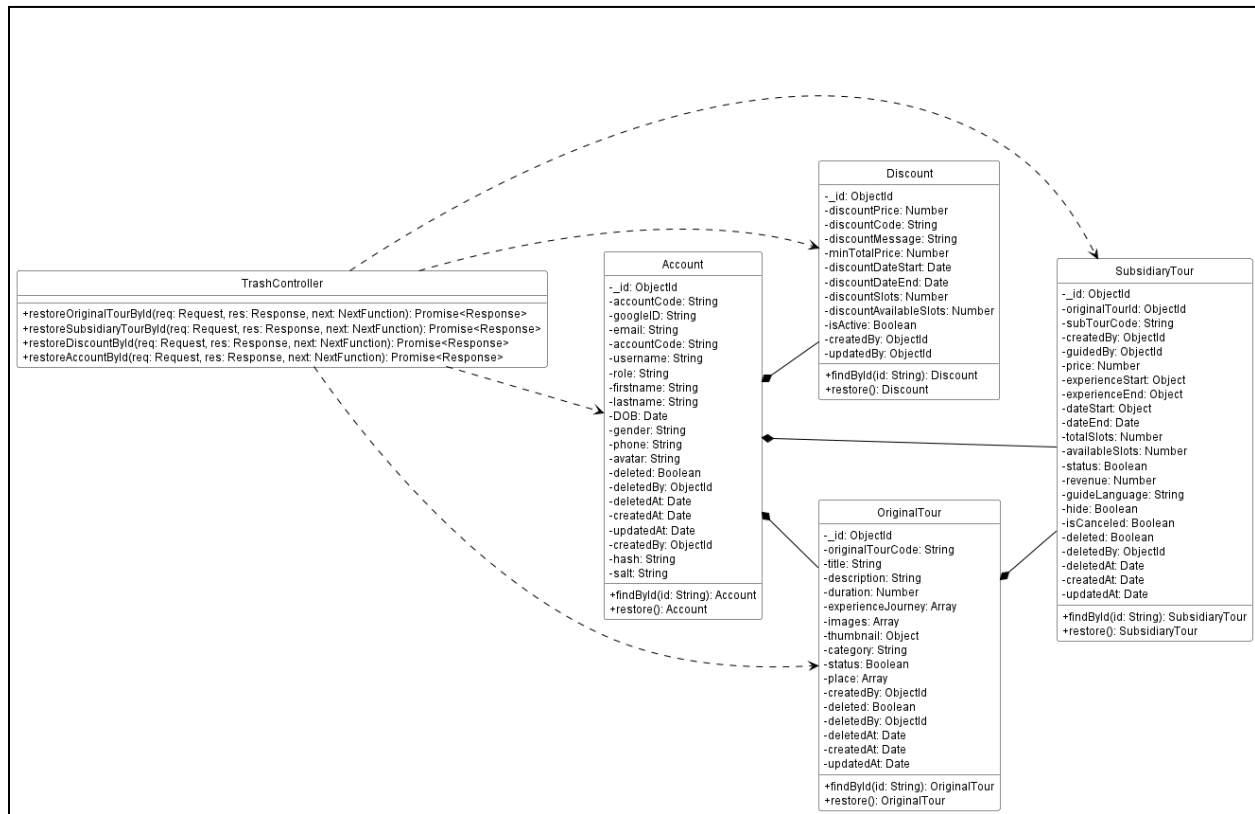


b. Sequence Diagram(s)

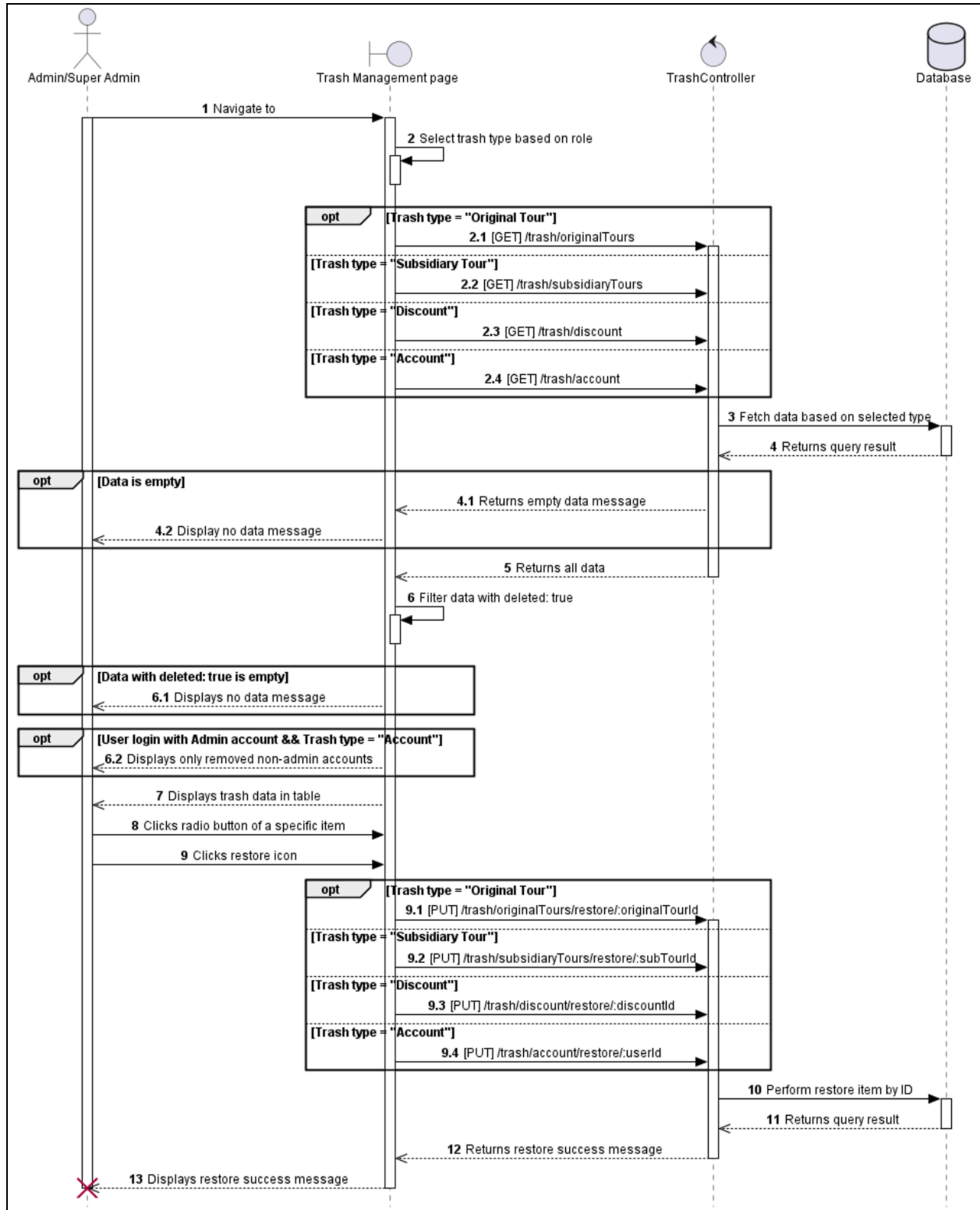


3.78 Restore

a. Class Diagram

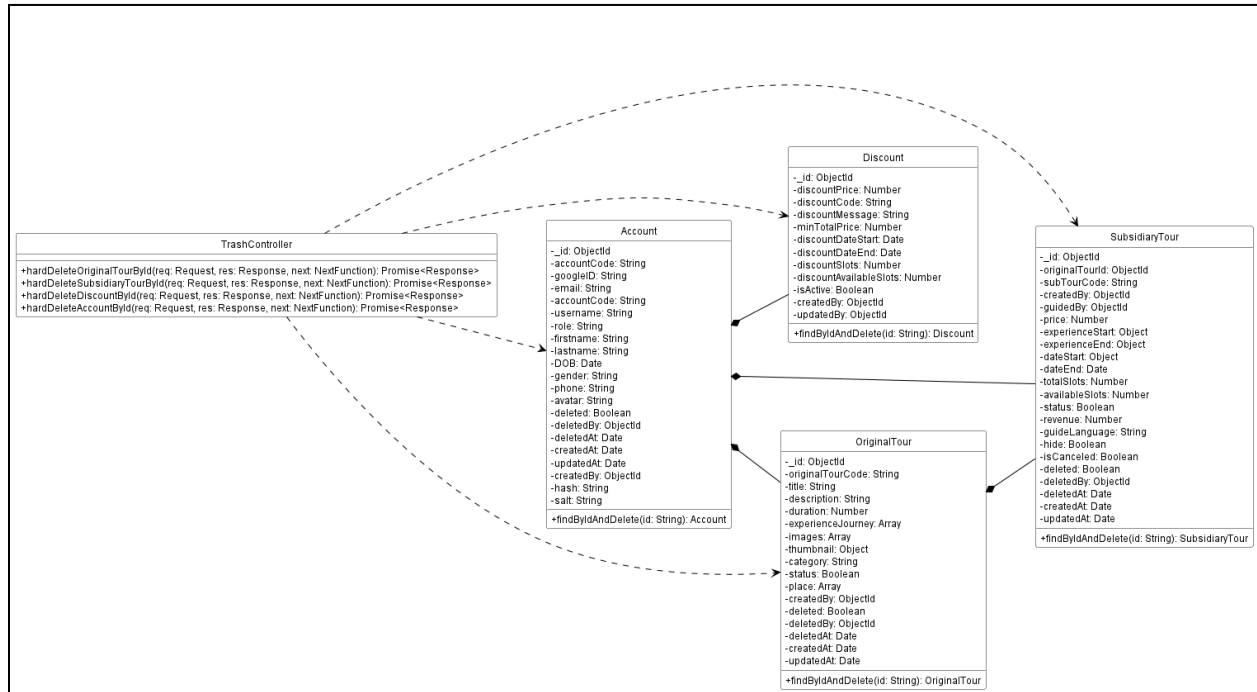


b. Sequence Diagram(s)

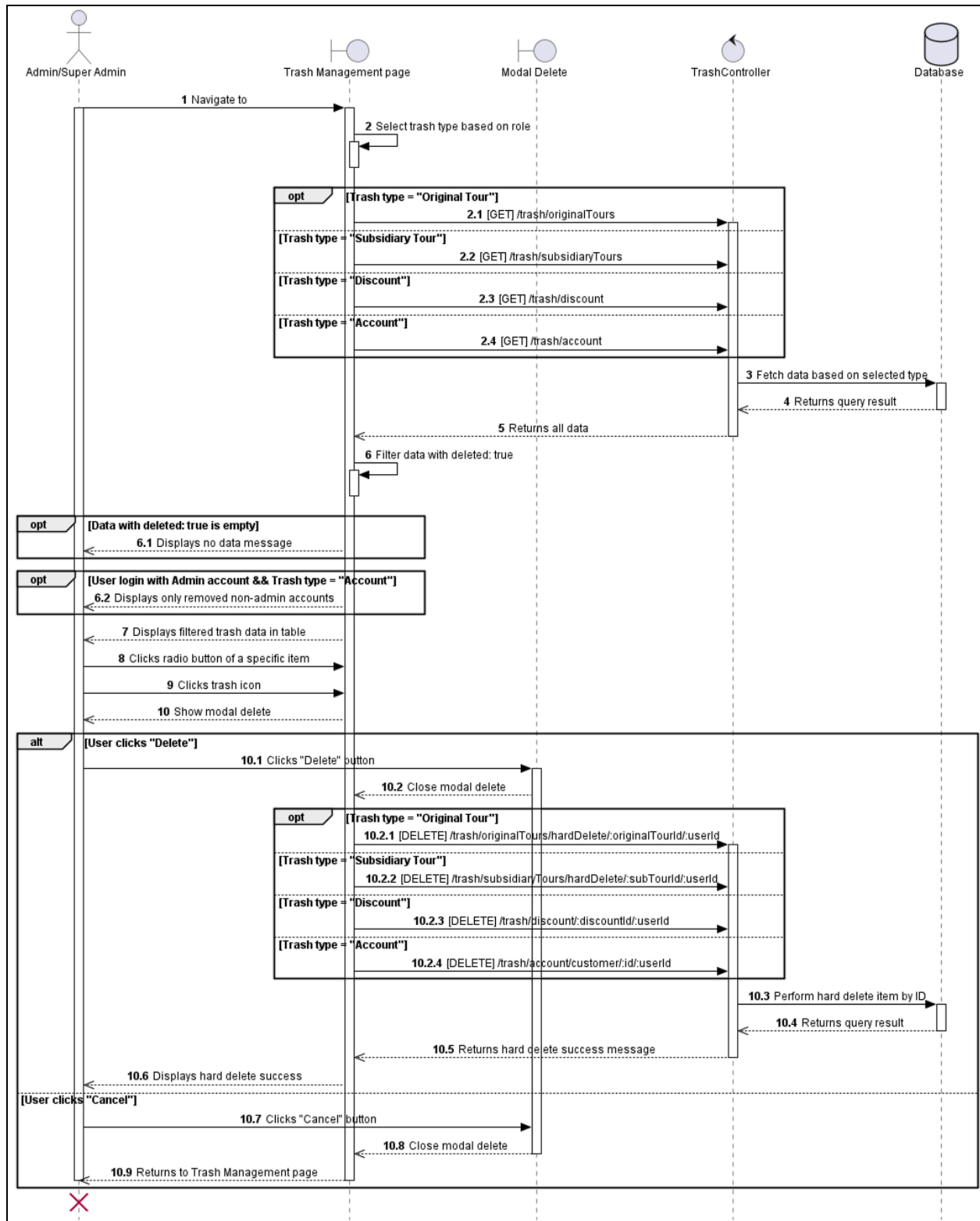


3.79 Hard Delete

a. Class Diagram

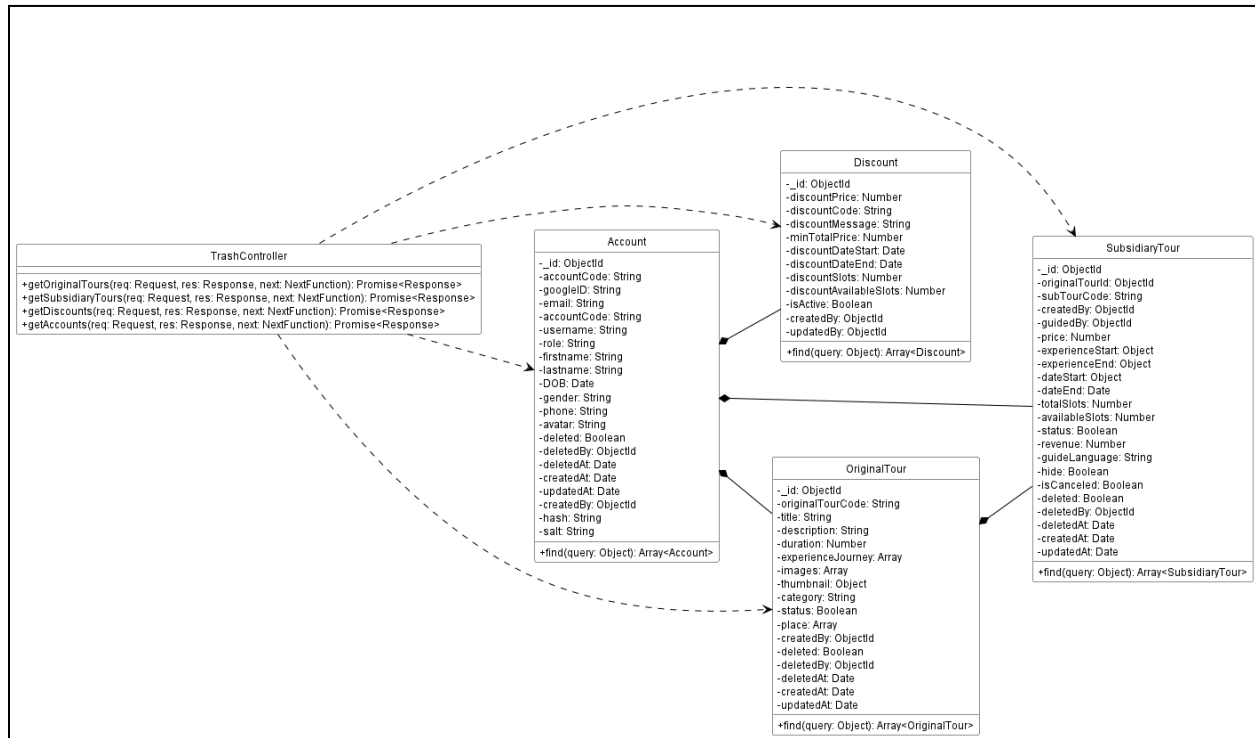


b. Sequence Diagram(s)

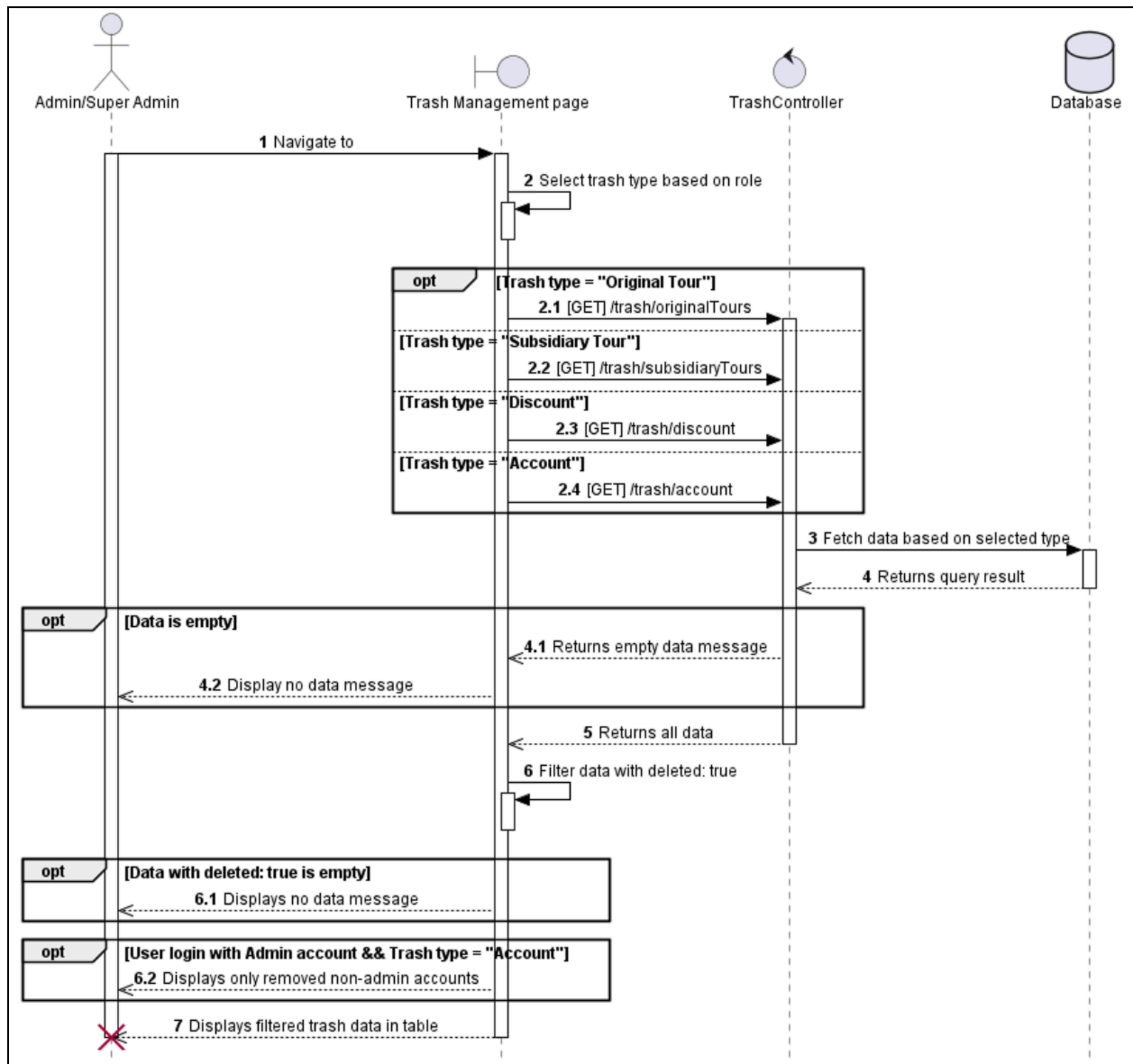


3.80 Filter Trash

a. Class Diagram

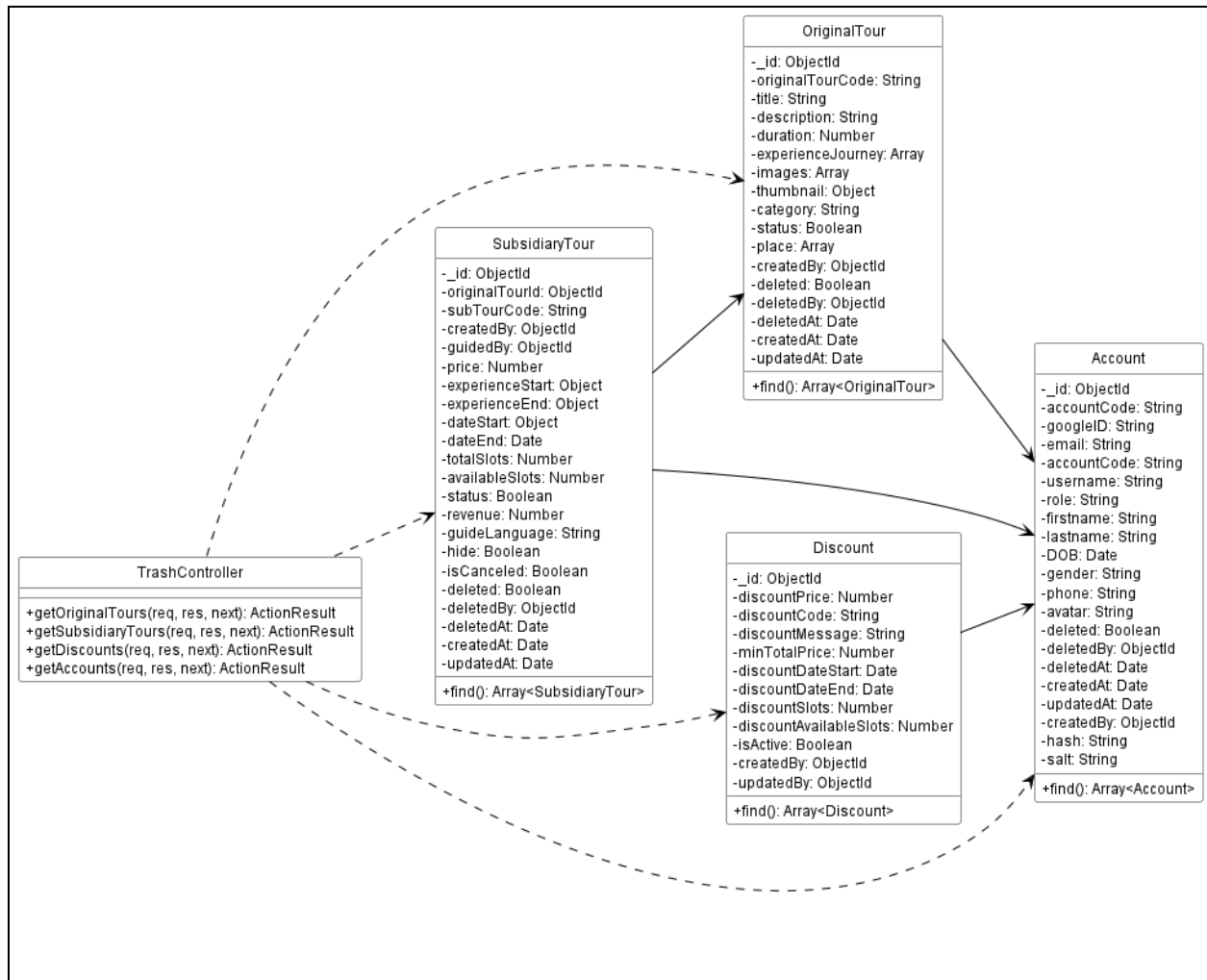


b. Sequence Diagram(s)

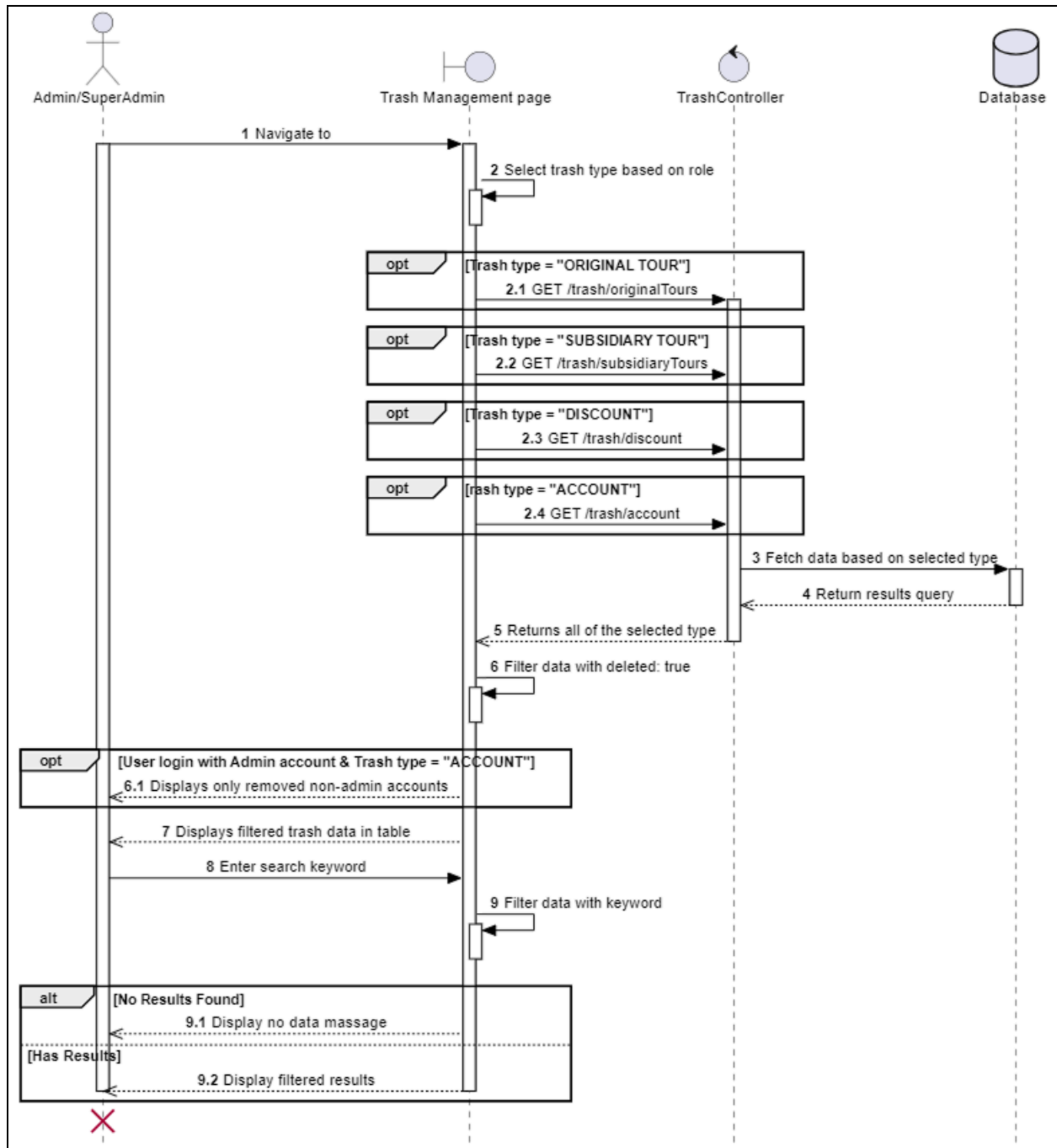


3.81 Search Trashes

a. Class Diagram

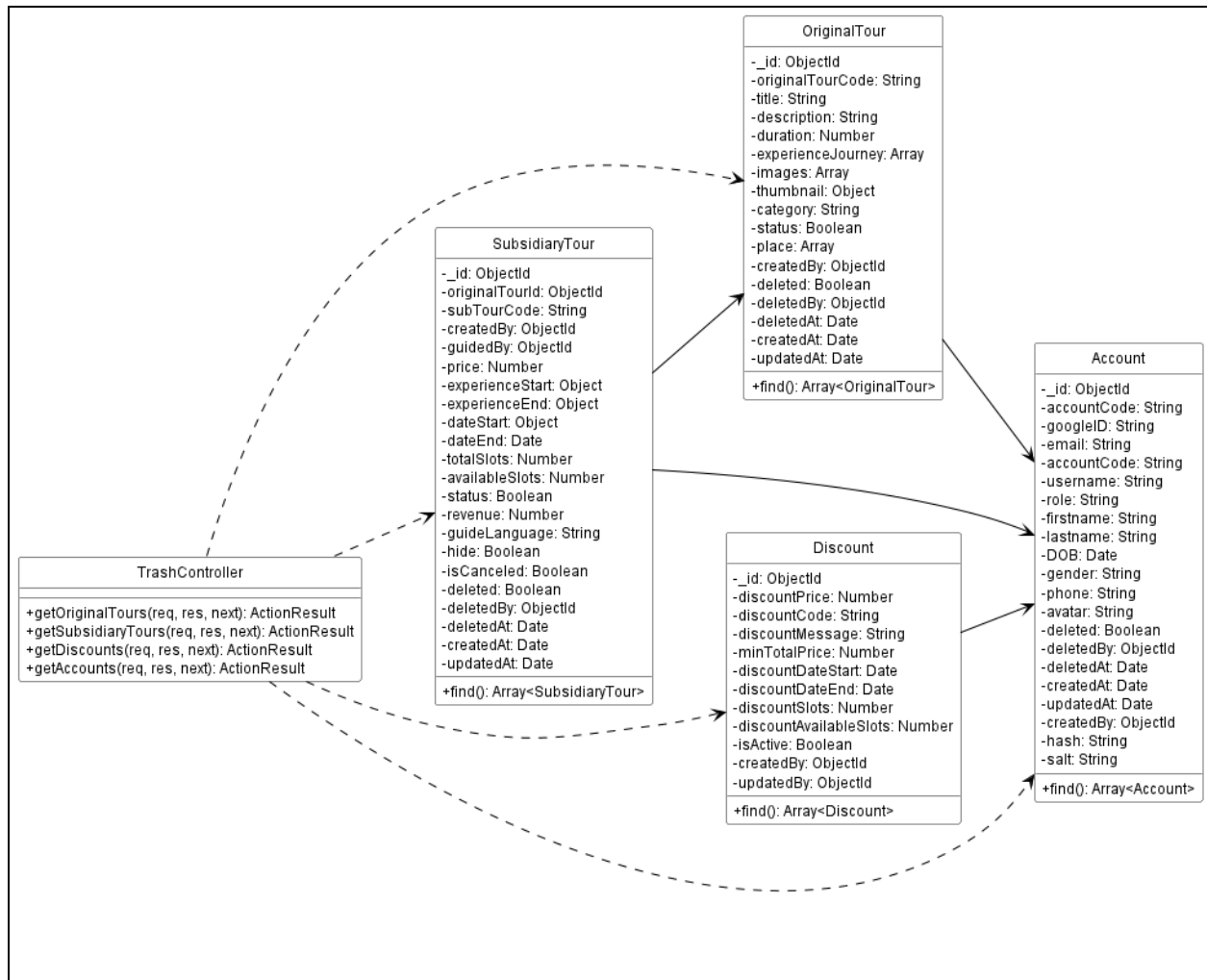


b. Sequence Diagram(s)



3.82 Sort Trashes

a. Class Diagram



b. Sequence Diagram(s)

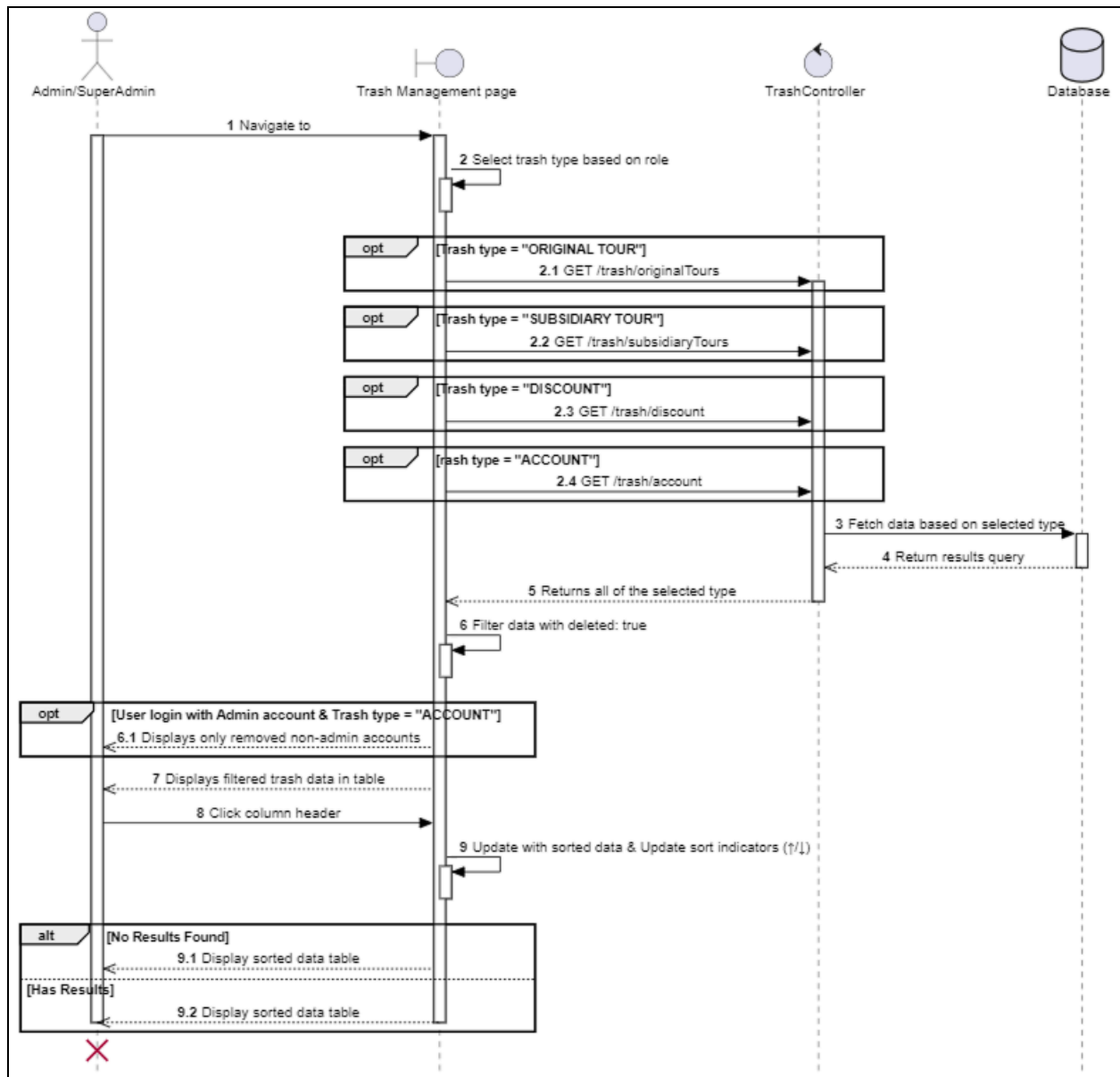
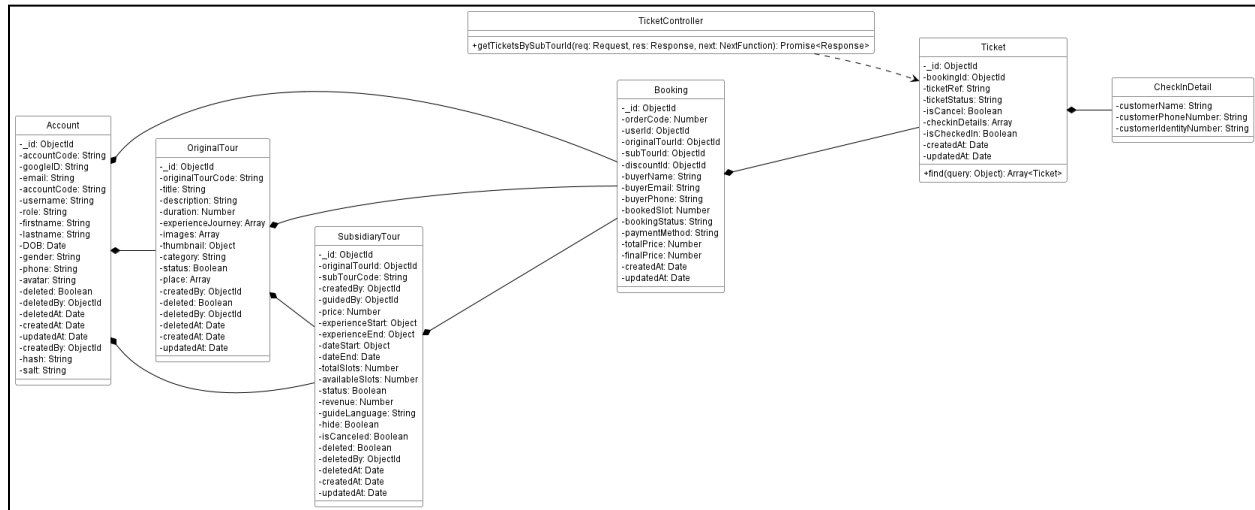


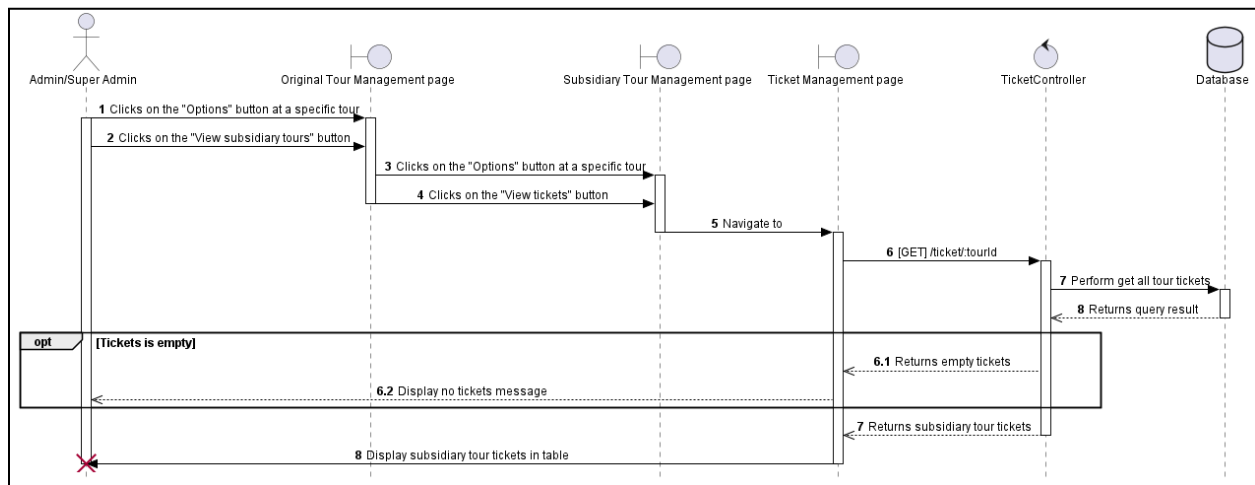
Figure 43. Create Booking (4)

3.83 View All Tour Ticket

a. Class Diagram

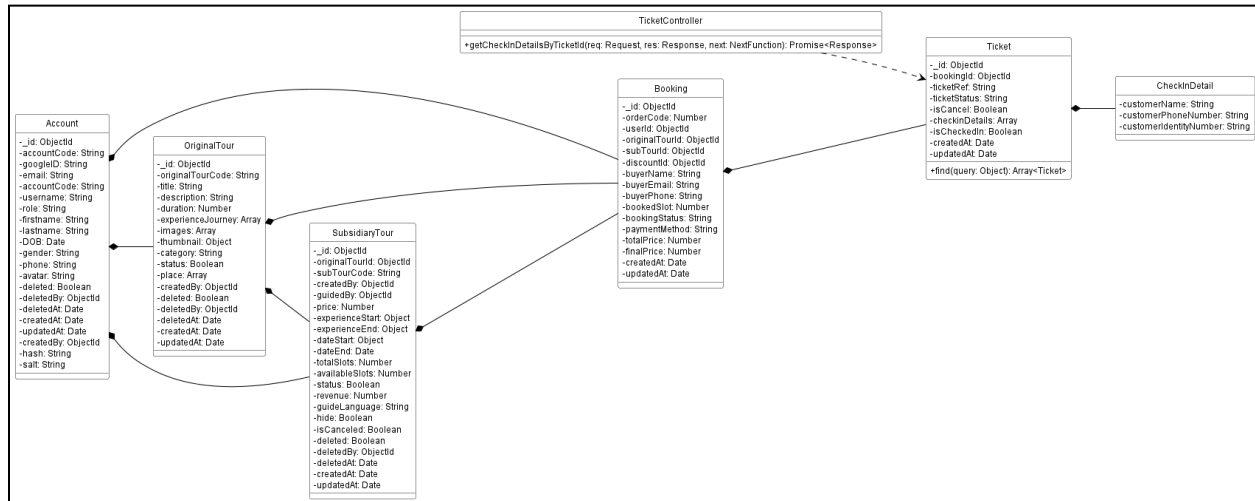


b. Sequence Diagram(s)

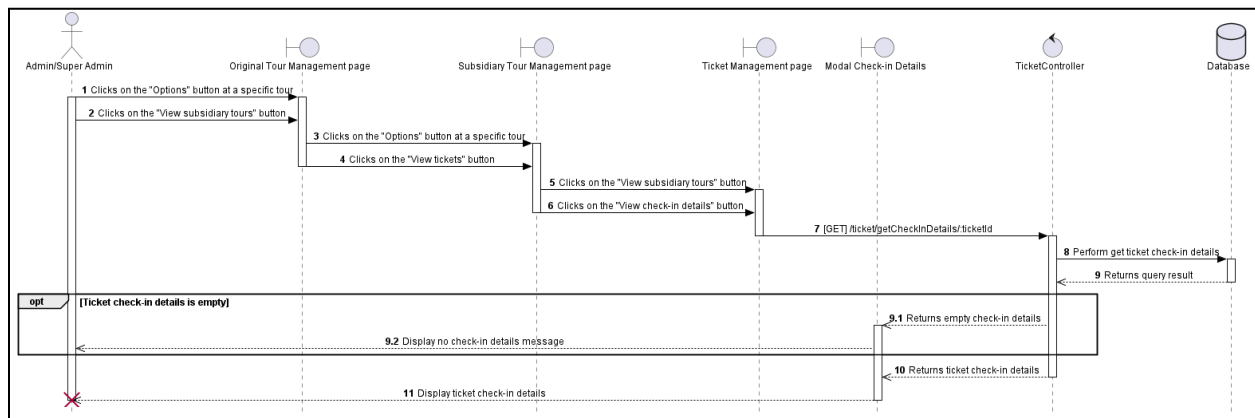


3.84 View Ticket Check-in Details

a. Class Diagram

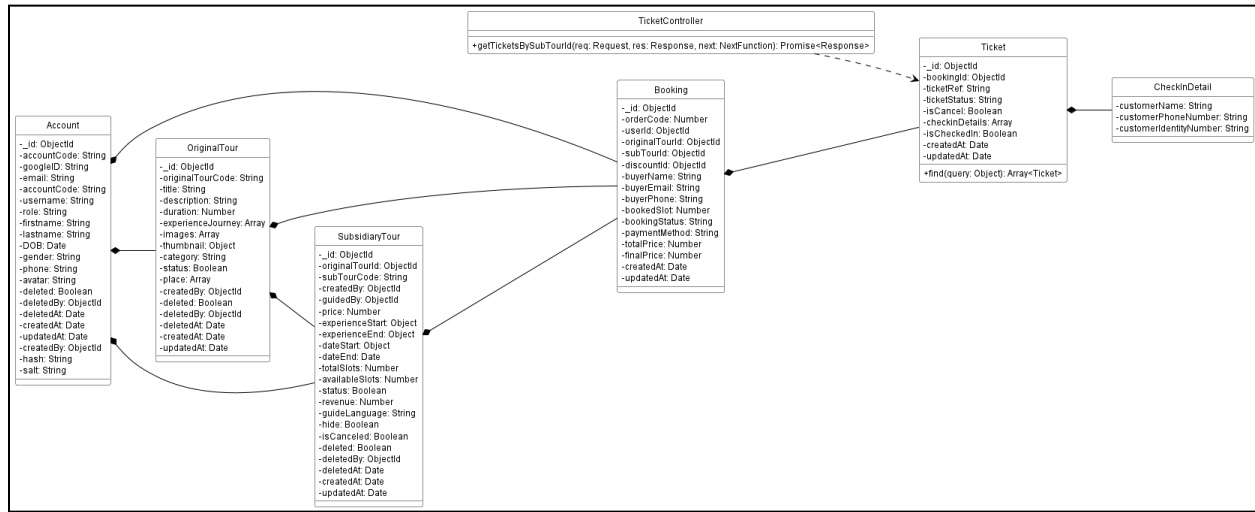


b. Sequence Diagram(s)

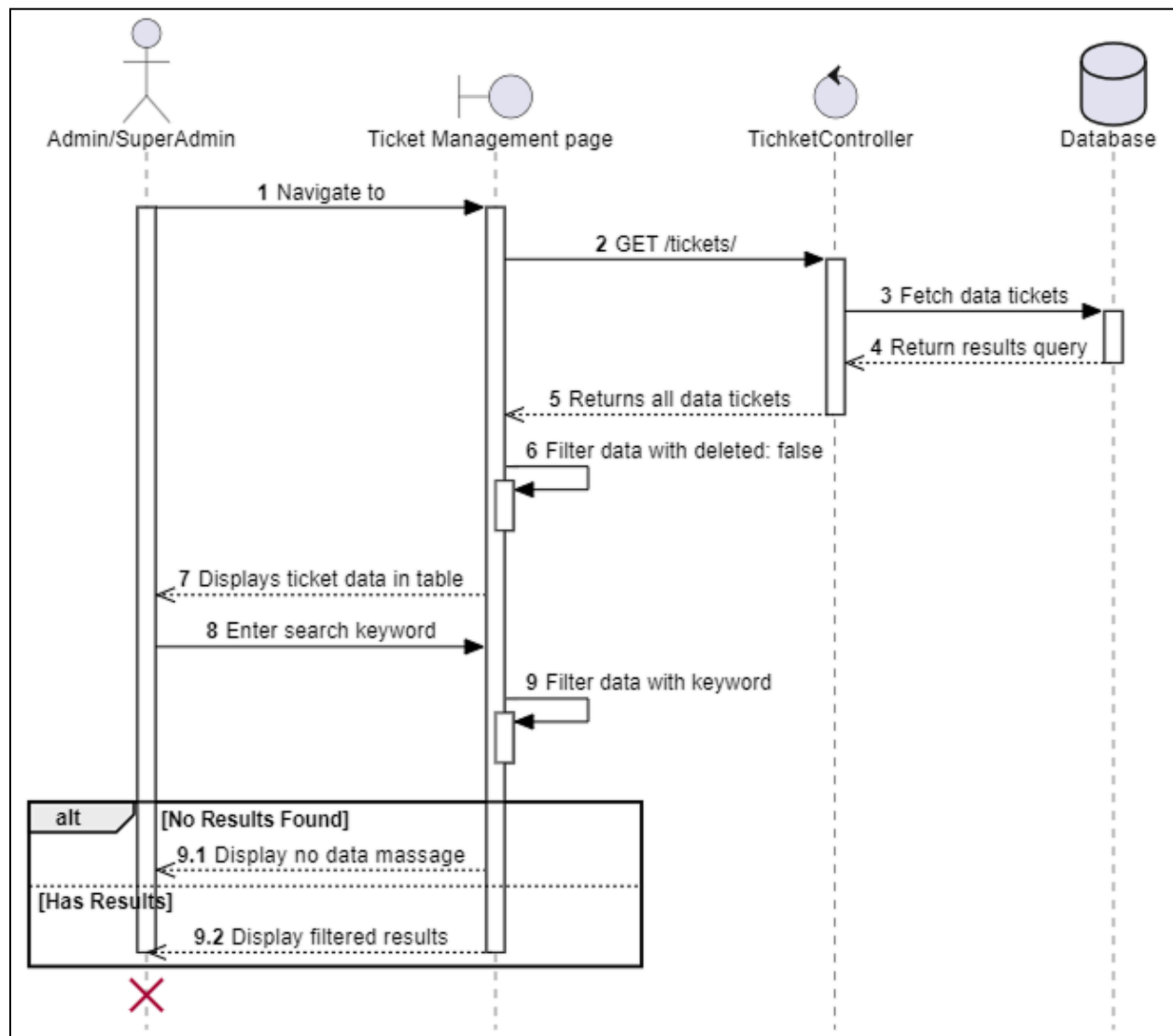


3.81 Search Tickets

a. Class Diagram

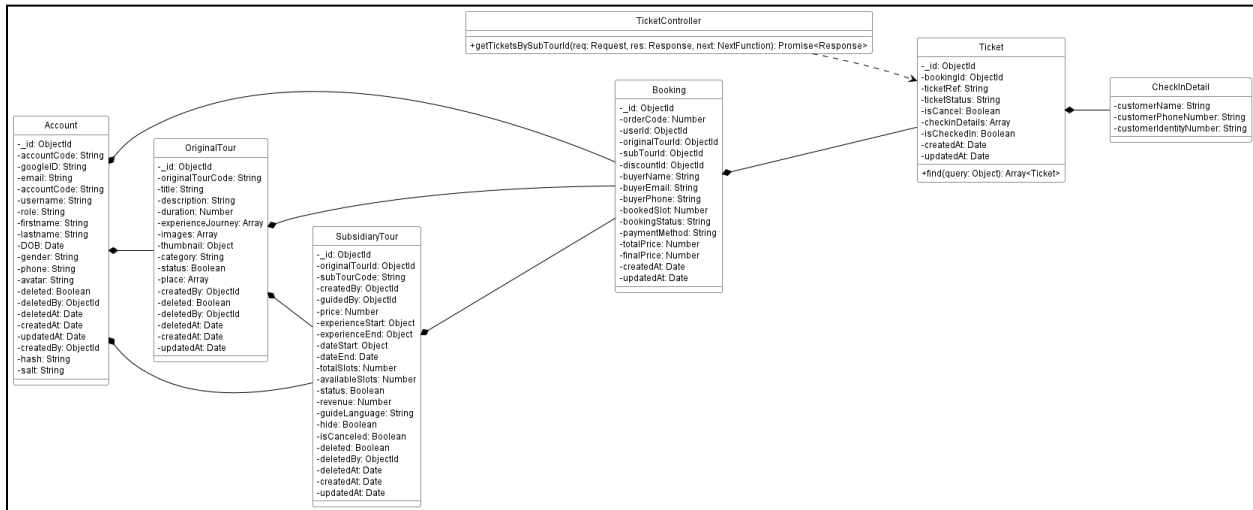


b. Sequence Diagram(s)



3.82 Sort Tickets

a. Class Diagram



b. Sequence Diagram(s)

